MIL-M-24100C
22 February 1991
SUPERSEDING
MIL-M-24100B
2 January 1974
(See 6.7)

MILITARY SPECIFICATION
MANUALS, TECHNICAL: FUNCTIONALLY ORIENTED MAINTENANCE MANUALS (FOMM) FOR ELECTRONIC, ELECTROMECHANICAL, AND ORDNANCE EQUIPMENT, SYSTEMS, AND PLATFORMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Coverage. This specification sets forth the content and format requirements for Functionally Oriented Maintenance Manuals (FOMM) (see 6.5.7) and revisions (see 6.5.45), and changes thereto, necessary for the installation, operation, organizational-level (see 6.5.35), intermediate-level (see 6.5.26), and depot-level maintenance repair (see 6.5.42), and parts (see 6.5.38) support of equipment (see 6.5.12), systems (see 6.5.53), and subsystems (see 6.5.54), without the services of manufacturer's representatives. Maintenance manuals are the fundamental operational and maintenance data base for the military operational and service forces, and are baseline documents for training of personnel.

1.1.1 Concept. The FOMM consists fundamentally of illustrations supported by narrative, vice narrative supported by illustrations as in conventional manuals. A FOMM is a complete operational and maintenance technical data base for the equipment or system it supports. FOMM manuals present data in a disciplined manner that shows clearly the functional and physical relationships that exist in an equipment or system.

1.1.1.1 FOMM illustrations consist of logically arranged functional block diagrams, blocked schematics, with keyed text, and maintenance dependency charts (MDC). These illustrations use symbols and distinctively shaped line art blocks to convey maximum information in a minimum of space.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC TMSS

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1.1.1.2 Shading techniques are used to denote the physical and functional relationships of systems and hardware. Demarcation between shadings provide psuedo-lines between functional levels and hardware assemblies such that they do not conflict or confuse the hard lines which represent the technical data presented on illustrations by reducing clutter. The MDCs provide a comprehensive method for determining equipment and system performance status and give positive direction for fault isolation.

1.1.2 Conceptual objectives. FOMM manuals are engineered to meet the following unique objectives:

a. To provide a state-of-art technical manual that presents all information necessary to install, operate, maintain, and repair military equipment and systems in the most comprehensive and effective manner and to maximize the utilization of space and to promote ease of use.
b. To present all technical data with reliance on all pictorial information supported by minimum narrative that is keyed to the illustration such that only required descriptive information needs to be read.
c. To present a troubleshooting concept that defines all parameters necessary for proper operation (positive troubleshooting) such that any fault may be determined between its last good indication and its first bad indication (as opposed to symptoms and potential problem determinations normally used in standard technical manuals).
d. To utilize a diagramatic concept (dependency chart) to assure that an equipment or system can be 100 percent checked out as functioning properly in all modes of operation through the exercise of all operating controls and switch positions. (This has normally not been possible in standard technical manuals through the use of signal flow block diagrams because of space limitations.)
e. To simplify diagrams by using a shading technique to denote hardware without hard lines that conflict with or confuse circuitry or signal flow.
f. To present all information in a manner that a technician will normally troubleshoot, diagnose, and isolate potential problems.
g. To eliminate foldouts, by employing a two facing page landscape concept that provides up to 11 by 33 inches of data, because of their difficulty of use and expense in printing, binding, and handling.
h. To present all data in information packages such that all needed materials to perform an action is included within the package. The information packages employs a referencing technique intended to ensure that the technician is never dead-ended.
i. To provide a reduced size reproduction of complex system troubleshooting information for use by technicians in limited space or poor environments.
j. To provide information in such a manner that it can be generated on electronic displays by computer access.

1.1.3 The FOMM is arranged to separate (normally by separate Support Volumes and Troubleshooting/Repair Volumes) the data required to troubleshoot and repair the equipment from that information required to install, operate and

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understand the functional operation of the equipment. This approach addresses the "Technicians Handbook" concept which is intended to provide the qualified technician during troubleshooting and repair in one volume only, that information he requires to troubleshoot the equipment and isolate a fault to a unit of hardware without being encumbered by baseline data needed for other purposes. Further, all information required for the maintenance and repair of the equipment is arranged into hardware data packages and indexed sequentially by unit, assembly, and subassembly number. Thus all data required to troubleshoot, maintain, and repair a given unit or assembly is in a single work package that can be used without reference to other parts of the manual. Hardware data packages can be bound separately, when advisable to facilitate use (see 3.1.3). For example, to limit thickness (see 3.6.2.1) or so that the FOMM can be placed at the location of the respective assembly when equipment units are installed in different rooms or compartments. In addition, this specification covers the requirements for an approximate 5 by 10-1/2 inch sized reprinting of the "Troubleshooting/Repair Volume" so that a technician can carry this smaller volume in his pocket.

1.1.4 Issues. This specification covers the content, format, quality assurance, and preparation requirements for the following issues of a technical (see 6.2):

a. Preliminary technical manual (PTM) (see 6.5.41)
b. Final manual (see 6.5.15)
c. Technician's pocket manual (see 6.5.55)
d. Supplement manual (see 6.5.51)
e. Permanent change (see 6.5.39)
f. Interim change (see 6.5.24)
g. Complete revision (see 6.5.45.1)
h. Update revision (see 6.5.45.3)

1.1.5 Data items. The following manual data items are described in this specification and shall be delivered as specified (see 6.2):

a. Book plan (see 6.5.4).
b. Review draft copy (RDC) (see 6.5.44).
c. Final reproducible copy (camera-ready) (see 6.5.16).
d. Photolithographic negatives (see 3.9.4).
e. Replenishment materials (see 6.5.43).
f. Certification checkoff list (see 3.10.6).

1.1.6 Classification. FOMM manuals shall be one of the following types as specified (see 6.1):

Type I - Equipment manuals (all types).
Type IS - Service test equipment manuals (see 6.5.46).
Type IX - Experimental equipment manuals (see 6.5.14).
Type II - System/subsystem manuals.
Type IIS - Service test system manuals.
Type IIX - Experimental system manuals.

1.2 Applicability to services. Certain paragraphs in this specification are not applicable to all services. When a paragraph applies to fewer than all the using services, the paragraph text is prefixed with the letter (A) Army, (CG)

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U.S. Coast Guard, (F) Air Force, (MC) Marine Corps, and (N) Navy, as applicable. The service designation for a numbered paragraph applies to all subparagraphs unless otherwise indicated. (F) Throughout the specification the term "manual" equates to "volume".

1.3 Applicability to types of manuals. The basic data requirements for each manual type are depicted in figure 1. Also, certain paragraphs in this specification are only applicable to certain manual types. These paragraphs are prefixed by the manual type (for example I, IS, IX) to which they apply. Paragraphs not so designated apply to all manual types. Unless otherwise indicated (see 6.2), the designation for a numbered paragraph shall apply to all of its subparagraphs.

1.4 Security classification. Marking, handling, and production of classified material shall be in accordance with DOD Instruction 5220.22-M (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (See 6.2).

SPECIFICATIONS

MILITARY

MIL-M-008910	Manuals, Technical, Illustrated Parts Breakdown, Preparation of.
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-F-17655	Field Changes and Field Change Kits; General Specification for.
MIL-D-23140	Drawings, Installation Control, for Shipboard Electronic Equipment.
MIL-P-24534	Planned Maintenance System: Development of Maintenance Requirement Cards, Maintenance Index Pages and Associated Documentation.
MIL-M-24747	Technical Repair Standards (TRS) Manual; Preparation of
MIL-T-31000	Technical Data Packages, General Specification for.

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MIL-M-38784	Manuals, Technical: General Style and Format Requirements.
MIL-P-38790	Printing Production of Technical Manuals: General Requirements for.
MIL-M-38807	Manuals, Technical: Illustrated Parts Breakdown, Preparation of.
MIL-M-85337	Manuals, Technical: Quality Assurance Program; Requirements for.

STANDARDS

MILITARY

MIL-STD-12	Abbreviations for Use on Drawings, and in Specifications, Standards and Technical Documents.
MIL-STD-17-1	Mechanical Symbols (Other than Aeronautical, Aerospacecraft and Spacecraft Use), Part 1.
MIL-STD-17-2	Mechanical Symbols for Aeronautical, Aerospacecraft and Spacecraft Use, Part 2.
MIL-STD-335	Manuals, Technical: Repair Parts and Special Tools List.
MIL-STD-1364	Standard General Purpose Electronic Test Equipment.
MIL-STD-2073-1	DoD Material Procedures for Development and Application of Packaging Requirements

FEDERAL INFORMATION PROCESSING STANDARDS

FIPS PUB 24	Flowchart Symbols and Their Usage in Information Processing

HANDBOOKS

MILITARY

MIL-HDBK-131	Identification Markings for Fasteners.
MIL-HDBK-300	Technical Information File of Ground Support Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-6094.

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(Copies of the Federal Information Processing Standards (FIPS) are available to Department of Defense activities from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-6094. Others must request copies of FIPS from the National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161-2171.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

DEPARTMENT OF THE NAVY

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-5001049	Piping System Symbols and Abbreviations

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Code 202.2, Portsmouth, NH 03801.)

PUBLICATIONS

DEPARTMENT OF DEFENSE (DOD)

Cataloging Handbook H4/H8	Commercial and Government Entity (CAGE)
DOD 5220.22-M	Industrial Security Manual for Safeguarding Classified Information (USDP).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-9372.)

DEPARTMENT OF THE NAVY

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

16-1-523	Avionics Preferred Standard Test Equipment.

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0967-LP-000-0120	Electronics Installation Maintenance Book, Circuits.
0969-LP-019-7000	Electrical Test Equipment Application Guide.
S9AA0-AA-SPN-010/	General Specifications for Ships GEN-SPEC of United States Navy.
S9086-00-STM-000/ NSTM	Naval Ship Technical Manual (NSTM).

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S9086-CZ-STM-000/ CH90	Inspections, Tests, Records, and Reports
ST000-AA-IDX-10/	Portable Electrical/Electronic Test Equipment PEETE (PEETE) Index for Support Requirements of Shipboard Electronic, Electrical, IC, Weapons and Reactor Systems.

(Application for copies should be addressed to the Standardization Documents Ordering Desk, Building 4D 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

DEPARTMENT OF THE AIR FORCE

TO 00-5-1	Air Force Technical Order System.
TO 33K-1-01	Calibration Standards and Calibration Equipment.

(Application for copies should be addressed to Headquarters, Air Force Logistic Command DPO, Wright-Patterson Air Force Base, OH 45433.)

MARINE CORP

TM-6625-15/1	Electronics Test Equipment Listing.

(Application for copies should be addressed to Commander, Marine Corps Logistics Bases, Code 876, Albany, GA 31704-5000.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents in the solicitation (See 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Y14.15	Electrical and Electronics Diagrams. (DoD adopted)
Y32.4	Graphic Symbols for Plumbing Fixtures for Diagrams Used in Architecture and Building Construction. (DoD adopted)
Y32.10	Graphic Symbols for Fluid Power Diagrams.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Y32.2.4	Graphical Symbols for Heating, Ventilating, and Air Conditioning.

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Y32.2.6 Graphical Symbols for Heat-Power Apparatus.

Y32.2.3 Graphical Symbols for Pipe Fittings, Valve, and Piping.

(Application for copies should be addressed to the American Society of Mechanical Engineers, Inc., 345 East 47th Street, New York, NY 10017.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

200-75 Electrical and Electronics Parts and Equipments Reference Designations for. (DoD adopted)

315-86 Graphic Symbols for Electrical and Electronics Diagrams. (DoD adopted)

991-86 Logic Circuit Diagrams. (DoD adopted)

(Application for copies shall be addressed to the Institute of Electrical and Electronics Engineers, Inc., 345 East 47th Street, New York, NY 10017.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The following are the requirements for development of FOMMs for electronic, electromechanical, and ordnance equipment, systems, and platforms.

3.1.1 Technical accuracy. Information in the manual shall accurately represent the system, subsystem, equipment or function being described. Technical inaccuracies or inconsistencies shall be considered justification for rejecting a technical manual.

3.1.2 Style and level of writing, and depth of coverage. The style and level of writing and depth of coverage shall be in accordance with MIL-M-38784 and as specified in 3.1.2.1 through 3.1.2.4.

3.1.2.1 Style of writing. The style of writing shall be factual, specific, concise, comprehensive, and unambiguous. Descriptive writing shall be third person, present tense, active voice, and indicative mood. Instructions and procedures shall be second person, present tense, active voice, and imperative mood.

3.1.2.2. Types I, IS, II, and IIS levels of writing. The level of writing and development of text for operational and service test use shall be in

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accordance with MIL-M-38784 and the following:

a. As a general guide, the level of writing should be a reading level of 9 for a high school graduate having specialized training as a technician in military training courses.
b. Operating instructions shall be procedural and written to the level of an operator having previous experience in the operation of similar or related equipment.
c. The level of writing for other portions of the manual shall be to that of a technician having previous maintenance experience with similar or related equipment.

3.1.2.3 Types IX and IIX levels of writing. Unless otherwise specified (see 6.2), IX and IIX manuals will be used by highly skilled and experienced military personnel for operational evaluation. Type IX manuals shall be written to the level of an engineer familiar with the equipment involved. Type IIX manuals used for operational evaluation shall be written to the same level as type IIS manuals. Inclusion of mathematical equations, derivations, or notations shall be limited to those required to understand the operation and maintenance of the equipment.

3.1.2.4 Depth of coverage. Unless otherwise specified (see 6.2), manuals shall provide all information necessary to maintain the system, subsystem, or equipment through organizational-, intermediate-, and depot-level maintenance (see 3.6.2.2).

3.1.3 Arrangement (A, CG, MC, N). A FOMM shall be arranged into two volumes, a "Support Volume" and a "Troubleshooting/Repair Volume," except as noted in 3.6.2. The Support Volume shall be designated Volume 1 and shall include all user support information. Additional accident prevention information is provided in NAVSEA publication S9AA0-AA-SPN-010/GEN-SPEC, Section 602h. The Troubleshooting/Repair Volume shall be designated as Volume 2 and shall include all data necessary to troubleshoot, align, and repair the equipment or system. Separately-bound subdivisions of the manual shall be designated as parts. Covers, title pages, and other front matter shall be included in each part as specified in 3.2.1.1 and 3.2.1.2 of this specification. Each separately bound volume and part shall have a unique manual number supplied by the Government.

3.1.3.1 Arrangement (F). There shall be a separate support manual and a troubleshooting/repair manual, each with a separate publication number. These manuals may be further divided, if necessary. In such cases, a dashed number added to the basic technical order number will be assigned to each division by the Government.

3.1.3.2 Support volume. The Support Volume (see 6.5.52) subject matter shall be arranged in the order shown, and conform to the paragraphs below:

a.	Front matter	(see 3.2.1)
b.	General information	(types I, IS, and IX - see 3.2.2.1.1) (types II, IIS, and IIX - see 3.2.3.1.1)
c.	Safety precautions	(types I, IS, and IX - see 3.2.1.13) (types II, IIS, and IIX - see 3.2.3.1.2)

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d.	Operating instructions	(types I, IS, and IX - see 3.2.2.1.2) (types II, IIS, and IIX - see 3.2.3.1.3)
e.	Theory of operation	(type I and IS - see 3.2.2.1.3) (type II and IIS - see 3.2.3.1.4)
f.	Maintenance procedures and Standards	(type I and IS - see 3.2.2.1.4) (type II and IIS - see 3.2.3.1.5)
g.	Plan for maintenance and overhaul procedures	(type I - see 3.2.2.1.4.7)
h.	Alignment	(type II - see 3.2.3.1.6)
i.	Installation	(types I, IS, and IX - see 3.2.2.1.5) unless otherwise specified (see 6.2) (types II, IIS, and IIX - see 3.2.3.1.7) unless otherwise specified (see 6.2)
j.	Parts list (separate chapter) when specified	(types I, IS, and IIX - see 3.2.2.1.6) (see 6.2) (types II, IIS, and IIX - see 3.2.3.1.8)
k.	User activity comment sheet (N)	(all types - see 3.2.2.1.7)

3.1.3.3 Troubleshooting/repair volume. The troubleshooting/repair volume (see 6.5.56) subject matter shall be arranged in the order shown below and shall conform to the referenced paragraphs:

a.	Front matter	(see 3.2.1)
b.	Family trees	(types I, IS, and IX - see 3.2.2.2.1.2, figures 2 and 3)
c.	Turn on/check out chart	(types I, IS, II, and IIS - see 3.2.2.2.1.4)
d.	Interconnecting cabling/piping diagram	(types I, IS, and IX - see 3.2.2.2.1.3 and figure 4) (types II, IIS, and IIX - see 3.2.3.2.1.5)
e.	Overall function (system) information	(types I, IS, and IX - see 3.2.2.2.1.5) (types II, IIS, and IIX - see 3.2.3.2.11)
f.	Major function (subsystem) information	(type I - see 3.2.2.2.2) (type II - see 3.2.3.2.2)
g.	Hardware information	(type I - see 3.2.2.2.3) (type II - see 3.2.3.2.3)
h.	Built-In Test Equipment (BITE) Automatic Test Equipment (ATE) fault isolation information	(type I - see 3.2.2.2.4) (type II - see 3.2.3.2.4)
i.	User activity comment sheet (N)	(all types - see 3.2.2.1.7)

3.1.3.3.1 Relationship of hierarchical data (types I, IS, IX, II, IIS, and IIX) (See 6.5.23). FOMM manuals shall be comprised of three functional levels of hierarchical data: (1) hardware, overall, and major levels. Whenever any functional level cannot be illustrated on a single page showing inputs and outputs, one or more functional illustration packages shall be developed so that the functional path may be easily traced in a left-to-right top-to-bottom format thereby simulating much larger diagrams. Subfunction diagrams shall be styled and formatted in the same manner as the top level diagram to which it relates. The equipment/system troubleshooting/repair volume shall be

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arranged and organized in accordance with figure 5.

a. Column 1 applies when it is possible to present the function/hardware information (see 6.5.22) package in three hierarchical levels without subfunctional divisions.
b. Column 2 applies when it is necessary to expand the overall function diagram (see 6.5.36) because space limitations prevent full identification of all major functions.
c. Column 3 applies when it is necessary to subdivide a major function diagram to fully depict the functional details.
d. Any level may be subfunctional as necessary, including hardware levels as basically illustrated in columns 2 and 3.

3.1.4 Types I, IS, IX, II, IIS, and IIX data requirement charts. The data and diagrams required for the specified type of volume shall be in accordance with figure 1.

3.1.5 Warnings, cautions, notes and icons.

3.1.5.1 Warnings and cautions. WARNING and CAUTION statements shall be included for emphasis in all applicable installation, operating, maintenance, and repair procedures and data in accordance with MIL-M-38784 and the following:

a. A WARNING statement shall be used to call particular attention to a step of a procedure which, if not strictly followed, could result in personal injury or death.
b. A CAUTION statement shall be used to call particular attention to a step of a procedure which, if not strictly followed, could result in equipment being damaged or destroyed.
c. The use of WARNING or CAUTION statements shall be limited to significant cases only, as overuse may result in operators ignoring them.
d. When a WARNING or CAUTION statement is two paragraphs or longer, the format (for example, indentation or italics) should make it clearly identifiable.

3.1.5.2 Notes. Notes shall be as follows:

a. NOTES shall not be used with, or in lieu of, procedural steps.
b. NOTES shall be used solely in narrative or illustrative nonprocedural data. NOTES shall not be used in lieu of WARNING or CAUTION statements, nor shall they be used to modify an action.
c. NOTES shall be in accordance with MIL-M-38784.

3.1.5.3 Icons. Icons shall be in accordance with MIL-M-38784.

3.1.6 Accessibility and referencing of data. Cross-referencing techniques in text and illustrations shall provide rapid access to all maintenance data (equipment and equipment circuits) so that the technician can efficiently complete the maintenance action without becoming dead-ended.

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3.1.7 Maintenance. Unless otherwise specified (see 6.2), type I equipment manuals, type II system/subsystem manuals, and type III platform system manuals shall include all information and data necessary to perform all organizational-, intermediate-, and depot-level maintenance as defined by the maintenance philosophy established for the equipment or system. No manual shall omit maintenance and repair or applicable logistics information for any equipment or component designated military-service or agency repairable, regardless of the location of the designated repair point (field, station, ship, tender, or shore facility) unless the contract requires that the omitted information be included in a separate document that is either provided or made available to the Government, and reference is made thereto.

3.1.8 General format. The FOMM shall consist of illustrations supported by narrative, and shall be a complete technical data base for the equipment/system or platform it supports. FOMM's shall clearly indicate the functional and physical relationships that exist in an equipment or system. FOMM illustrations (see 3.7.4) shall consist of logically arranged functional block diagrams and blocked schematics (see 6.5.2) with keyed text (see 6.5.28) and Maintenance Dependency Charts (MDC's) (see 6.5.36), in addition to any other augmenting type of illustrations and data required to install, operate, and maintain the equipment/system. These illustrations shall use symbols and distinctively shaped line art blocks to convey maximum information in a minimum amount of space. Shading techniques shall be used to denote the physical and functional relationships in most functional diagrams. The MDC's and other logic diagrams shall provide a comprehensive method for determining equipment/system performance status and give direction for fault isolation.

3.1.9 FOMM fault-isolation presentation techniques.

3.1.9.1 Troubleshooting and repair. The FOMM shall present troubleshooting, diagnostics, and repair data on an integrated functional basis (how it works) rather than a topical basis (by subject divisions, such as individual chapters for theory, equipment arrangement, testing and troubleshooting, alignment, maintenance, and repair). The purpose for developing an integrated functional approach is to:

a. Separate all the data required to isolate and repair a problem from the data that is needed to install, operate, perform periodic maintenance, and provide detailed theory and logistics information. (This approach permits the qualified technician to repair and maintain equipment without referencing through other chapters to find needed information.)
b. Present data so that the technician can trace a problem from how it affects the functioning of the equipment, to the assembly or piece part that is causing malfunction.
c. Show all of data necessary to assure proper equipment/system operation on an increasing time line (X and Y axes) such that an equipment fault can be isolated between the last good indication and the first bad indication.
d. Trace how the equipment or system functions without being distracted by how the hardware is physically packaged [unit (see 6.5.57), assembly (see 6.5.1) and subassembly (see 6.5.49)].

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3.1.9.2 Information packages. Troubleshooting and repair information shall be divided into functional and hardware information packages, each of which contains all information necessary to accomplish a given task. The functional information packages shall be arranged in sequence from overall function information to major function information. The functional information packages shall provide the technician with all of the information required to systematically isolate a fault to a hardware unit. Once the fault has been isolated, the hardware information package for that unit shall be accessed to obtain maintenance and repair data. The hardware information packages shall be arranged hierarchically by system, subsystem, equipment, unit, assembly, and subassembly (as applicable) by reference designation numbers. Thus, for a given corrective action, all data required to troubleshoot, maintain, or repair a given unit or assembly shall be in a single work package that can be used without referencing other parts of the manual. In addition, this specification covers the requirements for a technician's pocket manual which shall be a reduced size reprint of the troubleshooting/repair manual.

3.1.9.2.1 Illustrative functional data. The illustrative functional data (see figure 6) shall be the primary informational components for trouble-shooting and repair of equipment. Functional diagrams shall be formatted from left-to-right, top-to-bottom, with inputs and outputs clearly identified on each page or combination of facing pages. Any diagram can be subfunc-tionalized by expanding an internal functional entity into an intermediate subfunction information package using the same input/output/subfunction concept as prescribed by its top level diagram. It is intended that a technician isolating a fault to a subfunction or item of hardware will rarely return to a higher level diagram to that which he/she ultimately is working.

3.1.9.2.2 Keyed text. The keyed text used to describe the functional blocked areas of the diagram shall describe the overall purpose of the major function, subfunction, or assembly as applicable. Such text shall precede the keyed text descriptions. Reference shall be made to chapter 3 of the support volume when background information is required.

3.2 Detailed requirements.

3.2.1 Front matter. The front matter shall consist of the following, as applicable:

a. Cover, title page, backbone, and back cover (A, CG, F, MC) (see 3.2.1.1).
b. Cover/title page (N) (see 3.2.1.2).
c. Warning page (A, MC) (see 3.2.1.3).
d. List of effective pages (see 3.2.1.4).
e. Promulgation page (CG, MC, N) (see 3.2.1.5).
f. Change record (MC, N) (see 3.2.1.6).
g. Table of contents (see 3.2.1.7).
h. List of illustrations (see 3.2.1.8).
i. List of tables (see 3.2.1.9).
j. Functional index (see 3.2.1.10).
k. Foreword (see 3.2.1.11).
l. Description of symbols, codes, abbreviations, and shading (see 3.2.1.12).

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m. Safety summary (CG, F, MC, N) (see 3.2.1.13).
n. Explanation of MDC usage (see 3.2.1.14).

3.2.1.1 Cover/title page. (A, CG, F, MC). Covers/title pages, shall be prepared in accordance with MIL-M-38784 and 3.7.12. The publication number, classification, and distribution statement (see 6.5.11) shall be provided by the Government. The type of publication shall be identified as a "FOMM Technical Manual for" and shall include the title "Support Volume" or "Troubleshooting/Repair Volume," as applicable (see figure 6).

3.2.1.2 Backbone and backcover. Backbones and back covers shall be prepared in accordance with MIL-M-38784.

3.2.1.3 Warning page (A, MC). The warning page shall be prepared in accordance with MIL-M-38784.

3.2.1.4 List of effective pages. The list of effective pages shall be prepared in accordance with MIL-M-38784.

3.2.1.4.1 Identification of writing contractor. The list of effective pages shall also include the name and address of the technical manual contractor/ subcontractor (if different from the manufacturer), and publication contract number.

3.2.1.4.2 Table of source data. The list of effective pages shall include a table of source data which summarizes all source data referenced throughout the manual (see 3.2.2.2.3.3.1.10.1) with the appropriate revision number or letter (see figure 7).

3.2.1.5 Promulgation page [CG, N (not all commands)]. A promulgation page shall be the first right-hand page following the list of effective pages in volume 1 of the manual. The contractor shall request the promulgation page from the Command or agency when submitting the manuscript for acceptance.

3.2.1.6 Change record (MC, N). The change record page shall be prepared in accordance with MIL-M-38784 and shall be inserted in all volumes. The change record page shall be on a separate sheet and shall not be printed on the reverse side.

3.2.1.7 Table of contents. A table of contents shall be prepared in accordance with MIL-M-38784 for the data included in the support volume.

3.2.1.8 List of illustrations. A list of illustrations shall be prepared in accordance with MIL-M-38784 for the illustration included in the support volume.

3.2.1.9 List of tables. A list of tables shall be prepared in accordance with MIL-M-38784 for the tables included in the support volume.

3.2.1.10 Functional index. A functional index in accordance with figure 8 shall be prepared for troubleshooting/repair volumes. This index shall be prepared as a chart which lists both the functional and hardware breakdown of the equipment versus types of data. Page numbers for each specific type of data shall be entered in the body of the chart. In addition to the table of

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contents, the support volume shall contain a functional index covering all parts of the troubleshooting/repair volume. The troubleshooting/repair volume shall contain a functional index for the entire FOMM. If the troubleshooting/repair volume is subdivided into parts, each part shall contain a functional index for all parts.

3.2.1.10.1 Relationship to family trees. The functional index shall be directly related to the family trees (see 3.2.2.2.1.2) by corresponding functional identification numbers and hardware numbers.

3.2.1.10.2 Clarity. The functional index shall clearly differentiate between the information package groupings (overall function data from major function data, major function data from hardware data, and unit 1 hardware data from unit 2 hardware data) by such means as spaces, bold or shaded lines, indentures. In addition, it shall include the location of all data required to interpret the technical data (for example, explanation of special symbols, circuit codes, integrated circuits).

3.2.1.11 Foreword. Information shall be provided in a foreword to the FOMM that explains the contents and use of the manual. The foreword shall include:

a. An explanation of the scope and intent of the manual.
b. The types and arrangement of data included.
c. User instructions that explain how the FOMM is used to troubleshoot and repair the equipment or system (see figure 9).
d. Any additional information specified in MIL-M-38784 that will assist the user in efficient use of the manual.

3.2.1.12 Description of symbols, codes, abbreviations, and shading. The following charts shall be included with the foreword in the troubleshooting/repair volume/manual (for equipment and systems, see figure 10:

a. A chart that lists and defines each of the functional entity identifier codes (see 6.5.21) used in the manual.
b. A chart that illustrates and describes the symbols unique to the manual.
c. An illustrated chart that defines the multishading technique used in the manual.
d. A chart that defines all nonstandard abbreviations used in the manual.

3.2.1.13 Safety summary (CG, F, MC, N). A safety summary in accordance with MIL-M-38784 shall be prepared for each volume/manual/part and shall be included in the front matter.

3.2.1.14 Explanation of maintenance dependency chart (MDC) usage. The front matter of each FOMM volumes that containing MDC's shall include instructions on their use. A chart shall accompany the instructions that describes all symbols used in the MDC's (see figure 11).

3.2.1.14.1 Explanation of logic diagram usage. When logic diagrams are used in addition to or in lieu of MDC's, instructions shall be provided for use of these diagrams; together with descriptions and illustrations of any special symbols used.

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3.2.2 Equipment manuals (type I). Equipment manuals shall contain the data specified on figure 1.

3.2.2.1 Support volume (type I). The support volume shall be arranged in the following order, and shall conform to the referenced paragraphs:

a. Front matter (see 3.2.1).
b. Chapter 1 - General information (see 3.2.2.1.1).
c. Chapter 2 - Operation (see 3.2.2.1.2).
d. Chapter 3 - Theory of operation (see 3.2.2.1.3).
e. Chapter 4 - Maintenance procedures and standards (see 3.2.2.1.4).
f. Chapter 5 - Installation (see 3.2.2.1.5).
g. Chapter 6 - Parts list (see 3.2.2.1.6).
h. User activity comment sheet (N) (see 3.2.2.1.7).

3.2.2.1.1 Chapter 1 - General information. The contents of this chapter shall be such that command-level, supervisory personnel, and other users having an interest in the equipment can easily and rapidly determine the purpose, physical and functional characteristics, and the operational capabilities of the equipment.

3.2.2.1.1.1 Introduction. The introduction shall explain the purpose, scope, supersession data, and applicability of the technical manual; including all models, serial numbers, and configurations covered. Warranty/guarantee information shall be included, as applicable. The interface relationship of the technical manual to other referenced publications, and the relationship of the equipment to referenced systems or other equipments, shall also be included.

3.2.2.1.1.2 Equipment description. The equipment description shall explain the intended use (why, where, when, how, and with what), capabilities, and limitations of the equipment. The text covering physical description or structural arrangement shall be brief. The text shall avoid unnecessary or repetitious details that are better illustrated. Equipment units shall be clearly illustrated and identified. If the technical manual covers more than one equipment model or configuration, a table defining the differences between equipments and models covered shall be included. Appropriate assembly and detail drawings shall be included. Title blocks and relevant notes may be included on the drawings. Pictorial illustrations for different models need not be included if the differences are minor and can be indicated on the basic illustration.

3.2.2.1.1.3 Equipment illustrations. For installations consisting of more than one unit/assembly, a pictorial diagram representing the equipment, or all units comprising the equipment, shall be included and shall be designated figure 1-1 (see figure 14). The pictorial diagram shall show the major units of the equipment, relative size of each unit, basic interconnections between units, and their relationship with other equipments. The diagram shall be a full left-hand page or two facing pages, and shall be assigned the page number 1-0.

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3.2.2.1.1.4 Reference data. Reference data, equivalent to but not restricted to the following, shall be included in tabular form (see figure 13):

a. Description (nameplate data) which identifies manufacturer, type, model, and component identification number (CID) (see 6.5.7), if applicable.
b. Functional characteristics; such as power requirements (overall and units, including power factor), horsepower, pressure, capacity, modes of operation, power output, frequency, pulse characteristics, and sensitivity, including tolerances where applicable.
c. Capabilities and limitations; such as pounds of thrust, knots, turning radius, minimum and maximum ranges, degree of coverage, resolution, and accuracy.
d. Rated outputs; such as wattages, voltages, horsepower, and gallons per minute.
e. Environmental characteristics; such as ambient temperatures, heat dissipation per unit, humidity limits, and coolants (air flow in cubic feet per minute, water/oil in gallons per minute per unit of chilled water).

3.2.2.1.1.5 Equipment, accessories, and documents supplied. A tabular listing of all equipment and documents supplied shall be included. The equipment, units, and accessories (such as hydraulic oil lines, waveguides, special cables, special tools, test equipment, miscellaneous parts, and Government furnished items) which form a part of, or are supplied with, the subject equipment shall be listed. The table shall include the following:

a. Column 1 - Quantity. This column shall contain the quantity of each unit and accessory supplied with the equipment.
b. Column 2 - Item name or nomenclature. This column shall contain the official name (for example, pump, winch) or nomenclature (name and designation) of each component, unit, or accessory.
c. Column 3 - CID, RIC, or unit number. This column shall contain the Component Identification Designator (CID), Repairable Identification Code (RIC), or unit number of each component (see 6.5.6), unit, or accessory.
d. Column 4 - Overall dimensions. This column shall detail the crated (if available) and uncrated height, width, and depth (inches or centimeters) of each component, unit, or accessory.
e. Column 5 - Weight and volume. This column shall detail the crated and uncrated weight and volume (in cubic feet or meters) of each component, unit, and accessory.

3.2.2.1.1.6 Equipment and publications required but not supplied. A tabular listing of all test equipment, equipment, tools, materials, and publications required but not supplied shall be included. Selection of test equipment shall be made from the following publications:

(CG, N) Select from NAVAIR 16-1-523, NAVSEA 0969-LP-019-7000, ST000-AA-IDX-10/PEETE, and MIL-STD-1364, as applicable.
(F) Select from MIL-HDBK-300 and Technical Order 33K-1-01.
(MC) Select from USMC TM-6625-15/1.

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The list shall contain the following:

a. Column 1 - Category. This column shall contain the generic name of each piece of equipment required.
b. Column 2 - Recommended equipment. This column shall contain the Government nomenclature or identification of the recommended equipment. (N) When applicable, this column shall also contain the subcategory code (a four-digit group of numbers) included in NAVSEA ST000-AA-IDX-010/PEETE and used to identify a measurement requirement. Unless otherwise specified (see 6.2), NAVSEA ST000-AA-IDX-010/ PEETE shall be used as a guide for the selection and application of test equipment required for shipboard use. If a measurement requirement is established which cannot be satisfied by the subcategory codes listed in this guide, MIL-STD-1364, or alternative test procedures cannot be developed utilizing the listed subcategory codes, a request for assistance should be directed to the contracting officer.
c. Column 3 - Alternate. This column shall contain the model number of the standard or substitute standard general purpose electronic test equipment, listed in MIL-STD-1364, which can be used to satisfy the measurement parameters.
d. Column 4 - Equipment test parameters. This column shall specify the range of test parameters which must be satisfied by the test equipment (it shall not define the range(s) of the test equipment model).
e. Column 5 - Application. This column shall indicate the intended use of the equipment (for example, scheduled maintenance, troubleshooting, corrective maintenance, or installation).

3.2.2.1.1.7 Field changes, modification work order alterations, and factory changes. A table of authorized changes shall be included as follows:

a. Change 1 - Change number. This column shall list the change number for each field and factory change considered and included in preparation of the volume/manual.
b. Change 2 - Nomenclature. This column shall list the equipment nomenclature or component identifier and serial numbers of equipments affected by the change.
c. Column 3 - Description. This column shall contain a brief statement identifying the change and its purpose.

3.2.2.1.2 Chapter 2 - Operation. Operating instructions shall include all procedures needed to use the equipment efficiently and effectively to accomplish its designated task. These operating instructions shall be in sufficient detail to allow operators, not trained on this equipment, to independently and safely operate the equipment without additional training or explanation. The introduction to this chapter should explain the intended function and application of the equipment so that the operators will know exactly what they should expect to accomplish with the equipment. The introduction shall also describe each operator's relationship to the equipment and other operators, and shall identify those units having controls and indicators which the operator is expected to use in the performance of assigned duties. The chapter shall be supported by illustrations which identify and locate all operator controls, protective devices, jacks, and

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indicators. Procedural instructions shall be included to provide routine operating procedures, emergency procedures, manual procedures, automatic procedures, safety precautions, operating limits, and equipment limitations. Additionally, the coverage shall address the relationships of controls to each other (as applicable) and their affect upon the equipment operation. Operating information shall be presented by a combination of five basic formats:

a. Operator's data table (see 3.2.2.1.2.1).
b. Operating procedures (see 3.2.2.1.2.2).
c. Turn-on/checkout procedures (see 3.2.2.1.2.3).
d. Operator's maintenance instructions and schedules (see 3.2.2.1.2.4).
e. Standard log forms (model set) (see 3.2.2.1.2.5).

3.2.2.1.2.1 Operator's data table. The operator's data table shall include the operating parameters and functional characteristics of the equipment (see figure 14). The table shall include a listing of the following:

a. Operating controls and indicators of the equipment.
b. Description of the purpose of each control or indicator.
c. Normal operating position of the control to aid in the emergency use of the equipment by an untrained operator.

The operator's data table shall be accompanied by an illustration of the related unit. When the size of the illustration precludes direct reading of the functional designation of controls, an alphanumeric coordinate method or leader callout method and location table shall be used for locating the operating controls.

3.2.2.1.2.2 Operating procedures. Operating procedures shall include the following:

a. Safety precautions. A description of hazards associated with equipment operation shall be included in the operating procedures action. The safety precautions shall cover, but not be limited to, such items as hazards due to radiation, noise, high voltage, jet exhaust, and moving machinery. Further, it shall include information on any hazardous material safety feature(s) bypassed by the battleshort operating mode including hazards to personnel and equipment, and the anticipated time to failure if the bypassed condition occurs.
b. Operator turn-on. This procedure shall include all steps necessary to bring the equipment from OFF through STANDBY condition to full ON operation.
c. Modes of operation. Procedures shall be provided for each mode of operation (for example, manual, automatic, local, remote, search, and track). The use and relative advantage of each mode shall be described.
d. Operation under interfering or offensive countermeasures conditions. A description of the sources and effects of interferences shall be included. A description of the equipment's interference and anti-jamming reduction features, the advantages of each feature, and the operating procedures to be followed in all possible situations shall also be provided. Supporting illustrations (such as indicator

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displays and waveforms) shall be included to provide typical observations of interference and jamming for evaluation by the operator.

e. Operator turnoff. This procedure shall include all steps necessary to bring the equipment from full ON operation to the OFF condition.
f. Battle-short or emergency operation. This procedure shall cover operating the equipment during emergency conditions (for example, cooling air failure, lube oil failure, and loss of cooling water). Maximum allowable time for operation of equipment before a catastrophic failure can be expected to occur shall be given. Emergency operator maintenance instructions shall be included. A CAUTION shall be provided to return the equipment to proper operation when the emergency is over.
g. Emergency turnoff. This procedure shall cover turning the equipment off during an emergency (for example, fire, water, smoke, hazard to personnel, loss of coolant or normal power).

3.2.2.1.2.2.1 Method of presentation. Operating procedures shall be presented in tabular form and shall be in clear concise, step-by-step procedures. They shall include the following:

a. A short explanation of the operation to be performed.
b. Initial safety requirements (actions, inspections, and reference to emergency turnoff procedures).
c. Connections of any accessory equipment not permanently connected.
d. Instructions for obtaining or confirming the presence of all critical inputs; such as power, coolant, air, signal, and air conditioning.
e. Procedures for setting controls and making adjustments which must be accomplished by the operator prior to equipment turn-on.
f. Procedures for determining operational readiness and the acceptable indications expected from built-in indicators; such as meters, lamps, gauges, cathode ray tubes, and recorder readouts.
g. Milestones in the operational status of the equipment that are identified and included by brief statements such as "the generator is now in STANDBY."
h. Visual or aural observations/indications that occur as a result of an operator action; such as boom lowering, sweep rotation, or blower motor running.
i. WARNINGS and CAUTIONS which immediately precede those procedures where failure to strictly follow all directions would be likely to result in personal injury or death, or equipment being damaged or destroyed. NOTES shall not be used (see 3.1.5.2).
j. Pictorial diagrams that support the procedure by identifying and locating all operating controls and indicating devices, as well as normal in-use positions or indications.
k. Operator's checks and adjustments in proper sequence.
l. Operator's maintenance actions and schedules.

3.2.2.1.2.3 Turn-on/checkout procedures. Procedures for determining operational readiness and the acceptable indications expected from built-in indicators (such as meters, lamps, gauges, cathode ray tubes, and recorder readouts) shall be described. Milestones in the operational status of the equipment shall be identified by brief statements, such as "the generator is now in STANDBY," "the boom is lowered," and "blower motor is running."

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Checkout procedures shall be limited to those that can be performed at operating positions. Duplication of the turn-on/checkout chart specified in 3.2.2.2.1.4 is acceptable.

3.2.2.1.2.4 Operator's maintenance instructions and schedules. These instructions shall define maintenance tasks and schedules to be performed by the operator. The maintenance tasks shall be limited in scope so that they will not be: dangerous; in conflict with other operational commitments; beyond the normal operator's technical training and the maintenance technician's responsibility, or compromise the operation of the equipment. Normally, these tasks will be restricted to minor adjustments, cleaning, filter replacement, servicing, and lubrication. Identify the use of miniature electronics repair/automatic test equipment (2M/ATE) and procedures to I level as maintenance policy.

3.2.2.1.2.4.1 Method of presentation. Maintenance procedures shall be presented in tabular form and shall be in clear, concise, step-by-step procedures. They shall include the following:

a. A short explanation of the task to be performed.
b. Initial safety requirements (actions, inspections, and reference to emergency turnoff procedures).
c. Separately identified and defined steps for each task and major subtask.
d. A ten step limit for each identified task or subtask.
e. WARNINGS and CAUTIONS preceding those procedures where failure to strictly follow all directions would be likely result in personal injury or death, or damage to the equipment NOTES shall not be used (see 3.1.5.2).
f. Illustrations that identify and locate all maintenance points including essential details.
g. A reference to standard log sheet, and a discussion of the log covering its utility and criteria.

3.2.2.1.2.5 Standard log forms (model set) (N). Unless otherwise specified (see 6.2), a model standardized log form shall be developed and included at the end of chapter 2 for equipment requiring the periodic recording of data for monitoring performance and accomplishing trend analysis.

3.2.2.1.2.5.1 Applicability. Operating machinery and equipment similar, but not limited, to the following types of equipment shall include the standard log form:

a. High-pressure air compressors.
b. Engines.
c. Air conditioning plants.
d. Carbon dioxide scrubbers.
e. Carbon monoxide-hydrogen burners.
f. Steam-vapor distilling plants.
g. Turbine generators.
h. Electronic equipment cooling water and dry-air systems.
i. Boilers.
j. Propulsion plant.

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3.2.2.1.2.5.2 Waiver of requirement. Guidance regarding standard log form requirements should be obtained from the procuring activity; a sample form is illustrated on figures 15, and listed in NAVSEA S9086-CZ-STM-000/CH 90. When it is deemed that the development of such a form would be of no value to the Government for the equipment being covered, the contractor shall request a waiver for standard log form requirements in writing at least three months prior to submittal of the review manuscript. Waiver requests shall include a written rationale.

3.2.2.1.2.5.3 Criteria. The model standardized log form shall serve as an operating log for all ships. However, it is not intended that this form should in any way prevent individual ship ships from accommodating their specific needs by adding or deleting readings as needed on a case by case basis. The basic criteria to be utilized in developing the model log shall be the following:

a. Analyzing the various equipment readings and pinpointing those that provide critical information.
b. Reviewing the results of the above analysis to ensure that only necessary parameters are included; thus minimizing the required number of log readings.
c. Formatting the log forms so that the log keeper and the reviewing officer can quickly tell when problems, either short- or long-term are developing.
d. Eliminating duplicate readings where feasible.

3.2.2.1.2.5.4 Format. The following considerations shall also apply to the development of the log form:

a. Log forms shall be developed in a columnar format to simplify use, with adequate space provided for identification, data, and checkoff. The time line should normally be on an hourly or daily basis, and may be arranged on a vertical (left side) or horizontal (top) axis, depending on the number of monitoring points.
b. Log sheets shall be 8-1/2 by 11 inches in size. When practical, the reverse side of the page may also be used for recording readings.
c. Monitoring points should indicate the normal values for each point, as well as the maximum and minimum tolerances for normal operation.
d. Readings that would endanger personnel or damage equipment shall be plainly indicated in boldface type, or other methods that are effective.
e. Instructions for the use of the log forms shall be included when necessary. Codes (such as H = High; L = Low; B = Bubbles) should be defined in the instructions or in a legend.
f. Indicators that provide information to the operator during a danger reading (such as shutdown, cutback, or an open valve) in the instructions or by a separate related column.
g. Space should be provided to indicate the make, size, model, application, and location of the equipment, or other relevant information.
h. Space should be provided for signing the watch, normally on the reverse side.

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i. Space should also be provided, if required, for certification by the watch officer, engineering officer, or others, normally on the reverse side.

3.2.2.1.2.5.5 Training summary. A training summary that explains how to use the log and the significance of the monitoring points and readings (see figure 16) shall be included in chapter 2.

3.2.2.1.2.5.6 Request for number. The contractor shall request a log form stock number from the contracting activity so that copies of the log form can be reprinted and placed in stock.

3.2.2.1.3 Chapter 3 - Theory of operation. This chapter shall contain all of the information needed to educate the manual user on the general principles of operation that go beyond the actual functioning of the equipment included in the troubleshooting/repair volume. The data shall provide enough information about operational doctrine so that the technician and other users will understand enough of the mission requirements to manipulate the equipment effectively. Examples of this type of information are gunfire control problems, sonar or radar principles, basic logic on which operation is based, concepts of communications systems, navigation theory, and machinery operation. In addition, this chapter shall expand on the information provided in the keyed text of the troubleshooting/repair volume when it is necessary to explain a new technology to the user. Illustrations such as patterns, nomographs, and simplified block and schematic diagrams shall be maximized to minimize necessary text. Chapter 3 shall include the following, as applicable:

a. Simplified schematic diagrams (see 3.2.2.1.3.1).
b. Integrated circuits and microminiature capsules (see 3.2.2.1.3.2).
c. Simplified piping diagrams (see 3.2.2.1.3.3).

3.2.2.1.3.1 Simplified schematic diagrams. Diagrams shall show, by means of graphic symbols, the electrical connections and functions of a specific circuit arrangement. (N) Diagrams shall be arranged functionally to show the operation of the circuits in accordance with NAVSEA 0967-LP-000-0120.

3.2.2.1.3.2 Integrated circuits and microminiature capsules. Simplified schematics [(N) in accordance with NAVSEA 0967-LP-000-0120] or logic diagrams showing input and output connections, truth tables, and waveforms as applicable, supported by a brief description of overall operation, shall be included for circuits within different types of nonrepairable integrated circuits and microminiature capsules. Schematic and wiring diagrams shall be supplied in the troubleshooting/repair volume for all circuit boards and modules designated as repairable. Manufacturer's data sheet information may be included if it meets both the technical contents of this requirement, and the format and legibility requirements of this specification.

3.2.2.1.3.3 Simplified piping diagrams. Diagrams (hydraulic, pneumatic, or fluid) shall show the interconnection of components by piping, tubing, or hose, and sequential flow in the system. Components such as pumps, heat exchangers, valves, gauges, shall be clearly identified.

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3.2.2.1.4 Chapter 4 - Maintenance procedures and standards. This chapter shall contain preventive maintenance procedures and performance test instructions to be accomplished on a scheduled or condition monitoring basis. The applicable chapters of NAVSEA S9086-00-STM-000/NSTM and MIL-P-24534 may be used as baselines for developing the procedures. These procedures shall be correlated with the installation checkout requirements specified in 3.2.2.1.5.9, so that any maintenance or performance test procedure also required for checkout may be properly referenced from chapter 5. The installation standards summary sheet (see 3.2.2.1.5.9.3.2) shall also include space for any maintenance or performance standard test result that is recorded by the installer. The recorded information will provide a reference standard to the technician when troubleshooting, when needed for installation acceptance certification, or for certifying proper overhaul. It is intended that the engineering effort required to develop preventive maintenance data be expended only once and that the data, where applicable, be based on the Logistic Support Analysis (LSA) and the Planned Maintenance Subsystem (PMS) (see 6.5.40) data, and only be reformatted as necessary. The LSA and PMS data shall be integrated with other technical data required by this specification. Chapter 4 shall contain the following information:

a. Introduction (see 3.2.2.1.4.3).
b. Scheduled maintenance action index (see 3.2.2.1.4.4).
c. Preventive maintenance procedures (see 3.2.2.1.4.5).
d. Scheduled performance tests (see 3.2.2.1.4.6).
e. Plan for maintenance of mechanical components (see 3.2.2.1.4.7).
f. Plan for maintenance of modules, printed circuit boards, assemblies, and components (see 3.2.2.1.4.8).
g. Identification of overhaul/depot-level maintenance (see 3.2.2.1.4.9).

3.2.2.1.4.1 When PMS is being developed separately under the governing contract, this chapter shall contain that information necessary to augment PMS. That is, PMS provides Maintenance Requirement Cards (MRC's) (see 6.5.31), figure 17, which define the minimum tests, performance standards, and schedules necessary to determine acceptable operation and reliability. This chapter shall provide subordinate reference standards (test values) necessary to isolate potential degradations when MRC performance standards are not met so that the equipment can be brought up to acceptable operation.

This chapter shall be arranged in a sequence that parallels the functional MRC tests (irrespective of scheduled periodicity) and schedule recommendations based on the Maintenance Index Pages (MIP's). Reference may be made to the MRCs and MIPs if all applicable shipboard and shore systems will have the PMS installed. Subordinate performance standards and repair procedures shall be provided in order to allow rapid fault isolation of degradated circuitry, or partial mechanical failure, or wear and tear. Detailed coverage shall be based on the applicable requirements of paragraph 3.2.2.1.4.2.

3.2.2.1.4.2 When PMS is not being provided as a separate contractual requirement, this chapter shall contain that additional information necessary to develop the PMS data. This chapter shall contain preventative maintenance procedures and performance test instructions to be accomplished on a scheduled or condition monitoring basis.

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3.2.2.1.4.3 Introduction. The introduction shall explain the purpose, scope, and arrangement of the scheduled maintenance data. When a preventive maintenance procedure is critical to the operation of the equipment and the schedule for servicing is mandatory, this information shall be written as a CAUTION statement.

3.2.2.1.4.3.1 Inspection work cards/maintenance requirement cards precedence (F, N). The following statements shall be included, as applicable:

a. (F) "Where conflict arises between procedures contained in the Inspection Work Cards and those in the manual, the procedures in the Inspection Work Cards shall take precedence."
b. (N) "Where conflict arises between procedures contained on the Maintenance Requirement Cards (MRC's) for this equipment and those in this manual, the procedures on the MRC's shall take precedence."

3.2.2.1.4.4 Scheduled maintenance action index. This index shall include a listing of all scheduled performance tests and preventive maintenance procedures. The index shall be tabulated as follows:

a. Column 1 - Periodicity. This column shall contain an alphanumeric list of all maintenance actions contained in this chapter. The following periodicity symbols, as appropriate, shall be used in the order of increasing periodicity as listed below:

Interval	Symbol	Interval	Symbol
Daily	D	Annually	A
Weekly	W	Overhaul cycle	C
Monthly	M	As specified (explain circumstances)	R
Quarterly (3 months)	Q		
Semiannually	S		

NOTE:
An R periodicity shall be preceded by a recommended calendar periodicity such as, DR, WR, MR, which may mean "perform as recommended provided that the equipment is operated under full load".

b. Column 2 - Maintenance action. This column shall name the maintenance action which corresponds to the periodicity symbol in column 1.
c. Column 3 - Reference. This column shall state the paragraph number that contains the procedure listed in column 2.

3.2.2.1.4.5 Preventive maintenance procedures. These procedures shall include the information required to inspect, clean, and lubricate the equipment, and shall contain the following:

a. Safety precautions.
b. A listing of tools, parts, materials, and test equipments required.

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c. Instructions for dilution and use of cleaning solvents. Solvents shall be identified by Government specification numbers and National Stock Numbers.
d. Instructions for properly maintaining all safety devices and interlocks, with WARNING and CAUTION statements provided as necessary.
e. Instructions for lubrication at place of installation; also types and quantities of lubricants to be applied. Lubricants shall be identified by Government specification numbers and National Stock Numbers. Specific lubricants for arctic or tropic environments shall be included.
f. Instructions for tightening or securing mechanical components as a part of routine maintenance to retain equipment design hardness against mechanical shock in a hostile environment. Such instructions shall contain appropriate cautions and lists and locations of critical fasteners.
g. The minimum rating or skill level of the technician expected to perform the preventive maintenance tasks.
h. Procedures for obtaining access to subassemblies or subcomponents.
i. Instructions for in-place balancing and noise reduction.
j. Inspection procedures for parts which deteriorate due to cycles of use, age, or climatic conditions.
k. Illustrations to identify lubrication points or other pertinent data.
l. Any other information pertinent to these procedures.

3.2.2.1.4.6 Scheduled performance tests. These tests shall contain step-by-step procedures to verify whether the equipment is operating within standards during all phases of operation. The tests shall contain the following:

a. Safety precautions.
b. A listing of tools and test equipments identified by type, manufacturer, and model number, or by assigned military designation.
c. The title of the test to be performed.
d. The minimum rating or skill level of the technician expected to perform the test.
e. Preliminary setup data required to perform the test.
f. Detailed procedures for accomplishing the test. Procedures requiring lengthy and identical setup data may be presented in detail in one procedure and referenced in succeeding procedures.
g. Values or conditions, with tolerances, that indicate normal operation.
h. Troubleshooting references at the end of each test procedure for location of the troubleshooting or corrective actions to be used if the test values are not within tolerances. Where causes of equipment failure can be predicted with reasonable confidence, reference may also be made directly to the appropriate maintenance procedure.
i. Illustrations to support the test instructions.

3.2.2.1.4.7 Plan for maintenance of mechanical components. Instructions shall provide, on a comprehensive and systematic basis, effective and efficient procedures for performing maintenance of those mechanisms designated as needing periodic overhaul. When mechanical assemblies, such as recorder drive trains, gearboxes, clutches, or transmissions, involve disassembly/assembly procedures, replacement of bearings, gears, or retaining rings, and

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do not require more elaborate tooling than a press for parts removal, such data and procedures shall be included in the troubleshooting/repair volume/ manual with the respective hardware information. Where the procedure would be more extensive, or is designated as depot maintenance, it shall be located in the support volume/manual. When the extent of the overhaul data justifies it, or depot maintenance data are not to be supplied to the organizational- or intermediate-level maintenance activity, the plan for maintenance of mechanical components shall be separately bound, designated as Chapter 6 or 7, as applicable, and entitled "Overhaul" or "Depot-Level Maintenance for . . ." When specified, overhaul data will be prepared in accordance with MIL-M-24747.

3.2.2.1.4.7.1 Representative considerations. Considerations to be weighed in the development of maintenance procedures shall include the following hardware characteristics:

a. Reliability, maintainability, simplicity, and ruggedness.
b. Transient and static operating conditions, shock, vibration, wear, corrosion, erosion, fatigue, operating temperature, moisture, roll, pitch, air blast, and other environmental forces, as applicable, which will affect the continued satisfactory performance of each part.
c. Accessibility, ease of removal, and replacement of parts requiring frequent removal.
d. Safety.
e. Methods of calibration and adjustment.
f. Rotatables (replacement of assemblies for subsequent overhaul and reissue versus spot overhaul and repair).
g. Organizational and intermediate test and repair capabilities.

3.2.2.1.4.7.2 Schedule. A recommended schedule for all maintenance actions (inspections, tests, adjustments, reconditioning, overhaul, and acceptance criteria) shall be included. The schedule shall ensure reconditioning/ replacement before the degradation results in major failure.

3.2.2.1.4.7.3 Reconditioning procedures. Procedures shall be included for reconditioning replaceable parts that are within the scope of the maintenance philosophy. The procedures shall provide detailed maintenance instructions at all maintenance levels to accomplish the reconditioning or replacement of each part subject to continuing degradation.

3.2.2.1.4.7.4 Illustrations. Overhaul and depot-level maintenance procedures shall rely heavily on engineering data and engineering drawings, supported by sectional and exploded views of complex assemblies and mechanisms. However, diagrams common to the troubleshooting/repair volume shall be included in the hardware data packages and may be repeated if necessary to maintain continuity. It shall be assumed that depot-level maintenance activities will be authorized to use the entire technical manual, but the organizational- and intermediate-level activities may not have access to depot-level material. Therefore, these procedures shall not alter the self-sufficiency concept of the organizational-and intermediate-level maintenance data.

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3.2.2.1.4.8 Plan for maintenance of modules, printed circuit boards, assemblies, and components. Unless a separate manual is acquired to cover the testing, maintenance, and repair of items designated as being Government depot-level repairable, comprehensive instructions shall be provided as follows:

a. Construction details for items such as special jigs, fixtures, extenders which must be fabricated.
b. A list of specialized test equipments, such as Government-available module testers, power supplies, and laboratory-type equipment.
c. Test procedures, including the use of programs and other specialized test equipments and facilities.
d. Use of 2M/ATE like contractor assisted support services (CASS), USM/465 and (TPS) for removal and replacement of integrated circuits and microminiature components. (identify circumstances for I level).
e. Calibration and alignment procedures.
f. Refurbishment.
g. Special handling and packaging for storage and reshipment.

3.2.2.1.4.8.1 Illustrations. The following illustrations shall be included unless they are contained in the troubleshooting/repair volume. If they are contained in the troubleshooting/repair volume, then proper reference shall be made to their identification and location.

a. Schematic diagrams (see 3.2.2.2.3.3.1).
b. Wiring diagrams for hardware modules and assemblies (see 3.2.2.2.3.3.6).
c. Parts location diagrams (see 3.2.2.2.3.3.5.1).
d. Auxiliary and test equipment interconnection diagrams.
e. Chart of signal and test specifications.
f. Detailed views of mechanical components.
g. Illustrations of special maintenance procedures.

3.2.2.1.4.9 Identification of overhaul/depot-level maintenance. When testing, maintenance, repair, and reconditioning procedures are restricted to depot-level maintenance facilities, this information shall be divided into a separate chapter following the parts list (chapter 6) or the installation information (chapter 5), as applicable, and shall be identified as chapter 7 or chapter 6, as applicable. Unless otherwise specified (see 6.2), this information shall be bound separately and shall include applicable front matter. Depot-level parts may be included in the parts list (chapter 6), provided that these parts are coded accordingly. When not included in the parts list, depot-level repair parts shall be listed separately (see 3.2.2.1.6.4.3) in the overhaul/depot-level maintenance chapter. When a separate depot-level manual is required (see 6.2), it shall include the information required by MIL-M-24747.

3.2.2.1.5 Chapter 5 - Installation. Drawings and information concerning installation shall be provided in this chapter. The following types of information shall be provided:

a. Installation drawings (see 3.2.2.1.5.1).
b. Site information (see 3.2.2.1.5.2).
c. Installation reference publications (see 3.2.2.1.5.3).

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d. Installation tools and materials (see 3.2.2.1.5.4).
e. Unpacking, depreservation, represervation, and repacking (see 3.2.2.1.5.5).
f. Foundation preparation (see 3.2.2.1.5.6).
g. Input requirements (see 3.2.2.1.5.7).
h. Installation procedures (see 3.2.2.1.5.8).
i. Installation checkout (see 3.2.2.1.5.9).

In addition, this chapter shall reference the performance standards in chapter 4 of the manual (see 3.2.2.1.4.6) for use in determining that the equipment, after installation, satisfies the operational requirements.

3.2.2.1.5.1 Installation drawings. Installation drawings or legible reproductions of accepted drawings in accordance with MIL-D-23140, or MIL-T-31000 (if applicable), shall be provided as follows:

a. Pictorial system diagram.
b. Outline and mounting dimensions diagram.
c. Interconnecting wiring and cabling/piping diagram or wire running lists.
d. Summary list of all installation materials required.
e. Input and output interface data.

In addition, when applicable, reference shall be made to the cognizant agency cable running sheets. The reference shall be a tabulation of drawing numbers for the cable running sheets that are required to accomplish the installation.

3.2.2.1.5.1.1 Arrangement. Installation drawings shall be divided into sheets and arranged on facing pages so that foldouts can be avoided.

3.2.2.1.5.2 Site information. These data shall supplement the installation drawings. If all site information is contained on the installation drawings, reference shall be made to the applicable drawing(s) by figure number.

3.2.2.1.5.3 Installation reference publications. Reference shall be made to user-available general publications required to complete the installation.

3.2.2.1.5.4 Installation tools and materials. Supplemental information to the summary list of installation materials shall be included. If no supplemental information is required, reference shall be made to the applicable drawing(s) by figure number.

3.2.2.1.5.5 Unpacking, depreservation, represervation, and repacking. This information shall supplement the installation drawings regarding unpacking, depreservation, represervation, and repacking. Step-by-step procedures and icons in accordance with MIL-M-38784 shall be used to ensure personnel safety and to prevent damage to the equipment with particular attention to hazardous material. The following list shall be used to determine the information to be provided:

a. Unpacking procedures.
b. Depreservation instructions.

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c. Recommended disposal procedures for packaging and packing materials, where applicable.
d. Methods of anchoring, blocking, and bracing.
e. Specialized cushion insert data.
f. Identification of repairable items, including their packaging and packing procedures.
g. Specialized multiple-use and reusable container data.
h. Identification of hazardous material, sensitive, or fragile components/gases/liquids. Anecdotes for contact with hazardous material shall be included.
i. Need and requirements for special environmental storage conditions.
j. Special handling procedures.

Supporting illustrations shall be provided to clarify procedures. Lists of parts that could require replacement due to loss or damage (for example, attaching hardware, shock mounts, and humidity indicators) shall be included with the appropriate illustration. When instructions for reshipment or storage are required, step-by-step procedures for preservation, if applicable, and packing shall be included and illustrated. Where packing is simply the reversal of unpacking, a statement of this fact is sufficient.

3.2.2.1.5.6 Foundation preparation. This information shall supplement the installation drawings. If all foundation preparation information is contained on the installation drawings, reference shall be made to the applicable drawing(s) by figure number.

3.2.2.1.5.7 Input requirements. A summary of the input data contained on the installation drawings shall be included. Parameters with tolerances shall be included with each input listed. Examples of inputs are as follows:

a. Power.
b. Ventilation.
c. Dry air (waveguide pressure).
d. Gyro information.
e. Fluid cooling.
f. Steam.
g. Freon.

3.2.2.1.5.8 Installation procedures. The following types of supplemental information not found on the installation drawings shall be included:

a. Instructions required to assemble units.
b. Instructions required to mount units, including bolting and bracing diagrams and data on shock mounts.
c. Instructions for making electrical, plumbing, transmission line, and all other interface connections (external) to the equipment.
d. Instructions for interconnecting units comprising the equipment.
e. Servicing procedures, such as initial lubrication.
f. Instructions for bonding and grounding.
g. Instructions on specialized rigging and hoisting.

3.2.2.1.5.9 Installation checkout. Step-by-step procedures to demonstrate that the equipment operates correctly and within tolerances shall be provided.

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These procedures shall provide for equipment checkout in three test phases:

a. Phase 1 - Installation inspection and preenergizing procedures.
b. Phase 2 - Turn-on and preliminary tests.
c. Phase 3 - Installation verification test.

3.2.2.1.5.9.1 Phase 1 - Installation inspection and preenergizing procedures. Inspection procedures in the form of checklists shall be provided to verify the following:

a. The equipment (both primary and auxiliary) has been properly installed. All cables, antennas, waveguides, transmission lines, dehydrators, coolant lines, and piping are in accordance with plans and specifications.
b. The applicable test equipments listed in chapter 1 of the manual are available at the site, are operating correctly, and have been calibrated.
c. (N) The Allowance Parts List (APL) is available, and the Coordinated Shipboard Allowance List (COSAL) includes the equipment data.
d. The authorized field changes, modifications, alterations, and mandatory retrofits have been accomplished.
e. The rotating devices are free from obstruction.
f. The equipment is accessible for maintenance.
g. The pre-energizing servicing procedures, including lubrication, have been accomplished.
h. The equipment can be started safely.

3.2.2.1.5.9.2 Phase 2 - Initial turn-on and preliminary tests. Procedures for the first equipment energizing shall be included. This may be accomplished by reference to the applicable portions of the operating procedures in chapter 2 of the manual. Step-by-step procedures for testing the equipment's electrical supply circuits, including distribution panels, switches, breakers, relief valves, and interlocks, shall be included. Procedures for testing the piping, electrical cables, wire rope, stays, proper installation of transmission lines and waveguides, hanger spacing, torquing of connectors, pressure testing, flow rates, standing wave ratio, and attenuation checks shall also be included.

3.2.2.1.5.9.3 Phase 3 - Installation verification test. Complete instructions for testing the equipment in all modes of operation shall be included. Where applicable, reference to the scheduled performance tests in chapter 4 of the manual (see 3.2.2.1.4.4) shall be made. Procedures shall include checking gauges, meters, alarms, and other sensing devices for proper operation and calibration. The test shall verify that all inputs are within tolerance. As applicable, resistance tests, voltage standing wave ratio tests, and insertion loss tests shall be provided to verify the proper installation of antennas and antenna-to-equipment waveguide runs. Transducer impedance and source level checks shall be provided to verify proper installation of transducers, domes, and cables. Preliminary setup data shall be included in each procedure. When it is required that an alignment be accomplished prior to performing a test, the alignment shall be included or, if lengthy, a reference be made to the alignment procedures in the troubleshooting/repair volume/manual.

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3.2.2.1.5.9.3.1 Testing procedures. Testing procedures shall be presented in a logical order as follows:

a. Energize the equipment.
b. Test the first units (normally power supplies) which must be operating properly. When test results are within the required tolerances, include reference to the next logical test. Include a reference to the corrective maintenance or troubleshooting data to be used if test results are not within tolerances.
c. Enter installation and reference standards on the installation standards summary sheet (see 3.2.2.1.5.9.3.2). Include instructions for installers to actually test and enter reference standard values, and certify proper installation on the installation standards summary sheet.

3.2.2.1.5.9.3.2 Installation standards summary sheet. The installation standards summary sheet shall provide spaces for recording the results of all installation verification tests (see figure 18). Each space shall be identified by the step or paragraph number which provides the instructions for accomplishment. Each space shall contain the respective unit of measurement (for example, amperes, decibels, volts). Waveforms shall be included to show the points on the pattern where the measurement is to be taken. In addition, critical installation data shall be included, such as the length of the transmission line.

3.2.2.1.6 Chapter 6 - Parts list. When a separate parts list is required (see 6.2) in addition to the parts data required by 3.2.2.2.3.3.5, the parts list shall be prepared in accordance with the requirements of the contracting activity as detailed in 3.2.2.1.6.1 through 3.2.2.1.6.4.8.

3.2.2.1.6.1 Separate parts list (N). When specified in the contract or order, a separate illustrated parts breakdown shall be prepared in accordance with MIL-M-008910.

3.2.2.1.6.2 Separate parts list (A). When specified in the contract or order, a separate parts list shall be prepared in accordance with MIL-STD-335.

3.2.2.1.6.3 Separate parts list (F). When specified in the contract or order, a separate parts list shall be prepared in accordance with MIL-M-38807.

3.2.2.1.6.4 Equipment parts list (CG, MC, N). The parts list shall catalog and identify all shipboard, tender, and shore-based repair parts, including attaching hardware. This chapter shall include the following:

a. Introduction (see 3.2.2.1.6.4.1).
b. List of major components or units (see 3.2.2.1.6.4.2).
c. Parts list (electrical, mechanical, hydraulic, or pneumatic equipment) (see 3.2.2.1.6.4.3).
d. Parts list arrangement (electronic equipment) (see 3.2.2.1.6.4.4).
e. List of common items (see 3.2.2.1.6.4.5).
f. List of attaching hardware (see 3.2.2.1.6.4.6).
g. List of manufacturers (see 3.2.2.1.6.4.7).
h. Parts location illustrations (see 3.2.2.1.6.4.8).

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3.2.2.1.6.4.1 Introduction. The introduction shall contain an explanation of the scope and arrangement of the parts list. The following types of information shall be included:

a. Models of equipments and the serial numbers of the equipments covered.
b. Explanation of any special notes and symbols.
c. Explanation and instructions for using the list of common item descriptions.
d. Explanation and instructions for using the list of attaching hardware.
e. Explanation and instructions for using the parts list.
f. Explanation and use of the list of manufacturers.

3.2.2.1.6.4.2 List of major components or units. A list of the major components or units shall be included as specified in 3.2.2.1.6.4.2.1 and 3.2.2.1.6.4.2.2.

3.2.2.1.6.4.2.1 List of major components (electrical, mechanical, hydraulic, or pneumatic equipment). A list of major components in tabular form shall be included as follows:

a. Column 1 - APL/CID number. This column shall list the APL/CID number for each major component.
b. Column 2 - Quantity. This column shall list the quantity of components.
c. Column 3 - Name. This column shall list the official name of the major components.

3.2.2.1.6.4.2.2 List of major units and assemblies (electronic equipment). A list of the major units and assemblies in tabular form shall be included as follows:

a. Column 1 - Unit number. This column shall list the unit numbers for each major unit and assembly.
b. Column 2 - Nomenclature. This column shall be subdivided into the following:

    (1) Name of unit. This column shall list the official name of the unit or assembly.
    (2) Designation. This column shall list the type designation of the unit or assembly.

c. Column 3 - Page number. This column shall list the number of the first page of the parts listing for the major unit or assembly.

3.2.2.1.6.4.3 Parts list (electrical, mechanical, hydraulic, or pneumatic equipment). Except when reference designators are assigned to parts in accordance with ANSI Y32.16, the parts list shall be prepared in accordance with 3.2.2.1.6.4.3.1. When equipment part designators are assigned in accordance with ANSI Y32.16, the parts list shall conform to the requirements of 3.2.2.1.6.4.4.

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3.2.2.1.6.4.3.1 Parts list. Parts listings shall be prepared in tabular form as follows:

a. Column 1 - Figure and index number or coordinates. This column shall list the figure and index numbers or coordinates which show the location of the part.
b. Column 2 - Name and description. This column shall provide the designation name of the part and descriptive data to identify the parts of the equipment. Descriptive information shall include the military identification number of the part whenever available. Parts that do not have a military identification number shall include physical characteristics (material, grade, series, dimensions, specification, and any other information necessary to order replacement parts from the manufacturer of the part without going to the prime contractor).
c. Column 3 - Quantity. This column shall list the quantity of parts required.
d. Column 4 - Commercial and Government Entity (CAGE) Code. This column shall provide the original item manufacturer's CAGE code in accordance with cataloging handbook H4/H8. Identification numbers are not required for common hardware items such as screws, washers, and O-rings that are available from many sources. However, these items shall be identified by a military specification (MS) or other military type number.
e. Column 5 - Original manufacturer's part number. This column shall list the part number assigned by the original manufacturer of the part and, if desired by the contracting activity, the engineering drawing and index number. Part numbers are not required for common hardware items that are available from many sources.

3.2.2.1.6.4.4 Parts list arrangement (electronic equipment). The parts list shall be divided and arranged by major units in numerical sequence (for example, unit 1 with its parts will precede unit 2 with its parts). Replaceable mechanical parts of repairable assemblies, such as gears, spacers, brushes, and nonprovisioned parts subject to wear or breakage, such as handles, slides, and locks, shall be assigned mechanical part (MP) reference designators if not so assigned and included in the parts list. Parts attached to the unit shall be listed first in alphanumeric order, followed by unit assemblies with parts and subassemblies with parts, also listed in alphanumeric order as follows:

Unit 1	1
(cabinet parts)	1AT1
	1B1
	1C1
	1CR1
	1R1
	and so forth
Assembly	1A1
(assembly parts)	1A1AT1
	1A1B1
	1A1C1
	1A1CR1

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	1A1R1 and so forth
Subassembly (subassembly parts)	1A1A1 A1A1AT1 1A1A1B1 1A1A1C1 1A1A1CR1 1A1A1R1 and so forth
Unit 2	2 and so forth

3.2.2.1.6.4.4.1 Tabulation. Parts listings shall be prepared in tabular form with columns headed as follows:

a. Column 1 - Reference designation. This column shall provide the reference designations of all parts listed in sequential order. MP numbers shall be assigned to mechanical parts subject to replacement, such as handles, slides, gears, and pulleys, that are included in the APL but not assigned MP numbers on the engineering drawings. With the exception of screws, nuts, bolts, and other attaching hardware, every functioning part in the equipment shall be assigned and have a reference designator. The unit numbering method of assigning reference designations, as specified in ANSI Y32.16, shall be used to identify units, assemblies, subassemblies, and parts. The parts list shall be divided and arranged by major units in numerical sequence. When reference designations have been cancelled for more than two consecutive items, only the first and last of the designations are to be listed, separated by the word "thru" (for example, "3A1R69 thru 3A1R99 not used").

b. Column 2 - Notes. This column shall contain equipment reference information such as serial number, model number, and configuration data.

c. Column 3 - Name and description. This column shall include descriptive data to identify the parts of the equipment and assist in determining substitutes. Such information shall include the name, electrical or mechanical characteristics, MS or military identification number of the item, and when applicable, attaching hardware. Common parts, such as washers, springs, nuts, and bolts, shall be identified only by their MS or military identification numbers. Those parts not having MS or military identification number shall also include physical characteristics that identify the parts within the set [for example, the manufacturer's part number and CAGE code; or the equipment contractor's part number, CAGE code, Federal Supply Code Number, National Stock Number (NSN), and drawing number]. Replaceable mechanical parts that are assigned an MP number as specified in the preceding subparagraph (a), shall include the manufacturer's part number or engineering drawing number along with the name and description of the item. The statement "same as . . ." or "equivalent" shall not be used for describing identical parts. Identical parts that are used more than two times, but less than five times, shall include the complete description or conform to the

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requirements for identical parts used more than five times. For identical parts that are used more than five times in the equipment, the complete item description shall be given in the list of common item descriptions and reference made thereto by the item number. Attaching hardware, including the quantity required, shall be identified by the assigned letter code (see 3.2.2.1.6.4.6). For example, "C(4)" would be the third listed piece of attaching hardware in which four pieces are used. When nonstandard parts have been approved and there are multiple sources, each source shall be cited in the description. If selected values for critical parts are required, sufficient information, such as criteria for selection and range of values, shall be provided to permit the repair activity to make the selection. For each part, the part number of the actual item identified by the manufacturer shall be used unless the part is physically modified by the equipment contractor.

d. Column 4 - Parts location reference. This column shall reference the parts location illustration normally contained in the troubleshooting/ repair volume (see 3.2.2.2.3.3.5.1) by assembly reference designation number. If the parts location illustration follows the parts list (see 3.2.2.1.6.4.8), reference shall be by figure number, followed by the item number or locating coordinate numbers in parentheses.

3.2.2.1.6.4.4.2 Mechanical parts list cross-reference table. When the number of mechanical parts included in the parts list exceeds 25, a separate mechanical parts list cross-reference table shall be included. This table will provide part number-to-reference designator cross-referencing for all mechanical parts. The table shall be arranged in a logical sequence so that piece and assembly parts identified by manufacturer's part number can be conveniently cross-referenced to corresponding reference designations.

3.2.2.1.6.4.5 List of common items. This list shall include the description of all multiple parts (over five applications). The description shall contain the same information specified in 3.2.2.1.6.4.4.1. Like parts shall be grouped and arranged in alphabetical order. Item numbers shall be assigned consecutively; for example:

Item number	Description
1	CAPACITOR, FIXED, CERAMIC: DI-ELECTRIC 3PF, 500 VDCW; MIL TYPE CCZ1UJ030C
2	CAPACITOR, FIXED, GLASS: DI-ELECTRIC 5,100PF, ±1% 300 VDCW; mfr 86262, dwg 231B743H15
3	RESISTOR, FIXED, COMPOSITION: 3,000 ohms, ±5% 1/2W; mfr 42384, dwg 446D9
4	RESISTOR, FIXED, COMPOSITION: 4,000 ohms, ±5% 1/2W; MIL TYPE RC20GF402J

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3.2.2.1.6.4.6 List of attaching hardware. A list of standard attaching hardware shall be included and identified in accordance with MIL-HDBK-131, and further identified by military standards (AN/JAN, MS) numbers; for example:

Letter Code	Name and Description
A	SCREW, CAP, HEX HEAD, DRILLED HEAD CRES: MS51100-8 1/4-28, UNF-2A, 1 in. lg
B	WASHER, FLAT, STEEL, ROUND: 0.750 in. OD, 0.312 in. ID, 0.066 in. thick

3.2.2.1.6.4.6.1 Exception. Items used fewer than five times need not be included in the list of attaching hardware, provided that item is completely identified in the parts list following the part identification that is being attached.

3.2.2.1.6.4.7 List of manufacturers. The list of manufacturers shall contain the names, addresses, and code symbols of all manufacturers supplying the items for the equipment as referenced in the parts list. The list shall be presented in numerical sequence by code number. CAGE codes shall be in accordance with cataloging handbooks, H4/H8.

3.2.2.1.6.4.8 Parts location illustrations. Parts location illustrations shall be provided to locate all parts listed in the parts list. When the parts location illustration is not included in the troubleshooting/repair volume, it shall be included at the end of the parts list. (Parts location illustrations located in other parts of the manual may be referenced.) Standard attaching hardware items (such as nuts, bolts, washers, screws) need not be called out or illustrated (except on exploded views) unless they are referenced in a procedure. Illustrations required for repair and replacement in the troubleshooting/repair volume need not be repeated, but shall be referenced.

3.2.2.1.7 User activity comment sheet (N). Three copies of the user activity comment sheet (similar to figure 19) shall be included after the last page of all separately bound volumes and parts. Copies of the user activity comment sheet will be provided by the Government.

3.2.2.2 Troubleshooting/repair volume (type I). The troubleshooting/repair volume shall present information in a systematic top-down hierarchy from the most general (basic) information to more detailed information to assist qualified technicians in rapid fault isolation, repair, and alignment. This volume shall maximize the use of standard symbols to represent data such that information is presented with maximum utilization of space. When no standard symbol exists, a unique symbol may be generated for the purpose. When practical, mechanical items, such as, switches, relays, gears, assemblies, etc., should be used in a manner that provide a likeness to the actual hardware for ease of understanding. Figure numbers shall be used and all hardware shall be identified by the symbols used in the family tree (see 3.2.2.2.1.2.1). The troubleshooting/repair volume shall be arranged in the following order and shall conform to referenced paragraphs.

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a. Front matter (see 3.2.1).
b. Overall function information (see 3.2.2.2.1).
c. Major function information (see 3.2.2.2.2).
d. Hardware information (see 3.2.2.2.3).
e. BITE/ATE fault isolation information to include 2M repair (see 3.2.2.2.4).
f. User activity comment sheet (N) (see 3.2.2.1.7).

3.2.2.2.1 Overall function information. Basic information, such as functional and hardware arrangement, interconnection, and turn-on/checkout of the equipment, as well as overall function and alignment information. Chapter 7 shall be arranged as follows:

a. Introduction (see 3.2.2.2.1.1).
b. Family trees (see 3.2.2.2.1.2).
c. Interconnecting cabling/piping diagram (see 3.2.2.2.1.3).
d. Turn-on/checkout chart (see 3.2.2.2.1.4).
e. Overall function information package (see 3.2.2.2.1.5).

3.2.2.2.1.1 Introduction. The introduction shall explain the use and function of the basic equipment and provide overall information.

3.2.2.2.1.2 Family trees. Two family trees shall be provided; a functional family tree and a hardware family tree.

3.2.2.2.1.2.1 Functional family tree. A functional family tree (as shown on figure 2) shall be provided which shows the functional arrangement of the equipment and the arrangement of the functional information packages. The blocks in the functional family tree shall be assigned titles and identification numbers (number within an oval) and shall correspond to the functional index and functional information packages. The first major function shall normally be power distribution.

3.2.2.2.1.2.2 Hardware family tree. A hardware (equipment) family tree shall be provided which shows the breakdown structure of the equipment into units, assemblies, and subassemblies as shown on figure 3.

3.2.2.2.1.3 Interconnecting cabling/piping diagram. An illustration (see figure 4) of all electrical, hydraulic, and pneumatic interconnections shall be provided. The diagram(s) shall identify all cables, pipes, waveguides, and connecting devices by reference designation. A tabular listing of parts information shall be included for all items not included in other parts listings.

3.2.2.2.1.4 Turn-on/checkout chart. A combined turn-on/checkout chart shall be supplied (see figure 20). The chart shall be similar in appearance to the overall function-level MDC except that only the sensory indications shall be included in the body of the chart. The heading of the turn-on/checkout chart shall identify the major function information which shows the respective indicator. A specification table shall augment the sensory indications so that the user can judge whether the indications are within the desired range (for example, "antenna rotates at 4 r/min"). Non-critical information, such as the speed of a blower, need not be included.

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3.2.2.2.1.4.1 Checkout procedures. Checkout procedures shall be limited to those that can be performed at the operating positions. Turn-on and checkout steps shall be the same steps and in the same sequence as those contained in the overall function-level MDC. The procedures in the left-hand columns shall be in the normal operational sequence, and shall account for the complete exercising of all functions within the equipment that result in a normal operating display.

3.2.2.2.1.5 Overall function information package. Overall function information, containing overall equipment alignment and adjustment procedures and all diagrams, text, test procedures, test parameters, and interfacing data that are required to isolate any potential equipment problem to a major function (see 3.2.2.2.2), intermediate subfunction (see 3.2.2.2.1.5.1.1), or hardware information package (see 3.2.2.2.3.3), shall be provided as an overall equipment information package. This information package shall include the following as a minimum:

a. Overall function diagram (see 3.2.2.2.1.5.1).
b. Keyed text (see 3.2.2.2.1.5.2).
c. Maintenance dependency chart (see 3.2.2.2.1.5.3).

3.2.2.2.1.5.1 Overall function diagram. The overall function diagram shall be the fundamental element of the overall information package. It shall be supported by necessary troubleshooting procedures, diagrams, test and specification data. The overall function diagram shall identify all major functions, the interrelationship and system interfacing relationships between the various major functions that comprise the equipment (see figure 21). The major functions shall normally be power distribution. Overall function diagrams shall be limited to a single page or two facing pages. The overall function diagram shall have a one-to-one correspondence with the major functions and intermediate functions (if any) shown on the functional family tree. Each major and intermediate subfunction shall be assigned an identification number (number within an oval) that shall correspond to the functional family tree (see 3.2.2.2.1.2.1). The overall function diagram shall show the input and output of the equipment. Intermediate subfunction diagrams can be used to expand sections between the input and output using the requirements criteria for the overall function diagram.

3.2.2.2.1.5.1.1 Intermediate subfunction diagram. Intermediate subfunction diagrams shall be provided whenever it's top level diagram cannot include all major function information on two facing pages. Intermediate subfunction diagrams shall be in a format of the overall function diagram and its supporting data (see 3.1.9.2.1). When the total number of major functions on the overall function diagram becomes too numerous to be shown clearly on a single page or two facing pages, two or more of the major functions may be subfunctionalized logically combined into one or more subfunctions to simplify presentation of the overall function diagram. A subfunction diagram shall identify all major functions and the interrelationship between the various major functions that comprise the intermediate function. The hierarchical information level presented by the intermediate function diagram is the same as that presented by the overall function diagram (see figure 5, column 2). However, to provide a smooth transition from information package to information package during the top-down troubleshooting effort, the intermediate function information package that contains the diagram shall be

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located immediately preceding the major information packages it defines (see 3.2.2.2.2.2).

3.2.2.2.1.5.1.2 Presentation. The overall function diagram (and intermediate subfunction diagrams) shall be arranged so that a user can readily understand and identify the major functions performed by the equipment (see 3.3). Each major function shall be represented by a single-level indicator. Each signal that interfaces between major functions shall be identified with a signal nomenclature name or code. Signal lines shall also have arrowheads that show the direction and nature of the information flow. Mechanical linkage interfaces between major functions shall also be shown. Each front panel switch or control, as well as indicators that are involved with the major function and the major function MDC, shall be identified by front panel markings, and shall be shown in a white background area. Gray shades shall be used to define the equipment boundaries.

3.2.2.2.1.5.2 Keyed text. Keyed text shall be in accordance with 3.4 and 3.1.9.2.2 and shall describe the operation of each of the major functions as it relates to the others and to the overall equipment operation.

3.2.2.2.1.5.3 Maintenance dependency chart. The MDC shall contain all data necessary to isolate a potential fault to a major function (see figure 22).

3.2.2.2.2 Major function information. Major function information (see 6.5.34) shall provide all diagrams, text, test procedures, test parameters, and interfacing data that are required to isolate a potential equipment problem that has been isolated to a major subfunction diagram or hardware information package. Major function information and intermediate function information (if any) shall be provided as follows:

a. Introduction (see 3.2.2.2.2.1).
b. Intermediate subfunction information package (see 3.2.2.2.2.2).
c. Major function information package (see 3.2.2.2.2.3).

3.2.2.2.2.1 Introduction. The introduction shall explain the major function (see 6.5.32) diagrams, their supporting keyed text and their MDC's or other troubleshooting diagrams.

3.2.2.2.2.2 Major function information package. Major function information, containing major function alignment and adjustment procedures (not included in the overall equipment alignment procedures), and all diagrams, text, test procedures, test parameters, and interfacing data that are required to further isolate any potential equipment problem in the major function to a hardware information package, module, or piece part, shall be provided in the major function information package. Major function information packages shall be identified and arranged in a sequence by the same identification numbers (number within an oval) used to identify the major function and subfunctions on the functional family tree and overall function and subfunction diagrams. This information package shall include the following as a minimum:

a. Major function and subfunction diagram (see 3.2.2.2.2.2.1).
b. Keyed text (see 3.2.2.2.2.2.2).
c. Maintenance dependency chart with signal specification and fault logic diagram (see 3.2.2.2.2.2.3).

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If the equipment maintenance philosophy is such that the requirement for acquiring hardware information packages is waived because the hardware item is designated nonrepairable by the Government, the major function information packages shall provide the following, as required:

a. Necessary repair/alignment instructions (see 3.2.2.2.2.3.4).
b. Necessary parts data (see 3.2.2.2.2.3.5).

3.2.2.2.2.2.1 Major function diagram. One major function diagram (see figure 23) shall be prepared for each functional block shown on the overall function diagram. The diagram shall show the interrelationships between the functional entities as established on the blocked schematics. Major function diagrams can apply to analog circuits, logic (digital) circuits, mechanical circuits, and power distribution, hydraulic, etc., or combinations thereof, and equipment ground/return. The functional bounds of the diagram shall be decided by the function performed and shall not be limited by the physical subdivisions of the equipment. Additionally, the functional bounds shall be chosen to minimize the number of interfaces to other major functions. Major function diagrams shall be limited to a single page or two facing pages. All interfacing functions and connections shall be included. When the total number of functional entities on a major function diagram becomes too numerous to be shown clearly within the one-page or two facing page limitation, two or more functional entities (see 6.5.21) nested within the major function may be logically combined into one or more subfunctions (see figure 24). Interfacing points between the nested subfunction and its major function shall be clearly indicated. The functional entities that comprise the subfunction are then shown on a subfunction diagram (see 6.5.50). The hierarchical information level presented by the subfunction diagram is the same as that presented by the major function within which the entities are nested (see figure 5, column 3).

3.2.2.2.2.2.1.1 Subfunction diagram. Subfunction diagrams (See figure 24 one of the subfunctions of figure 23) shall be constructed to simplify presentation of the major function diagram. Each subfunction diagram shall be limited to a single page or two facing pages. Subfunction diagrams shall be developed by the same rules used for major function diagrams. Interfacing points to the major function diagram within which the subfunction is nested shall be clearly indicated. Subfunction diagrams can be further subfunctionalized as necessary to maintain the two-facing page limitation.

3.2.2.2.2.2.1.2 Presentation. The major function diagrams (and subfunction diagrams) shall be constructed to illustrate the operation and interrelationships of the functional entities. The details of the diagrams shall be arranged so that major signals flow (see 6.5.48) from left to right to the maximum extent possible, and feedback signals flow from right to left. Power circuits shall normally be included on the power distribution diagram(s). Relay coils and electronic switches shall be grouped on the diagrams in a manner consistent with the equipment design or construction and by the logic operations performed. Such groupings may be by control functions, mode of operation, physical limits of cabinets or assemblies, and so forth. A functional name shall be shown for each relay coil and electronic switch. Mechanical linkage from a relay coil to its contact sets shall be used to signify the direction of motion for the moveable contact. Contacts of relays and switches shall be identified by the contact number/letter code which

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appears on the respective schematic. Lamps, indicators, and switches shall be labeled by reference designation and by the assigned panel. General and specific notes shall be used prolifically to clarify any conditions which are not apparent by observation of the diagram to minimize the need for supporting text. Signal specification identification shall be included prominently in the signal flow lines or test points in accordance with 3.3.11.

3.2.2.2.2.2.1.3 Symbology. Symbology shall conform to the examples shown on figure 10 and the following:

a. Analog circuits. Each functional entity on a blocked schematic, composed of piece parts (as opposed to integrated circuits), shall be represented by a line art triangle, truncated triangle, rectangle, or a square. Other electronic symbols in accordance with IEEE-315-86 may be used to improve understanding. See figure 11 for definition and use of the shapes for electronic/electrical diagrams.
b. Logic circuits. The distinctively shaped logic symbols as shown in IEEE 991-86 need not be used on the functional diagrams if they can be represented by normal functional block techniques. (i.e., a line art rectangular box shall be shown which represents the level of functions or group of functional entities. The line art box shall contain the same name or circuit identifier code as that used on the corresponding schematic located in the hardware functional information package. Inputs and outputs shall be from left-to-right, top-to-bottom. The size of the rectangle shall be chosen to accommodate the number of signal leads shown.
c. Mechanical circuits. Mechanical functional entity shall normally be represented by line art boxes, but may be line art illustrations. Electrical circuits in combination with mechanical components shall be as specified for analog circuit.
d. Power distribution diagrams. Power distribution diagrams shall depict the distribution of primary alternating current power, secondary power, and direct current power from the input plugs, terminal boards, breakers, switches, or fuse boxes to the various assemblies, subassemblies, and/or modules of the equipment. The major power distribution diagrams shall be generated as necessary to depict the entire power distribution within the equipment. It should normally be subfunctionalized by specific ac and dc power distribution (see figure 25). Whenever two or more power distribution diagrams are generated, an overall power distribution diagram shall also be generated to depict the overall power distribution in the equipment. The requirements for the symbology used on the power distribution diagrams shall be the same as those used for any of the functional diagrams. The following rules apply in the preparation of the power distribution diagrams:

(1) Motors, transformers, regulators, power supplies, assemblies, subassemblies, modules, power line devices, such as fuses, circuit breakers, switches, and relay contacts, connections, including plugs, jacks, terminal boards in the distribution path, metering circuits and indicators shall be shown and identified.
(2) Voltages and tolerances shall be included as required.
(3) Grounds, commons, neutrals, and return lines shall be shown.

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The symbology used shall be as that appearing on the associated schematic diagram.

(4) Whenever practical, power paths shall be displayed from left to right and top to bottom.

e. Equipment ground/return/shield diagrams. Equipment ground/return/shield diagrams shall be prepared for equipment which would require excessive diagram space to illustrate each equipment's grounding, signal returns, and shielding termination on functional block diagrams and power distribution diagrams. The symbology used shall be the same as that on the associated blocked schematic diagrams.

3.2.2.2.2.2.1.4 Interface with other functions. Energy, signals, or data entering or leaving a function must be interconnected and consistently identified to establish an interface between diagrams. The output of a function shall terminate in a functional area (enclosed by a shaded area or rows of dots) representing the major function diagram on which the signal is to be used. The input to a function shall originate from a functional area, representing the major function diagram from which it came. The interfacing input and output major function areas shall be identified by the major function name and function identification number used on the functional family tree and overall function diagram. Whenever possible, interface points shall be selected at connectors or other identifiable tie points where measurements can be made. These connectors or tie points shall appear on both interfacing functional diagrams. Except for cases involving mechanical motion, such as in relays, interfaces internal to single components shall be avoided. When interface tie-in is to other equipment, all signal nomenclature and other pertinent interface considerations shall be included.

3.2.2.2.2.2.1.5 Interconnections. Plugs, jacks, cables, piping, and junction boxes which provide interconnection between the units, assemblies, and so forth, shall be shown and identified. Point-to-point wiring between circuit elements (see 6.5.5) within an assembly need not be shown. However all interconnections between assemblies shall be identified.

3.2.2.2.2.2.1.6 Hardware identification. Each identifiable unit, assembly, or subassembly shall be identified as follows: major hardware units shall be identified in the upper left-hand corner of the hardware area with the unit reference designation, (nomenclature can be added if necessary for clarity); assemblies and subassemblies shall be identified in the upper left-hand corner of their respective hardware area by their reference designations.

3.2.2.2.2.2.1.7 Functional entity arrangement. Functional entities shall be grouped and enclosed in a functional area to illustrate their functional relationships. Each functional area shall be assigned a name that denotes the function (for example, "divide by 10 counter, AGC network, servoampliifier"). When individual parts function independently, the individual part shall be enclosed in a separate block and shall be properly identified. If blue shading is used, each major function diagram will contain only one level of blue shading. Functional blocks shall not overlap hardware boundaries unless the hardware shadings (gray) are cut back so that the functional area becomes a white or blue block.

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3.2.2.2.2.2.1.8 Functional entity identification. Each functional entity symbol shall contain the same circuit identifier code or name as the schematic entity which is being represented. Symbols representing a single part or device shall be identified by assigned reference designations.

3.2.2.2.2.2.2 Keyed text. Keyed text shall be in accordance with paragraphs 3.1.9.2.2 and 3.4 and figure 23 and shall describe the "what" and "when" of each functional entity.

3.2.2.2.2.2.3 Maintenance dependency chart/fault logic diagram. Unless it serves no justifiable purpose, an MDC or fault logic diagram shall be prepared for each major function/subfunction diagram and shall contain data necessary to isolate a potential fault to a subfunction, hardware, or piece part mounted on the mainframe. MDC development shall be in accordance with 3.5. Trouble-shooting information for major functions which contain large amounts of digital circuitry may consist of fault logic diagrams to substitute for or augment the MDC (see 3.2.2.2.2.2.3.1). When an automatic test program is used with built-in performance monitoring/fault localization circuits, MDC's or fault logic diagrams are required for the performance monitoring/fault localization circuitry and test programs. MDC's or fault logic diagrams are not required for functions in which fault localization is accomplished by BITE or ATE. When extensive Built In Test Equipment (BITE) or Automatic Test Equipment (ATE) is utilized, fault isolation using such equipment shall be provided in a section following the hardware information (see 3.2.2.2.4).

3.2.2.2.2.2.3.1 Fault logic diagrams. Fault logic diagrams shall be based on a fault indication observed during troubleshooting (see figure 26). The diagram shall comprise a branching series of questions pertaining to fault isolation. Each question shall pertain to a further observation or measurement, and shall result in a "yes" or "no" answer, thereby progressively narrowing the possible functional area of the fault. Tolerance values shall be presented in those instances where a definitive "yes" or "no" is not obtained. This progression and elimination shall isolate the functional area of the equipment containing the fault, and then refer the user to the portion of the manual containing the information needed to complete the fault isolation and repair. Each diagram shall include or make reference to information establishing the initial conditions required for the fault isolation procedure.

3.2.2.2.2.2.3.2 Combined fault logic and maintenance dependency diagram. The combined fault logic and maintenance dependency diagram provides a faster method of troubleshooting than either the fault logic diagram or MDC used alone (see figure 27). The diagram shall be composed of a branching series of questions pertaining to fault isolation. Each question shall pertain to observations or measurements that can result in a "yes" or "no" answer. Each "no" answer will direct the user to a simplified MDC of the circuit(s) that produced the observations or measurements. The simplified MDC and its associated signal specifications will direct the user to the fault location (module, components, or piece part).

3.2.2.2.2.2.4 Repair/alignment instructions. Whenever the maintenance requirements are organizational only and no hardware repair/alignment is acquired, repair and alignment instructions shall be provided for the major functions. Instructions shall be included for the removal, repair,

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adjustment, and replacement of assemblies, subassemblies, modules, or parts that have adjustments. Interacting adjustments shall be referenced to the equipment alignment procedures.

3.2.2.2.2.2.4.1 Repair illustrations (CG, F, MC, N). Repair illustrations shall be provided in accordance with 3.2.2.2.3.3.4.2.1.

3.2.2.2.2.2.4.2 Repair illustrations (A). Repair illustrations shall be provided in accordance with 3.2.2.2.3.3.4.2.2.

3.2.2.2.2.2.5 Parts data. Whenever the maintenance requirements are organizational only and no hardware parts data are required, parts data shall be provided for the major functions. Parts data shall consist of a parts location diagram and a parts data table. The parts data shall list and identify all replacement items.

3.2.2.2.2.2.5.1 Parts location diagram. Line art diagrams shall be used for parts location whenever possible. Expanded views shall be used with the main illustration to highlight areas of importance. A sufficient number of illustrations and views shall be used so that all reference-designated items can be easily located. Also, in equipments having modular construction, an illustration shall be provided to identify each module and its location. For attaching parts, not assigned reference designators, a table of attaching hardware shall be included.

3.2.2.2.2.2.5.1.1 Location method. All assemblies and items assigned a reference designation or name, including hardware parts referenced in disassembly and repair procedures, shall be located by means of leaders and callouts or an alphanumeric coordinate system for location.

3.2.2.2.2.2.5.2 Parts data table. Parts data shall be listed in tabular form with columns headed as follows:

a. Column 1 - Reference designation. This column shall list the items by reference designation in alphanumeric sequence.
b. Column 2 - Locating coordinate or figure reference and locating procedures. This column shall provide the locating coordinate of the item (if grid method is used) or item number when leaders and callouts are used. A figure reference column shall be provided to eliminate ambiguity when more than one parts location diagram is required.
c. Column 3 - Name and description. When a separate parts list is provided as specified in 3.2.2.1.6, this column need only provide the generic name of the item. When a separate parts list is not specified (see 6.2), this column shall contain descriptive data to identify the parts of the equipment. The parts list shall include, but not be limited to, such designated repair items as electrical-electronic circuit elements, valves, bearings, special-threaded fittings, transformers, and power circuit breakers, as well as replacement items such as contact tips, coils, O-rings, gaskets, shims, gears, and brushes. Parts shall be identified by part or identifying number (PIN), contractor's number, or manufacturer's number (drawing number). The contractor and manufacturer may be identified by name or CAGE code. The quantity of the attaching

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hardware, as shown on the illustration, shall be added in parenthesis after the description. Common hardware items such as nuts, bolts, and washers that are within the mechanical assemblies also shall be listed. Common hardware items of electronic chassis need not be listed unless they are referenced in a procedure. Hardware items with special characteristics (for example, stainless steel, nonmagnetic, and extra hardness) shall be listed.

3.2.2.2.3 Hardware information. Hardware information for the trouble-shooting/repair volume (see figure 5) shall be extensively detailed and consist of the following:

a. Introduction (see 3.2.2.2.3.1).
b. Integrated circuit data (see 3.2.2.2.3.2).
c. Hardware information packages (see 3.2.2.2.3.3).

3.2.2.2.3.1 Introduction. The introduction shall provide a description of the schematics, keyed text, MDCs, alignments and repairs, procedures, parts, and wiring data.

3.2.2.2.3.2 Integrated circuit data. The technical data for each type of IC module shall be provided (see figure 11). This data shall include, but not be limited to such things as:

a. Type and designation.
b. The base configuration, indicating the key or index, and power and return pin numbers.
c. Logic diagrams for all two-state IC's.
d. Truth tables for all two-state IC's.

3.2.2.2.3.3 Hardware information packages. Hardware information packages shall be prepared for units, assemblies, and subassemblies or combinations thereof. Hardware information packages can be combined on one or more pages, provided the data are clearly arranged in reference designation sequence. When three or more common assemblies are contained in the equipment/system, a common hardware information package shall be included and identified (see

3.6.3.2.2.1). Hardware information package shall consist of the following, as applicable:

a. Blocked schematic diagrams (see 3.2.2.2.3.3.1).
b. Keyed text (see 3.2.2.2.3.3.2).
c. Maintenance dependency chart* (see 3.2.2.2.3.3.3).
d. Alignment and repair* (see 3.2.2.2.3.3.4).
e. Parts data* (see 3.2.2.2.3.3.5).
f. Wiring diagrams (see 3.2.2.2.3.3.6).

*MDC's, repair data, and parts data are not required for assemblies contractually designated as non-repairables. However, schematic, logic, and other relevant information needed for understanding functions and required theory of these items is required (see figure 28) for hardware information package layout.

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3.2.2.2.3.3.1 Blocked schematic diagrams. One or more blocked schematic diagrams shall be provided for each unit, assembly, and subassembly as needed. Identical assemblies need be illustrated only once, but shall include all assigned reference designators (See 3.6.3.2.2.1). A schematic for each drawer or chassis and all other enclosures shall be provided.

3.2.2.2.3.3.1.1 Presentation. The blocked schematic shall be arranged to produce an easy-to-understand functional layout. Circuit elements shall be grouped into functional entities (see figure 28). The arrangement of the elements within functional entities shall assist in circuit comprehension for a user trained in the related subject. The location of functional entities and devices on the page shall cause the direction of signal flow to be obvious and, when possible, from left to right. When possible, feedback signal flow shall be from right to left. Voltage and resistance information shall be included on the diagrams of vacuum tube and transistor circuits as an aid to fault localization and isolation. The voltage or resistance value at each pin or test point shall be placed next to the pin or test point on the diagram or on a separate voltage and resistance table. Significant signal specifications shall be included prominently in the signal flow lines or test points in accordance with 3.3.11.

3.2.2.2.3.3.1.2 Symbology. Schematic symbols for discrete electrical or electronic circuits parts shall be in accordance with IEEE 315-86. Schematic symbols for logic integrated electronic circuits shall be in accordance with the distinctive shape requirements of IEEE 991-86. Identification of terminals on relays, transformers, potentiometers, switches, and so forth, shall be in accordance with ANSI Y14.15. Symbols for mechanical, hydraulic, and pneumatic devices shall clearly portray how the parts relate and operate to produce the motion or energy transformation required. To indicate signal flow on fluid power piping diagrams, symbols in accordance with ANSI Y32.10 may be used. Except when a pictorial presentation is not feasible or practical, mechanical symbols shall conform to MIL-STD-17-1 (except for aeronautical/space, naval ships, and architectural/building), MIL-STD-17-2 (for aeronautical/space), Drawing 803-5001049 (for naval ships), or ANSI Y32.4 (for architectural/ building), as applicable.

3.2.2.2.3.3.1.3 Functional entities. Circuit elements and functional devices (see 6.5.20) shall be grouped into functional entities according to the following rules:

a. Circuit elements and functional devices that work together to perform a basic operation shall be grouped as a single functional entity (for example, amplifiers, emitter followers, or multivibrators).
b. The symbols representing portions of integrated circuit elements working together to perform a basic function shall be grouped to form functional entities (for example, amplifiers, gates, and flip-flops). Discrete circuit parts (see 6.5.10) associated with an integrated circuit element such as pull-up resistors or integrating capacitors, shall be grouped within the integrated circuit functional entity.
c. Basic groupings shall consist of the least number of circuit elements or functional devices that perform a discrete operation.
d. Parts of mechanical, hydraulic, and pneumatic devices, such as gears and pumps, shall be grouped so that the functional entity portrays a discrete operation (see foldout 1).

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3.2.2.2.3.3.1.3.1 Electrical power circuits. Circuit elements and functional entities involved with the control and distribution of electrical power within an assembly, such as decoupling filters, voltage dividers, and voltage regulators, shall be grouped within a composite power entity (C-PWR-1). When possible, the composite power entity shall be positioned in the lower left side of the schematic (see figure 28).

3.2.2.2.3.3.1.3.2 Functional entity identification. Each functional entity shall be enclosed in a functional area (see figure 28). The entity identifier codes that describe the functional circuits usually consist of three or more letters and a number(s) arranged as shown on figure 10. The list on figure 11 does not include all possible combinations of coding. New codes may be generated to meet specific needs. To the maximum extent possible, the functional entity identifier codes shall be placed in the upper left-hand corner of their respective functional areas. An encircled keyed text-reference number shall precede the identifier codes. A single circuit element or functional device which constitutes the complete functional entity shall be identified by its assigned reference designation. Solenoids of all electrical-mechanical devices, such as relays, valves, and brakes, shall be identified by a functional name in addition to any reference designation (see foldout 1).

a. The first letter of a functional entity code shall define the general characteristics of the functional entity.

C - composite
H - hydraulic
I - integrated circuit
L - magnetic
M - mechanical
N - network (passive)
P - pneumatic
Q - transistor
V - electron tube
U - microwave
X - semiconductor diode

b. The second group of letters shall define the purpose of the functional entity.

AMP - amplifier
BUF - buffer
EF - emitter follower
FF - flip-flop
GAT - gate
OSC - oscillator
VD - voltage divider

c. The number shall differentiate between similar classifications of a functional entity used more than once in an assembly.

Q-OSC-1
Q-OSC-2
and so forth.

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3.2.2.2.3.3.1.3.3 Treatment of logic circuits. Schematic symbols for integrated electronic logic circuits (see foldout 2) shall be in accordance with the distinctive shape requirements of IEEE 991-86. Logic functional entities shall be grouped into two functional levels. The first functional level shall be assigned a function name such as, DECODER or DIVIDE BY 10 COUNTER. The assigned name and keyed text number shall appear in the upper-left corner of this area. The distinctive-shaped symbols of the logic circuits denoting the logic circuit shall appear within the boundaries of the functional area and be assigned a circuit identifier code. Flow charts for digital devices shall be provided to support the explanation of machine instructions and test programs, and shall conform to FIPS PUB 24.

3.2.2.2.3.3.1.3.3.1 Integrated circuits and microminiature capsules (N). Simplified schematic data in accordance with NAVSEA 0967-LP-000-0120 or logic diagrams showing input and output connections, truth tables, and waveforms as applicable, supported by a brief description of overall operation, shall be included for circuits within different types of nonrepairable integrated circuits and microminiature capsules. Manufacturer's data sheet information may also be included if it contains the technical contents of this requirement, and meets the format and legibility requirements as specified herein.

3.2.2.2.3.3.1.4 Mechanical devices. Diagrams shall show sufficient detail to indicate the operational sequence and arrangement of a mechanical device, including the electrical control circuits (see foldout 1). Nomenclature, symbols, part identification numbers, and necessary descriptive data shall be shown as required. Gears, shafts, clutches, levers, mechanical ratios or number of teeth, and direction of rotation, and so forth, shall be given.

3.2.2.2.3.3.1.5 Electronic tube filaments. Tube filaments shall not be treated as functional entities. Tube filaments (except those filaments which are an integral part of the cathode) shall be grouped and positioned just above or to the right of the composite power entity (see 6.5.8). Tube filament groupings shall show each filament pin connection and identify each filament by its associated tube reference designation.

3.2.2.2.3.3.1.6 Power circuit distribution lines. Power circuit distribution lines shall terminate with a break symbol (---) inside the composite power entity. Power, ground, and return lines shall not be shown connected to the functional entities by lines, but connections shall be represented by use of symbols for grounds and returns, and by break symbols for supplied power. Distribution power to integrated circuits may be indicated by statements in the general and specific notes. The voltage (V) shall be indicated at the break symbol of each distribution line (---- +5V). When the voltage data are not definable because of varying circuit parameters, then the distribution line shall be identified at the break symbol by its polarity and identifying letter(s) [for example, +V(A), +V(B), -V(A)].

3.2.2.2.3.3.1.7 Hardware identification. Each unit, assembly, or subassembly which has a military nomenclature or reference designation shall be identified by means of gray shading, and as follows:

a. The reference designation followed by the name of the assembly shall appear in the upper left-hand corner of the hardware area.

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b. The words REF DESIG PREFIX followed by the reference designation shall appear in the lower right-hand corner of the hardware area.

3.2.2.2.3.3.1.8 Circuit element data. In addition to the assigned reference designation, the following types of data for circuit elements or functional devices shall be included:

a. Resistors - resistance value, tolerance, and wattage.
b. Capacitors - capacitance value, tolerance, voltage rating, and polarity (if applicable).
c. Coils, motors, and other wire-wound devices - value, tolerance, and resistance when significant.
d. Power transformers - the measured voltage across the secondary windings.
e. Pumps - volume per unit time.
f. Breakdown diodes - zener voltage level.
g. Diodes and transistors - type number (JAN designation, if available).
h. Integrated circuits. IC element symbols shall contain an IC reference designation. Pin numbers shall be positioned next to the input and output lines, external to the symbol. The description of inputs and outputs of flip-flops, dividers, and so forth, shall be identified within the symbol (for example, C, D, J, K, Q). Identification of noninverting and inverting inputs to operational amplifiers shall be identified by plus (+) and minus (-) signs within the symbol.

3.2.2.2.3.3.1.9 Unused terminal information. Spare relay contacts, connector pins, terminal connections, other connective-type devices, switch contact sets, or spare leads, and so forth, which are available for future use shall be shown either in the schematic area or listed in tabular form on the same or facing page.

3.2.2.2.3.3.1.10 Notes and source data blocks. A portion of the diagram, preferably in one of the four corners of the schematic, shall contain such notes as are required, and the source data of the diagram. Source data references used throughout the manual shall not include revision indicators. Source data with the appropriate revision indicators shall be summarized in the table of source data on the list of effective pages. General notes shall be of the "unless otherwise specified" variety (for example, "resistance values are in ohms", "diodes are type 1N270"). Flag notes shall explain special or unusual situations shown on the diagram (for example, "R10 is 15K for serial numbers 22 and up, C13 is selected per receiver alignment procedures"). The source data block shall identify the specific engineering documents (and their issue) used to create the blocked schematic.

3.2.2.2.3.3.1.10.1 Multiple-use assembly table. A multiple-use assembly table shall be provided in the notes area when identical assemblies are used in the equipment. The table shall indicate the hierarchical location of the assemblies.

3.2.2.2.3.3.1.11 Referencing between assembly and subassembly schematic diagrams. The destination of all assembly and subassembly jacks/plugs which connect to a different unit than the one in which they are installed, or which connect to an assembly or subassembly in a different hardware information

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package, shall be provided so that direct access can be made to the proper package schematic, without reference to the functional index.

3.2.2.2.3.3.2 Keyed text. Keyed text shall be provided for each blocked schematic diagram in accordance with 3.4 and figure 28, and shall describe each functional entity included on the blocked schematic.

3.2.2.2.3.3.3 Maintenance dependency chart. An MDC (see figure 33) shall be supplied for each different assembly in accordance with 3.5 to permit fault isolation to the functional entity level (piece part). Certain assemblies may not require MDC's if they contain few functional entities (front panels, mainframes, etc.), or if the functional entities are covered on another MDC because related piece parts, such as servoamplifiers, are on different assemblies.

3.2.2.2.3.3.4 Alignment and repair. Overall alignment and repair instructions shall be included in accordance with the following paragraphs.

3.2.2.2.3.3.4.1 Alignment. Each assembly containing adjustment parts shall be provided with detailed alignment instructions. Interacting adjustments shall be referenced to the equipment alignment procedures.

3.2.2.2.3.3.4.2 Assembly repair. Instructions shall be provided for removal, repair, adjustment, and replacement of assemblies, subassemblies, and parts when the method or procedure is not obvious (for example, removal of other assemblies is necessary in order to gain access). Information such as mechanical adjustments, tolerances, clearances, wear limits, bolt-down torques (including bolt-head identification), and in-place balancing shall be included.

3.2.2.2.3.3.4.2.1 Repair illustrations (CG, F, MC, N). Exploded and sectional views giving details of mechanical assemblies (such as gear trains) shall be provided as necessary to supplement detailed removal, replacement, or disassembly instructions (see figure 29). Mechanical and electromechanical parts covered in the procedure shall be keyed to the illustration by a callout (index) number. These parts shall be tabulated (see figure 28) and described by reference designation and item name, or manufacturer or contractor part number. However, only one identification number may be used for each item. The illustration and procedures shall be provided on the same or facing page, when practical. Information on the use of special tools and test equipments supplied with the equipment shall be provided. Any WARNINGS or CAUTIONS which must be observed to protect personnel or equipment shall also be included in accordance with 3.1.5.

3.2.2.2.3.3.4.2.2 Repair illustrations (A). Exploded and sectional views giving details of mechanical assemblies (such as gear trains) shall be provided to supplement detailed piece part removal, replacement, or disassembly instructions. Each part covered in the procedures shall be keyed to the illustration by callout (index) number. Views and procedures shall be provided on the same or facing page. Any WARNING or CAUTION statements required to protect personnel or equipment or both shall be included as specified in 3.1.5. A statement shall be made that the special tools, parts, and special equipment sets supplied with (or issued for use with) the material

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are listed in the Basic Issue Items List and Items Troop Installed or Authorized List, or Repair Parts and Special Tools List, as applicable; and that common tools are authorized in the Table of Organization and Equipment, Modified Table of Organization and Equipment, Table of Distribution and Allowance, or Modified Table of Distribution and Allowance. These items need not be described or explained. However, the proper use of certain tools in these sets may be illustrated in connection with assembly, disassembly, or adjustment of the material when appropriate. The use of specially designed tools and equipments issued for the purpose of maintaining or adjusting the material shall be briefly described.

3.2.2.2.3.3.5 Parts data. Parts data shall be provided for each government designated repairable assembly (including mainframe assemblies, cabinets, and other enclosures) of the equipment. Parts data shall consist of a parts location diagram and a parts data table. The parts data table shall list and identify all repair parts. Mechanical parts that are subject to replacement (although not provisioned as spare parts, such as handles, slides, plates, and covers) and included in the provisioning documentation shall be assigned an MP number, be included in the parts data table, and be identified on the respective parts location diagram.

3.2.2.2.3.3.5.1 Parts location diagrams. Parts locations diagrams shall be provided in accordance with 3.2.2.2.2.2.5.1.

3.2.2.2.3.3.5.1.1 Printed circuit boards. Printed circuit boards shall be illustrated foil side up. When printed wiring appears on both sides of the board, both sides shall be illustrated. Parts mounted on the board shall be outlined in black solid (front) or dashed (rear) lines (even though mounted on the reverse side of the board), and their connections to the printed wiring clearly illustrated. If insufficient room exists, separate illustrations with top and bottom views shall be provided. Each part shall be labeled with the applicable reference designation. To facilitate parts location, a locating grid and corresponding guide chart shall be provided when more than 30 items are mounted on a board. Boards of flat packs shall be similarly presented.

3.2.2.2.3.3.5.2 Location method. Illustrations locating 15 or more items shall use a coordinate system for parts location. Leaders and callouts, index numbers, or reference designators can be used to augment the grid system for clarity. Assemblies and parts assigned a reference designation or name, including hardware parts referred to in disassembly and repair procedures, shall be located by means of an alphanumeric coordinate system and leaders and callouts. Exploded views shall include an index table with the following columns: index number (when applicable), zone (locating coordinates), part name, quantity, and reference designation. If index numbers are used, each train of parts on an exploded view shall be consecutively numbered beginning at the left of the illustration, or in the order of disassembly/assembly. The coordinate locators shall appear on all four sides of the parts location diagram. Zone size shall be as required to clearly locate all reference-designated parts.

3.2.2.2.3.3.5.3 Parts data table. A parts data table shall be provided as specified in 3.2.2.2.2.2.5.2.

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3.2.2.2.3.3.6 Wiring diagrams. A pictorial diagram (see foldout 3) or wire list (see figure 30) showing point-to-point wiring shall be prepared for each chassis, back plane, and front panel assembly. Identification of all terminals for relays, transformers, switches, and so forth, shall be in accordance with ANSI Y14.15. Wiring diagrams for printed circuit boards are not required. Entries in wire lists shall be arranged in alphanumeric order. Wiring diagrams or wire lists shall be located with the appropriate hardware information package. Wire lists shall always be separated from the corresponding parts data table (see 3.2.2.2.2.2.5.2).

3.2.2.2.4 BITE/ATE fault isolation information. When the maintenance philosophy of the equipment requires extensive use of computer or microprocessor-based BITE or ATE to isolate faults, the manual shall include but not be limited to, a Built-In Test (BIT) functional block diagram, BIT fault indication tables, numeric fault code tables, and fault isolation procedures/logic diagrams. This material shall be included in the overall or major function information package, as applicable. If more appropriate, the material shall be included in the performance monitoring fault isolation unit hardware package.

3.2.3 System manuals (type II). System manuals shall contain the data designated on figure 5. System manuals shall provide system-oriented instructions for operation, maintenance, installation, and performance standard testing. Detailed equipment data should be provided by reference to the equipment manuals. However, equipment-level technical data in accordance with 3.2.2 shall be included to augment the requirements specified here for equipment or component not covered by an equipment technical manual. For systems composed of units or components not covered by individual manuals, separate hardware information packages in accordance with 3.2.2.2.3.3 shall be provided for all major units, assemblies, subassemblies, and components with all information diagrams and procedures necessary to isolate any potential failure to a unit, assembly, subassembly, module, circuit board, or piece part as defined by the maintenance philosophy for the item.

3.2.3.1 Support volume (type II). The support volume for systems/subsystems shall be arranged in the following order and shall conform to the referenced paragraphs.

a. Front matter (see 3.2.1).
b. Chapter 1 - General information (see 3.2.3.1.1).
c. Chapter 2 - Safety precautions (see 3.2.3.1.2).
d. Chapter 3 - Operation (see 3.2.3.1.3).
e. Chapter 4 - Theory of operation (see 3.2.3.1.4).
f. Chapter 5 - Maintenance procedures and standards (see 3.2.3.1.5).
g. Chapter 6 - Alignment (see 3.2.3.1.6).
h. Chapter 7 - Installation (see 3.2.3.1.7).
i. Chapter 8 - Parts list (see 3.2.3.1.8).
j. User activity comment sheet (see 3.2.2.1.7).

3.2.3.1.1 Chapter 1 - General information. Chapter 1 shall describe the system in general physical and functional terms as follows:

a. Introduction (see 3.2.3.1.1.1).
b. Physical arrangement (see 3.2.3.1.1.2).

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c. System equipment (see 3.2.3.1.1.3).
d. Associated system equipment (see 3.2.3.1.1.4).
e. Reference data (see 3.2.3.1.1.5).

3.2.3.1.1.1 Introduction. The introduction shall define the system and its relationship with other systems. The mission of the system shall be stated. The text shall be supported by a diagram(s) showing the interrelationships of the system equipments (see figure 31). The major functional relationship of inputs and outputs to and from related systems shall be indicated.

3.2.3.1.1.2 Physical arrangement. System areas and compartments shall be described, and the system equipment and units contained in the areas shall be listed. The physical arrangement description shall be supported by the following illustrations:

a. Outboard and inboard profile drawings of the vehicle, aircraft, ship, building, or station showing compartment, room, or area locations and identifying equipment units comprising the system (see figure 32).
b. Separate illustrations of each compartment, room, or area, identifying the listed system equipment (see figure 33). Other equipments in the compartments, rooms, or areas need not be listed in the text or called out on the illustrations if they do not directly affect the operation or maintenance of the subject system.

3.2.3.1.1.3 System equipment. Each of the equipments comprising the system shall be illustrated, identified, and described. Descriptions of operator-attended equipment shall include statements as to the nature and purpose of units, controls, and indicators. The text shall be supported by illustrations. All equipment shall be shown in relative scale, though not necessarily illustrated proportionally. Each equipment shall be separately illustrated with significant features called out for proper support of the text used to relate equipment to each other.

3.2.3.1.1.4 Associated system equipment. Descriptions and illustrations of associated system equipment shall be limited to the major units thereof. The descriptions shall be more condensed than those of subject system equipments; otherwise, the same requirements are applicable. In the description, emphasis shall be placed on those associated system equipment that constitute operational or functional interfaces with the subject system. Such units shall be included in the system illustrations.

3.2.3.1.1.5 Reference data. Reference data shall include a list of the equipments comprising the system and their official designations. A list of common names and abbreviated or informal nomenclature, as well as system characteristics and a list of referenced publications shall also be included:

a. Capabilities. A summary of system capabilities shall be provided. The summary shall include data such as gallons per minute, transfers per hour, boom capacity, related ranges, resolution, accuracy, data handling capability, and number of channels. Such data shall be presented in tabular form.
b. Reference publications. A list of manuals that pertain to the system and system equipment, and other documents of interest (such as training manuals and manuals of associated system equipment) shall be

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included in a list of publications. This list shall include the title and publication number of the referenced publications.

c. Tools and test equipments. A list of all special tools and test equipments for system-level maintenance tests shall be included. Selection of test equipments shall be in accordance with 3.2.2.1.1.6. Special tools are defined as those tools not listed in the GSA Supply Catalog. [Copies of this catalog may be found in offices of the Defense Contract Administration Services Management Area (DCASMA).] An illustration and description of special items required shall be provided for identification. Information shall be presented in tabular form. For each item, the description or table shall include the following:

(1) Official name or nomenclature.
(2) Part or identifying number (PIN).
(3) A brief description of the item's use with a reference to the procedure(s) requiring its use.

3.2.3.1.2 Chapter 2 - Safety precautions. Chapter 2 shall describe the hazards associated with system operation and maintenance. To permit wide and unrestricted use, chapter 2 should contain only unclassified information as follows:

a. Introduction (see 3.2.3.1.2.1).
b. Electromagnetic radiation hazards and precautions (see 3.2.3.1.2.2).
c. Hazards to divers (see 3.2.3.1.2.3).
d. System hazards and precautions (see 3.2.3.1.2.4).
e. WARNINGS, CAUTIONS, NOTES and ICONS (see 3.2.3.1.2.5).

3.2.3.1.2.1 Introduction. This introduction shall be directed to system supervisory personnel, and shall include the following:

a. Purpose, scope, and organization of the system safety instructions.
b. Basic safety concepts.
c. Basic responsibilities for safety.

3.2.3.1.2.2 Electromagnetic radiation hazards and precautions. Radiation hazards to personnel and the precautions to be taken shall be described. The hazards of radiation to flammable or explosive materials shall also be described. The descriptions shall include the following:

a. Locations of radiation hazardous areas.
b. Minimum safe distance on the axis of beam radiation.
c. Precautions to be taken when entering areas of radiation hazard (such as the wearing of copper-screen goggles).
d. The effects of radiation on flammable or explosive materials (such as the induction of radio frequency (RF) currents in metals, sparking, and the consequent possibility of igniting flammables or detonating explosives). The text shall be supported by one or more illustrations identifying the areas of radiation hazards and the location of antennas.

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3.2.3.1.2.3 Hazards to divers. When applicable, the description of hazards to divers shall include, but not necessarily be limited to, the following:

a. The nature and intensity of sound energy in water as related to divers.
b. Precautions to be taken when working near operating equipment.
c. Illustrations identifying the hazardous areas and locations of sonar domes, propellers, water intakes, and so forth.

3.2.3.1.2.4 System hazards and precautions. Descriptions of system hazards and precautions, addressed to system personnel and referenced to particular system equipments, shall be included. The descriptions shall be organized to be consistent with the operation of the system. The descriptions shall supplement and extend equipment safety instructions to the system level by warning of potential hazards that can be present during operation or maintenance.

3.2.3.1.2.4.1 Operational safety summary. A summary shall be included which emphasizes the proper use of equipment controls and describes the hazards to operators. It shall include a listing of hazards to persons in areas remote from the operation. This summary shall also recommend operator precautions. An emergency procedure shall be included which highlights the controls that control immediate braking or deenergizing of the system.

3.2.3.1.2.4.2 Maintenance safety summary. A maintenance safety summary shall emphasize the proper use of controls, describe the hazards to maintenance personnel, potential damage to the equipment, and recommended precautions.

3.2.3.1.2.4.3 Hazardous components. Hazardous components, including radioactive devices and elements used with the system, shall be identified and briefly described. The description shall summarize the general handling precautions for such components. The description of a hazardous component shall include brief statements as to the purpose, manner of functioning, nature of built-in safety devices, and nature of the hazardous element. It shall also indicate the relative sensitivity of the component to mechanical shock, vibration, electromagnetic and radioactive radiation, and electrostatic discharges.

3.2.3.1.2.5 Warnings, cautions, notes and icons. WARNINGS, CAUTIONS, NOTES blocks and ICONS shall be included on applicable pages, as specified in MIL-M-38784 and 3.1.5.

3.2.3.1.3 Chapter 3 - Operation. Chapter 3 shall be divided into two sections as follows:

a. Section 1 - Operator instructions (see 3.2.3.1.3.1).
b. Section 2 - Conditions of readiness (see 3.2.3.1.3.2).

3.2.3.1.3.1 Section 1 - Operator instructions. Operator instructions shall describe system operating situations, modes, and procedures. The descriptions shall be detailed to the level required for an understanding of the operational interfaces of the system equipments and associated systems. Illustrations shall be included when necessary for clarity (see figure 14). The various operating modes shall first be described to acquaint the operator

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with all equipment combinations that can be used to effect a given mode of operation.

3.2.3.1.3.1.1 Preoperational conditions and setup. These instructions shall list specific preoperational conditions presumed to be in effect prior to system operation. A system readiness checkoff list of significant switch positions and indicator status shall be tabulated. Controls provided for maintenance of self-tests shall be included in the checkoff list. For indicators such as dials, where a band of readings is possible, upper and lower limits shall be delineated. The initial conditions of associated system equipments that directly affect system operation shall also be given.

3.2.3.1.3.1.2 Operating modes. The primary operating mode shall be discussed in detail, and alternate modes shall be treated as modifications of the primary mode. Operating procedures common to all modes shall be detailed under the primary mode and referred to under the alternate modes, with such modifications of procedures as may be necessary. Each mode shall be described in the logical sequence of major phases, events, options, supervisory commands, responsive actions, and the following:

a. Those equipment operational controls and indicators having system significance shall be explained in the description. When controls must be actuated and indicators observed in a sequence to achieve system operation, the descriptions shall cite each control and indicator with a number to indicate the sequence.
b. WARNINGS and CAUTIONS shall be used to emphasize the safe operation of controls where improper use could result in hazards to personnel or damage to the equipment. Each control shall be followed by a brief description of its effects (equipment actuation or display indication, or both) at the operator station and at remote stations. The primary mode description shall be supported by both general and detailed illustrations.
c. Operational phases which involve operator judgment shall be illustrated by operational diagrams. The diagrams shall indicate the conditions that must be favorable prior to an operator action or, if unfavorable, indicate the alternate action. Illustrations displaying such items as dials, gauges, status lights, which indicate the favorable or unfavorable conditions, shall be included. Special procedures to be followed during equipment failure (separate from emergency procedures) shall also be described.
d. Operational conditions that involve physical danger or hazardous material shall augment WARNINGS and CAUTIONS by use of ICONS in accordance with MIL-M-38784.

3.2.3.1.3.1.2.1 Normal operation. The duties of system operators shall be described in terms of general responsibility and specific step-by-step procedures for operating the system in all of the primary modes. System controls and indicators provided only for maintenance and nonsystem application shall be omitted.

3.2.3.1.3.1.2.2 Emergency operation. Step-by-step procedures shall be provided for emergency operation of the system. If specially designated controls have been provided for emergencies, a short statement shall be

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included describing how they modify or otherwise affect normal system operation. Emergency procedures shall be supported by illustrations.

3.2.3.1.3.1.2.3 Special operations. Special operations such as test checkout, training, or evaluation exercises shall be described. Supporting illustrations shall include block and pictorial diagrams.

3.2.3.1.3.2 Section 2 - Conditions of readiness. This section of the operation chapter shall be provided for guidance purposes and shall represent engineering considerations. It shall list conditions of readiness requirements prior to the operation of the system. Activities or conditions within rooms or compartments or areas outside the system shall be described as necessary.

3.2.3.1.3.2.1 Conditions coverage (N). An orientation for system personnel shall be included. The conditions of readiness presented shall include the following:

a. Watch condition.
b. General quarters.
c. Emergency condition.

Coverage shall include, but not be limited to, the following:

d. A block diagram showing signal paths of the conditions of readiness (see figure 34).
e. Block diagrams and descriptions of interior communications links between compartments and areas, including telephone circuits, announcing systems, and closed-circuit television.
f. Presumptions relative to system status at specific times (for example, from watch condition to general quarters, or from general quarters to emergency condition).

3.2.3.1.3.2.1.1 Watch condition. The condition of all system equipments and the personnel situations (how many of what class/type of personnel to be stationed where) for watch condition shall be described. Control settings shall be described and illustrated.

3.2.3.1.3.2.1.2 General quarters. The general quarters description shall detail the condition of all system equipments and the personnel situations. The control settings shall be described and illustrated.

3.2.3.1.3.2.1.3 Emergency condition. The condition of all system equipments and the personnel situations for emergency condition shall be described. The control settings shall be described and illustrated.

3.2.3.1.4 Chapter 4 - Theory of operation. This chapter shall contain all theory of operation necessary to inform the supervisory personnel and technicians of the general principles of operation beyond the actual functioning of the system. Examples of this information are missile trajectory solutions, computing principles, basic logic upon which the system operation is based, tactical data system operations, concepts of missile handling, missile system operation, and navigation theory. In addition, this chapter shall be used to expand on the information provided in the keyed

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text of the troubleshooting/repair volume. Illustrations such as patterns, nomographs, maps, and simplified diagrams shall be used to reduce narrative.

3.2.3.1.5 Chapter 5 - Maintenance procedures and standards. Chapter 5 shall contain all system scheduled test procedures, together with necessary explanations and illustrations. These procedures shall be correlated with the installation checkout requirements of 3.2.3.1.7.9, such that any maintenance or performance test procedure also required for checkout may be properly referenced in chapter 7 (see 3.2.3.1.7 for installation data). The installation standards summary sheet (see 3.2.3.1.7.11) shall also include space for any maintenance or performance test result that is recorded by the installer. The recorded information will provide a reference for the technician while troubleshooting, or when needed for installation acceptance certification. (Scheduled system maintenance may be planned in accordance with a schedule and procedure set forth on planned maintenance cards. For guidance in establishing procedures, planned maintenance documents applicable to a particular generic family of equipment may be obtained from the contracting activity.) When the contract or purchase order requires the preparation of system MRC's (see 6.2) and all system installations may not be serviced under the planned maintenance system, the contractor shall include the preventive maintenance procedures in the volume/manual in order to make it a complete data base, requirements of 3.2.2.1.4.1 and 3.2.2.1.4.2 also apply. System test procedures shall be based on measurements that ascertain proper performance and operation of two or more equipments in the system. Individual equipment tests are required only when a system test can not ascertain proper operation of the equipment. Chapter 5 shall include:

a. Introduction (see 3.2.3.1.5.1).
b. Scheduled maintenance action index (see 3.2.3.1.5.2).
c. Scheduled test procedures (see 3.2.3.1.5.3).

3.2.3.1.5.1 Introduction. The introduction shall provide an explanation of the purpose, scope, and arrangement of the scheduled maintenance material. When a preventive maintenance procedure is critical to the operation of the system and a schedule for servicing is mandatory, this information shall be conspicuously written as a CAUTION statement.

3.2.3.1.5.1.1 Introductory statement (N&MC). The following statement shall be included: "The scheduled maintenance instructions in this manual are cancelled when the Planned Maintenance Subsystem (PMS) is implemented for this system at your site or aboard your ship or station, as applicable." (See 6.5.40 for PMS definition.)

3.2.3.1.5.1.2 Introductory statement (F). The following statement shall be included: "The procedures in the Inspection Work Cards shall take precedence over the procedures in this manual."

3.2.3.1.5.2 Scheduled maintenance action index. This index shall include a listing of all required scheduled performance tests, and shall be tabulated in accordance with 3.2.2.1.4.4.

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3.2.3.1.5.3 Scheduled test procedures. Detailed procedures for setting up and performing complete system tests shall be presented. Each test shall be numbered and titled to clearly define the action and the output to be tested.

3.2.3.1.6 Chapter 6 - Alignment. Chapter 6 shall present the corrective adjustment procedures and support information necessary to restore either electrical or mechanical alignment or both between various system equipments. All values and tolerances shall be included. The alignments shall be cross-referenced to the respective fault isolation procedures and diagrams in the system troubleshooting/repair volume. Alignment procedures shall include reference to equipment publications when additional procedures are required at the equipment level. The alignment procedures shall be presented in step-by-step sequence. All system alignment and adjustment procedures shall be provided without reference to any other technical manual.

3.2.3.1.7 Chapter 7 - Installation. Installation drawings and information that are necessary to install and check out the system, and that are not contained in the equipment technical manuals, shall be included in this chapter. System installation data shall include, but not be limited to, the following:

a. Utilities lists (see 3.2.3.1.7.1).
b. System interconnecting diagrams (see 3.2.3.1.7.2).
c. Cable run diagrams (see 3.2.3.1.7.3).
d. System piping diagrams (see 3.2.3.1.7.4).
e. System cable interconnection checks (see 3.2.3.1.7.5).
f. Active system tests (see 3.2.3.1.7.6).
g. System component installation procedures (see 3.2.3.1.7.7).
h. Installation drawings (see 3.2.3.1.7.8).
i. Installation checkout (see 3.2.3.1.7.9).
j. Test procedures (see 3.2.3.1.7.10).
k. Installation standards summary sheet (see 3.2.3.1.7.11).

Reproduction of appropriate installation drawings in accordance with MIL-T-31000, cognizant agency cable running sheets, or drawings included in station or ship information books may be provided if informationally complete, and if the quality meets the requirements of this specification (see 3.7.1.1).

3.2.3.1.7.1 Utilities lists. The utilities required (for example, air, water, power, steam, or freon), and the quantities of each, shall be presented as a tabular list for each system, compartment, and area.

3.2.3.1.7.2 System interconnecting diagrams. Interconnecting block diagrams shall be presented in accordance with 3.2.3.2.1.5, and shall reference, or disassembly the diagram in the troubleshooting/repair volume.

3.2.3.1.7.3 Cable run diagrams. When the system is to be installed in a specific class or type of vehicle, aircraft, ship, or station, isometric diagrams shall be used to indicate the location of all cable runs between compartments, rooms, or areas. When the system manual does not cover specific installation location, idealized diagrams may be provided. Each cable run diagram shall indicate (by floor, deck, room, or compartment and frame

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identification) the location of all cables shown in the interconnecting diagrams.

3.2.3.1.7.4 System piping diagrams. When the system is to be installed in a specific class or type of vehicle, aircraft, ship, or station, isometric diagrams shall be used to indicate the locations of all system piping runs between compartments, rooms, or areas. When the system manual does not cover specific installation locations, idealized diagrams can be provided. Each piping run diagram shall indicate by floor, deck, room, or compartment and frame identification, the location of all pipes, valves, fittings, tanks, and other items.

3.2.3.1.7.5 System cable interconnection checks. Continuity and megger check procedures shall be provided to verify the proper installation of all system cables. These checks shall be conducted with all power off, and all equipments completely shut down and isolated from power sources.

3.2.3.1.7.6 Active system tests. Active system test procedures required to verify the proper installation and operation of the system shall be included. References may be made to applicable tests and procedures in chapter 5. Procedures for complete setup, testing, shutdown, and data analysis shall also be provided.

3.2.3.1.7.7 System component installation procedures. Complete step-by-step instructions shall be provided for installation of system components not covered in any of the equipment manuals for the equipments comprising the system. The following types of supplemental information not provided in the equipment technical manuals shall be included:

a. Instructions required to assemble components.
b. Instructions required to mount components, including boring and bracing diagrams and data on shock mounts.
c. Instructions for making electrical, waveguide, plumbing, and all other interface connections between equipments, components, and other systems.
d. Servicing procedures, such as initial lubrication and adjustments.
e. Instructions for bonding and grounding.

3.2.3.1.7.8 Installation drawings. The following drawings shall be included:

a. Pictorial diagrams.
b. Outline and mounting dimension data.
c. Interconnecting wiring and cabling diagrams, or cable running sheets.
d. Primary power distribution.

3.2.3.1.7.9 Installation checkout. Step-by-step checkout procedures shall be included to demonstrate that the system operates correctly and within tolerances in accordance with 3.2.2.1.5.9. These procedures shall provide for system checkout in three test procedures:

a. Phase 1 - Installation inspection and pre-energizing procedures (see 3.2.2.1.5.9.1 a, b, and d through h).

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b. Phase 2 - Initial turn-on and preliminary tests (see 3.2.2.1.5.9.2, with reference to chapter 3 of the system manual vice chapter 2 of the equipment manual).
c. Phase 3 - Installation verification test (see 3.2.2.1.5.9.3, with reference to chapter 5 of the system manual vice chapter 4 of the equipment manual).

3.2.3.1.7.10 Test procedures. Testing procedures shall be presented in a logical order as follows:

a. Energize the system.
b. Perform the test.
c. When the test results are within the required tolerance, the procedure shall direct the technician to the next logical test. When the test results are out of tolerance, a reference to the corrective maintenance or troubleshooting data shall be included. Reference shall be made to troubleshooting diagrams except where precise causes of failure can be predicted; in which case reference may be made directly to an alignment or repair procedure.

3.2.3.1.7.11 Installation standards summary sheet. An installation standards summary sheet shall be provided in accordance with 3.2.2.1.5.9.3.2 for system manuals.

3.2.3.1.8 Chapter 8 - Parts list (A, CG, MC, N). A system parts list shall be included in the system manual. The system parts list shall be prepared in accordance with 3.2.2.1.6, and shall cover only those interfacing equipment, units, assemblies, components, and parts that are installed between equipments andare neither included in the parts list of the equipment technical manual nor in the parts list required for the troubleshooting/repair volume.

3.2.3.2 System troubleshooting/repair volume (type II). The system troubleshooting/repair volume shall be in one or more parts/sections in accordance with 3.6.2.1.2, shall be arranged in the following order, and shall conform to referenced paragraphs.

a. Front matter (see 3.2.1).
b. Overall function information (see 3.2.3.2.1).
c. Major function information (see 3.2.3.2.2).
d. Hardware information (see 3.2.3.2.3).
e. BITE/ATE fault isolation information (see 3.2.3.2.4).
f. User activity comment sheet (N) (see 3.2.2.1.7).

3.2.3.2.1 Overall function information. Overall function information shall provide troubleshooting information which will permit fault diagnosis to a major function of the system. This information shall include the following:

a. Introduction (see 3.2.3.2.1.1).
b. Overall function diagram (see 3.2.3.2.1.2).
c. Keyed text (see 3.2.3.2.1.3).
d. Maintenance dependency chart (see 3.2.3.2.1.4).
e. System interconnection diagrams (see 3.2.3.2.1.5).

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3.2.3.2.1.1 Introduction. The introduction shall describe the general content and purpose of the overall function information.

3.2.3.2.1.2 Overall subfunction diagram. The overall function diagram shall identify all major functions and the interrelationship between the major functions that comprise the system.

3.2.3.2.1.2.1 Intermediate subfunction diagram. One or more intermediate subfunction diagrams shall be prepared in accordance with 3.2.2.2.1.5.1.1.

3.2.3.2.1.2.2 Presentation. The overall function diagram and intermediate subfunction diagram(s) shall be arranged so that a user can readily understand and identify the major functions performed by the system (vice equipment) as specified in 3.2.2.2.1.5.1.2. Each equipment that is involved in system operation shall be shown inside a major function area. When one equipment is used in more than one major function, it shall be represented in each functional presentation.

3.2.3.2.1.3 Keyed text. Keyed text shall be in accordance with 3.4 and shall describe the operation of each major function, and the relationship of the equipment involved.

3.2.3.2.1.4 Maintenance dependency chart. The MDC shall be in accordance with 3.5 and shall contain data which will aid a technician in isolating a potential fault to one of the major functions. Selection of signal specifications and indications shall be based on front panel and easily accessible indicators. The use of external test equipment shall be held to a minimum. When an automatic test program is used with built-in performance monitoring/fault localization circuits, MDC's or fault logic diagrams are required for the performance monitoring/fault localization circuitry and test programs. MDC's or fault logic diagrams are not required for functions in which fault localization is accomplished by BITE or ATE. When extensive BITE or ATE is utilized, fault isolation using such equipment shall be provided in chapter 12 (see 3.2.3.2.4).

3.2.3.2.1.5 System interconnection diagrams. Interconnection diagrams shall be provided which show each equipment or interfacing component as a block, or pictorially. Cables running between equipment shall be identified by cable number. The illustration shall indicate junction boxes, switchboards, components, and so forth, from which interconnections enter or leave. The area location (room or compartment) of all hardware items shall be indicated.

3.2.3.2.1.5.1 Wire identification. Wires used in the system interconnection, and all spare wires within any given cable, shall be shown by schematic diagram or wire list. Source destination connection numbers/letters and signal name or other identification shall be included for each wire.

3.2.3.2.2 Major function information. Each major function information package for the system shall provide troubleshooting information which will permit fault isolation to a given equipment or equipment unit. This information shall include the following:

a. Introduction (see 3.2.3.2.2.1).
b. Major function block diagrams (see 3.2.3.2.2.2).

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c. Keyed text (see 3.2.3.2.2.3).
d. Maintenance dependency chart (see 3.2.3.2.2.4).

3.2.3.2.2.1 Introduction. The introduction shall describe the general content and purpose of the major function information.

3.2.3.2.2.2 Major function block diagrams. One major function diagram shall be prepared for each function block shown on the overall function diagram. The diagram shall show the interrelationships between the equipments (units). Only those units involved with the major function need be shown. Identification of the overall equipment shall be indicated so that the unit can be related to its equipments. Interfacing components, junction boxes, switches, and other items involved with the major function shall be included. Presentation shall be similar to the overall block diagram. Points of entry and exit between equipments, however, shall be indicated by cable, connector, and pin numbers. Signal names shall be included and shall be consistent from diagram to diagram.

3.2.3.2.2.3 Keyed text. Keyed text shall be in accordance with 3.4 and shall describe the operation of each equipment or unit shown on the major function diagram. The description shall also relate the equipment to its mode of operation or effect on the major function.

3.2.3.2.2.4 Maintenance dependency chart. The MDC shall be in accordance with 3.5 and shall contain data sufficient to isolate a potential fault to an equipment or unit, provided that there is an equipment panel having fault isolation information. When no such panel exists, data sufficient to isolate the potential fault to the component shall be provided.

3.2.3.2.3 Hardware information. Each hardware information package for the system shall provide maintenance information for interfacing units used for system operation, and for which there are no technical manuals. This information package shall include the following:

a. Introduction (see 3.2.3.2.3.1).
b. Blocked schematics (see 3.2.3.2.3.2).
c. Keyed text (see 3.2.3.2.3.3).

3.2.3.2.3.1 Introduction. The introduction shall describe the general content and purpose of the hardware information.

3.2.3.2.3.2 Blocked schematics. Blocked schematics in accordance with 3.2.2.2.3.3.1 shall be supplied for all interfacing components, switches, voltage regulators, filters, line drivers, transformers, junction boxes, and other items used for system operation and for which there are no technical manuals. If these items can be incorporated schematically on the system major function diagrams, separate blocked schematics are not required.

3.2.3.2.3.3 Keyed text. Keyed text in accordance with 3.4 shall be included to describe functional operation.

3.2.3.2.4 BITE/ATE fault isolation information. This chapter shall provide the fault isolation material required to isolate a fault when fault localization is accomplished through extensive use of computer or microprocessor-based

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BITE or ATE. The material provided shall consist of, but not be limited to, BIT functional block diagrams, BIT fault indication tables, numeric fault code tables, and fault isolation procedures/logic diagrams. If the hardware is a major operational and maintenance activity, a separate BITE/ATE fault isolation information package can be developed, maintained, and located after the hardware information package.

3.3 Diagramming techniques.

3.3.1 Functional diagrams. Functional block diagrams shall portray the functional hierarchical subdivisions by blocked areas as required by the following types of diagrams:

a. Overall and intermediate function block diagrams. Each major function indicated on the functional family tree (see 3.2.2.2.1.2.1) shall be included on the overall or intermediate subfunction diagram by a coded or shaded block. Function shades shall be indicated within hardware by a white block that is formed by masking out the functional area from the gray shading.
b. Major function and subfunction diagrams. Functional entities shall be portrayed by line art shapes (see 3.2.2.2.2.2.1.3). A single shape or group of line art shapes which work together on a single piece of hardware shall be enclosed in a blocked area and given a functional title.
c. Blocked schematic diagrams. Each functional entity shall be blocked to show functional significance (see 3.2.2.2.3.3.1.3).

3.3.1.1 Review draft copies (RDC). Functional blocks in RDCs shall be enclosed in a dotted border (except for functional entities, which shall be line art symbols).

3.3.1.2 Preliminary technical manuals and final manuals. Functional blocks in preliminary and final manuals shall appear on a white background. Functional subordinates within the block shall be enclosed in a dotted border. Except when more than two levels are required to convey functional information, functional blocks shall be white. The first level of functional information shall be shown by demarcation (cutting back) gray levels of hardware. Where no hardware is involved for a functional presentation, the functional block shall be outlined by a series of dots. Second level functional information shall be demarcated by a series of double dots. When three or more levels of functions will not be clearly indicated by further series of dots, all multifunctional areas (for final manuals) should be shown by blue shading in an equivalent manner as that created for gray shades (hardware levels). Hardware boundaries shall be indicated by dot-dash lines, with the number of dots corresponding to the first level, second level, etc of containment.

3.3.2 Hardware diagrams. Hardware (units, assemblies, subassemblies, or parts thereof) schematic diagrams shall always be enclosed by blocks, with the hardware identity shown in the upper left-hand corner of the hardware block.

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3.3.2.1 Review draft copies (RDC). Hardware diagrams for RDCs shall be by dash-dot-dash lines, with the number of dots denoting hardware level of containment (one dot indicating first shade of gray, two dots indicating second shade of gray) (see figure 23).

3.3.2.2 Preliminary and final manuals. In preliminary and final manuals, hardware shall be represented by levels of gray shading. When a unit, assembly, and subassembly are shown on the same diagram, the lightest shade of gray shall represent the unit; the next darker shade of gray shall represent the assembly; and the next darker shade of gray shall represent the subassembly. A diagram treating a single assembly shall use the lightest shade of gray to represent the assembly and darker shades for subassemblies. Consistency of gray shades shall not be maintained for hardware levels between different diagrams, so that the lightest gray shades can be used. The functional blocks (see 3.3.1.3) shall be white and be defined by cutting back and eliminating the gray shading. Hardware blocks shall be shaped or broken as necessary to accommodate this requirement.

3.3.2.2.1 Mechanical devices. Diagrams shall include sufficient detail to indicate the operational sequence and arrangement of a mechanical device including the electrical control circuits (see foldout 1). Nomenclature, symbols, PIN's, and necessary descriptive data shall be shown as required. Gears, shafts,clutches, levers, mechanically driven switches, motors, synchros, and so forth, shall be shown in functional arrangement. Gear ratios or number of teeth and direction of rotation, and so forth, shall be given.

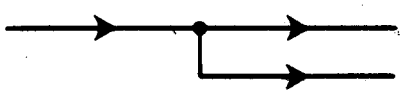
3.3.2.3 Signal flow precedence. Direction of signal flow shall be given priority over hardware layout; accordingly, hardware blocks shall be shaped, repeated or broken as necessary to accommodate the requirements for signal flow to be shown as a flow from left to right.

3.3.3 Keyed text reference number. Each functional blue-shaded area (functional entities, parts, and other blocked areas that need explanation) shall be assigned an encircled Arabic key number just prior to the name, functional entity identifier code, or reference designation. The key numbers shall be assigned sequentially in direct relationship to the functional signal flow.

3.3.4 Energy, signal, or data flow. The nature and direction of energy, signal, or data flow shall be coded by use of special arrowheads superimposed on signal lines in addition to appropriate flagged annotations (see figure 10 and IEEE315-86 for a listing of flow coding). (The generation of new coded arrowheads or symbols to meet specific needs beyond those covered is acceptable.) Flow lines shall be connected in a manner that illustrates subordination or coordination of flow importance as indicated below.

Subordinate junctions shall be shown as follows:



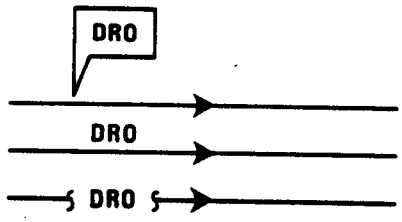


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Coordinate junctions shall be shown as follows:

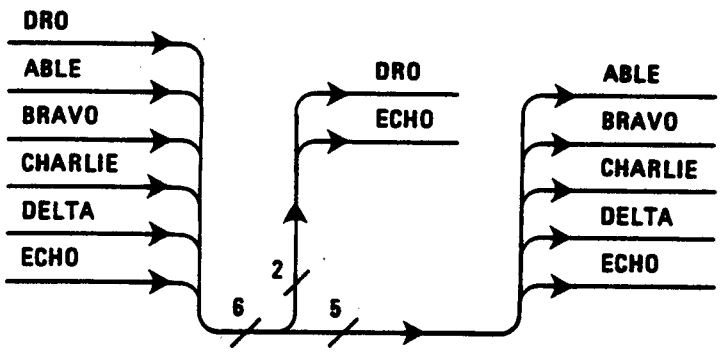


3.3.5 Identification of data flow. Meaningful signal names or codes shall be provided for each connection into and out of blocked schematic diagrams. Signal names shall be located within the body of the diagrams as necessary to identify the signal paths. Such names shall be placed within a flag. Where there is no room for signal flags, the signal name or code may be placed above the line or in line with the signal line, as shown:



Names of particular signals shall be consistent from one diagram to the next.

3.3.5.1 Cabling of data flow. In order to save space on diagrams when large numbers of signal flow lines are present, cabling shall be used. Each signal line shall be identified by its signal name at its source and destinations. These cables Shall be arranged as illustrated below:



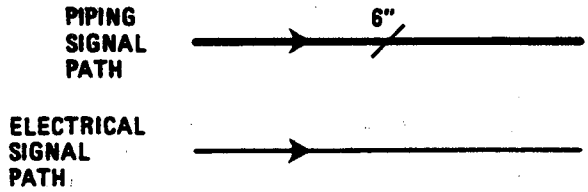
The cables shall curve at every 90-degree corner in order to avoid confusion with single signal lines, which make right angles at corners. Distribution from the cable shall be shown by means of a curved breakoff line. The number of signals leaving a cable will not always add up to the number of signals entering the cable because some of the signals have multiple destinations.

3.3.6 Line weight. Significance of signal paths shall be indicated by line weight. Heavyweight lines represent a major signal path. The ratio between heavy line weight and normal line weight shall be not less than 3 to 1. Line

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weights on a diagram of one hierarchical level are not necessarily the same (for the same signal) on a different hierarchical-level diagram.

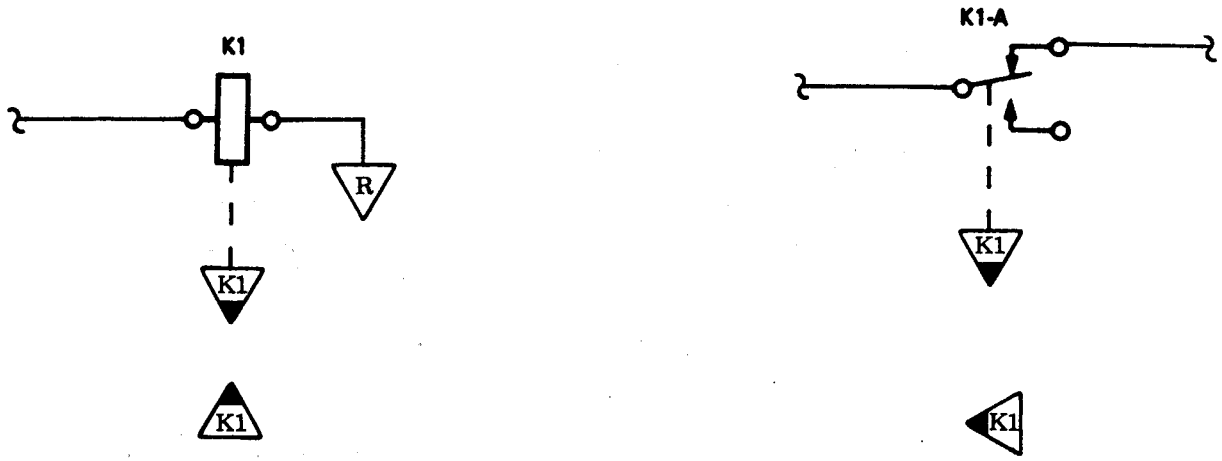
3.3.6.1 Line weights in system manual. Piping and electrical signal paths shall be indicated by line weight. Heavyweight lines shall represent piping signal paths, and lightweight lines shall represent electrical signal paths:



The ratio between heavyweight lines and lightweight lines shall be not less than 3 to 1.

3.3.7 Test points. where possible, test points shall be shown outside of functional areas. The marked nomenclature of front panel test points shall appear on a white background.

3.3.8 Mechanical connections. Mechanical connections shall be represented by dashed lines unless pictorially illustrated. However, relay contact sets that are separated from the operating solenoid, sections of a multisection switch separated from each other, a motor illustrated remote from the driver device, and so forth, may have the mechanical connection broken and illustrated with triangle flags pointing in the direction of connection to avoid crossing lines as follows:



The lower triangle flag set shall point in the direction away from the contact set, switch section, and so forth, to the activating device (for example, solenoid or front panel control).

3.3.9 Shielding. Shielding shall be shown on major function diagrams, subfunction diagrams, and hardware schematic diagrams. The ground return symbol shall identify the end of the shield where the return is accomplished.

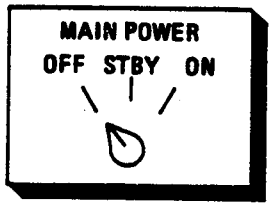
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3.3.10 Adjustments, controls, and indicators. Adjustments, controls, and indicators shall be identified pictorially to indicate the means of adjustment, control, or indication:

METER,  SCREWDRIVER,  HEXNUT,  LAMP,  KNOB 

The marked nomenclature of the adjustment, control, or indicator and its appropriate symbol shall appear in the hardware area representative of the hardware upon which it is located. The relationship of an adjustment, control, or indicator to its proper part shall be illustrated by means of a dashed-line connection.

3.3.10.1 Front panel adjustments, controls, or indicators. Front panel adjustments, controls, or indicators shall appear on a white background outlined by a bold line with shadowing on two adjacent sides as shown below:



3.3.11 Signal specifications. Significant signal specifications and test parameters shall be indicated on (or by reference to the respective signal specifications associated with MDC;s) all functional block and schematic diagrms so that there is a direct correlation between signal flow lines and test points withoutsearching through the related MDC's (for example, signal flow lines and test points shall include a reference such as "SS14," encircled by a cartouche located at a prominent point). Whenspace is a factor, the reference may intersect the signal flow lines.

3.4 Keyed text (troubleshooting/repair volume). Each functional block and hardware schematic diagram shall be supported by keyed text with any required augmenting illustrations (timing diagrams, and so forth). Keyed text shall be referenced by encircled numbers from corresponding numbers in the functional blocked areas of the diagrams (see 3.3.3). Text shall be located on the same or facing page as the diagram it describes. It shall be in columnar format and in numerical sequence. When it is necessary to include the text elsewhere in the information package, the keyed text symbol shall be included on the same or facing page and referenced to the location in the information package where it is included (normally a back-up page).

3.4.1 Coverage. Text shall comply with 3.1.2 and provide the following: "what it is," "what it does," and "how and when it is done." When space is restrictive, sentences need not be complete if a clause can adequately convey the technical meaning.

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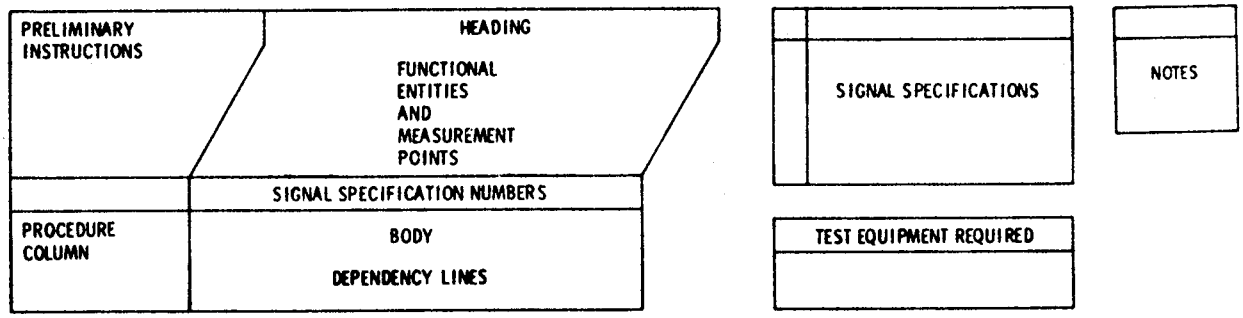
3.4.2 Supplementary data. Truth tables, timing diagrams, equations, and so forth, shall be used to augment the keyed text.

3.5 Maintenance dependency charts production

3.5.1 Layout. Each MDC shall consist of the following areas:

a. Procedure column (see 3.5.1.1).
b. Heading (see 3.5.1.2).
c. Signal specification numbers (see 3.5.1.3).
d. Body (see 3.5.1.4).
e. Signal specification table (see 3.5.1.5).
f. Test equipments required (see 3.5.1.6).
g. Notes (see 3.5.1.7).

A typical layout with the relative locations of the seven basic areas is shown in the following illustration:



3.5.1.1 Procedure column. This column shall consist of three types of procedural information:

a. Preliminary instructions (see 3.5.1.1.1).
b. Procedural steps (see 3.5.1.1.2).
c. Checkout steps (see 3.5.1.1.3).

Procedural columns shall be located in the left-hand area of the MDC. For the purpose of the troubleshooting analysis, the completion of the listed procedural steps shall result in the completed check-out of the functional divisions presented by the corresponding schematic or block diagram. The steps shall exercise applicable positions of all switches and operating controls. The procedures shall take maximum advantage of built-in self-test features. They shall also include improvised checkout steps covering portions of the equipment for which no external symptoms are developed incident to malfunction, and for which no self-test facility exists.

3.5.1.1.1 Preliminary instructions. The top portion of the procedural column adjacent to the heading shall be labeled "Preliminary Instructions" and shall contain a listing of all initial switch positions, control settings, personnel or equipment actions, etc., which must be accomplished or verified before proceeding with subsequent procedural and checkout steps.

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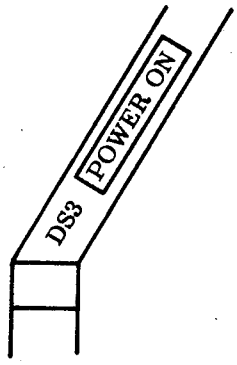
3.5.1.1.2 Procedural steps. Procedural steps shall consist of those procedures which must be performed to bring the equipment or system into full operational status. They shall be numbered sequentially and placed flush left in the procedural column. Each procedural step shall be enclosed between bold horizontal lines across the body of the chart to indicate those events that are associated with the particular step. Procedures required to energize and exercise the equipment or system shall be complete. WARNING and CAUTION notices shall be placed immediately prior to the written procedure to which they apply when actions might be dangerous to personnel, or damage equipment.

3.5.1.1.3 Checkout steps. Checkout steps shall consist of those procedures which can be exercised as options on the part of the technician. For example, they shall include those steps necessary to check out and troubleshoot fault detection circuits, feedback circuits, built-in-test features, and so forth. Checkout steps shall be lettered sequentially, indented, and located within the numbered procedural step which established the operational configuration of the equipment or system required for the checkout step.

3.5.1.2 Heading. This column shall identify the hardware locations of the functional entities and the associated event measurement or observation points which correspond to symbols in the body of the MDC. Entries in the heading corresponding to functional entity symbols (●) shall be identified by functional name, alphanumeric code, or by reference designations, as appropriate. Entries corresponding to event symbols (A) shall provide a checkpoint at which that event occurs, and shall have an assigned number in the signal specification number row.

3.5.1.2.1 Event entries. Requirements concerning event (see 6.5.11) entries are as follows:

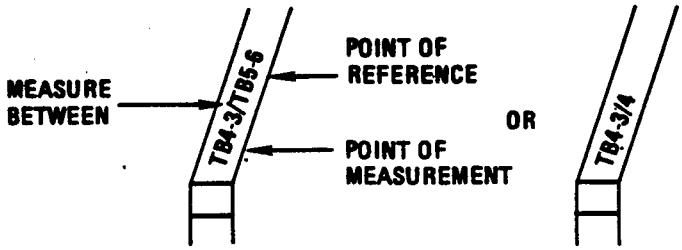
a. For event entries to be observed on the equipment or system, the point of observation shall be illustrated as follows:



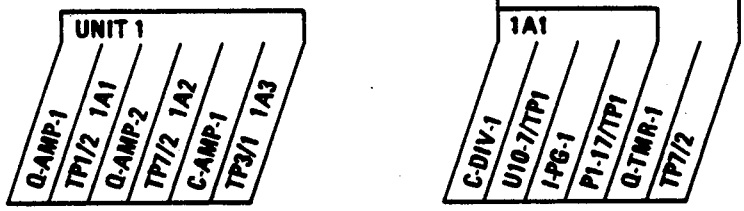
When the indication can be seen from outside the equipment, the panel nomenclature (if any) for the indicator shall be illustrated by a black-line box around the panel nomenclature.

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b. For events to be measured, the points of measurement shall be listed as follows:



3.5.1.2.2 Physical location of entered items. The physical location of the functional entity or event shall be identified in the heading at the top of each column. Assembly numbers or short colloquial names shall be used as a reference key. When adjacent columns refer to items in the same assembly, the location identifier can be so placed that it applies to more than one column, as follows:



3.5.1.3 Signal specification numbers. Events requiring specific data shall be assigned a signal specification number for the particular MDC. These numbers provide a reference to the detailed signal data contained in the signal specification table. Numbers shall be assigned sequentially, left to right, to all events requiring an entry in the signal specification table.Events that are from other MDC's need not carry the original signal specification number as long as the originating MDC and measurement point are identified in the heading.

3.5.1.4 Body. The body of the MDC shall consist of a series of horizontal lines upon which are represented the dependencies or circuit actions that result from the performance of a procedural step. Each line shall represent a short series path from an event point (test point or other) to an indication or second event point through one or more functional entities. Such horizontal lines, representing short serial paths, are called dependency lines. These dependency lines shall be arranged from top to bottom in the order of increasing dependency.

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3.5.1.4.1 Body symbology. Three basic symbols shall be used in the body of the chart, as follows:

a. Event box



Indicates input or output signal or indication. Nomenclature in the event box specifies the type of action and availability of the event. To assist in determining the accessibility of the event listed in the MDC, front panel observables shall be indicated by the use of a black background and white lettering.

b. Dependency marker



Indicates dependency upon an input or previous event.

c. Functional entity dot



Indicates a precisely definable circuit element, functional entity, replacement module, or any other group or device.

3.5.1.4.1.1 Variations. Variations of the three basic symbols may be used to increase the versatility of the MDC. Sample variations are shown below:

a. Adjustable functional entity



Indicates that the output of functional entity is adjustable.

b. Partial functional entity dot



Indicates a precisely definable circuit element, functional entity, replaceable module, or any other group or device which requires more than one test action to prove that the entity is totally good.

c. Replay contact



Variation of the partial functional entity dot used to identify relay contact position.

Dot above slanted line indicates contact must be good when solenoid is deenergized.

Dot below slanted line indicates contact must be good when solenoid is energized.

d. Contracted event



Indicates that more than one (N) distinct event occurs. The individual measuring points shall be specified in the signal specification table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:05Z
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e. Combination dependency



Indicates the number (M) of the total (N) distinct event occurs. The individual measuring points shall be specified in the signal specification table.

f. Or dependency



Indicates that a functional entity requires only one of the dependencies shown to produce the described event output.

g. Feedback dependency

Indicates that a function of the output event is supplied as an input dependency for the structure shown.

3.5.1.4.2 Dependency structure. A dependency structure consists of a series of dependency lines. Each dependency line is constructed using the three basic symbols. The simplest dependency line would appear as follows:

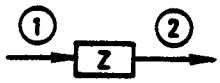


This dependency line indicates that the event (A) is dependent upon the proper operation of the functional entity (●) and the availability of some previous event (▲).

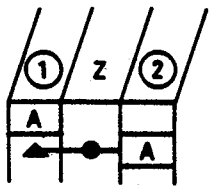
3.5.1.4.3 Mapping methods and techniques. Mapping methods and techniques for signal flow through a functional block or schematic diagram shall be as specified in 3.5.1.4.3.1 through 3.5.1.4.3.14.

3.5.1.4.3.1 Simple dependency line. The mapping of a single functional entity with one input and a single output shall result in a simple dependency line as follows:

BLOCK DIAGRAM



MDC



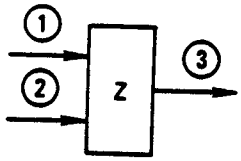
The mapping shows that the output event at (2) is dependent upon the functional entity Z and the input event at (1).

3.5.1.4.3.2 Multiple input dependency line. A functional entity with more than one input and a single output shall be mapped as a multiple input

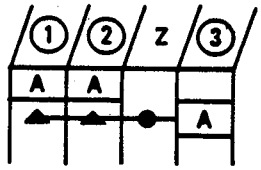
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dependency line as follows:

BLOCK DIAGRAM



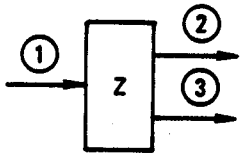
MDC



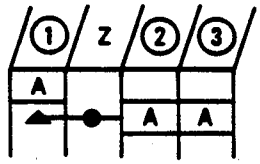
The mapping shows that the event at (3) is dependent upon the functional entity Z and the input events at (1) and (2).

3.5.1.4.3.3 Multiple output dependency line. A functional entity having one input and multiple outputs shall be mapped as a multiple output dependency line as follows:

BLOCK DIAGRAM



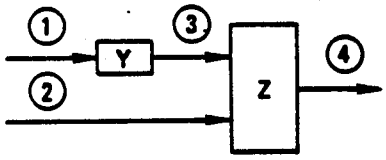
MDC



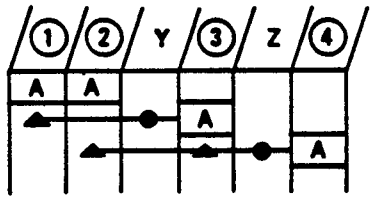
The mapping shows that the output events at both (2) and (3) are dependent upon the functional entity Z and the input event at (1).

3.5.1.4.3.4 Dependency chain. A dependency chain shall consist of a series of interrelated dependency lines. The following example shows a dependency chain consisting of two dependency lines:

BLOCK DIAGRAM



MDC

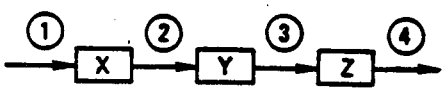


The last dependency line in the example shows that the output event at (4) is dependent upon the functional entity Z and the events at (2) and (3).

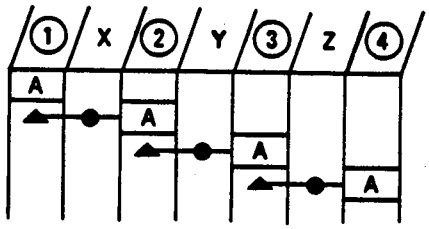
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3.5.1.4.3.5 Series relationships. Series relationships shall be shown as follows:

BLOCK DIAGRAM

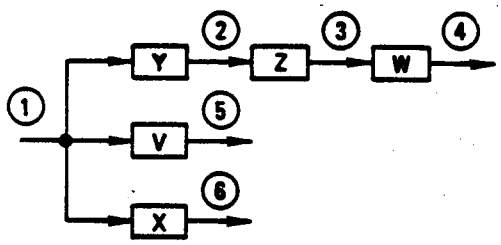


MDC



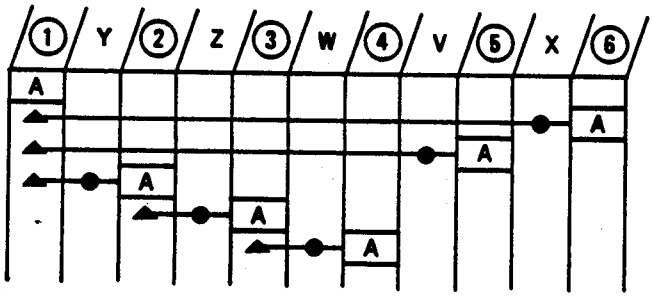
3.5.1.4.3.6 Serial parallel relationships. The preferred illustration for serial parallel relationships shall be as follows:

BLOCK DIAGRAM

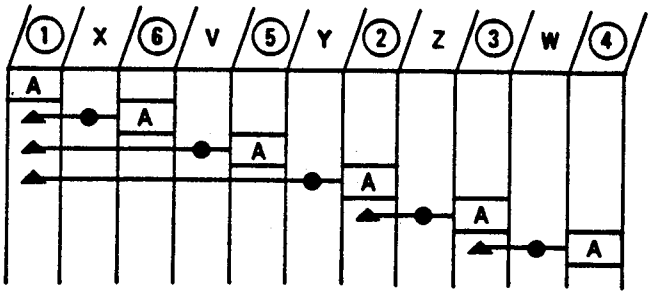


MDC

Desirable



Undesirable

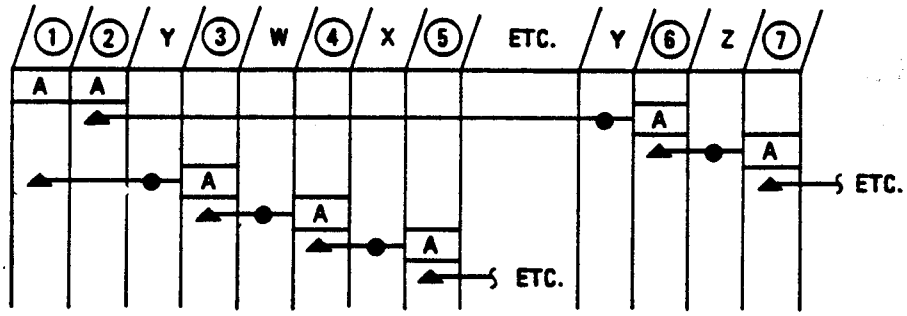


The desirable presentation shows clearly by its structural layout that there are three separate parallel dependency chains. It also shows that one of the dependency lines has several serial relationships. The undesirable presentation wrongly infers by its structural layout that there is only one serial dependency chain.

3.5.1.4.3.7 Horizontal line usage. Normally, only one dependency line shall be used for each horizontal segment of the chart. However, if height is a problem on an MDC, more than one dependency line may be placed on a particular

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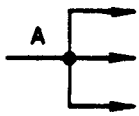
horizontal segment if care is exercised. The dependency chains shall be separated by ample space to avoid confusion. An example of this technique is as follows:



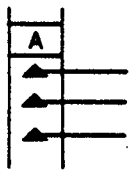
In this manner, each dependency chain is easily identifiable.

3.5.1.4.3.8 Parallel divergent branches. When a path diverges into a number of paths, it shall be depicted by an event/dependency column using a single event and the appropriate number of dependency markers. This situation is shown as follows:

BLOCK DIAGRAM

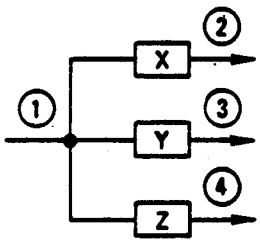


MDC

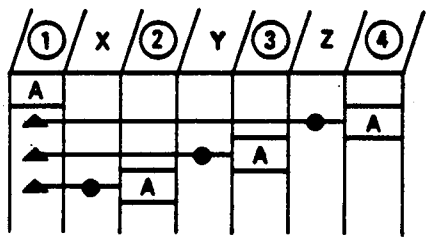


The following example utilizes this method:

BLOCK DIAGRAM



MDC

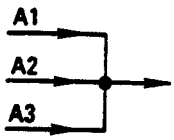


3.5.1.4.3.9 Parallel convergent branches. When paths converge into a single path, they shall be depicted by an event/dependency column using a composite

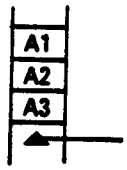
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event and a single dependency marker. This situation is shown as follows:

BLOCK DIAGRAM

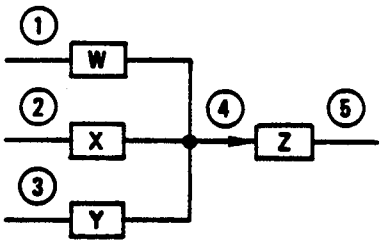


MDC

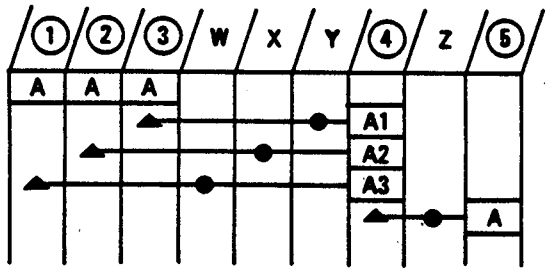


The following example illustrates this rule:

BLOCK DIAGRAM

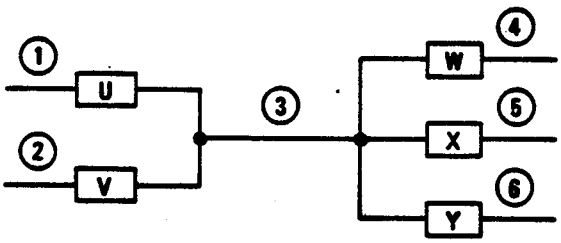


MDC

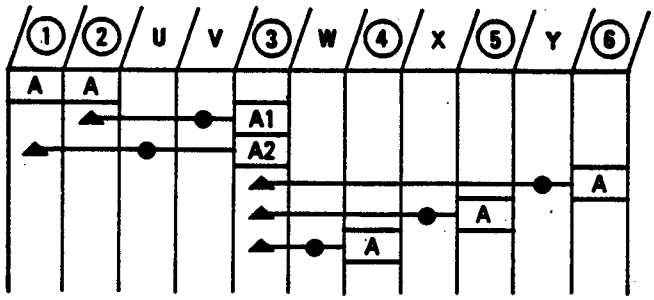


3.5.1.4.3.10 Convergent and divergent branches. When both convergent and divergent branches exist, they shall be illustrated as follows:

BLOCK DIAGRAM



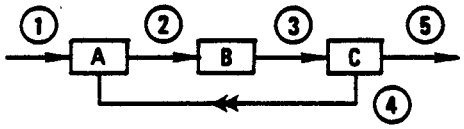
MDC



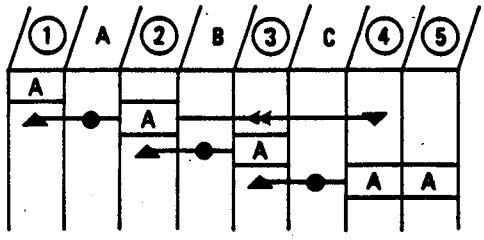
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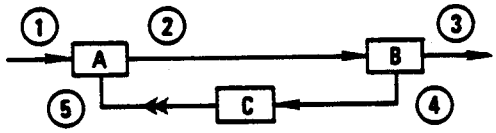
3.5.1.4.3.11 Feedback circuits. Feedback circuits shall be illustrated as follows:

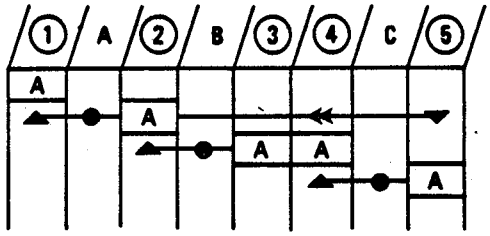
BLOCK DIAGRAM



MDC

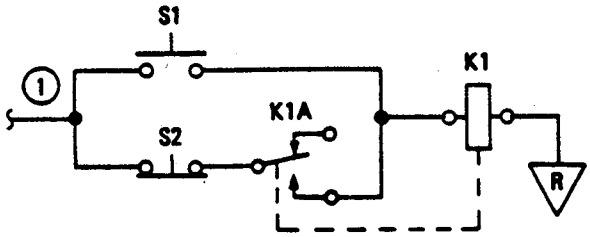




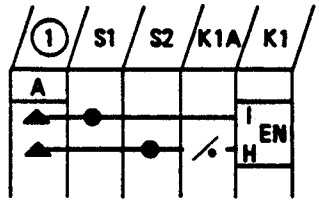


3.5.1.4.3.12 Relay holding path. A relay circuit held energized by a path separate from the initial energizing path shall be mapped as follows:

CIRCUIT



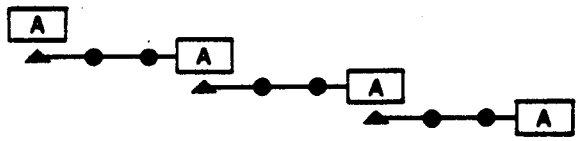
MDC



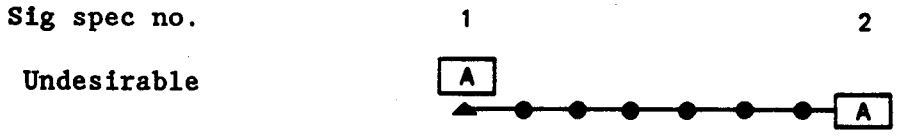
3.5.1.4.3.13 Fault isolation. To simplify fault isolation, dependency charts shall not be constructed with a large number of dots (functional entities) depicted on the same horizontal line. Desirable and undesirable examples are as follows:

Sig spec no. 1 2 3 4

Desirable



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3.5.1.4.3.14 Time delays. Protective time delays, which occur after an operational step is initiated, shall be illustrated by placing a band of screen shading horizontally across the page immediately after the entry of the time delay device. Time is assumed to begin with the action that initiates the step in which the time delay occurs. The length of time delay shall be indicated within the shaded band. When the time delay is not directly related to the step, special notations may be added within the time delay band (for example, "2 seconds after motor exceeds 3,000 rpm").

3.5.1.5 Signal specification table. The signal specification table shall list the description, specifications, characteristics, tolerances, and so forth, for every event in the body of the MDC assigned a signal specification number. The description shall be a voltage, photograph of the waveform, or other measurable/ observable characteristic which is associated with each of the specifications. The test equipment settings shall be included as part of the signal specification to ensure that waveforms on oscilloscopes will agree with those illustrated. When photographs of waveforms are supplied, they shall be retouched to ensure that the pertinent technical information is visible and will e reproduced when copied. (Refer to figure 23 for a typical signal specification table.) A signal specification shall be given for each significant change in signal character, at points of signal divergence, and at points of signal convergence. A WARNING notice shall be included at the heading of the signal specification table when the table indicates that voltages over 70 volts must be measured using external test equipments.

3.5.1.6 Test equipments required. A table of all recommended test equipments needed to perform required maintenance tasks shall be included with each MDC (see figure 23). Selection of test equipments shall be in accordance with 3.2.2.1.1.6.

3.5.1.7 Notes. The notes portion of each MDC shall consist of general and flag notes as required (see figure 23).

3.5.1.7.1 General notes. General notes shall be provided as required, and shall be of the "unless otherwise specified" type (for example, "all voltages are ±10 percent").

3.5.1.7.2 Flag notes. Flag notes shall be provided as required, and shall explain any special or unusual situations on the MDC (for example, "duty cycle can vary greatly as a function of the relative phase of the input signals").

3.6 Format. The format of every FOMM shall be based on the functional complexity, the user's environment (shipboard, aircraft, flight line, depot, and so forth), and the maintenance philosophy for the overall equipment or

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system. Maximum use shall be made of illustrations. Text shall be used to present required data to augment illustrations. Keyed text in accordance with 3.1.9.2.2 and 3.4 shall be used to describe the diagrams/illustrations and each and every functional block (which shall be identified by key number). When optional chapters are not specified, chapter numbering shall be changed to maintain numerical sequence. Normally a two volume concept shall be used (i.e., the first volume shall be topically arranged by subject matter and the troubleshooting volume shall be arranged functionally. In consonance with troubleshooting methodology, the troubleshooting volume shall be formatted such that information can be accessed and utilized with little reference to the support volume and indexes. No procedures shall be dead-ended and all illustrations/text shall provide information to locate a continuing procedure until a potential fault is isolated and repair effected.

3.6.1 General requirements. Style and format for the support volume shall conform to MIL-M-38784 and 3.6.2.1.1. Graphic symbols and reference designations for mechanical, electronic, and electrical diagrams shall be in accordance with ANSI Y14.15, Y32.4, Y32.10, Y32.16, ASME, Y32.2.4, Y32.2.6, Y32.2.3, and IEEE 991-86. Abbreviations used in text and on illustrations shall be in accordance with MIL-STD-12.

3.6.2 Volume breakdown. Unless otherwise specified (see 6.2), FOMM's shall be divided into volumes. The first volume shall be entitled "Support Volume" and the second volume shall be entitled "Troubleshooting/Repair Volume." However, if the requirements of the support volume can be accomplished on twenty 11 by 17 inch pages or less, the FOMM shall consist of a single volume: for type I manuals, it shall be entitled "FOMM Technical Manual for . . ." for type II manuals, "FOMM System Technical Manual for . . . ."

3.6.2.1 Subvolume breakdown

3.6.2.1.1 Support volume (A, CG, MC, N). Each support volume shall contain one or more chapters. The support volume shall not exceed 1500 printed pages (750 sheets). When the thickness exceeds 750 sheets, the volume shall be broken down into parts by removing chapters and binding the parts separately. The number of chapters bound in each part is dependent on how many chapters need to be removed to maintain the maximum number of sheets per part. Under no circumstances shall a chapter be subdivided except where the chapter exceeds approximately 750 sheets. Therefore, all volumes shall contain a finite number of chapters. The priority of chapter removal shall be as follows:

a. Overhaul/depot level maintenance.
b. Parts list.
c. Installation.
d. Maintenance procedures and standards.
e. Theory of Operation.
f. Operation.

3.6.2.1.2 Troubleshooting/repair volume (A, CG, MC, N). When this volume exceeds approximately 300 pages (150 sheets, 11 by 17 inch page size), the troubleshooting/repair volume shall be bound into 2 or more parts. The first subdivision shall contain functional maintenance information and shall be designated Volume 2, Part 1. The second subdivision shall contain the

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hardware maintenance information. When it is necessary to further subdivide the troubleshooting/repair volume, the functional and hardware maintenance information shall be divided into additional parts and bound separately as needed. When the functional information package must be divided, the overall functional information shall be separated from the major functional information into parts. If the major functional information must be further divided, each additional part shall be comprised of one or more major functions with complete subfunction information included. All additional parts shall be sequentially numbered.

3.6.2.1.3 Manual breakdown (F). The manual breakdown shall conform to 3.6.2.1.1 and 3.6.2.1.2, but shall be broken into chapters rather than parts. Each chapter shall have a separate title, title page, list of effective pages, and so forth. Page numbers shall not be consecutive between chapters. In such cases, a dashed number added to the basic technical order number shall be assigned to each chapter.

3.6.2.2 Divisions of volume/manual by level of maintenance. Unless otherwise specified (see 6.2), the FOMM shall be divided into volumes for issue to the operator, organization and intermediate levels of maintenance designated. Data required for each volume/manual shall be as shown on figure 35. When the maintenance philosophy specifies depot-level repair and separate documentation is not acquired under the contract or is not available to the Government for this purpose, the FOMM shall include these data in the support volume chapter on "Overhaul/Depot Level Maintenance (see 3.2.2.1.4.7).

3.6.2.2.2 Maintenance levels (F, N). Unless otherwise specified (see 6.2), all FOMM's shall include all organizational- and intermediate-level data necessary for designated repair through all levels of maintenance except overhaul.

3.6.2.3 Single-volume manual. A single-volume FOMM shall be prepared to 8 1/2 by 11 inch size. When a single-volume FOMM is supplied, the support data (see 3.2.2.1) shall be presented preceding the troubleshooting/repair data. Front matter data requirements for the entire manual shall be combined logically in the front of the manual.

3.6.3 Arrangement of information (troubleshooting/repair volume

3.6.3.1 Information packaging. Diagrams, related text, MDC, repair data, and parts data as applicable shall be located on one page (11 by 17 inch) when space permits. When two or more pages (11 by 17 inch) are required, the diagram shall appear on the right-hand page, or across the two-page spread, with the keyed text appearing on the same or facing page. The other required information (MDC, etc.) shall be included on the same or facing page or, when necessary, on immediately succeeding pages. When schematic diagrams carry beyond the two-page spread, they shall be identified by sheet numbers (for example, sheet 1 of 4, or sheet 2 of 4), and appropriate sheet number references shall be included on the diagrams.

3.6.3.1.1 Space utilization. Blank spaces and pages shall be avoided unless necessary for logical information packaging. Sizing techniques, consistent with the typography requirements of 3.7.2 and the legibility requirements of 3.9.3, shall be used to meet the requirements of 3.6.3.1, even though final

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chart, table, or illustration size may vary between pages of the manual. It is appropriate to refer to reference keyed text detailed elsewhere within the information package to reduce space (see figure).

3.6.3.2 Order of information packages

3.6.3.2.1 Functional information packages. Functional information shall begin with the overall equipment/system information, followed by the major subfunction information packages. When intermediate functions are generated, they shall precede the major functions represented on that intermediate subfunction. Major function packages shall be arranged in numerical sequence by their keyed text numbers, as assigned on the overall or intermediate block diagram.

3.6.3.2.2 Hardware information packages. Hardware information packages shall be arranged in a top-down breakdown order by reference designations (Unit 1, 1A1, 1A1A1, 1A2, 1A2A1, etc. When space permits, hardware data for two or more assemblies may appear on the same or facing page.

3.6.3.2.2.1 Common hardware assemblies. When an assembly, subassembly, module, printed circuit board, or major component is used two or more times in one or more units, a single hardware information package may be used if it is identified by all assigned reference designators. The common hardware information package shall show all interface connections. When an equipment/system contains three or more common hardware assemblies, the common hardware information packages shall be grouped together and be clearly identified, referenced, and interfaced (see 3.2.2.2.3.3). All diagrams having common assemblies shall reference the common hardware information package. The common hardware information package shall identify all hardware assemblies in which its used.

3.6.4 Numbering. Numbering in FOMM's shall conform to the requirements specified in 3.6.4.1 through 3.6.4.2.4.

3.6.4.1 Support volume. Numbering for paragraphs, pages, tables, and illustrations shall be in accordance with MIL-M-38784. Procedural steps shall be numbered sequentially with Arabic numerals. Substeps shall be lettered (a, b, c, d, and so forth). Carryover lines in procedural steps shall return to the margin of the text page.

3.6.4.2 Troubleshooting/repair volume. With the exception of the appendices and checkout steps on MDC's and turn-on/checkout charts, Arabic numerals shall be used throughout.

3.6.4.2.1 Pages. Troubleshooting/repair volume pages shall be numbered in accordance with MIL-M-38784. Pages of the troubleshooting/repair volume shall be numbered sequentially within the title page as page 2-1. Appendix pages shall be numbered sequentially within each appendix with Arabic numerals prefixed with the letter designation of the appendix (for example, page B-6 is the sixth page of appendix B). Pages having one side blank shall carry two page numbers on the printed side. The page number of the blank page shall be followed parenthetically by the word "blank" [for example, 2-71/(2-72 blank) or (2-71 blank)/2-72]. Whenever the troubleshooting/repair volume is divided

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into more than one part, the page number shall also identify the part (for example, volume 2, part 1, page 6 shall be identified as 2-1-6; volume 2, part 2, page 43 shall be identified as 2-2-43).

3.6.4.2.2 Procedural steps. Procedural steps used to present step-by-step instructions shall be numbered in sequence with Arabic numbers. However, the indented checkout steps listed on the MDC and turn-on/checkout chart shall be lettered sequentially (a, b, c, d, and so forth).

3.6.4.2.3 Appendices. Appendices shall be identified by capital letters in an alphabetical order in accordance with MIL-M-38784.

3.6.5 Illustration titles. Illustration titles shall be assigned to all figures in the support volume and to all pages of the troubleshooting/repair volume.

3.6.6 Running heads and feet. Complete running heads and feet shall be included on all pages except title pages or pages otherwise blank. Blank pages which back up classified pages shall be marked with the security classification of the backed up page.

3.6.6.1 Running heads

3.6.6.1.1 Security classification. The security classification, including unclassified pages, of classified manuals shall be at the top center of each page in bold face type in accordance with DOD 5220.22-M.

3.6.6.1.2 Technical manual identification number. The technical manual identification number, as assigned for each volume and part, shall be in bold face type at the top center of each page (page unit) direct below the security classification.

3.6.6.1.3 Binding edge. When specified, the binding edge shall indicate the equipment or subject to which the manual applies and relate to the prime title. Appropriate abbreviations may be used.

3.6.6.1.4 Outer edge. When specified, significant reference information such as chapter, section or subject titles or paragraph number or figure number, as appropriate, shall be used or added. Appropriate abbreviations may be used. Troubleshooting volumes shall identify the information package title including the appropriate family tree designator enclosed in an elipse.

3.6.6.2 Running feet

3.6.6.2.1 Page number. Page numbers shall be located at the lower outer edge ending at the outside margin and shall be in bold face type. Even numbers, including zero, shall be assigned to left-hand pages and odd numbers to right-hand pages.

3.6.6.2.2 Issue indicator. When specified, the issue indicator of basic manuals, revisions and the change designator for change pages shall be located at the outer edge of all pages on the same line as, and 1/2 inch to the inside of, the page number. When specified, the word "Original" shall be included on basic pages.

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3.6.6.2.3 Security classification. The security classification, including unclassified pages, of classified manuals shall be at the bottom center of each page in bold face type in accordance with DOD 5220.22-M.

3.6.6.2.4 Troubleshooting/repair volume titles. The titles of diagrams and text shall be placed in the lower outer corner of each page.

3.6.6.2.4.1 Overall information package pages. The title for each page shall identify its general subject matter.

3.6.6.2.4.2 Major function information package pages. The title shall correspond with the functional family tree and include the appropriate function/subfunction designator in an elipse.

3.6.6.2.4.3 Hardware information package pages. The title shall correspond with the hardware family tree and include the hardware unit/assembly designator and name.

3.7 Production. The production requirements for all phases of FOMM development shall be in accordance with the following paragraphs.

3.7.1 Reproduction. Reproduction quality for review draft copies (RDCs), preliminary manuals, and final manuals shall be as follows.

3.7.1.1 Hard copy. The FOMM pages shall be produced by the offset process or an equivalent process. Letters, lines, and symbols shall be of uniform contrast throughout the final printed manual. The final manual shall have clear-cut, uniform differentiation of the shades of gray. Blurred or smudged printing or dropout of characters, lines, letters, or failure to meet the other requirements listed herein shall because for nonacceptance from the contractor or Government activity responsible for printing. For RDCs and preliminary manuals, reproduction in black and white shall be accomplished by any process that provides clear readable copy with white background. Final manuals shall meet the requirements of MIL-P-38790 for good reproduction.

3.7.1.1.1 Presentation. Presentation shall be as follows:

a. Support volumes shall be prepared in 8-1/2 by 11 inch size, or for reduction to that size. The maximum printing area (image area) shall be 7 by 10 inches. When an illustration will exceed one page, the illustration will be divided (unless the usefulness of the illustration will be adversely affected), planned for presentation on facing pages, and be numbered FIGURE X-X (sheet 1 of 4), FIGURE X-X (sheet 2 of 4), and so forth. Additional sheets may be used for succeeding pages if required. Foldouts shall be avoided.
b. Unless otherwise specified, the troubleshooting volume shall be prepared in the 11 by 17 inch size, or 8-1/2 by 11 inch size, or for reduction to these sizes. The maximum printing area shall be 10 by 15-3/4 inches or 7 by 10 inches respectively. The 11 by 17 inch volume shall be bound on the 11 inch dimension. All 8-1/2 by 11 inch volumes shall be landscaped and bound on 8-1/2 inch dimension. Foldouts shall be avoided. Schematic and block diagrams shall be prepared strategically within the two-page spread, providing that sufficient space is

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provided for keyed text. When there is insufficient area for the entire text, text shall be provided for all "key" numbers on the diagram, but text for a key number may be continued in available space within the respective information package, provided the direct reference to the location is made. Whenever possible, continued text shall be in the same location, preferably on the following page.

3.7.2 Typography and lettering. Typography and lettering for the final manual shall be in accordance with table I. RDCs and preliminary manual illustrations and diagrams may be hand-lettered, but shall be legible when reduced to final size.

3.7.2.1 Typography and lettering for technician's pocket manual. Final type size for pocket manuals shall be no greater than a 50 percent reduction of the size appearing in the associated troubleshooting/repair manual, but no upper case character shall have a printed final size of less than 0.035 inch.

3.7.3 Photolithographic negatives. For each page of a manual and the covers/title page, the following kinds of fully composed, lithographic stable-base negatives are required:

a. Pages having shading. A negative that has been screened to provide the required tints (one for each gray tint and one for blue tint) plus the appropriate line art negative. The gray tint negative should normally be composited with the line art negative.

TABLE I. Typography and lettering*.

Use	Final Size	Case
Security classification	0.125 inch	Upper
Chapter/section titles	0.105 inch	Upper
Notes (word only)	0.090 inch	Upper
Cautions (word only)	0.090 inch	Upper
Warnings (word only)	0.090 inch	Upper
Figure titles	0.090 inch	Upper & lower
Diagrams and schematics	Not less than 0.070 inch with 20 letters per inch	Upper
Text, procedural data, and other large blocks of written matter	Upper case letters not less than 0.090 inch final size	Upper & lower
Covers and title pages:		
Nomenclature and title	18 to 24 point	Upper
Publication number	24 point	Upper
Security classification	24 point	Upper

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* When a technician's pocket manual (see 3.8.3) is specified (see 6.2), selection of type and size shall be based on the typography requirements specified in 3.7.2.1

b. Pages having only line art, halftones, or text. A lithographic negative with halftone negatives stripped in place or taped over a clear window on the basic page negative.

3.7.3.1 Negative registration and quality. Line negatives used to produce plates for the printing of each page shall be stable- base film and shall be pin-bar registered, free of defects, and opaqued free of pinholes and shaded area crop markings. Screen negatives shall not be opaqued for purposes of removing pinholes.

3.7.3.2 Negative identification. Negatives shall be identified so that no confusion can exist as to their relationship. Matching negatives shall be clearly identified to preclude any confusion or mismatch.

3.7.4 Illustration and artwork production. Line drawings and illustrations rather than photographs or renderings are preferred.

3.7.4.1 Illustrations. Line artwork is preferred. Continuous tone artwork (photographic renderings) shall only be prepared when it can be justified to represent a cost savings. Continuous tone artwork shall provide a clear definition of shapes, tonal qualities, and surface texture when reproduced. Artwork shall appear on a white background. Rubdown or stick-down materials used in preparation of the final artwork shall be permanently affixed to the artwork. No wax rubdown material should be used.

3.7.4.2 Line artwork. Final line artwork shall be prepared on stable-base materials. Line weights shall be of sufficient strength to reproduce sharply and clearly in the final reproduction size when reduced to final page size, and shall meet the requirements of 3.7.3. The artwork shall be prepared so that Ozalid or equivalent copies of the original artwork can be obtained, when necessary, early for training purposes. Rubdown or transfer symbols and lettering are acceptable provided there is no wax-backed material used on the rubdown or transfer symbols and letters. When transfer symbols are used, they shall be adhered in such a manner to prevent dropoff when copies are prepared for working or review prints.

3.7.4.3 Line density. The line density for graphic material and background area shall be uniform to achieve the highest possible contrast for better readability and reducibility. Parallel lines on wiring, function, and schematic diagrams shall in no case be less than 1/16 inch apart when reduced to final printed size.

3.7.4.4 Spacing. The density of the information presented shall be such that the spacing between lines and characters shall be not less than one-half the character height used.

3.7.5 Page size.

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3.7.5.1 Final manuals. Unless otherwise specified, the maximum final size for the manual or volume shall be 11 by 17 inches. The size of the troubleshooting/repair volumes/manuals shall be 11 by 17 inches, end-bound (short dimension), with no foldouts permitted; an 8-1/2 by 11 inch or 8-1/2 by 14 inch size of the troubleshooting/repair volume may also be used. The page size for multivolume manuals shall be as follows:

Volume/Manual	Page Size (Inches)
Support volume/manual	8-1/2 by 11
Troubleshooting/repair volume/manual	11 by 17 or, 8-1/2 by 11, or, 8-1/2 by 14, as specified
Technician's pocket manual	6 by 9 (50 percent reduction of 11 by 17)

3.7.5.2 Preliminary technical manuals. Page size shall be the same as for final manuals.

3.7.5.3 Review draft copies. Page sizes shall approximate final size.

3.7.6 Masks. Masks for the preparation of shaded areas shall be prepared. Mask material shall be Ulano Graphic Arts Supplies, Inc. Rubylith DM3, Amberlith D3M, or equivalent. One mask for each shade of gray and when required one mask for and each shade of blue shall be prepared. The area to be shaded shall be stripped away, and the area not to be shaded shall be left opaque. (Other methods that produce the same results are acceptable.)

3.7.6.1 Technician's pocket manual masks. A second set of masks shall be provided for making a 6 by 9 inch size printing of the troubleshooting/repair volume/manual.

3.7.7 Screens. Screens shall be provided for all gray and blue shaded areas, when required, and when specified either Rubylith, Amberlith or equivalent masks used in the screen making, shall be supplied. Screens shall be prepared from a 133 line-per-inch master screen. Techniques that are predicated upon precise dot dimensions are acceptable provided that adequate shade separation and the requirements of 3.7.1 can be obtained, and will be uniform throughout the manual. Dot orientation on screens shall be such that moire patterns are not evident in the final printed manuals.

3.7.7.1 Technician's pocket manual screens. Unless otherwise specified (see 6.2), a second set of screens shall be provided for making a 6 by 9 inch size printing of the 11 by 17 inch troubleshooting/repair volume and parts. The screens for this manual shall be 200 lines per inch.

3.7.7.2 Dot sizes. The printed dot sizes required of the screen negatives shall be as specified in table II. A deviation of 0.0005 inch is allowable for both gray and blue (when required) shading. For an individual illustration, deviations (if any) shall be in the same direction to ensure adequate shade separation. Except for the technician's reduced manual, no shaded area on diagrams shall have a width dimension of less than 1/8 inch when printed.

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TABLE II. Dot sizes required.

Level	Dot size (black)		Dot size (blue)	
	133-line screen (inch)	200-line screen (inch)	133-inch screen (inch)	200-line screen (inch)
First	0.0015	0.0010	0.0025	0.0017
Second	0.0025	0.0017	0.0035	0.0023
Third	0.0035	0.0023	0.0045	0.0030
Fourth	0.0045	0.0030	0.0050	0.0033
Fifth	0.0055	0.0033	0.0055	0.0036

3.7.8 Use of shading.

3.7.8.1 Shading for final manuals. When blue shades are required (see 3.3.1.2), functional areas shall be identified in accordance with 3.7.8.4.

3.7.8.1.1 Shading on schematic and block diagrams. Gray shades shall be used to define the physical hierarchy of the equipment or system (see 3.3.1.2). Blue shades may be used to define the three or more functional entities and functional groupings (see 3.3.1.2). When more than one shade of gray or blue is used, the first-level shade (lightest) shall represent the highest level in the hardware (gray) or functional (blue) hierarchy shown on a particular diagram. No attempt should be made to maintain the same shade between illustrations. Each diagram should be based on its own level of hardware and function.

3.7.8.1.2 Shading on maintenance dependency charts. Alternate columns in the body of the MDC may be a screened the first level gray or shading. A final-size screen negative (stable-base film), the line negative shall be supplied.

3.7.8.2 Shading for preliminary technical manuals. Only gray shading shall be used in preliminary manuals to define hardware boundaries. Coded dotted lines shall be used in preliminary manuals to identify second and higher levels of functional entities and boundaries (see 3.3.1.2). The gray shading shall meet the contrasting requirements of the final manuals (see 3.7.7.2).

3.7.8.3 Shading for review draft copies (RDC). Color shading is not required for RDCs. When shading is not used, hardware containment for review manuscripts shall be by means of dot-dash coded lines. The number of dots shall indicate the shade of gray to be used in the final manual. Dotted lines shall be used to define functional boundaries.

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3.7.8.4 Shading for type IS and type IX manuals. Shading shall be as specified in 3.7.8.2.

3.7.9 Printing direction pages.

3.7.9.1 Reproducible copy. When final photolithographic negatives and screens are not to be provided, a complete set of running sheets and explicit instructions (except for normal industry practices) for the preparation of masks, negatives, and screens shall be included with the final reproducible copy (FRC).

3.7.9.2 Negatives and screens. Unless otherwise specified, final photolithographic screens shall be provided (see 6.2). A complete set of running sheets and explicit instructions for registering, printing and production shall be included with the negatives.

3.7.10 Ink requirements. Ink requirements for printing shall be as follows:

a. Black ink - Sleight Helmuth Tri-dimensional Balance Process Black No. 6543, or equivalent.
b. Blue ink - color no. 169 sgB (strong green blue) as listed in National Bureau of Standards Circular 553 and supplement 2106, or equivalent.

3.7.11 Page stock. Unless otherwise specified (see 6.2), page stock shall be in accordance with table III.

TABLE III. Paper stock.

Type manual	All pages
Support volume/manual	Paper stock shall be in accordance with MIL-P-38790.
Troubleshooting/repair volume/ manual types I, II, III	Paper stock shall be 60-pound wove.
Types IS, IX, IIS, IIX preliminary technical manuals and RDCs	Any paper (compatible with printing or duplicating process and its intended use) which provides clear, readable copies with white background may be used.
Technician's pocket manual	Casco or Shorewood gloss enamel or equivalent substance weight, not less than 80 pounds per 1,000 sheets, basis 25 by 38 inches.

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3.7.12 Covers

3.7.12.1 Cover stock.

3.7.12.1.1 Support volume. Cover stock shall be in accordance with MIL-P-38790, except that 100-grain, 0.0035-inch, linear polyethylene covers may be used when approved (see 3.7.12.1.2). When polyethylene covers are to be used, they shall be translucent and shall not contain any printing or silkscreening.

3.7.12.1.2 Troubleshooting/repair volume. Cover stock shall be 100-grain, 0.035-inch, linear polyethylene, California Calendar Company Number CCC-84-590, or equivalent, translucent, and shall not contain any printing or silkscreening.

3.7.12.2 Cover presentation (A, CG, F, MC). Cover information shall be prepared by silk-screen process. Information to be printed on the covers shall not be stamped in gold or any other metal foil. The covers shall be approximately 1/2 inch longer and wider than the included pages. Manuals of 8 sheets (16 pages) or less need no covers, but shall be supplied in vinyl envelopes (see 5.1.3). Covers for RDCs are not required.

3.7.13 Binding

3.7.13.1 Binding of support volumes. Binding shall be in accordance with MIL-P-38790. Manuals shall be prepared in loose-leaf form. Unless otherwise specified (see 6.2), corrosion-resistant posts and screws, polypropylene binders, or snapring metal prong fasteners shall be used. Metal parts shall be treated to resist corrosion. Manuals shall be punched or drilled in accordance with MIL-P-38790. Page sizes and punching or drilling for change pages shall be the same as for the original manual pages. Fillers shall be used to build up the binding edge to the same thickness as the right-hand edge of the manual.

3.7.13.2 Binding of troubleshooting/repair volumes. Binding shall be bound for landscape presentation (see 3.7.1.1.1) on the left-hand edge (11 inch side) of the manual, except when 8 1/2 by 11 inch paper is used, it shall be bound on the 8 1/2 inch dimension. Manuals shall be prepared in loose-leaf form to facilitate insertion of change pages, and binding shall conform to 3.7.13.1.

3.7.13.3 Binding of technician's pocket manual. Spiral, GBC, or equivalent bindings shall be used along the 6 inch, left-hand edge of the manual.

3.7.13.4 Special binding. When a different binding method than that specified can be considered an improvement for the manual usage, the contractor may request the contracting activity to permit a deviation. Samples and specifications for the binding method shall be submitted for the contracting activity's approval.

3.7.14 Title page (N). Title pages (which may be a "self-cover" if no protective cover is required or used) shall be on white paper with black ink

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printing. The type of publication shall be identified as a "FOMM Technical Manual for" and shall include the title "Support Volume/Manual" or "Troubleshooting/Repair Volume/Manual," as applicable (see figure 8). The following information shall appear on the title page of all Navy technical manuals, changes, revisions and reprints:

a. Bar code.
b. Destruction statement, if applicable.
c. Distribution statement.
d. Security classification if applicable.
e. Technical manual stock number of basic issue.
f. Navy Technical Manual Identification Numbering System (TMINS) number.
g. Warning (export control) statement, if applicable
h. Publication date
i. NAVSEA logo (seal)

3.8 Issues. The issues of a technical manual shall conform to the requirements specified herein. Issues shall include the following, as applicable:

a. Preliminary manual (see 3.8.1).
b. Final manual (see 3.8.2).
c. Technician's pocket handbook (see 3.8.3).
d. Supplements (see 3.8.4).
e. Changes and revisions (see 3.8.5).

3.8.1 Preliminary technical manual (PTM). The PTM shall include all data required for the final manual and shall be validated in accordance with 4.3. The PTM shall reflect the review manuscript with all valid Government RDC comments incorporated and shall be superseded by the final manual. Unless PTMs are required as a deliverable item (see 6.2), PTMs shall only be produced to fulfill the need specified in 4.5.2.

3.8.2 Final manual. The final manual shall be prepared in full conformance with this specification and the requirements of the applicable contract or order. The production requirements shall be as specified herein (see 3.7) and in the applicable contract.

3.8.3 Technician's pocket manual. A 6 by 9 inch page size reprinting of the troubleshooting/repair volume shall be provided as specified (see 6.2). A unique publication number shall appear on the cover/title page. Other pages shall be a reduced-size copy of the troubleshooting/repair volume and maintain the technical manual identification number of the copy from which it was reproduced.

3.8.4 Supplements. A supplement manual shall be prepared to separate classified information from the body of the text, or to complement the data in an existing final or preliminary manual. Unless otherwise specified (see 6.2), the supplement shall conform to the support volume and troubleshooting/repair volume requirements (see 3.2.2.2) of this specification. Supplements shall be prepared in accordance with MIL-M-38784.

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3.8.5 Changes and revisions. Changes and revisions shall be prepared in accordance with MIL-M-38784. Changes and revisions shall be issued to correct all inaccuracies or omissions found during review or usage, or for contractor-generated design or production changes to the equipment or system. Necessary corrections and changes to technical manuals shall be issued throughout the life of the contract and guarantee period.

3.8.5.1 Changes. Changes to an existing manual shall be issued in the form of replacement pages as specified in MIL-M-38784 and shall be prepared in accordance with this specification. Pages shall be revised only when significant changes are required. The change shall include an instruction sheet (see 3.8.5.1.1.3) or transmittal page (see 3.8.5.1.2). Reproducible copy shall be furnished for altered or additional pages or both. Copy shall be furnished for both sides of the page on which the change is made. The reproducible copy for the change page and backup page shall be identified for ready reference by the user. This shall be accomplished by placing a vertical bar in the margin opposite the changed material. An explanation of the method used for identification shall be included in the instruction sheet or transmittal page. Changes shall be numbered in a single consecutive series for each publication.

3.8.5.1.1 Changes (CG, MC, N). Changes may consist of two types, interim or permanent, as specified in the contract or order (see 6.2).

3.8.5.1.1.1 Interim (temporary) changes (CG, MC). An interim (temporary) change is an expeditious method of promulgating a minor manual change by means of change pages. Interim changes shall be used when there is insufficient time available to publish a permanent change prior to need, or as specified in the contract (see 6.2). Interim changes shall be subject to replacement by the next issued permanent change. An interim change shall be identified, dated, and shall contain an instruction sheet in accordance with 3.8.5.1.1.3.

3.8.5.1.1.2 Permanent changes (CG, MC, N). Upon application, change numbers or letters shall be used to identify changes and will be assigned by the contracting acquiring activity. Permanent changes for the FOMM shall be in the same style and format as the basic manual and shall include all corrections issued in previous interim changes (unless they were covered in previous permanent changes), in addition to all other changes required (see 6.2). Change numbers and publication identification numbers shall be requested from the contracting activity when the RDC incorporating changes is submitted for review and acceptance. Changes shall be prepared in accordance with the following:

a. Changes made to any given page of the FOMM shall require a complete check and correction of associated pages that reflect the subject change.
b. Production of changes shall be prepared to the same style, format, and specifications as the basic manual, except that preliminary manual printing requirements may apply. When the manuals are available, the changes shall be inserted (by the supplier) into the manuals to be shipped with the equipment.

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c. A revised title page, including the original approval date of the publication and the approval date of the change, shall be prepared. Change numbers for the change shall be inserted in the lower right hand corner in accordance with MIL-M-38784. The publication identification number shall appear at the top of the page in accordance with MIL-M-38784 and shall remain the same.
d. When the RDC of the change is submitted for acceptance, the publication identification number and change number shall be requested from the contracting/acquiring activity.
e. When a change is made to an MDC, the entire chart shall be reprinted, including the backup page. Markings shall be the same as for all changed pages.
f. A new title page, list of effective pages (including the source data table), and index, as applicable, and the revised pages shall be prepared in accordance with MIL-M-38784 and as specified herein.
g. The numbering of procedural steps or a list of items changed, deleted, or added by a change shall be given the number of the preceding step or item, and suffixed starting with the letter "A".
h. Change pages shall be prepared and punched or drilled to the exact size and dimensions of the pages they replace.

3.8.5.1.1.3 Instruction sheets (CG, MC, N). A change (interim or permanent) shall include an instruction sheet which shall provide instructions for inserting new and revised pages, and disposing of superseded pages. The instruction sheet shall include the following, as applicable:

a. The identification number for SPAWAR publications shall appear in the upper left-hand corner of each page. The change shall be identified with a "T" (interim) or assigned a permanent identification number by the contracting activity as well as the publication number to which it applies.
b. The total number of pages included in the change shall be indicated on the first page for the instruction sheet.
c. A supersession notice (see 3.8.5.2.3) shall appear on the title page.
d. The security classification shall be located on all pages of classified interim or permanent changes and shall appear on the top and bottom center of each page.
e. A statement shall be included indicating when the change is in effect. If it supersedes an earlier interim or permanent change, a statement of that fact shall be given.
f. If an equipment change is involved (field or production change), a statement shall be included indicating the purpose of the change and the extent (serial numbers or conditions of application) to which it applies to the equipment population.
g. When the equipment nomenclature has been changed because of a production or field change to the equipment, the nomencla-ture references in the text and illustrations need not be modified merely to include the new nomenclature, providing that information equivalent to the following is included in the general information section:

"Unless otherwise specified, all references in the manual to Radar Set AN/SPS-10 apply equally to Radar Set AN/SPS-10A."

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h. If an equipment change does not affect the entire equipment population, the change shall be prepared so that it will describe both the affected and unaffected equipment.
i. When the change applies to the entire equipment population, the change shall cover only the modified equipment. A note shall be included in the first paragraph of the instruction sheet to the effect that maintenance support activities shall not dispose of the superseded pages until it is established that all of the equipment population has been modified.
j. Instructions shall be concise and clear, illustrations and diagrams corrected, as applicable, and specific data to be changed shall be set off in quotation marks.
k. The instructions shall be followed by a statement that the instruction page shall be inserted in the manual immediately following the front cover/title page.
l. For changes pertaining to equipment modifications, the user shall be instructed not to insert a change in a manual until the modification has been accomplished.
m. Printed matter shall be arranged on the page so that all copy is readable when bound in the manual.

3.8.5.1.2 Changes (A, F). Minor changes, such as editorial corrections, changes in terminology, or updating of references, shall be held until the page requires substantial revisions. The transmittal page changes shall include, in tabular form, a list of pages to be removed from the basic publication and new or revised pages to be added, a supersession notice (if applicable), and any special instructions.

3.8.5.1.3 Numbering and marking changes. All pages shall include its change number preceding the page number in accordance with MIL-M-38784.

3.8.5.1.4 Pen-and-ink corrections (CG, MC, N). Pen-and-ink corrections are not acceptable unless the correction is promulgated by a message, letter, or a published article and change is followed up by the issue of a permanent change.

3.8.5.1.4.1 Pen-and-ink corrections (F). Pen-and-ink corrections shall be avoided.

3.8.5.2 Revisions. A revision is a second or subsequent edition of a manual or volume of a manual which normally supersedes the preceding edition. When a revision is procured to cover a separate equipment model or a different system application, and the basic issue is not to be superseded, the revision shall be identified as a non-superseding updated revision. The non-superseding update revision shall be classified as "original," have a unique Government identification number on the cover (A, CG, F, MC) and all pages, and shall not include a supersession notice. Revisions shall conform to the contract requirements, MIL-M-38784, and this specification.

3.8.5.2.1 Updated revision (A, CG, MC, N). An updated revision (see 6.5.45.3) shall be developed when the number of pages required to supply corrected and additional information for the manual is more than 25 percent of the total pages of the manual but less than 50 percent. (This does not

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include previously issued permanent change pages, but does include outstanding interim change pages.) Blank pages shall not be included as part of the basis for determining the 25 percent figure. An updated revision is required when the current manual is adequate both in format and accuracy, but does not completely cover the equipment because of production or field changes or modification made to the equipment after the manual was issued. Material for the updated revision shall be prepared in the same manner as replacement pages of the permanent change. The final reproducible copy of the manual shall be such that it can be reprinted in its entirety and issued as a revised manual as opposed to being assembled by the user.

3.8.5.2.2 Complete revision. A complete revision (see 6.5.45.1) shall be developed when the number of pages required to correct and provide additional information for the manual exceeds 50 percent of the total pages of the manual and when the existing style and format is adequate. (This does include previously issued and outstanding interim and permanent change pages.) A complete revision shall be a completely rewritten manual incorporating all previous corrections and changes, and shall comply with all content and format requirements of this specification.

3.8.5.2.3 Supersession notice. Unless otherwise specified by the Government when a revision is issued, the following supersession notice shall be included on the title page: "This publication supersedes (Air Force, Army, Coast Guard, Marine Corps, or Navy as applicable, volume/manual number), date (day, month, year)."

3.9 Deliverable data items. The deliverable data items of the FOMM's shall be as specified (see 6.2), and shall conform to the requirements specified herein. Data items shall include the following, as applicable:

a. Outline/book plan (see 3.9.1).
b. Review draft copy - RDC (see 3.9.2).
c. Final reproducible copy - FRC (see 3.9.3).
d. Photolithographic negatives (see 3.9.4).
e. Replenishment materials (see 3.9.5).
f. Certification checkoff list (see 3.9.6).

3.9.1 Book plan. The book plan shall consist of a narrative/outline and a dummy manual. The narrative/outline shall be submitted to the contracting activity for acceptance prior to development of the manuscript manual. If rejected, the unacceptable portions shall be corrected and resubmitted. The accepted narrative/outline shall become an addendum to this specification governing the preparation of the technical manual to the extent that it agrees with the requirements of the contract. The dummy manual shall be based on the accepted narrative/outline, maintained by the contractor during the technical manual development process, and submitted to the contracting activity for acceptance when development is completed. Acceptance of the book plan (narrative/outline and dummy manual) shall not waive any requirements within the scope of the contract.

3.9.1.1 Narrative/outline. The narrative portion of the narrative/outline shall indicate the planned technical manual development approach, identify which options are permitted by the contract and specification, and specify options the contractor is electing to pursue. The outline portion shall be

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based on the development of a table of contents and list of illustrations in accordance with 3.2.1.7 and 3.2.1.8, and a functional index in accordance with 3.2.1.10. The outline shall conform to the requirements of MIL-M-38784 and the following:

a. The FOMM volume structures shall be reflected by the outline.
b. The outline requirements of MIL-M-38784 shall apply to the FOMM support volume.
c. The outline of the Troubleshooting/Repair Volume shall reflect the diagrammatic structure of the volume content requirements.
d. A preliminary list of test equipment, required by 3.2.2.1.1.6, shall be included as part of the outline.
e. A summary of the major points of the quality assurance program requirements as specified (see 4.2) shall be included.
f. A synopsis of the validation approach to the manual content requirements shall be included based on a plan for use as supplied by the Government and a maintenance engineering or logistics support analysis of the equipment. The synopsis shall indicate coverage on a comprehensive and systematic basis of the most effective and efficient method of performing necessary maintenance. The synopsis shall be correlated to the logistics support analysis reports (if available) by direct referencing.

3.9.1.2 Milestone schedule. As an addendum to the book plan, the contractor shall furnish a milestone schedule for development of the publication for the individual contract. Milestones to be identified shall include, but are not limited to, the following:

a. Book plan submission.
b. RDC drafting - 25, 50, 75, and 100 percent completion points.
c. Schematic preparation.
d. Parts listing.
e. Editing.
f. Composition.
g. RDC submission.
h. Validation.
i. Reproducible preparation completion.

3.9.1.3 Dummy manual. After acceptance of the narrative/outline, a dummy manual shall be developed for the troubleshooting/ repair volume in accordance with the content and format requirements of this specification. The dummy manual shall portray the planned manual coverage by volume, division, etc., and shall reflect the approved outline. Each page of the troubleshooting/ repair volume shall be represented in the dummy manual. A layout page shall be prepared initially for each information package. As the content of each page in the manual is developed to the point where significant information can be shown, a layout page shall be prepared for the dummy manual. Each page of the dummy manual shall reflect the content of its corresponding RDC page. Text, procedures, illustrations, tables, and MDCs shall be keyed to the particular areas of the prepared page layout. When RDC pages are prepared to final format, they may be used in the dummy manual. (When the dummy manual is completed by having all draft pages inserted, it shall be formulated for submittal as the review draft copy.)

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3.9.1.4 Updating. The book plan shall be kept up-to-date throughout the development of the technical manual review draft and until the draft is accepted. Significant changes to the book plan shall be submitted to the Government for acceptance. Improved data presentation shall be a prime justification in preparing changes to the outline and dummy manual. Dummy manual pages shall be available for in-process review conferences. The dummy manual shall be complete when submitted for Government acceptance.

3.9.1.5 Acceptance. The outline/book plan shall be submitted to the contracting activity for acceptance prior to development of the manuscript technical manual. If rejected, the unacceptable portions shall be corrected and resubmitted. The accepted outline/book plan shall become an addendum to the specification governing the preparation of the technical manual to the extent that it agrees with the requirements of the contract. Acceptance of the outline/book plan by the Government shall not waive any requirements within the scope of the contract.

3.9.2 Review draft copy (RDC). Unless otherwise specified (see 6.2), the RDC for review shall be prepared and submitted to the designated activity for acceptance prior to the preparation of the final reproducible copy. The RDC shall contain all front matter, text, tables, and illustrations needed to meet the content requirements of this specification. The RDC shall be proofread, edited, validated, and collated in the same manner as the final manual prior to submittal. Production of the RDC shall be formulated from the dummy manual and be in accordance with the contract and as specified herein, as follows:

a. Each page of the RDC shall be presented on one side of the paper and shall contain the exact wording and content intended for the reproducible copy. The production method may be the most economical method at the option of the contractor.
b. Reproduction of pencilled illustrations are acceptable if their technical content, clarity, correctness, and adequacy meet specification requirements. Illustrations submitted for review shall be legible.
c. Shading or color coding is not required in the RDC but functional boundaries shall be illustrated by dotted lines. Hardware boundaries shall be indicated by dot-dash lines, with the number of dots corresponding to the level of containment (first level, second level, and so forth).

3.9.3 Final reproducible copy (FRC). FRC, including original artwork (complete in all respects for photographic reproduction), shall be prepared as specified herein. Artwork prepared for manuals under contract or order shall be the property of the contracting activity and shall be delivered as the contracting activity directs. Text pages may be typed (using a typewriter or word processor), reproduced mechanically or by a computer typesetting system. Unless otherwise specified (see 6.2), the text in the support volume shall be double column, flush left column format. If a computer typesetting system is used, text may be justified.

Spacing and lettering shall conserve space without lessening usability or clarity of material. Letters, lines, and symbols shall be of a uniform contrast throughout the text material and shall not touch.

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3.9.4 Photolithographic negatives. Unless otherwise specified (see 6.2), photolithographic negatives, including screens (see 3.7.7), shall be prepared as specified herein (see 3.7.3).

3.9.5 Replenishment materials. Replenishment materials consisting of one copy of the final manual, the photolithographic negatives (including the line negatives and screen negatives) and the printer's assembly instructions used for the printing of the manual shall be supplied (see 3.7.9).

3.9.6 Certification checkoff list. A signed and completed checkoff list shall be prepared (see figure 36) for review manuscripts, preliminary and final manuals, and revisions. The checkoff list shall be submitted to the contracting activity with RDC and shall be modified to agree with the manual and shall indicate the degree to which the RDC conforms to this specification. A similar list shall be prepared for all permanent changes to the manual and shall certify the extent to which the content requirements conform to this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspec-tion, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality assurance program requirements (CG, MC, N). The contractor shall document and maintain a written quality assurance program as required in accordance with MIL-M-85337 to assure that the manual will accurately and adequately reflect the equipment or system covered, and shall include the necessary installation, operation, maintenance, repair, and parts support procedures for satisfactory equipment performance.

4.3 Validation/verification

4.3.1 Validation/verification (A). Validation/verification and the requirements thereof shall be in accordance with MIL-M-38784 and MIL-STD-335.

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4.3.2 Validation/verification (F). validation and verification shall be in accordance with Technical Order 00-5-1.

4.3.3 Validation/verification (CG, MC, N). Validation and verification shall be in accordance with MIL-M-85337.

4.4 Quality assurance program acceptance. Acceptance of the contractor's quality assurance program by the Government in no way relieves the contractor of the final responsibility to furnish manuals as specified. Acceptance shall not preclude additions, refinements, or changes by the contractor to the quality assurance program when evidence indicates that the program does not or will not meet the requirements of the contract or this specification. When additions, refinements, or changes to the program are determined necessary by the contractor, acceptance shall be obtained from the cognizant Government activity prior to implementation.

4.5 Approval and acceptance (MC,N). All publications shall be submitted for approval and acceptance to the contracting activity or representatives thereof. Deliverable items listed in 3.9 shall be submitted for approval or acceptance with sufficient time (or as scheduled in the contract) to allow an adequate review by the Government. Deliverable items will not be accepted by the Government until the contractor has complied with all conditions of approval.

4.5.1 Publication or change number. The contractor shall request a publication or change number from the contracting activity.

4.5.2 Preliminary technical manuals (PTM). When there is insufficient time (less than approximately 6 months) to permit acceptance and still provide printed final manuals with the delivered equipment, permission to supply PTM shall be requested from the contracting activity. Permission shall also be obtained from the contracting activity prior to shipping the PTM when any required data are missing. Prior to shipment with any equipment, a review copy of the PTM shall be submitted to the cognizant Government representative for approval and acceptance. The delivery of PTM does not relieve the contractor of any contractual requirements pertaining to delivery of complete, adequate, and accurate final manuals. PTMs shall be updated to represent production equipment prior to submittal with the hardware.

4.5.2.1 Replacing PTMs with final manuals. When the contractor is responsible for printing and distribution, a self-addressed, contractor-furnished postcard containing information equivalent to the following notice shall be attached to the title page of all PTMs which accompany the equipment:

> "IMPORTANT NOTICE: This is a preliminary technical manual for (insert nomenclature of equipment), publication (insert number). A copy of the FINAL manual will be forwarded directly to you when printed. Return this card IMMEDIATELY, indicating your activity and mailing address."

4.5.3 Review draft copies (RDC) for manuals, changes, and revisions. Validated RDCs of the formal manual, changes, or revisions shall be submitted to the contracting activity for acceptance. A minimum of 6 weeks from receipt of material shall be allowed for Government review.

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4.5.4 Rejection criteria. Contractor failure to meet all contractual requirements, including those specified in this and other referenced specifications, shall be cause for rejection of the deliverable items. The following rejection requirements shall also apply:

a. Book plan. Failure to conform with the Government-accepted book plan. (Acceptance by the Government of a book plan which includes a waiver of contract requirements does not waive such requirements for the manual. Waivers of such requirements shall require a contract modification.)
b. Technical accuracy and inconsistencies. Failure of all information to exactly represent the equipment or system being described and to be consistent with descriptive terms.
c. Omission of data. Omission of any applicable data elements required by this specification.
d. Validation. Failure to provide adequate evidence of validation (see 6.5.58).
e. Workmanship. Failure to conform to the requirements of 3.7 and MIL-M-38784. (Inclusion of significant typographical and grammatical errors.)
f. Reproducible copy, photolithographic negatives, and shading screens. Failure to include changes or corrections required as conditions of acceptance. Inadequate legibility or to meet the requirements of MIL-P-38790.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

5.1 Packaging requirements. Unless otherwise specified (see 6.2), the packaging requirements for delivery of book plans, review draft copies, preliminary manual, final reproducible copy, technical manuals, replenishment materials, and certification checkoff list shall be in accordance with this specification.

5.1.1 Classified material. Classified material shall be packaged and marked in accordance with the requirements of DOD Instruction 5220.22-M (see 6.2).

5.1.2 Single and multivolume manuals. Single copies and multivolume technical manuals shall be packaged to preclude damage in transit. Multivolume technical manuals shall be furnished as complete sets except for bulk manuals shipped for stock. Stock copies of identical volumes shall be packed and shipped in common containers.

5.1.3 Manuals of 16 pages or less. Manuals consisting of 16 pages (8 sheets) or less shall be enclosed in a clear, durable, vinyl envelope not exceeding 15-1/2 by 13 inches in size. The title page and publication number of the manual shall be printed on or be visible through the envelope. The bulk quantity of manuals shall be individually enclosed in the vinyl envelopes.

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5.1.4 Replenishment material (N). The replenishment materials shall be packaged in accordance with 5.2.2.2. The container(s) shall be identified (see 5.4.1) and marked "FOR STORAGE". The container with a transmittal letter shall be forwarded to:

Director
Naval Publications and Printing Services
Eastern Division
Building 4, Section D
700 Robbins Avenue
Philadelphia, PA 19111

5.2 Preservation. Unless otherwise specified (see 6.2), preservation shall be level C as follows:

5.2.1 Manuals

a. Accompanying Equipment. Manuals, which accompany equipment shipments, shall be unit packed in a transparent, waterproof plastic bag, minimum 4 mil thick. Closure shall be heat sealing. Technical manuals shall not be placed within any flexible, sealed barrier enclosing components. The copy(s) of the manual shall be placed in the shipping container housing the main unit. Packing lists shall indicate which container contains the technical manual(s) and shall also state the approximate location therein. For ease of removability, the location of the manual shall be such that it is readily accessible when the container is opened and no subsequent damage is sustained by the flexible barrier enclosing the equipment. Multivolume manuals, each volume unit packed as specified herein, shall be assembled as a complete set and contained in a fiberboard box and closed meeting the requirements of the applicable carriers rules and regulations.
b. Bulk. Manuals when shipped for stock shall be furnished unwrapped. Multivolume manuals shall not be furnished in sets. Manuals shall be packed for shipment in accordance with 5.3. Containers shall be marked "FOR STOCK" (see 5.4.1 for container marking information. Correction material (changes or temporary corrections) and field change bulletins (see MIL-F-17655) shall be packages separately as separate line items.

5.2.2 Data items (see 1.1.4 and 3.9). Each data item shall be contained in a transparent or opaque unit pack (wrap, bag, envelope, or mailing tube) and sealed. Flexible item(s) in flat unit packs shall be provided with paperboard or fiberboard stiffeners to prevent item and unit pack damage.

5.2.2.1 Original artwork. When original artwork is mounted on illustration boards, it shall have a protective flap. The illustration boards shall have instructions, registration marks, and so forth, as required. Original artwork shall be packed flat (never folded) except that large sheets may be rolled if this method is authorized by the agency receiving the artwork.

5.2.2.2 Photolithographic negatives and masks. Photolithographic negatives including screens and masks shall be packed flat (never folded or rolled) and double-packaged. A printing instruction sheet shall be packed with the

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negatives, placed unfolded in an envelope clearly marked with the publication number and titled printing/set-up instructions, and packed on top the negatives. A sheet of tissue paper shall be inserted between each negative. The interior material shall be waterproof and free of any chemical substances that would discolor or otherwise degrade the negatives. The exterior package shall be a standard commercial carton at least equal to Interstate Commerce Commission standards, and of sufficient strength to protect the negatives against the forms of damage that frequently occur during shipping (see 5.4.1 for marking requirements).

5.2.2.3 Certified checkoff list. Certified checkoff lists shall be packaged in standard manila envelopes.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2).

5.3.1 General requirements for levels A, B and C. Containers selected (see 5.3.2), shall be of minimum weight and cube consistant with the protection required, of uniform size, and contain identical quantities of identical material.

5.3.2 Levels A, B and C containers. Material shall be packed in exterior shipping containers for the level of packing specified, in accordance with table VII of MIL-STD-2073-1, Appendix C, and herein. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option.

5.3.2.1 Caseliners, closure and gross weight

5.3.2.1.1 Caseliners. Unless otherwise specified (see 6.2), level A shipping containers shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.2.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and domestic or fire retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.3 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.2.2 Shipments in multiple containers. For equipment shipped in multiple shipping containers, the technical manuals shall be placed in the container housing the main unit of the equipment. Packing lists, invoices, or bills of lading shall indicate the number of the container that encloses the technical manuals (see 5.4.1), the publication identification number, nomenclature, and so forth. For ease of removability, the location of the manual(s) within the container shall provide immediate accessibility when the container is opened. The shipping container housing the manuals shall be marked "MANUALS ENCLOSED".

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5.3.3 Commercial. Material shall be packed for shipment in accordnace with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be a minimum of two, 3-inch by 4-inch nominal wood skids laid flat, or a skid or sill type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage and storage.

5.4 Marking levels A, B, C and commercial.

5.4.1 Container marking information. Interior packs and shipping containers shall be marked including bar coding for shipment, stowage, and storage in accordance with MIL-STD-2073-1, Appendix F. In addition to the sender and address information, the exterior of each container shall show the following, as applicable:

a. Publication identification numbers.
b. Stock number (N).
c. Equipment nomenclature.
d. Review draft copy.
e. Reproducible copy.
f. For storage.
g. For stock.
h. Manuals enclosed.
i. Number of containers in shipment.
j. Replenishment materials.
k. Original artwork.
l. Negatives

5.4.2 Packing list. A copy of the letter of transmittal, the packing list, or bill of lading shall be placed inside the carton or container. When a shipment consists of multiple containers, the packing list material shall be enclosed in the first container and shall identify the material packed in each container.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. FOMM's are technical manuals used to support equipment and system training installation, operation, maintenance, repair, and parts provisioning. FOMM's are also used for purposes of on-the-job training and classroom instruction. The arrangement of data is specifically designed so that technicians can effectively understand, troubleshoot, and repair equipment or systems for which they have had little formal training. It is specifically designed to employ a separate volume organized on a functional basis to provide troubleshooting information to paralled the method of fault isolation and repair methodology by which military forces are trained. Further, it is self-contained requiring limited indexing, figure numbering, paragraphing, and referencing to meet these objectives. This concept is unique to FOMM. Support data is provided in a separate volume organized topically as are standard technical manuals.

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6.1.1 Types I and II manuals are final manuals that are provisioned to military services for operational use of equipment and systems.

6.1.2 Types IS and IIS manuals are manuals supplied to military services for evaluating the applicability of equipment or systems for military use. Service test manuals are supplemented by final manuals when such equipment/system is acquired for service use.

6.1.3 Types IX and IIX manuals are technical manuals containing minimum required data to provide operation and maintenance of research and development (R&D) models of equipment and systems. These manuals are not intended for operational use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. When a new technical manual is required to support an equipment or system, the manual type shall be specified. (see 1.1.3 and 1.1.5).
c. The equipment or system models, configurations, and components to be covered (see 1.1). Type I manuals should be acquired to support an equipment (or set) including all its units and assemblies as delivered on the contract. Type II manuals should be acquired for a specific system installation (aircraft, ship) or subsystem (NTDS, fire control, IFF) wherein each equipment is supported by its own maintenance manual, rather than for a broad group of dissimilar installations when the individual equipment manuals do not provide the necessary integrating information.
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
e. Whether the equipment or system is intended for operational (field use), service test, or experimental (R&D) - (N). (If the equipment or system is considered to be experimental but is intended for service test, the manual should be specified as service test.
f. Size of manual (see 3.7.5.1). The troubleshooting volume is 11 by 17 inches (see 3.7.1.1.1.6). When an equipment or system is of such a nature the troubleshooting information can be presented or well accommodated on a smaller sheet or cannot be physically accommodated in its user environment, the 8 1/2 by 11 inch or 11 by 14 inch sheet manual shall be specified.
g. When the depth of coverage is other than that necessary to maintain the equipment or system through organizational (O), intermediate (I), and depot (D) level maintenance as defined by the maintenance philosophy (see 3.1.2.4, 3.1.7, and 3.2.2.1.4.9). When the maintenance philosophy involves maintenance of the hardware at the "O", "I", and "D" levels, the manual shall be prepared to cover all levels of maintenance unless a separate manual is under procurement to cover I and or D levels.
h. What issues are required (see 1.1.3).
i. What data items are to be delivered (see 1.1.4 and 3.9).
j. When the level of writing for types I, IS, II and IIS manuals is other than as specified (see 3.1.2.3).

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k. When the level of writing for types IX and IIX manuals is other than as specified (see 3.1.2.3).
l. Applicable publication number, classification, and distribution statement (see 3.2.1.1).
m. Whether or not installation data/chapter is required. (Installation chapters are required whenever the military services are expected to install or reinstall the equipment or system.) Whenever a system is only supposed to be installed and reinstalled by the shipbuilder or manufacturer, installation information will not be required in the FOMM (Installation data must be available in other forms such as drawings, procedures, etc.
n. If a model standardized log form is required (log forms are only required for machinery when it is necessary to maintain operating records) (see 3.2.2.1.2.5).
o. When separate parts lists are required (see 3.1.3.2. and 3.2.2.1.6). For complex equipment with an extensive number of repair parts, it is desireable to include a separate parts list in a support volume to preclude detail piece part descriptions in the troubleshooting volume which would encumber its ease of use for troubleshooting. Separate parts lists are not normally used for equipment with a limited number of repairable parts.
p. When other then NAVSEA 0969-LP-019-7000, MIL-STD-1365; NAVAIR 16-1-523; Air Force MIL-HDBK-300, or to 33K-1-01, or USMC TM-6625-15/1 (as applicable) is to be used for selection and application of test equipment (see 3.2.2.1.1.6).
q. When MRCs are to be prepared and delivered (see 3.2.2.1.4.1) separately under a separate line item, or other contract, or order.
r. When depot level maintenance information is limited to the depot, it shall be bound separately (see 3.2.2.1.4.9).
s. When a single volume FOMM is to be supplied (see 3.6.2.3). [Most FOMM manuals are prepared for two volume issue. However, a single volume may only be supplied for equipment or systems minor in scope and complexity, and where only a few (less than 1000) repairable piece parts are involved and the troubleshooting and repair is straight forward and relatively easy.]
t. Photolithographic negatives and screens shall always be specified unless the government has established a contractual relationship with the preparer to design and fabricate the masks necessary for the government to develop negatives and screens (see 3.9.4).
u. Specify a preliminary technical manual when it is necessary to field equipment for test and evaluation, or when an extended period of use is required prior to government verification (normally applies to complex equipment and systems only) (see 3.8.1).
v. To augment the final manual, a technician's pocket manual, which is a reduced size reprint of the troubleshooting and repair volume, shall be specified when the convenience of use will be beneficial in the using environment, that is submarines, flight decks, cramp quarters, etc. (see 3.8.3). No cost other than printing, handling, and shipping costs are justified on its procurement).
w. Supplement manuals (see 3.8.4) shall be specified when:

(1) The classified information can be confined to the supplement manual such that its basic technical manual will be rated as "unclassified".

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(2) It augments a technical manual to provide for a different model and can be justified as a cost effective method of promulgation.
(3) It can be justified as a practical method of promulgation.

x. Changes shall be specified only when they pertain to technical manual corrections resulting from a hardware change or modification. They may also be specified if it has been determined that significant omissions of technical data or information has been identified. Do not specify a change to correct minor or editorial corrections. When changes are justified on the above basis, the change shall include the correction of all outstanding temporary or interim changes and may include minor changes (see 3.8.5). Whenever the number of change pages is estimated to exceed approximately 25% of the total number of pages in the manual, an update revision shall be specified (see 3.8.5.2).
y. When the total number of pages requiring change exceed approximately 25% of the total number of pages in the manual, a revision shall be specified (see 3.8.5.2).
z. A book plan/dummy manual shall be specified whenever a technical manual is required for a new equipment or system (see 3.9.1). Also see 6.2.b.
aa. A review draft copy (RDC) shall always be specified for every new technical manual (see 3.9.2). For change procurement, unless changes to the technical manual have been pre-approved (TMDER, ECP, etc) by the government, RDC shall be specified.
ab. Double column text shall be specified (see 3.9.3). Unless the presentation of the equipment or system to which the text pertains is more effective, then a single column text shall be specified.
ac. Replenishment material shall always be specified for delivery to the government for subsequent use in reprinting (see 3.9.5). (No costs other then handling and shipping costs are justified on the procurement of replenishment material).
ad. Certification check-off list (see 3.9.6) shall be specified for delivery with all review draft copy (see 3.9.2).
ae. Unless a greater time is required by the acquiring activity, six working weeks shall be specified for review of the review draft copy.
af. When preservation, packing, and marking are other than as specified (see 5.1).
ag. For classified equipment, attach a DD Form 254 specifying the following:
(1) Security classification.
(2) Downgrading and declassification notification.
(3) Areas requiring security protection.

6.3 Technical manual acquisition. This specification must be listed on the Contract Data Requirements List (DD Form 1423) in order to acquire the technical manuals described by this specification, except where DoD FAR Supplement 27.475.1 exempts the requirement for DD Form 1423.

6.4 Industrial security manual for safeguarding classified information. The Industrial Security Manual for Safeguarding Classified Information, DOD Instruction 5220.22-M, should apply.

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6.5 Definitions. The following paragraphs define words or phrases used throughout this specification. (SGM)

6.5.1 Assembly. A number of parts or subassemblies or any combination thereof, joined together to perform a specific function or to fulfill a specific purpose.

6.5.2 Blocked schematic. A diagrammatic presentation in which the circuit elements have been grouped (blocked) into functional entities in order to present a clearer picture of the assembly operation.

6.5.3 Blue. Refers to the color of the functional areas in the final manual but is not normally required for most applications. Blue is a shading methodology that shows functional levels when the functional levels exceed 2 in number.

6.5.4 Book plan. Comprised of a narrative outline and dummy manual detailing the scope of the planned manual. It also includes the plan and schedule for manual development.

6.5.5 Circuit elements. Those electrical, electronic, or mechanical parts not subject to disassembly without destruction of their designated uses (for example, transistors, resistors, capacitors, transformers, or valves).

6.5.6 Component. A composite fabricated unit, generally complete within itself, performing a stated service when installed in its proper position within a ship or station (for example, controller, motor, or power voltage regulator).

6.5.7 Component identification number. A design identifier which gives the manufacturer's name and the characteristics for a particular piece of equipment. For purposes of this specification, the contracting activity shall specify the significant component identification to be used.

6.5.8 Composite entity. A group of functional entities that work together to perform a larger function, such as a series of flip-flops connected as a ripple counter.

6.5.9 Dependency chart. See 6.5.30 for maintenance dependency chart (MDC) definition.

6.5.10 Discrete parts. Individually distinct parts as opposed to integrated circuits (for example, resistors, transistors).

6.5.11 Distribution statement. A statement used in marking a technical document to denote the conditions of its availability for distribution, release, or disclosure. This statement will be provided by the acquiring activity upon request.

6.5.12 Equipment. One or more assemblies capable of performing a complete function and delivered as a manufactured unit and with its own nomenclature.

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6.5.13 Event. An action, or a presence of a voltage, signal, pressure, or other data at a defined observation or measurement point, that results from a turn-on, operational, or checkout procedure, or the presence of an initiating signal, data, etc.

6.5.14 Experimental equipment. An equipment designated to prove the feasibility of a concept.

6.5.15 Final manual. A completed book, accepted by the contracting activity, which complies with all requirements (including production requirements) of this specification, and has been validated, verified by the government to reflect the production equipment and is adequate for its intended use.

6.5.16 Final reproducible copy (FRC). An original or designated master of the complete and final technical manual material to be used as the master for reproduction or replication purposes. The specific media required shall be identified. The FRC includes all text pages, tables, illustrations, and drawings as well as original, integrally related artwork suitable for use in the development of printing plates and is complete in all respects for photographic or computer-aided reproduction. FRC is supplied final size. When negatives and screens are acquired, they are the reproducible copy.

6.5.17 Functionally Oriented Maintenance Manual (FOMM). An acronym for manuals prepared to this specification (see 1.1.1). It differs from standard manuals in that it organizes data on a basis of how an equipment or system operates and parallels military diagnostics, fault isolation, repair practices, and training.

6.5.18 Functional block/area. A block or area defined on a functional diagram to denote the function(s) being defined. Functional blocks are normally white and delineated by a hardware shading or dotted lines. Multilevel function blocks may be shown by blue shadings in the same manner that gray shading shows level of hardware. They may also be shown by differently separated dots.

6.5.19 Functional devices. One or more parts which perform a functional operation.

6.5.20 Functional entity. A group of parts connected together to perform a basic function (for example, amplify, oscillate, filter) and, in the event of a malfunction, provide a recognizable (visual, audible, or by means of a measurement) symptom.

6.5.21 Functional entity identifier code. A group of letters and a number which identify a particular functional entity. The first letter defines a general (physical) characteristic, the following letters define the purpose of the entity, and the number differentiates between identically defined entities on one assembly.

6.5.22 Hardware information. Detailed information contained in the troubleshooting/repair volume as designated by its hardware, unit, and sublevel identifiers. Hardware information packages consist of blocked

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schematics, keyed text, MDC, alignment, repair data, parts data, and troubleshooting information necessary to isolate a fault in an assembly, subassembly, module, circuit, or piece part.

6.5.23 Hierarchical data. An arrangement of information into levels from the most general to the most detailed. The troubleshooting/repair volume/manual uses this arrangement to illustrate the functional and physical packaging of an equipment. The basic levels consist of the following:

a. Overall information.
b. Major function information.
c. Hardware information.

6.5.24 Interim change (A, CG, F, MC). A formal expeditious method of promulgating a minor manual change by means of replacement pages or pen-and-ink corrections. (Interim changes are often referred to as temporary changes.)

6.5.25 Intermediate subfunction diagram. A block diagram constructed to the same rules as the overall function diagram. It fits into the hierarchical structure between the overall function diagram and the major function diagrams. It is only used when the number of major functions is too great to be presented on the overall function diagram.

6.5.26 Intermediate-level maintenance. The purpose of the intermediate-level maintenance action is to troubleshoot and repair down to the piece part. An activity designated for specific repair of an item deemed too dificult to be performed at the operational level or which may require equipment only available at the intermediate maintenance activity (IMA).

6.5.27 Issue. The technical manual (final or preliminary), revision or change thereto as promulgated for use or a particular purpose, such as test and evaluation.

6.5.28 Keyed text. A parallel reading descriptive text. Keyed text is not intended to be read in its entirety and needs to be read only when an element of the diagram needs explanatory information when reading diagrams. Keyed text is descriptive text provided for each functional blocked area (shaded area) on a particular diagram. Encircled numbers associated on the diagram are keyed to the appropriate text and vice versus and always appears on the same or facing page of the diagram except when referenced elsewhere. Keyed text is prepared in columnar format for each diagram in the troubleshooting/ repair volume and always appears on the same or facing page.

6.5.29 Landscape. A format that presents information along the long dimension of a page.

6.5.30 Maintenance dependency chart. A compression of functional blocks, circuits and circuit elements that are represented by dots (tabular arranged) and symbols that show the string of output events that result from given inputs or actions. A diagram showing the processing of energy/information through the functional grouping depicted on the related blocked schematic or functional blocked diagram.

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6.5.31 Maintenance requirement cards. A series of cards, part of the planned maintenance subsystem, used by the technicians for performing scheduled tests and preventive maintenance for given equipments and systems. When the planned maintenance subsystem is installed, the procedures on MRCs supersede the scheduled maintenance procedures in the technical manual.

6.5.32 Major function. Comprise internal functional operation (or modes of operations) that define how an equipment or system works (for example, transmit, receive, synchronization etc).

6.5.33 Major function diagram. A major function diagram is a detailed block diagram of a major function of the equipment or system. The diagram illustrates the interrelationships of all functional entities and devices, and the hardware levels of containment for that function.

6.5.34 Major function information. Comprises all data necessary to illustrate, describe, and troubleshoot a major function to isolate any fault of the function to an item of hardware. The major information package normally consists of a major function diagram, keyed text, MDC, etc.

6.5.35 Organizational-level maintenance. Represents the degree of maintenance that can be performed on a system on-site by the equipment operator or technician. It is limited by the training, tools, instruments, supply parts and the maintenance philosophy designated for the equipment.

6.5.36 Overall function diagram. A block diagram that depicts and identifies all interfaces between the major functions. In final manuals, it is a two-color diagram with the equipment represented by a gray shade and each major function represented by a blue shade.

6.5.37 Overall function information. Contains all information pertaining to the system or equipment required in its entirety to effectively isolate any potential fault to a major function or possible item of hardware. The overall information package contains such data as overall function and intermediate subfunction diagrams, intercabling diagrams, system checkout, MDCs, alignment data, etc.

6.5.38 Part. One piece (or two or more pieces together) which is not normally subject to disassembly without destruction.

6.5.39 Permanent change (N). A formal package to an existing manual which is issued in the form of replacement or additional pages. Permanent changes incorporate all outstanding Advance Change Notices (ACN's) and interim changes.

6.5.40 Planned maintenance subsystem. A formal methodology for maintaining equipment and systems promulgated by the DoD maintennce and military management system.

6.5.41 Preliminary technical manual (PTM). A manual which includes all data required for the final manual but has not been verified by the government and has not received final acceptance and may not comply with the production requirements for a final manual. PTMs are superseded by final manuals.

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6.5.42 Repair. Repair involves actions required to restore a assembly, subassembly or repairable component to a normal operating condition, but not necessarily to make it like new in physical appearances.

6.5.43 Replenishment material. Consists of one copy of the final manual and the photolithographic negatives, and screens, and instructions used for printing the manual.

6.5.44 Review draft copy (RDC). A copy of the manual used for review and coordination for technical accuracy and adequacy to evaluate the contractor's progress and assess compliance with applicable specifications and terms of the contract. (A) RDC equates to preliminary draft equipment publication (PDEP). (F) RDC equates to in-process preliminary technical manual (PTM). (N) RDC equates to review manuscript.

6.5.45 Revision. A second or subsequent edition of a manual complete with covers and all pages.

6.5.45.1 Complete revision. A completely re-written manual to the requirements of this specification.

6.5.45.2 Superseding revision. An update or complete revision which matches all configurations covered by the manual.

6.5.45.3 Update revision. An update revision is prepared in the same manner as permanent change pages and is promulgated when the number of change pages exceed approximately 25% of the total number of pages in the manual. However, the revised pages are not issued as a change package but are incorporated into the basic manual and then reprinted and distributed as a superseding technical manual, except for minor editorial and paragraph changes. An updated revision shall incorporate configuration modifications and all previous changes issued by temporary or interim changes to the previous manual.

6.5.46 Service test equipment. An equipment used for test under service conditions for evaluation of suitability and performance. It closely approximates the final design, has the required form, and employs approved parts or their interchangeable equivalents. It is often a commercial-of-the shelf item that is being tested for its military applicability.

6.5.47 Signals. The information/energy which may be sensed or measured at defined points within the equipment. Signals indicate different types of information depending upon the equipment involved. Examples are pressure, speed, temperature, voltage, etc.

6.5.48 Signal flow. The ordering of information/energy transfer through a system as shown by drawing layout, use of arrowheads, and descriptions in text. Signal flow does not automatically imply electrical current flow or physical motion.

6.5.49 Subassembly. Two or more parts that form a portion of any assembly. It is replaceable as a whole, but may likely contain parts that are individually replaceable.

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6.5.50 Subfunction diagram. A detailed block diagram of a nested part (subfunction) of a major function of the equipment or system. It presents the same hierarchical level of information as a major function diagram and is constructed to the same rules as a major function diagram. It is only used when the number of functional entities is too great to be presented within the two facing page limitation of a major function diagram.

6.5.51 Supplement. A separate publication which augments or complements a technical manual so that both jointly describe the configuration of the equipment and all necessary information for operation, maintenance, and repair.

6.5.52 Support volume. The first volume of a two-volume FOMM. This volume contains all information not generally required for troubleshooting, corrective maintenance, and repair. It is arranged topically by subject matter in the general size and format of a standard manual.

6.5.53 System. Comprises two or more equipment (sets) or components, each having its own identity and nomenclature, arranged and interconnected to perform a specific operation.

a. An electronic system (or subsystem) can be identified as an IFF system, ECM system, AEW system, ASW system, and so forth.
b. An electromechanical system can be identified as a propulsion system, replenishment system, degaussing system, and so forth.

6.5.54 Subsystem. A subsystem can be the same as a system when "system" is considered to be the basic weapon system or platform (for example, ship system, aircraft, missile).

6.5.55 Technician's pocket manual. A reduced-size copy of the complete troubleshooting/repair volume which augments the final manual. It is issued for use when the convenience of small size exceeds the disadvantages of marginal legibility.

6.5.56 Troubleshooting/repair volume. The second volume of a FOMM. It contains the drawings and troubleshooting data necessary for a qualified technician to locate and correct a fault in an equipment or system without reference, except for performance test procedures, to other documents. It is arranged in landscape format and organized on the basis of how an equipment or system operates, and parallels military diagnostics fault isolation, repair practices, and training.

6.5.57 Unit. A major assembly for a set or system, consisting of a collection of basic parts, subassemblies, and assemblies packaged together as a physically independent entity. A second hierarchial structure of a system or equipment. It is normally identified by a nomenclature and assigned a unit number by the manufacturer.

6.5.58 Validation. The process by which the contractor (or as directed by the DoD Component procuring activity) tests a technical manual for technical accuracy and adequacy, comprehensibility, and usability. Validation is conducted at the contractor's facility or at the operational site and involves

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the hands-on, unless otherwise agreed upon by the DoD Component, performance of operating and maintenance procedures including checkout, calibration, alignment, and scheduled removal and installation instructions.

6.6 Subject term (key word) listing.

Hardware information package
Integrated blocked schematic
Intermediate subfunction information package
Maintenance dependency chart
Major function diagram
Overall function information package
Support volume/manual
Troubleshooting/repair volume

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army-TM
Navy-SH

Review activities:

Army-AT, ME, CE, AV
Navy-CG, MC, EC
Air Force - 10

User Activity:

Navy-OS

Preparing activity:

Navy-SH
(Project TMSS-N197)

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DETAIL REQUIREMENT REFERENCE		SUPPORT VOLUME/ MANUAL													TROUBLESHOOTING/REPAIR VOLUME/MANUAL								
	TYPE I PARAGRAPH NO.	3.2.1	3.2.2.1.1	3.2.2.1.2	3.2.2.1.3	3.2.2.1.4	/	3.2.2.1.5	3.2.2.1.6	3.2.1.13	3.2.1	3.2.2.2.1.2	3.2.2.2.1.3	3.2.2.2.1.4	3.2.2.2.1.5.1	3.2.2.2.1.5.3	3.2.2.2.3.1	3.2.2.2.3.3	3.2.2.3.3.1	3.2.2.3.3.3	3.2.2.3.3.5	3.2.2.3.3.6
	TYPE II PARAGRAPH NO.	3.2.1	3.2.3.1.1	3.2.3.1.3	3.2.3.1.4	3.2.3.1.5	3.2.2.1.6	3.2.3.1.7	3.2.3.1.8	3.2.3.1.2	3.2.1	/	3.2.3.2.1.5	3.2.2.2.1.4	3.2.3.2.1.2	3.2.3.2.1.4	3.2.3.2.2.2	3.2.3.2.2.4	3.2.3.2.3.2	/	/	/	/
MANUAL TYPE	NUMBER OF MAJOR FUNCTIONS	FRONT MATTER	GENERAL INFORMATION	OPERATION	THEORY OF OPERATION	SCHEDULED MAINTENANCE	ALIGNMENT	INSTALLATION	PARTS LIST	SAFETY SUMMARY	FRONT MATTER	FAMILY TREES	CABLING/PIPING DIAGRAM	TURN-ON/CHECKOUT CHART	OVERALL FUNCTION DIAGRAM	OVERALL FUNCTION MOC	MAJOR FUNCTION DIAGRAM	MAJOR FUNCTION MOC	ASSEMBLY SCHEMATICS	ASSEMBLY MOC	ASSEMBLY PARTS DATA	WIRING DIAGRAM 6/
TYPE I	1	●	●	●	●	●	5/	5/	4/	●	●	●	●	●			●	●	●	●	●	●	●
	2 OR MORE	●	●	●	●	●	5/	5/	4/	●	●	●	●	●	●	●	●	●	●	●	●	●	●
TYPE I$ 3/	1	●	●	●		2/	5/	5/	4/ 1/	●	●	●	●	●			●		●		●		
	2 OR MORE	●	●	●	●	2/	5/	5/	4/ 1/	●	●	●	●	●	●		●		●		●		
TYPE IX 3/	1	●	●	●			5/	5/	4/ 1/	●		●	●				●		●				
	2 OR MORE	●	●	●			5/	5/	4/ 1/	●		●	●		●		●		●				
TYPE II		●	●	●	●	●	5/	5/	●	●	●		●	●	●	●	●	●	4/				
TYPE II$ 3/		●	●	●	●	2/	5/	5/	●	●	●		●	●	●		●		4/				
TYPE IIX 3/		●	●	●			5/	5/	●	●			●		●		●		4/				

1/ LESS ATTACHING HARDWARE (3.2.2.1.6.4.6).

2/ LESS REFERENCE TO TROUBLESHOOTING ACTIONS (3.2.2.1.4.4.h).

3/ PRODUCTION REQUIREMENTS IN ACCORDANCE WITH 3.7.8.6.

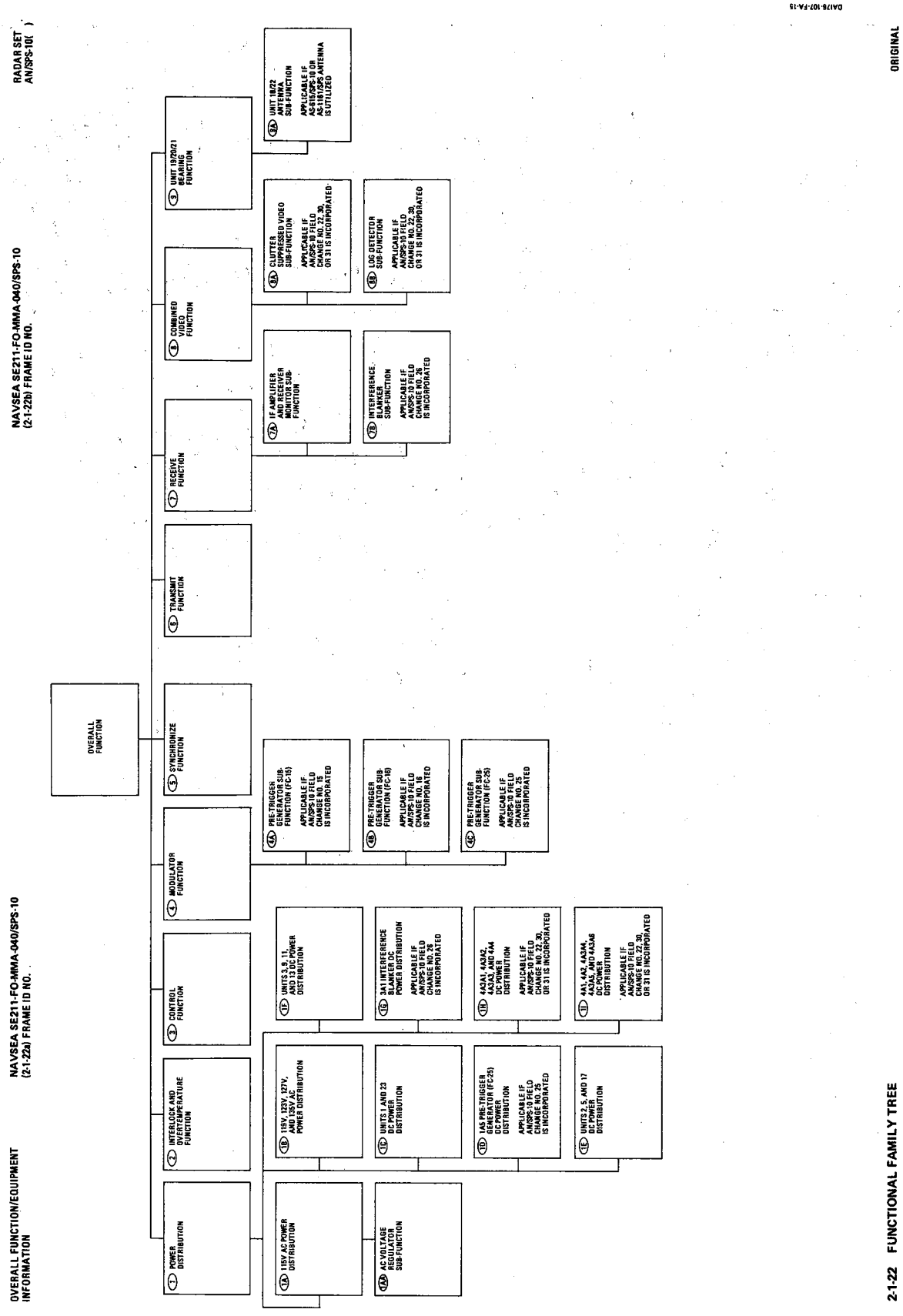
4/ WHEN SPECIFIED IN ORDER OR CONTRACT.

5/ UNLESS OTHERWISE SPECIFIED IN ORDER OR CONTRACT, ALIGNMENT AND INSTALLATION DATA SHALL BE INCLUDED.

6/ ONLY REQUIRED FOR CHASSIS, BACK PLANES, AND FRONT PANELS.

FIGURE 1. Type I and II volume/manual requirement

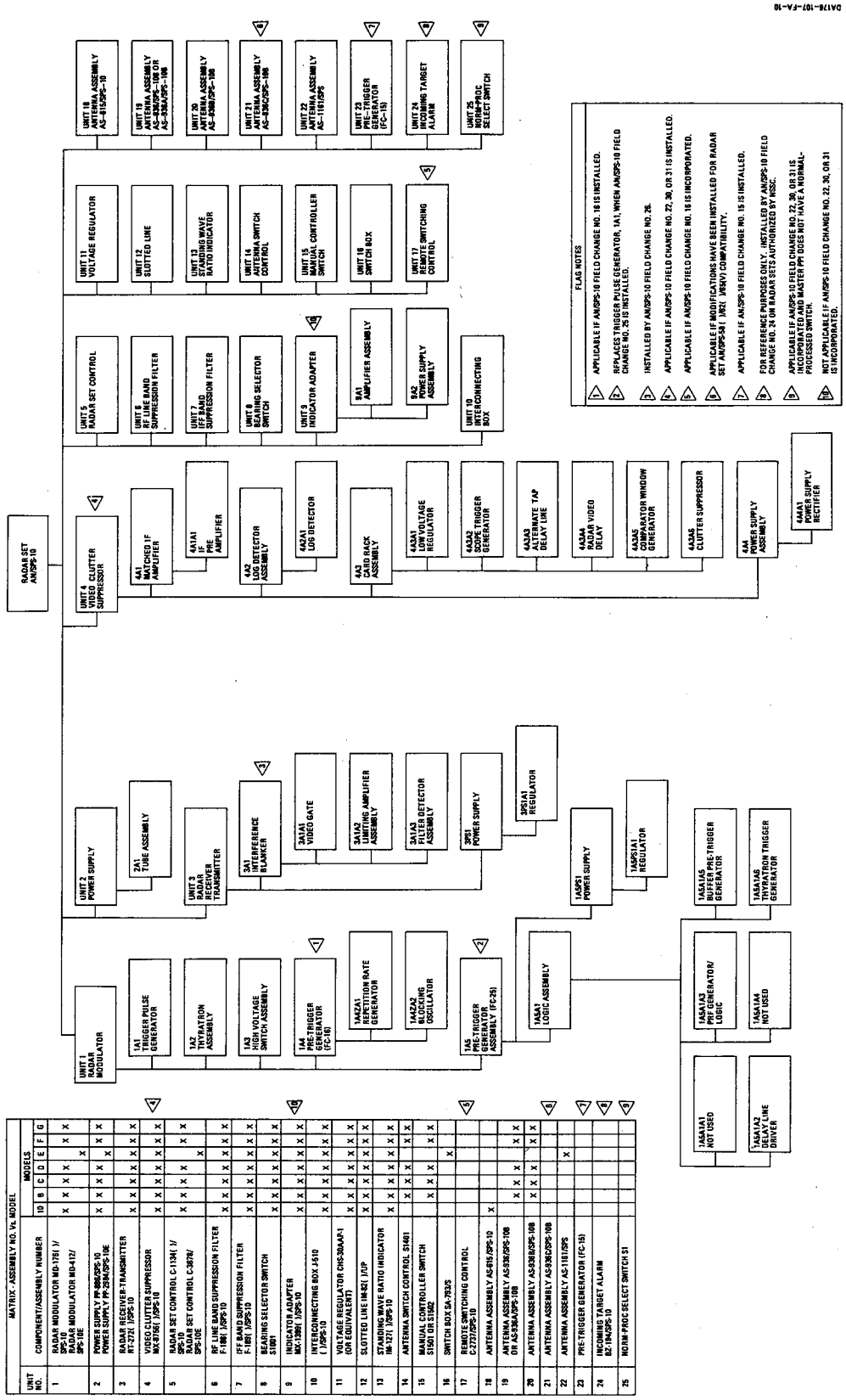
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FIGURE 2. Functional family tree

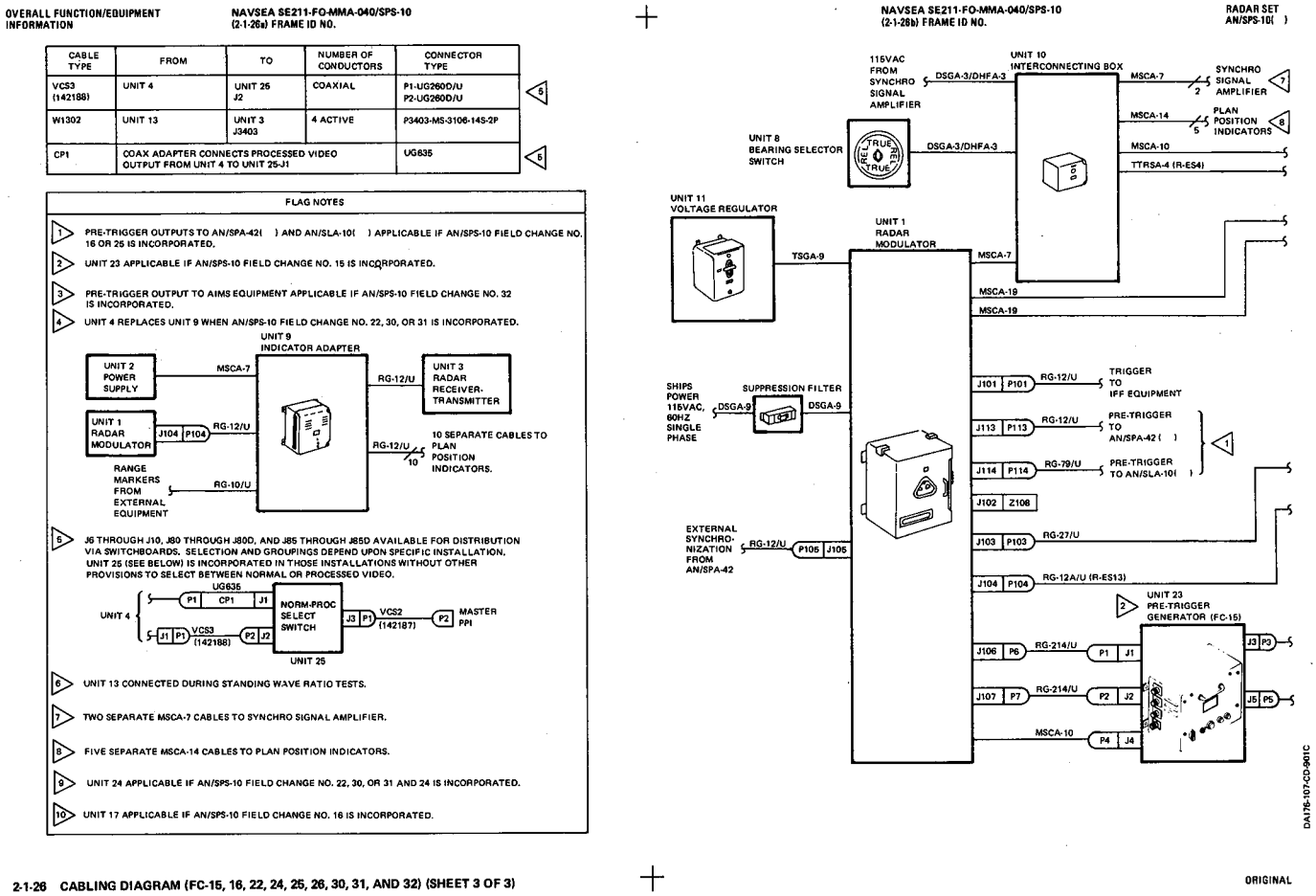
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FIGURE 3. Hardware family tree

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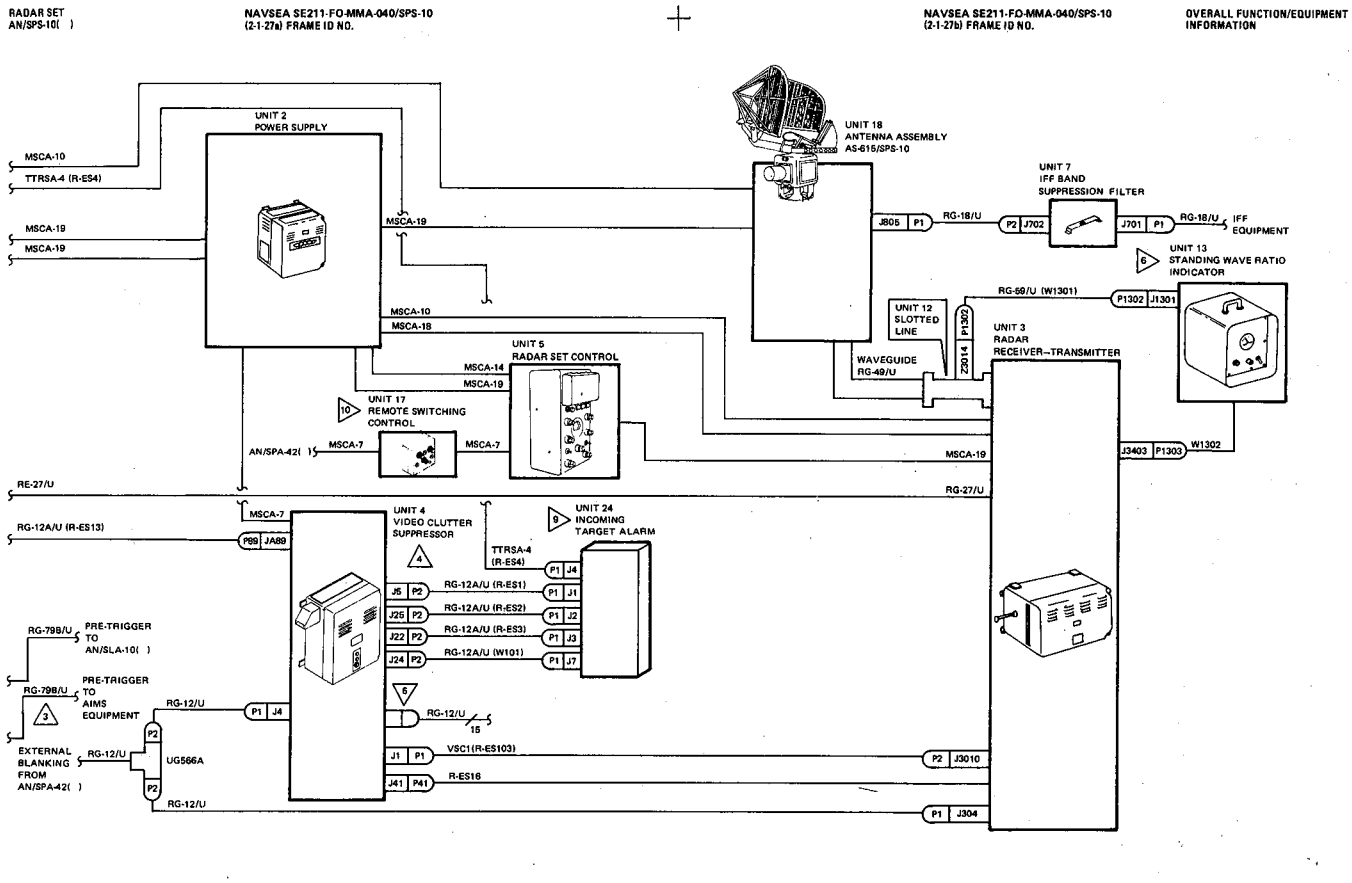
2-1-26 CABLING DIAGRAM (FC-15, 16, 22, 24, 25, 26, 30, 31, AND 32) (SHEET 3 OF 3)

ORIGINAL

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FIGURE 4. Interconnecting cabling/piping diagram (sheet 1 of 2)

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FIGURE 4. Interconnecting cabling/piping diagram (sheet 2 of 2)

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FRONT MATTER

NAVSEA SE211-FO-MMA-050/SPS-10 (2-2-6a) FRAME ID NO.

FOREWORD

This FOMM (Functionally Oriented Maintenance Manual) differs from conventional technical manuals in two broad areas: data arrangement, and method of presentation.

The FOMM is divided into two volumes. Data required to install, operate, and understand the general operational theory of the AN/SPS-10 series radars, and to provide contractor baseline preventive maintenance and parts information, is included in Volume 1, The Support Volume. Information required to troubleshoot and repair the radars is included in Volume 2, The Troubleshooting/Repair Volume. This arrangement enables the qualified technician to isolate and correct any radar failure without being encumbered by information needed for other purposes.

Volume 1 is arranged in sections similar to a conventional technical manual. Volume 2, however, is functionally arranged in information packages, rather than by chapters or sections. There are three information package levels, as follows:

1. The overall function/equipment data package. This contains all diagrams, test procedures, and test values (signal specifications) necessary to isolate a failure to a major function of the radar set.

2. The major function data packages. There is a data package for each radar set major function. Each contains the diagrams, test procedures, and signal specifications necessary to further isolate, to the hardware unit, assembly, or piece part, a failure isolated to a major function.

3. The hardware data packages. Each unit, significant assembly, and subassembly has a hardware data package. They are arranged in reference designator sequence for easy data access. Each contains the schematic, test, alignment, repair, and parts location data required to correct a failure isolated to that piece of hardware without a reference to other parts of the manual.

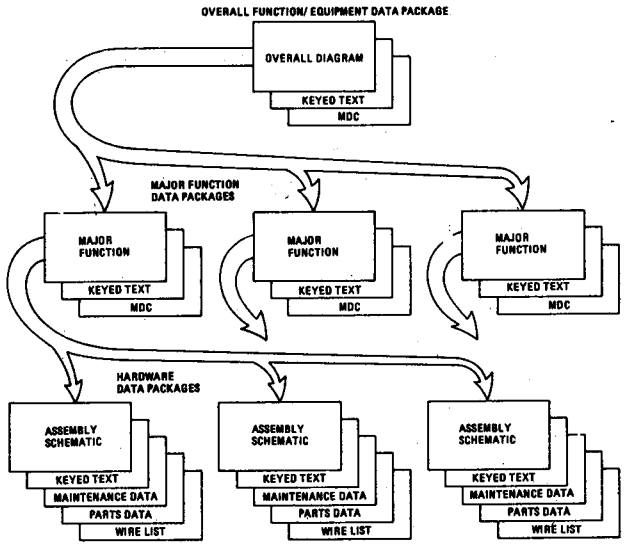
A Technicians Pocket Manual is provided for use by the technician in close quarters. It is a reduced-size reproduction of The Troubleshooting/Repair Volume for use where the regular size volume may be too bulky.

FOMM data presentation methods make it easier to find information and learn about the equipment. For example, shades of blue and grey are used on functional diagrams. Grey shades indicate different levels of equipment hardware (units, assemblies, subassemblies). Blue shaded boxes visually group components that work together to perform a basic function. A circled number (key) in the upper-left corner of each blue box refers to a numbered text paragraph (keyed text) on a facing page. The text describes the ctional operation of the circuit identified by the key number. Troubleshooting information is provided by a Maintenance Dependency Chart (MDC) where applicable. An MDC directs the technician through a logical sequence of signal tracing steps. Voltage or waveform information is provided for each circuit input and output, giving the indication or output from a circuit that must be present if its input is correct and the circuit is working properly.

The following chart illustrates the normal troubleshooting flow through the FOMM information package levels:

NAVSEA SE211-FO-MMA-050/SPS-10 (2-2-6b) FRAME ID NO.

RADAR SET AN/SPS-10( )



The FOMM troubleshooting method normally starts at the Overall Function/Equipment Data Package. A fault is isolated to a specific major function by following the instructions and procedures of The Overall MDC. Next, by referring to the Major Function Data Package and using the MDC for that function, the fault is isolated to a group of circuits on a particular hardware or, in some cases, to a piece part. Then, by referring to the particular schematic and related hardware package maintenance data, the fault can be isolated to a component.

Specifically, the data in this 2-volume FOMM is arranged as follows:

Volume 1 Support Volume

Section 1. General Information. This section contains information of special interest to command-level personnel. It identifies the units that make up the radar set, and provides the physical and functional characteristics of each unit.

Section 2. Operation. This section contains information of special interest to the operator. This includes operating procedures, descriptions of the operator controls and their usage, and how to interpret the presentations on the display.

Section 3. Theory of operation. This section contains background

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FIGURE 5. Troubleshooting/repaire volume hierarchial information (sheet 1 of 3)

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information and theory of operation of interest to a technician beyond the circuit descriptions and information provided in the Troubleshooting/Repair Volume.

Section 4. Scheduled Maintenance. This section contains contractor recommended maintenance procedures. It is not intended that this section be used by ships having the Planned Maintenance-subsystem (PMS) installed.

Section 5. Installation. This section contains installation drawings and other information regarding site selection, equipment cable and waveguide installation.

Section 6. Parts List. This section contains a complete parts listing plus a listing of common items, a listing of attaching hardware, and a listing of the manufacturers of parts.

Section 7. Overhaul. This section contains instructions for disassembly, cleaning, inspection, repair, reassembly, testing and packaging of those mechanisms designated as needing periodic overhaul.

Volume 2 Troubleshooting and Repair Volume

Family Tree. Two illustrations are provided: The hardware family tree which identifies the units that comprise the radar set; and the functional family tree that identifies the block diagrams that comprise the functions of the radar set.

How to Use the MDC. These illustrations and explanations are provided to explain the use of the maintenance dependency chart (MDC) in isolating faults to equipment functions of hardware components.

Cabling Diagrams. These diagrams identify the units and interconnecting cables of the Radar Set AN/SPS-10 series.

Overall Function Diagram. This diagram identifies all equipment functions and the signals that interface between functions.

Major Function Block Diagrams

These diagrams show the connections and circuits used to make up the Function and Power Distribution Diagrams. Blue shaded boxes are used to indicate which circuits work together, while grey shades are used to locate the hardware where the circuits are found.

Inside the blue boxes are line-art symbols that represent circuits, or functional entities, on the subassemblies. The shape of the symbols, and the codes inside them, tell what kind of circuits they are. All the special FOMM circuit symbols and codes used in this manual are identified in the Explanation of Circuit Identifier Codes and Explanation of Special Symbols. If there is only one component in a functional entity, the symbol will usually contain the name of that component.

This is how circuit identifier codes are used. The first letter gives you the category of the circuit:

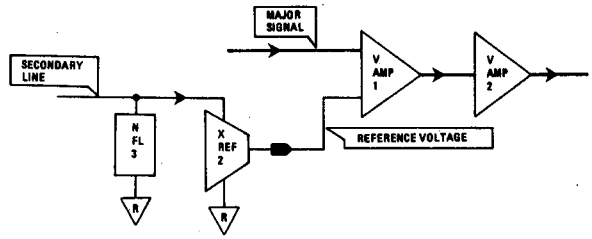
C - Composite
I - Integrated Circuit
N - Network
Q - Transistor
V - Vacuum Tube
X - Diode

The second grouping of two or three letters tells you what kind of circuit it is:

AMP - Amplifier
FL - Filter
REF - Reference

Finally, a number is assigned to permit identification of a particular circuit when more than one of the same type is used on a given subassembly.

For example:



In the example, you can see that there are at least two vacuum tube amplifiers on the subassembly (V-AMP-1 and V-AMP-2 are shown). There are also at least two other filter networks (N-FL-1 and N-FL-2); and at least one other diode reference voltage circuit (X-REF-1).

The example also shows how connections are identified by using flags, arrowheads and different thicknesses of lines. All interconnecting wires, plugs, jacks, terminal boards, adjustments, controls, and test points are shown and identified. Front panel controls and indicators are shown in white windows in a grey or blue shade. Controls and adjustments that are not front panel are-shown in boxes that are the same shade of grey as the subassembly where they are found. Hardware grey shades are identified by their designations in the upper left hand corner of their grey-shaded area. Some signals are generated on one Major Function and used on another. These are the lines that are shown connecting the Major Functions on the Overall Block Diagram. You can follow a signal from one function to another by tracing it to an interface box. This is a blue box with the name of the other function in the upper left-hand corner. The signal line is identified with a signal flag and broken inside the interface box. Its destination, or where it came from, is printed next to the break. You then go to the other function and find a corresponding interface from the first function. The signal is picked up, identified by its signal flag, and you can follow it to its point of connection.

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FIGURE 5. Troubleshooting/repaire volume hierarchial information (sheet 2 of 3)

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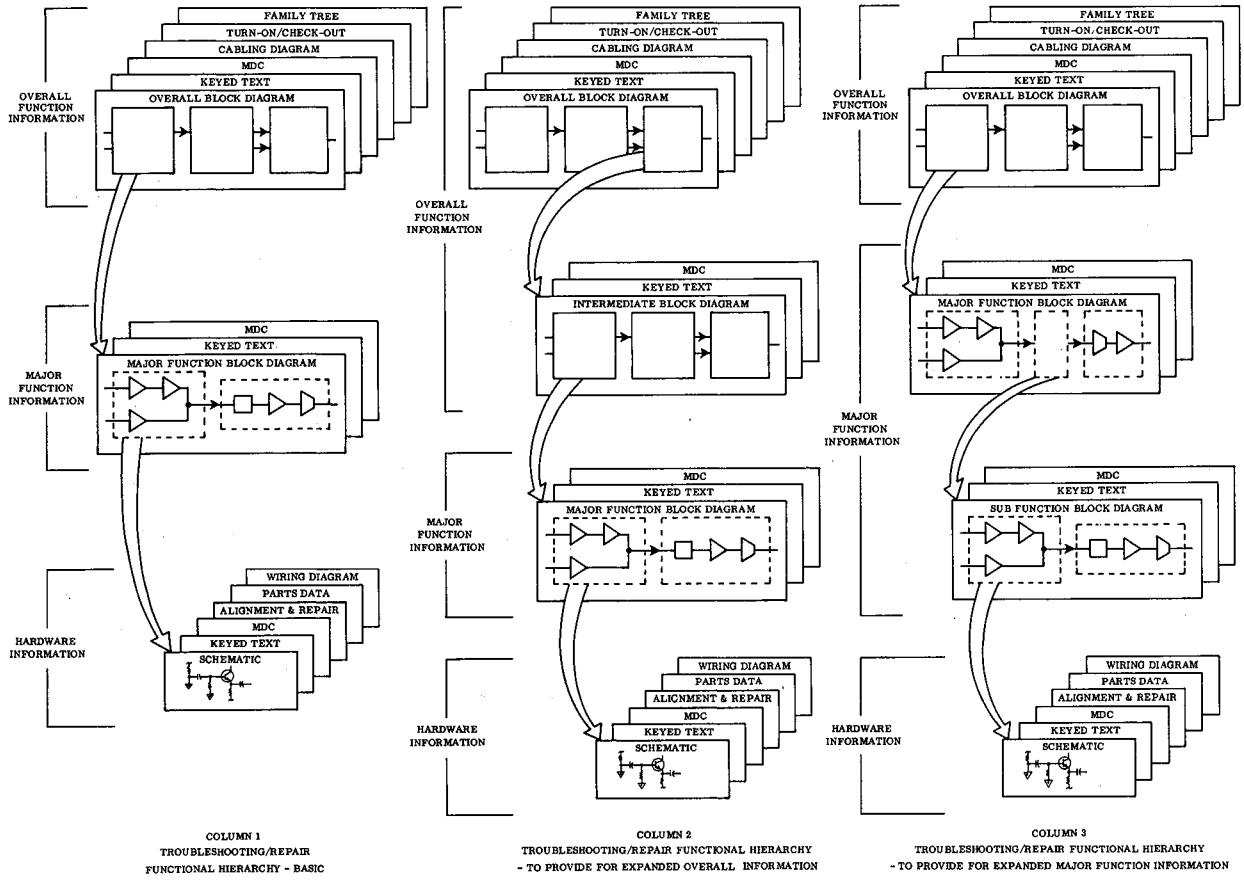


CHART 1 – TROUBLESHOOTING/REPAIR VOLUME ORGANIZATION

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FIGURE 5. Troubleshooting/repair volume hierarchial information (sheet 3 of 3)

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SE170-AN-MMF-020/MK 6 MODS 4B/C/E

VOLUME 2

FOMM
TROUBLESHOOTING/REPAIR VOLUME
FOR
DIGITAL DEAD RECKONING TRACER
AND DEAD RECKONING TRACERS
MK 6 MOD 4E AND MK 6 MODS 4B/4C

FUNCTIONAL INFORMATION
HARDWARE INFORMATION

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PUBLISHED BY DIRECTION OF COMMANDER, NAVAL SEA SYSTEMS COMMAND

1 OCTOBER 1987

SE170-AN-MMF-020



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FIGURE 6. Title Page

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SAFETY SUMMARY

The following are general safety precautions that are not related to any specific procedures and therefore do not appear elsewhere in this publication. These are recommended precautions that personnel must understand and apply during many phases of operation and maintenance.

KEEP AWAY FROM LIVE CIRCUITS

Operating and maintenance personnel must at all times observe all safety regulations. Do not replace components when the power is turned on. Under certain conditions, 115VAC is present when the power switch is off. Dangerous potential may exist when the power switch is off, due to the charge retained by certain capacitors. To avoid casualties, remove the input power plug and ground a circuit before touching it.

DO NOT SERVICE OR ADJUST ALONE

Certain adjustments must be made with the equipment open and power applied. For such events, use extreme caution to avoid contacting dangerous voltages. Under no circumstances should any person reach into or enter the enclosure for the purpose of servicing or adjusting the equipment except in the presence of someone who is capable of rendering aid.

RESUSCITATION

Personnel working with or near high voltages should be familiar with modern methods of resuscitation. Such information may be obtained from the Bureau of Medicine and Surgery.

The following warning appears in the text in this volume, and is repeated here for emphasis.

WARNING

HIGH VOLTAGE

is used in the operation of this equipment

DEATH ON CONTACT

may result if personnel fail to observe safety precautions. Learn the areas containing high voltage in each piece of equipment. Be careful not to contact high-voltage connections when installing or operating this equipment.

Before working inside the equipment, turn power off and ground points of high potential before touching them.

WARNING

ELECTROMAGNETIC RADIATION

DO NOT STAND IN THE DIRECT PATH OF THE ANTENNA WHEN THE POWER IS ON! DO NOT WORK ON THE WAVE GUIDES WHILE THE POWER IS ON!

High frequency electromagnetic radiation can cause fatal internal burns. It can literally "cook" internal organs and flesh. If you feel the slightest warming effect while near this equipment MOVE AWAY QUICKLY!

SOURCE DATA

NAVSEA 0967-071-5010	W/CHG 4	dtd 27 AUGUST 1968
NAVSEA 0967-071-5060	W/CHG 1	dtd 27 AUGUST 1968
NAVSEA 0967-103-8010	W/CHG 3	dtd 27 AUGUST 1968
NAVSEA 0967-103-8020	W/CHG 1	dtd 17 APRIL 1963
NAVSEA 0967-103-8060	W/CHG 1	dtd 27 AUGUST 1968
NAVSEA 0967-156-3010	W/CHG 2	dtd 8 AUGUST 1968
NAVSEA 0967-156-3040	W/CHG 1	dtd 1 MAY 1973
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NAVSEA 0967-441-5010	-	
NAVSEA 0967-445-7010	-	dtd MARCH 1972
NAVSEA 0967-484-7010	-	dtd 30 AUGUST 1972
NAVSEA 0967-484-7020	-	dtd 30 AUGUST 1972
NAVSEA 0967-LP-583-1010	-	dtd 1 JULY 1975
NAVSEA 0967-950-5010	-	dtd 13 JUNE 1961

AN/SPS-10 FIELD CHANGES 1 THRU 33
AN/SPS-10B FIELD CHANGES 1 AND 13
AN/SPS-10F FIELD CHANGES 10 THRU 12
AN/SPS-10, 10B, 10C, 10D, 10E, AND 10F MAINTENANCE REQUIREMENT CARDS dtd NOVEMBER, 1975

NSWSES LETTER W/ENCLOSURE 1 SER#4711		dtd 13 JANUARY 1977
MILITARY DATA SERVICE LETTER	SPS-58A/1-76	dtd 20 APRIL 1976
MILITARY DATA SERVICE LETTER	SPS-58A/2-76	dtd 21 APRIL 1976
MILITARY DATA SERVICE LETTER	SPS-58A/3-76	dtd 11 MAY 1976

AN/SPS-10 TECHNICAL MANUAL IN-PROCESS REVIEW MINUTES: dtd: 14-17 FEBRUARY 1977
30 AUGUST-2 SEPTEMBER 1977
28-30 MARCH 1978
6-7 SEPTEMBER 1978
AN/SPS-10 TECHNICAL MANUAL POST AWARD CONFERENCE MINUTES dtd 7-10 DECEMBER 1976
DUSTIN ASSOCIATES INC. ECR'S:

001	dtd 14 MAR 1977
002	dtd 6 MAY 1977
003	dtd 16 MAY 1977
004	dtd 25 MAY 1977
005	dtd 9 JUNE 1977
006	dtd 17 JUNE 1977

NAVSEC NORDIV LETTERS:

SER#243	dtd 4 APRIL 1977
SER#329	dtd 13 MAY 1977
SER#406	dtd 13 JUNE 1977
SER#432	dtd 22 JUNE 1977

AN/SPS-10 MICROFILM JOB NUMBERS:
26224-1 THRU 26224-5
26836
26944-1 THRU 26944-3
27000-1 THRU 27000-7
27077-1
27372-A THRU 27372-E
27377-A THRU 27377-E
28225
28336-A

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FIGURE 7. Table of Source Data

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FRONT MATTER

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-4a) FRAME ID NO.

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-4b) FRAME ID NO.

RADAR SET AN/SPS-10( )

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FIGURE 8. Functional Index (sheet 1 of 2)

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UNIT 19 ANTENNA ASSEMBLY AS-936, AS-936A/SPS-10B	271	271		272	271	271
UNIT 20 ANTENNA ASSEMBLY AS-936B/SPS-10B	275	275		276	279	279
UNIT 21 ANTENNA ASSEMBLY AS-936C/SPS-10B	280	280		281	281	281
UNIT 22 ANTENNA ASSEMBLY AS-1161/SPS	285	285		286	288	288
UNIT 23 PRE-TRIGGER GENERATOR (FC-15)	290	291	292	294		296
UNIT 25 NORM-PROC SELECT SWITCH	297			297		

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 8. Functional Index (sheet 2 of 2)

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SAMPLE PROBLEM

The example depicted on these pages illustrate the approach to use to isolate a fault. The Troubleshooting Procedure is organized into 4 major steps.

Step 1. Isolate the fault to a module, using the Turn-On/Check-Out Chart and Troubleshooting Flow Chart.

Step 2. Isolate the trouble to a major function using the Overall Function MDC.

Step 3. Isolate the fault to a subassembly, such as a printed circuit board, using the Major Function MDC.

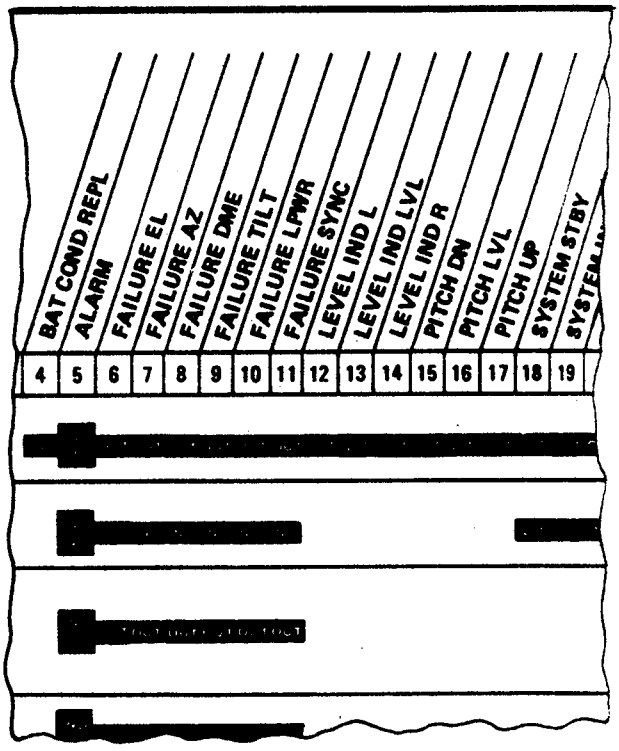
Step 4. Isolate the fault to a stage, circuit, or part, using the Hardware Level MDC.

Note: The Troubleshooting Flow Chart will isolate down to a Module, replacement of the module will repair the system. The module may then be troubleshot in the system now or at a later time.

SPEC. NO.	INDICATION SPECIFICATION
1	LIT IF AC POWER AVAILABLE
2	LIT IF ONLY DC POWER AVAILABLE
3	LIT IF BATTERY IS CHARGED
4	LIT IF BATTERY VOLTAGE IS LESS THAN 18 VOLTS
5	IND 1: SILENT IND 2: AUDIBLE SIGNAL
6	LIT WNEH BITE RESET PRESSED' GOES OUT WHEN BITE RESET RELEASED. PRESSED.
7	SAME AS SPEC NO. 6
8	SAME AS SPEC NO. 6
9	SAME AS SPEC NO. 6
10	SAME AS SPEC NO. 6
11	SAME AS SPEC NO. 6
	LIT FOR 12 SECONDS AFTER

TURN-ON/CHECK-OUT CHART

PRELIMINARY INSTRUCTIONS:
1. SET BAT SEL TO 32 IF BATTERY IS SILVER-ZINC OR TO 27 IF BATTERY IS LEAD-ACID.
2. SET LCL/RMT TO LCL.
3. SET MODE SELECT TO STBY-BAT CHG.
4. SET OFF/ON-RESET TO ON-RESET.
5. LEVEL SYSTEM IN ACCORDANCE WITH LEVELING PROCEDURE.
6. SET STATION IDENT TO ASSIGNED IDENTIFICATION.
7. SET AZ/EL CHAN TO 1-10.
8. SET DME CHANNEL SELECT TO ASSIGNED CHANNEL.
9. SET DME OFFSET TO +0.
10. SET ELEVATION GDNC CUTOFF TO 0.
11. SET ELEVATION OBST CLR TO 0.5.



INDICATION SPECIFICATION NUMBER

STEP 1
SET ALARM TO ON.

STEP 2
SET MODE SELECT TO DME.

STEP 3

FIGURE 9. Sample Problem (sheet 1 of 2)

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For the purpose of explaining the fault isolation process let us assume that the system has a problem with the Precision Surface Transponder Module (PSTM). The first thing to do when equipmant problems develop is to refer to the Turn-On/Check-Out Chart (it is also a good idea to verify conditions with the Turn-On/Check-Out Chart each time the system is used). Set the control as specified in the preliminary instructions and then proceed through the steps. At Step 2 a check of the failure indicators shows that there is a PSTM failure (DME failure indicator flashing). The next step in the fault isolation is to refer to the Troubleshooting Flow Chart.

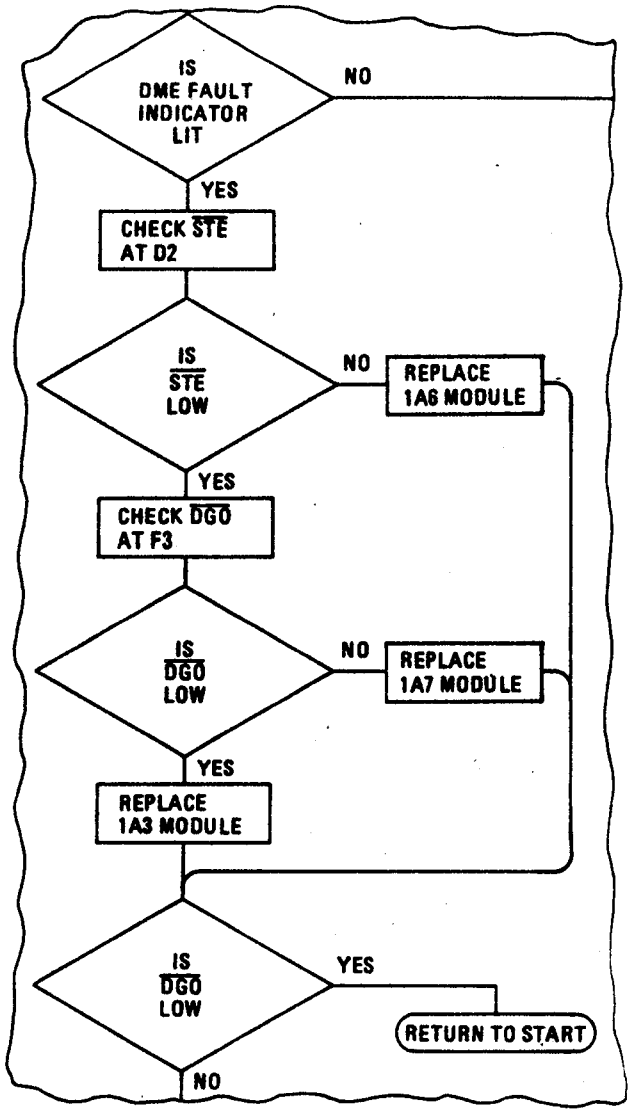
TROUBLESHOOTING FLOW CHART

PRELIMINARY INSTRUCTIONS:

PRIOR TO RADIATING IN THE TEST MODE INSERT SHORTING PLATE INTO THE WAVE GUIDE QUICK DISCONNECT ON 1A8. PRIOR TO APPLYING POWER, INSURE THAT ALL MODULES ARE INSTALLED.

1. SET BAT SEL TO 32 IF BATTERY IS SILVER-ZINC OR TO 27 IF BATTERY IS LEAD-ACID.
2. SET LCL/RMT TO LCL.
3. SET SPLITE SITE TOCOLOC.
4. SET SERV DMD TO OFF.
5. SET DME TO X.
6. SET AZ/EL TO 1-10.
7. SET ALARM TO ON. (MAY BE TURNED OFF IF ALARM IS SOUNDING).
8. SET EL OBST CLR TO 0.5.
9. SET EL GDNC CUTOFF TO 0.
10. SET CHANNEL IDENT TO ABC.
11. SET CHANNEL SELECT TO 00.
12. SET DME OFFSET TO +00.
13. SET MODE SELECT TO STBY-BAT CHG.
14. IN UNIT 1 SET AZ, EL SWITCHES TO ON AND TILT SENSOR SWITCH TO ON.
15. IN UNIT 1 SET SQUIT RATE SWITCH TO LOW.
16. SET ANGLE INFORMATION CHANNEL TO CHANNEL 5.

SPEC NO.	SIGNAL SPECIFICATIONS
	UNLESS OTHERWISE SPECIFIED 1. VALUES ARE ±10%. 2. LOW < +0.4V. 3. HIGH > +2.4V. 4. ALL MEASUREMENTS ARE REFERENCED TO G1.
	REQUIRED TEST EQUIPMENT MULTIMETER OR EQUIVALENT — AN/PSM-4
DC1	+5V AT A1
DC2	−5V AT B1
DC3	+12V AT C1
DC4	−12V AT D1
DC5	+24V AT E1
DC6	+50V AT F1



On the Troubleshooting Flow Chart, at the question box "Is the DME Failure Indicator Lit?", the answer is yes. Proceeding down the yes route we check $\overline{STE}$ at the front panel test point D2 and find it low. Proceeding on we check $\overline{DGO}$ at the front panel test point F3 and find it high.

The answer to question box "Is $\overline{DGO}$ Low?", is no. Following the no route we replace the 1A7 module with a spare 1A7 module. After replacement we check F3 and find $\overline{DGO}$ low. This repairs the system.

The failed 1A7 module (PSTM) can be troubleshot now or later. When ready to troubleshoot the failed 1A7 module, connect it to a working AN/TPN-30 system with test cables. Using the Overall Function MDC energize the system.

FIGURE 9. Sample Problem (sheet 2 of 2)

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Power Distribution Diagrams

Power Distribution Diagrams are special functions which show AC and DC power connections to all the equipment assemblies and subassemblies.

The AC power input is divided into two branches. One input branch will be available whenever the main bulkhead switch is ON, and will furnish the AC power for such items as the Antenna pedestal heater, interlock and overtemp circuits, etc. The non-fused low side for this branch is identified by the special symbol ▽N1▽ and a signal flag 115VAC(N1).

The second input branch will be energized, if the main bulkhead switch is ON and the Radar Control switch is set to STBY or RAD, and furnish the AC power required for normal Radar operation. The non-fused low side for this branch is identified by the special symbol ▽N2▽ and a signal flag 115VAC(N2).

WARNING

Voltages dangerous to life may exist between Terminal board connections
F02 ▽N1▽ and
F102 ▽N2▽ and
chassis ground.

Explanation of Signal Flow Lines

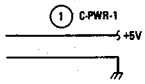
Functional entities are connected by means of lines which show the signal or energy flow through the equipment. Functional signal flow is shown from left to right wherever possible. Feedback signal flow is generally shown from right to left. Signal lines are connected to illustrate subordination and coordination of signal importance.

Schematic Diagrams

There is a schematic diagram for each assembly in the equipment covered by this manual. Schematic diagrams appear in reference designator order, i.e., first Unit 1 and its assemblies and subassemblies, then Unit 2 and its assemblies and subassemblies, etc. as identified on the family tree. Circuit elements or components that work together to perform a particular task (e.g., amplify, detect, rectify) are grouped in stages or circuits called functional entities. These entities are assigned circuit identifier codes, which are used to identify them on both the Maintenance Dependency Chart and the Major Function Diagram.

The functional entities are arranged to simplify the tracking of signal flow (usually from left to right). Signal lines are identified by means of signal flags, line thickness, and arrowheads.

Power inputs to a subassembly are handled differently than signals. Power and ground inputs are shown entering the schematic at the lower left-hand corner. They go into a power distribution functional entity called C-PWR-1. Filtering or decoupling circuits are shown connected to power inside this entity, but power and ground lines are not drawn directly to the signal processing functional entities. Instead, break-line and ground symbols are shown inside C-PWR-1, as in the example below:



At signal functional entities, break line and ground symbols are shown at points where power and ground connections would be made. Mechanical connections between decks of the same switch, or relay contacts and their solenoid, are shown by dashed lines.

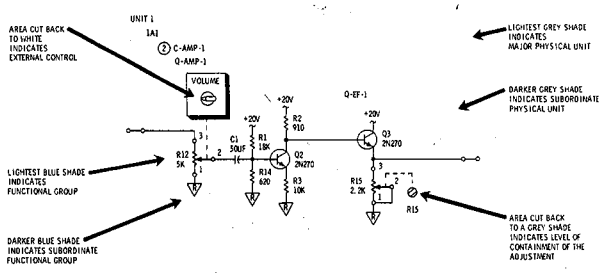


Figure 1.

Color Coding

Schematics and Functional Block Diagrams in this manual are color coded to aid the user in the ready and positive identification of the level of hardware he is on during each stage of the troubleshooting process. Refer to Figure 1 above for an explanation of the shaded areas found on a typical schematic.

Maintenance Dependency Charts

There is a Maintenance Dependency Chart (MDC) for each block diagram. All functional entities on a diagram, with their input and output measurement points, are listed in the MDC heading. The instructions on the left-hand side set up certain conditions in the equipment, and the body of the chart shows the resulting dependency structures. The concept behind a dependency structure is this: If the inputs are available, and a functional entity is working correctly, you will get the desired output. The Signal Specifications list the inputs and outputs for each step. Refer to "How to Use MDC's" for complete instructions.

Exploded Views

Exploded views are provided for five different configurations of the Antenna Assembly as an aid to overhaul and repair operations.

Parts Location Information

This information helps you locate parts and measurements points during troubleshooting. The first column, designated REF DESIG, lists the components by reference designator in alphanumeric order. The second column, designated LOCATING COORD, gives the locating coordinates of the part on the illustration. When more than one view is required in a separate coordinate system for complete illustration, the view is also identified. The third column gives the generic name of the component. For complete description of the component, refer to the complete parts list in the support volume. The fourth column is used for amplifying notes.

Wire Lists

A wire list is provided for each wired assembly. The tie points are arranged in alphanumeric order under the REF DESIG and FROM columns, with the other end of the wire or component listed under the TO column. The wire list is typical for all models.

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 10. Description of symbols, codes, and shadings (sheet 1 of 3)

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EXPLANATION OF SPECIAL SYMBOLS		
CATEGORY	SPECIAL SYMBOLS	DEFINITION
Signal Code Symbols		Functional Flow
		Reference Voltage
		Feedback Signal Path
		Signal Path to Energize Relay
		Major Signal Flow Line
		Secondary Signal Flow Line
Standard Symbols Used Only For Specific Purpose Indicated		Chassis Ground
	N1 N2	Non-fused, designated low side of AC line; not earth ground
		Screwdriver Adjustment
		Pushbutton Switch
		Toggle Switch
		Indicator Lamp
		Current Regulator
		Air flow actuated switch
Special Symbols Used On Diagrams		Circuit which changes the voltage or power level of the incoming signal. Circuit may consist of amplifier stages, emitter followers, etc. These circuits always contain one or more nonlinear active elements (vacuum tube or transistor).
		Circuit which generates a signal or processes an incoming signal in some manner other than to change signal voltage or power level. For example: oscillator, multivibrator, mixer, etc. These circuits always contain one or more nonlinear elements which may be active (vacuum tube or transistor) or passive (semi-conductor diode).

EXPLANATION OF SPECIAL SYMBOLS (CONTINUED)		
CATEGORY	SPECIAL SYMBOLS	DEFINITION
		Resistor, capacitor, inductor, or a network comprising any combination of these linear elements. For example: filter network, voltage divider, etc. Networks are generally shown by rotating the rectangle 90 degrees except where space is prohibitive.
		Composite: May be shown either horizontal or vertical. Used to encompass several individual amplifiers, etc. For example: an IF strip.
		Relay contacts: Unless otherwise specified, contacts are shown in de-energized position. Relay contacts are always depicted in such a manner that, when energized, relay swinging contact connects to stationary contact nearest to the relay solenoid. Relay solenoid.
	2 S1 C 1	Switch. Height may be extended to accommodate additional connections. 2 C 1 SCHEMATIC SYMBOL
	2	Indicates signal specification number.
	2	Function diagram identification number.

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 10. Description of symbols, codes, and shadings (sheet 2 of 3)

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RADAR SET AN/SPS-10( )

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-11a) FRAME ID NO.

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-11b) FRAME ID NO.

FRONT MATTER

EXPLANATION OF CIRCUIT IDENTIFIER CODES	
CATEGORY	DEFINITION
C	Composite circuit (one which has been subfunctionalized); Composite functional circuits containing one or more circuits falling into the categories listed below are preceded by the letter C.
I	Integrated circuits; Circuits containing IC's are preceded by the letter I.
M	Circuits containing mechanical items are preceded by the letter M.
N	Linear networks; Circuits containing several linear components (resistors, capacitors, etc.) arranged in a network are preceded by the letter N.
Q	Transistor circuits; Circuits containing transistors are preceded by the letter Q.
V	Vacuum tube circuits; Circuits containing vacuum tubes are preceded by the letter V.
X	Semiconductor circuits; Circuits containing semiconductor diodes are preceded by the letter X.

EXPLANATION OF CIRCUIT IDENTIFIER CODES (CONTINUED)			
CODE	DEFINITION	CODE	DEFINITION
AMP	Amplifier	MOT	Motor
ATR	Anti Transmit-Receive Tube	MXR	Mixer
		OCP	Overcurrent Protector
BLK	Blocking	OSC	Oscillator
BMV	Bistable Multivibrator	OVP	Overvoltage Protector
BO	Blocking Oscillator		
BRG	Bridge	PF	Pulse Forming
		PG	Pulse Generator
CAV	Cavity	PHT	Phantastron
CD	Charging Diode	PPA	Push-Pull Amplifier
CF	Cathode Follower	PSH	Phase Shifter
CLP	Clamp	PSP	Phase Splitter
CPL	Coupler	PWR	Power
CMP	Comparator		
CVD	Constant Voltage Drop	RCT	Rectifier
		REF	Reference
DET	Detector	REG	Regulator
DFA	Differential Amplifier	REL	Relay Solenoid
DIF	Differentiator	RPG	Ramp Generator
DLY	Delay	RES	Resistor
DSC	Discriminator	RPS	Regulated Power Supply
DUP	Duplexer	SCL	Scaling Circuit
DVR	Driver	SHP	Shaper
		SHR	Shunt Regulator
EF	Emitter Follower	SHT	Shutter
		SMT	Schmitt Trigger
FB	Feedback	SRG	Series Regulator
FF	Flip-Flop	SUM	Summing
FL	Filter	SUP	Suppression
		SW	Switch
GAT	Gate		
GR	Gear	TFR	Transformer
		THY	Thyratron
IND	Indicator	TR	Transmit Receive
ISO	Isolation Circuit	TST	Test
INT	Integrator		
INV	Inverter	VD	Voltage Divider
		VDR	Voltage Dropping Resistor
LIM	Limiter	VR	Voltage Regulator
MET	Meter		
MMV	Monostable Multivibrator		

ORIGINAL

2-1-11

Sample arrangement only. Type size does not conform to minimum specification requirements.

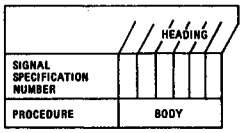
FIGURE 10. Description of symbols, codes, and shadings (sheet 3 of 3)

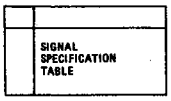
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HOW TO USE THE MAINTENANCE DEPENDENCY CHART

The Maintenance Dependency Chart is a symbolic representation of the signal flow through the equipment. It is based on the premise that in order to obtain the proper output from some entity, the entity must be operating properly and must have the correct inputs. The definition of an entity is a functional group of circuits identified by name or alphanumeric code on the schematics.

The MDC is composed of a Signal Specification Table and the chart itself. The chart is composed of four areas: The Procedure Column, the Heading, the Signal Specification Number Row, and the Body.





The Procedure Column specifies the actions required to place the equipment in the correct mode for testing.

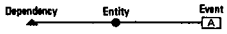
The Heading lists the entities and associated measurement or observation points which correspond to symbols in the body of the chart. When measurements are required, the heading lists the measurement point, a slash (/) and a second point, for example B9-40/B9-10. This is read "Measure at pin 40 of B9 with respect to pin 10 of B9".

Measurement points are assigned Signal Specification Numbers. These numbers cross reference the measurement point with the specification of the signal.

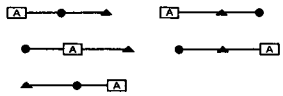
The Body of the chart is a map of signal/voltage flow through the circuitry of the equipment.

The Signal Specification Table describes the signal, voltage and/or indication that should be present at the corresponding measurement or observation point listed in the chart heading.

Three symbols are used in the chart body to depict the dependency structure of the circuitry involved.



In most cases, the dependency marker is to the left, the entity circle in the center, and the event symbol to the right on an MDC line. Occasions do arise where such an arrangement is not possible. Thus, some lines of an MDC may appear as:



The arrangement of the symbols does not alter the dependency structure or its meaning; the event [A] always requires the entity ● to be good, and relies on the input event aligned with the dependency marker ▲. Depending upon the circuit being charted, more than one of each of the symbols may appear on a given line. For example, should 2 dots and 2 event boxes appear on the same line, the line merely indicates that both entities must be good in order to produce both events. It does not imply that if the first entity is good the first event will be present.

☐ Event symbol indicates a signal. An event is an output which can be measured or sensed.

● Entity symbol identifies the functional circuits or devices used to produce an event.

▲ Dependency symbol indicates input signals to an entity required to produce correct events. An inverted (▼) dependency symbol is used in conjunction with a feedback arrow (↞) to denote a feedback loop.

Variation of the symbols are as follows:

Border of box around the event symbol denotes the ease with which the event may be observed. These boxes are defined as follows:

BOLD BORDER

[IND] Available signal or indication which may be recognized from outside the equipment, such as a visual display.

[RUN] Device will run and is visible or audible outside the equipment.

[LIT] Lamp is lit.

[OUT] Lamp is not lit.

[ROT] Device rotates and is visible from outside the equipment.

[DIS] Display is present.

[ON] Device is on.

[OFF] Device is off.

NORMAL BORDER

[A] Available signal which is within specification and requires use of test equipment for observation.

[A1] Signal meets first set of required conditions.

[A2] Signal available at same measurement point under a second set of conditions.

[NA] There are N signal outputs - from N different measurement points. Each measurement point is identified in the Signal Specification Table.

[EN] Relay solenoid energized.

[DEEN] Relay solenoid deenergized.

[ROT] Rotates.

Variations of the functional entity symbol are as follows:

● Indicates the entity must be performing its function properly for event ([A]) on same line to occur within specification.

Indicates the output of an entity is adjustable.

◐ Indicates an entity which is proven partially good if event ([A]) on same line occurs within specification. To prove the entity fully good, events on all dependency lines, of which the entity is a part, must be operating within specification.

Indicates a relay contact set that must be good in the de-energized state.

Indicates a relay contact set that must be good in the energized state.

▲ The dependency marker indicates that event directly above or below must be available (and within spec) for event on same line as dependency marker to occur (within spec), provided other represented functional entities along the line are performing properly.

ORIGINAL

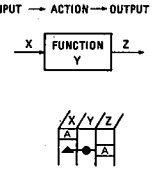
Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 11. How to use MDC (sheet 1 of 4)

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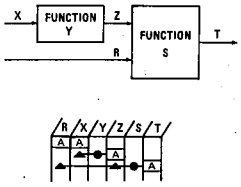
The following examples show development of various dependency lines:

(1) INPUT → ACTION → OUTPUT



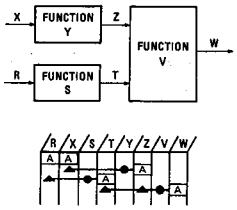
This example shows dependency of output event "Z" upon functional entity "Y" and input event "X".

(2) INPUT → ACTION → } ACTION → OUTPUT
INPUT →

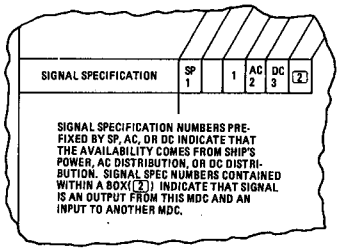


The last dependency line in this example shows dependency of output event "T" upon functional entity "S" and events "Z" and "R".

(3) INPUT → ACTION
INPUT → ACTION } ACTION → OUTPUT

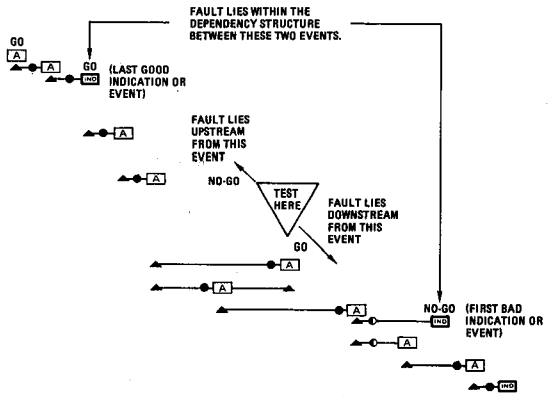


In this example, output events "T" and "Z" are developed as in example (2). The last dependency line shows dependency of output event "W" upon functional entity "V" and events "T" and "Z".



The Signal Specification Table cross references the event availabilities listed in the heading with the specification of the event. In many cases, photographs of the signal waveforms are used to describe the availability.

This example shows how to use the MDC to locate a faulty part.



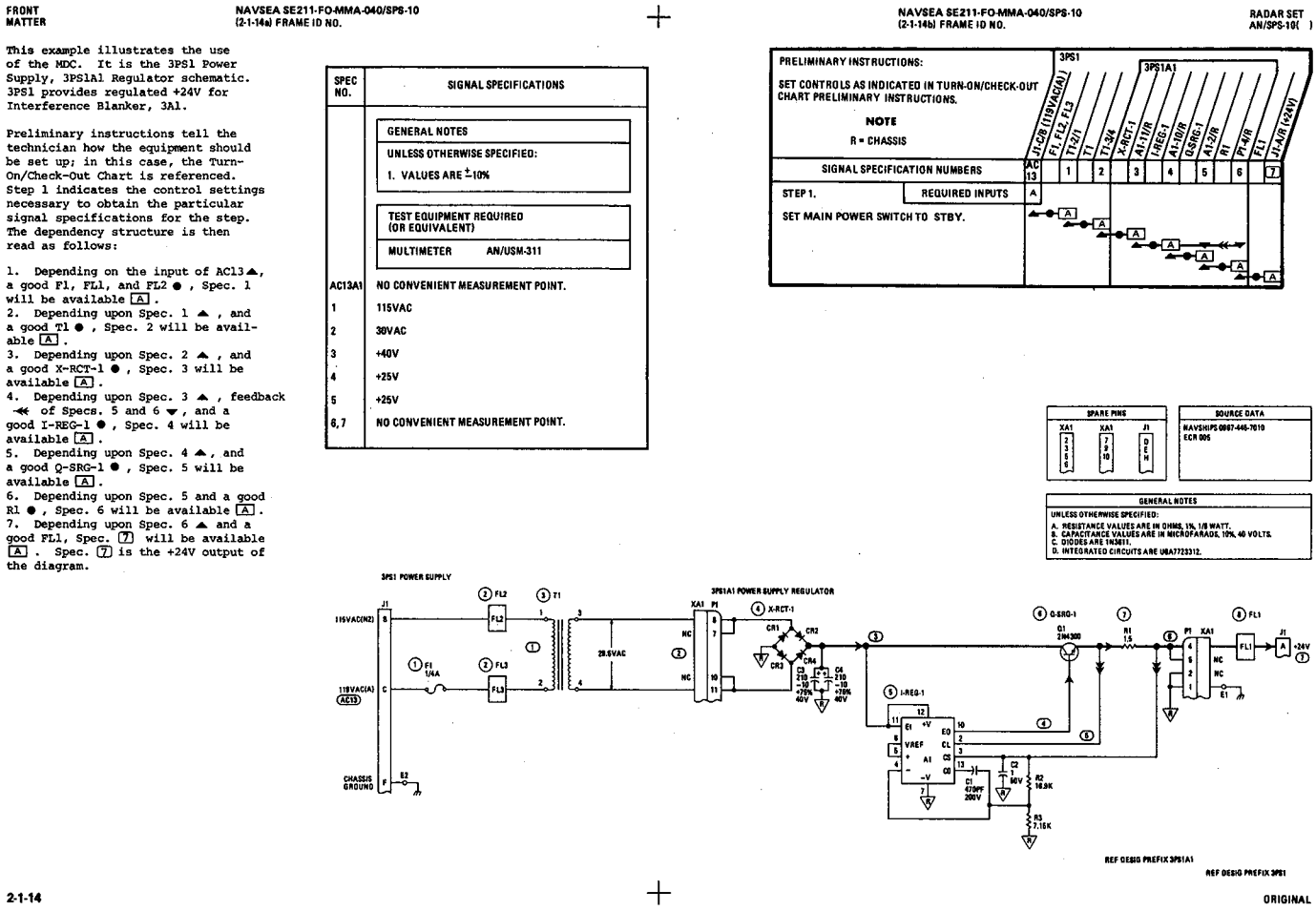
The chart excerpt above illustrates the fault isolation process using the MDC. The events are checked for reference and establish the first bad indication and the last good indication. The dependency structure between these two events identifies those entities which may have failed. To further narrow the area of concern, use split-halves troubleshooting process. The event next to be tested is located midway within the dependency structure defined by the last good and first bad indications. If the event is GO, the failed entity is shown within the dependency structure downstream from the event. If the event is NO-GO, the failed entity is shown upstream from the event. This procedure is repeated until a single dependency line is isolated whose input event is good and output event is bad. The failed entity is shown on this line, and the entity(s) identified by the dot(s) contain the fault.

The validity of MDC completeness is based upon two assumptions: (1) The wiring between functional entities, including connection of jacks and plugs, was checked at the factory and found to be accurate and complete. This is implied by the lines connecting the dependency marker, entity, and event symbol. (2) Any short circuit is attributed to the entity producing the event which is found to be bad (shorted). It should be kept in mind that an existing short may be in a module using an output and therefore replacing a module whose output appears shorted will not cure the problem.

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 11. How to use MDC (sheet 2 of 4)

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Check the source to verify that this is the current version before use.



Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 11. How to use MDC (sheet 3 of 4)

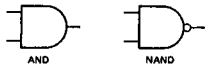
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:05Z
Check the source to verify that this is the current version before use.

INTEGRATED CIRCUITS USED IN THE AN/SPS-10 ( ) RADAR SET				
MIL SPEC./ PART NO.	GENERIC NO.	DESCRIPTION	IC NO.	PAGE NO.
	CA3045	5 TRANSISTOR ARRAY	1	
	HA2-2525-5	OPERATIONAL AMPLIFIER	2	
6905-7015-1	LM106H	VOLTAGE COMPARATOR/BUFFER	3	
	MC1520G	OPERATIONAL AMPLIFIER	4	
6905-7016-1	MC9601	RETRIGGERABLE MONOSTABLE MULTIVIBRATOR	5	
	SN5400J	QUAD 2 INPUT NAND GATE	6	
	SN54107J	DUAL JK FLIP-FLOP	7	
	SN54121J	MONOSTABLE MULTIVIBRATOR	8	
	SN54123J	DUAL RETRIGGERABLE MONOSTABLE MULTIVIBRATOR	9	
U6A7709312	UA709A	OPERATIONAL AMPLIFIER	10	
U5B771039X	UA710T	HI-SPEED DIFFERENTIAL COMPARATOR	11	
U6A7723312	UA723	PRECISION VOLTAGE REGULATOR	12	
U5R7723312	UA723L	PRECISION VOLTAGE REGULATOR	13	
6905-7017-1	UA741A	FREQUENCY COMPENSATED OPERATIONAL AMPLIFIER	14	
6905-7018-1	UA747A	DUAL OPERATIONAL AMPLIFIER	15	

INTEGRATED CIRCUIT DATA

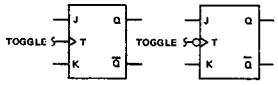
The logic convention used in this manual is the positive logic negation indicator system. With positive logic, the more positive logic level is called the 1-state, or high, and the less positive logic level is called the 0-state, or low. In the AN/SPS-10 ( ) a logic level is considered high when it is greater than +2.4V, and low when less than +0.4V.

The negation indicator system uses a small circle to denote negation. For example, the familiar AND gate symbol



becomes a NAND gate when the small circle is present on the output. The negation indicator on the AND gate output tells you that two high inputs produce a low output. The AND gate symbol is one of the basic symbols, and a NAND gate is simply an AND gate with a negation indicator. Similarily, a NOR gate symbol is composed of the OR basic symbol with the negation indicator at the output.

The negation indicator is used on rectangular symbols to show whether you need a high or a low pulse to perform a certain function.



You need a high pulse to toggle this flip-flop

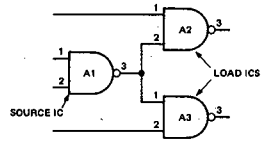
You need a low pulse to toggle this flip-flop

Refer to the IC Pin Configuration layout to assist with identifying measurement points and removal/installation component placement.

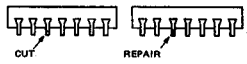
IC TROUBLESHOOTING NOTE

When troubleshooting integrated circuits in the AN/SPS-10 ( ) System you should give special consideration to the load of each IC. The load for most IC's is usually the input impedance of other IC's.

Refer to the diagram below. If the output of A1 is low when it should be high, the faulty IC is not necessarily A1. It is possible that a short circuit to ground exists in the input circuit of either A2 or A3.



Before you replace the source IC, use an X-ACTO knife to cut the output pin of the IC between the body of the IC and the printed circuit card. Bend the cut pin so that the two parts of the pin do not make contact. Measure the output of the IC again. If the output is correct, the problem is in one or more of the IC's that make up the load. Reconnect the output by soldering a small piece of wire on the IC pin across the cut. Use a heat sink to prevent heat damage to the IC.



To locate the faulty load IC, use the X-ACTO knife to cut the input pins of the loads, one at a time until the ground has been removed from the output of the source IC.

SERVICING CIRCUIT BOARDS

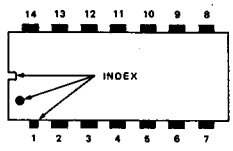
Printed circuit boards in this equipment have components mounted on one side of the board and conductive surfaces on both sides. Some important considerations are as follows:

1. Use a 37 to 47.5 watt chisel tip soldering iron with a tip diameter of 1/16 to 1/8 inch for parts other than integrated circuits. Use special integrated circuit soldering iron tip (Ungar No. 1235 or equivalent) to remove integrated circuits. Small diameter rosin core solder should be used when replacing parts.
2. Components may be removed by placing the soldering iron on the component leads on either side of the board and pulling the component straight away from the board. If heat is applied to the component side of the board, greater care is required to avoid damage to the components, especially semi-conductors. Heat damage may be minimized by gripping the lead with long nose pliers between the soldering iron and the component, thereby forming a heat sink. Use special heat sink when removing integrated circuits.
3. If a component is obviously damaged or faulty, clip the leads close to the component and then unsolder the leads from the board.
4. Excessive heat or force will destroy the laminate bond between the metal plated surface (conductor) and the board. If this problem should occur, the lifted conductor may be cemented down with a small amount of quick-drying acetate base cement having good insulating properties. Another method of repair is to solder a section of good conducting wire along the damaged area.
5. Before replacing a component, heat the remaining solder in the component hole and clean it out with a toothpick or "solder sucker". Sharp pointed metalic tools are not recommended since they may loosen eyelets in boards.
6. Tin and shape replacement component leads to fit existing holes.
7. Install the replacement component in the same position as the original.

IC PIN CONFIGURATION

IC'S MAY USE ONE OR MORE OF THE FOLLOWING INDEX METHODS:

1. KEY INDEX
2. DOT INDEX
3. SMALL PIN 1



TOP VIEW

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 11. How to use MDC (sheet 4 of 4)

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Check the source to verify that this is the current version before use.

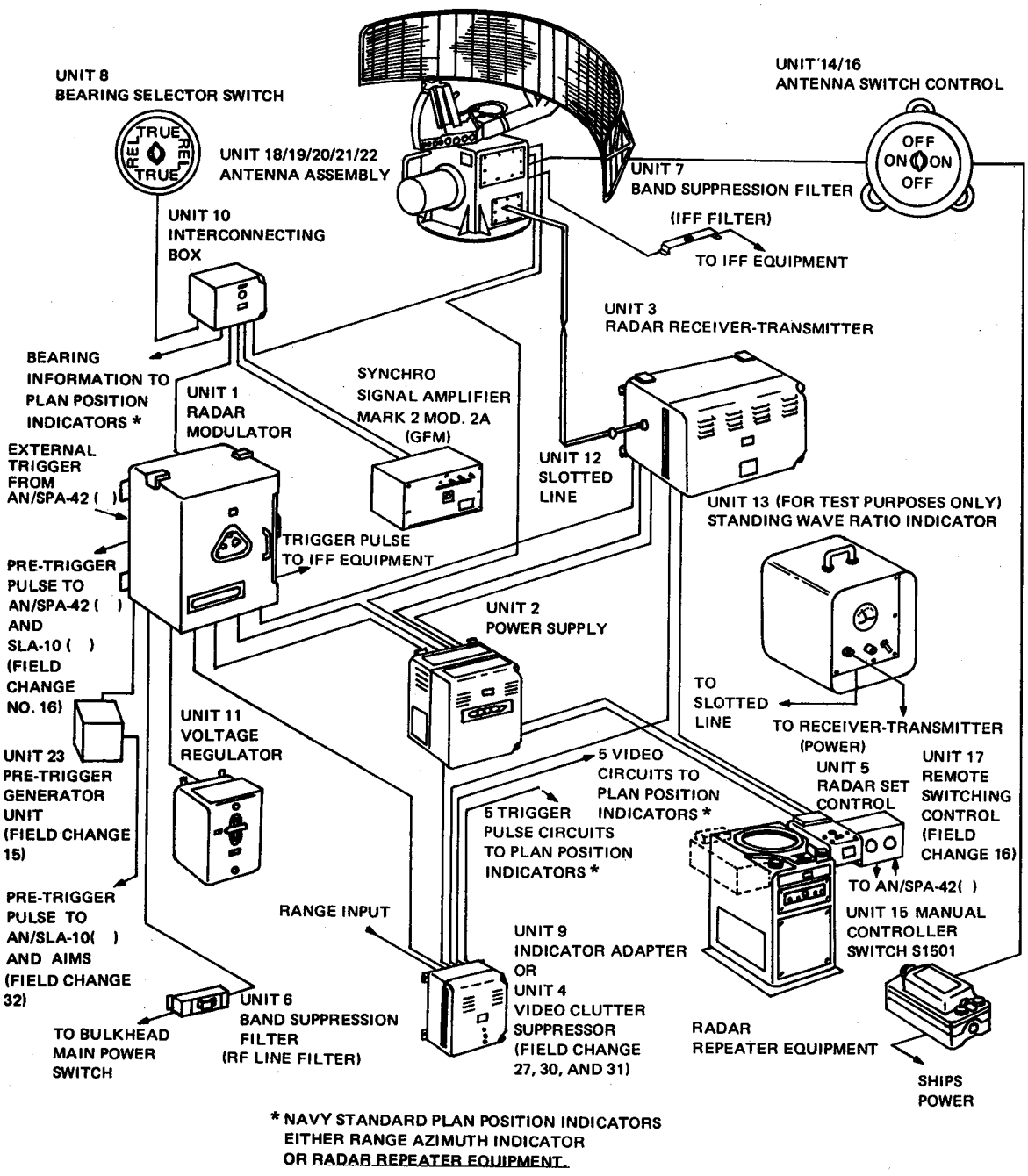


FIGURE 12. Relationship of units

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TABLE 1-1 FUNCTIONAL CHARACTERISTICS OF RADAR SET AN/SPS-10 ( )

CHARACTERISTIC	FUNCTION-VALUE
FREQUENCY RANGE	5450 TO 5825MHz
BEACON RECEIVER FREQUENCY	5450 ± 2MHz
TYPE OF FREQUENCY CONTROL	AMPLITUDE - MODULATED (PULSED) MAGNETRON OSCILLATOR
TYPE OF EMISSION	RADAR PULSE (0.25 OR 1.3 MICROSECONDS ±10%) BEACON PULSE (2.25 MICROSECONDS ±10%)
PEAK POWER OUTPUT	190KW TO 285KW
PULSE RATE (INTERNAL SYNC)	RADAR: 625 TO 650PPS NOMINAL BEACON: 305 TO 330 PPS
(REMOTE SYNC VIA AN/SPA-42( ))	RADAR: 627PPS ± 5% NOISE MODULATED
TYPE OF RECEIVER	SUPERHETERODYNE
INTERMEDIATE FREQUENCY	30MHz
BANDWIDTH	NARROW BAND: 1MHz; WIDE BAND: 5MHz
INTERFERENCE BLANKER	5MHz
INPUT POWER CHARACTERISTICS	115VAC ±10%, 60Hz, SINGLE PHASE
POWER INPUT	3.5KVA AT 0.9 POWER FACTOR
RADAR RESOLUTION (EFFECT OF INSTALLATION AND INDICATOR NOT CONSIDERED)	
BEARING	LESS THAN 1 DEGREE
RANGE: SHORT PULSE	50 YARDS
RANGE: LONG PULSE	275 YARDS
HUMIDITY	UP TO 95% RELATIVE
AMBIENT TEMPERATURE (EXCEPT ANTENNA)	0 TO 50 DEGREES C (32 TO 122 DEGREES F)
AMBIENT TEMPERATURE (ANTENNA)	-35 TO +60 DEGREES C (-31 TO +140 DEGREES F)
CONSTRUCTION (EXCEPT ANTENNA)	DRIP PROOF
CONSTRUCTION (ANTENNA)	WATERTIGHT
ANTENNA (AS-615/SPS-10)	
ROTATION	16 ±1 RPM
BEAM WIDTH, HORIZONTAL (RADAR)	APPROXIMATELY 1.5 DEGREES
BEAM WIDTH, HORIZONTAL (IFF)	APPROXIMATELY 6 DEGREES
BEAMWIDTH, VERTICAL (RADAR)	BETWEEN 12 AND 16 DEGREES
BEAM WIDTH, VERTICAL (IFF)	APPROXIMATELY 22 DEGREES
INPUT POWER CHARACTERISTICS	115VAC, 60Hz, SINGLE PHASE

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 13. Reference data table

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:05Z
Check the source to verify that this is the current version before use.

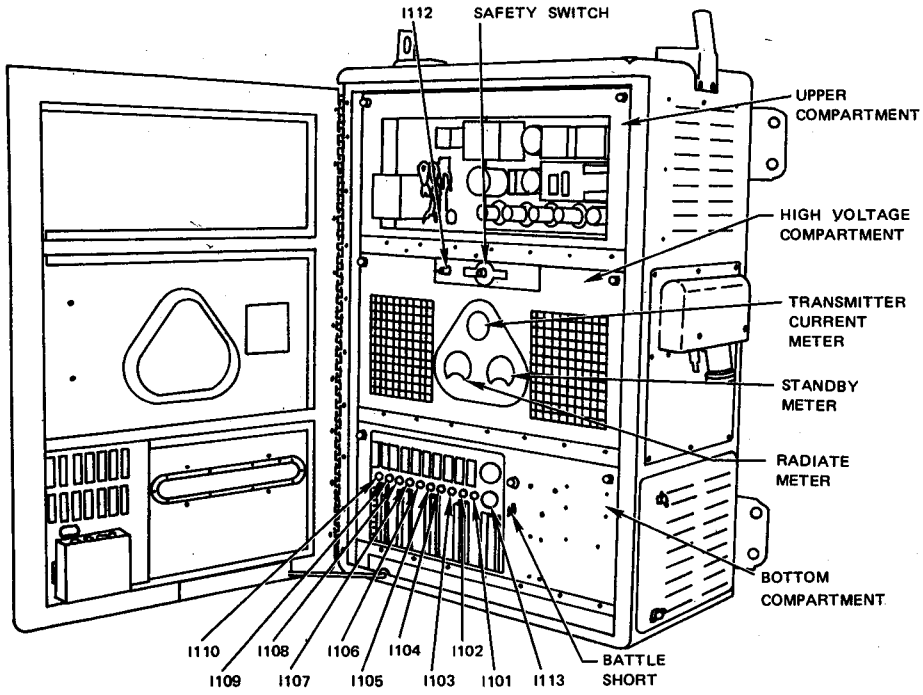


FIGURE 2-2. UNIT 1 RADAR MODULATOR

FIGURE 14. Operator data table (sheet 1 of 2)

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TABLE 2-1 UNIT 1 OPERATOR DATA (FIGURE 2-2( ))

CONTROL/INDICATOR	REF DESIG	FUNCTION	NORMAL STATUS
I101 through I110	I101 through I110	Blown fuse indicators: Light white to indicate a blown fuse.	OUT
I112	I112	Equipment ON indicator: Lights clear when equipment is on.	LIT
I113	I113	BATTLE SHORT indicator: Lights red when equipment is in BATTLE SHORT operation.	OUT
BATTLE SHORT	S104	ON position - places equipment in BATTLE SHORT operation by shunting the contacts of Undercurrent Relay K103.	OFF
Z106/S103	Z106/ S103	Safety switch: De-energizes main power relay K101 when in the OFF position.	ON
TRANSMITTER CURRENT METER	M103	Indicates Magnetron current. Graduated in POOR, NORMAL, and DANGER.	NORMAL
STANDBY METER	M102	Indicates STANDBY and RADIATE elapsed time in hours.	As applicable
RADIATE METER	M101	Indicates RADIATE operation elapsed time in hours.	As applicable

TABLE 2-2 UNIT 2 OPERATOR DATA (FIGURE 2-3( ))

CONTROL/INDICATOR	REF DESIG	FUNCTION	NORMAL STATUS
I201 through I1208	I201 through I208	Blown fuse indicators: Light white to indicate a blown fuse.	OUT
I209	I209	Equipment on indicator: Lights clear when equipment is on.	LIT
INTERLOCK BYPASS	S201	Bypasses Power Supply interlock when pressed.	OUT

FIGURE 14. Operator data table (sheet 2 of 2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:05Z
Check the source to verify that this is the current version before use.

*HIGH PRESSURE AIR COMPRESSOR

COMPRESSOR NUMBER _____

USS _____ (SSBN–) DATE _____

	SHUTDOWN		NORM *		00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
HPAC																												
TEMP CIRC WATER OUT	130[a]		95	110																								
PRESS 1st STAGE	83[b]		42	58																								
2nd STAGE	286[b]		195	225																								
3rd STAGE	1376[b]		975	1225																								
4th STAGE	ABOVE 5100[b] BELOW 4250[c]	4300	4500	4800																								
TEMP 1st STAGE SUCTION			90-110																									
1st STAGE DISCHARGE	430[d]		285	305																								
2nd STAGE SUCTION				135																								
2nd STAGE DISCHARGE	430[d]		315	345																								
3rd STAGE SUCTION			90	110																								
3rd STAGE DISCHARGE	430[d]		330	385																								
4th STAGE SUCTION			95	115																								
4th STAGE DISCHARGE	430[d]		370	410																								
FINAL DISCHARGE			85	120																								
FRAME OIL PRESS	BELOW 12[e] ABOVE 40[f]		15	35																								
LEVEL LO SUMP		AT FULLMARK																										
TEMP LO SUMP			85-120																									
LO LUBRICATORS - LOWER SIGHT GLASS LEVEL		MAINTAIN ABOVE ½																										
CIRC WATER PUMP DISCH PRESS	BELOW 15[i]	25	45																									
TEMP CIRC WATER IN			75	85																								
LUBRICATOR FLOW RATE EACH GLASS	ABOVE 3[g] BELOW 1[h]																											

* CIRCLE LOG VALUES WHICH ARE OUTSIDE NORMAL RANGE.
** SHIPS WITH ENGINE ROOM AFW COOLING SYSTEM - MEASURE TEMPERATURE ON DOWN STEAM SIDE OF THERMO CONTROL VALVE IF EXISTS.
a WILL SHUT DOWN AUTOMATICALLY.
b RELIEF VALVE LIFTS.
c WILL CAUSE WET AIR TO BE SUPPLIED TO BANKS. RESET BACKPRESSURE CONTROL VALVE.
d WILL SHUT DOWN AUTOMATICALLY. EXPLOSION MAY OCCUR AT HIGHER TEMPERATURE.
e INADEQUATE LUBRICATION, WILL SHUT DOWN AUTOMATICALLY.
f LUBE OIL BLOCKADE, MANUAL SHUTDOWN REQUIRED.
g WILL CAUSE EXCESS CARBON.
h INADEQUATE LUBRICATION.
i MANUAL SHUTDOWN REQUIRED. (FOR PROBLEMS SEE TECH MANUAL TABLE 4-2-1, SHEET 3).

TIME METER READING*

AT 24:00 _____

AT 00:00 _____

HOURS TODAY [ ]

*WHEN METER INSTALLED _____

FOR INGERSOLL-RAND MODEL NA 13.5 CFH, 4500 PSI AIR COMPRESSOR

CSS-14 HPAC LOG
REV 1

AS 34 0101 LF-L34 5100

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 15. Standard log form

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"TRAINING SUMMARY FOR THE INGERSOLL-RAND
MODEL NA 13.5 CFH, 4500 PSI HIGH PRESSURE
AIR COMPRESSOR (HPAC) LOG FORMS
FOR USE ON SSBN627, CLASS

The following summary discusses selected values on Air Compressor Log forms which is of concern to an operator. It will be noted on the log forms that some values outside the listed ranges need not be circled. These values are shown in black and may be either the upper or lower value shown for that particular range. The values which are recorded on the log form which fall outside the normal range red values are of concern and must be circled. The red values are the significant values and values circled on the log forms will show that a potential problem area is developing or has already developed.

TEMP CIRC WATER OUT- Values lower than the 95% F given for the normal range of the circulating water out temperature need not be circled. This value is dependent on the "circulating water" in temperatures which are recorded elsewhere on the new log form.

PRESSURE 4th STAGE- The lower value shutdown is given for this stage as it is an indication that back pressure regulating values in the air dryers are not functioning properly (see TECH MANUAL for H.P. Dehydrator and Filter - NASEA S6228-AB-MMA-000/NA, paragraph 1-3-4 and page 1-5). This value provides the required pressure for proper functioning of the ships air dryers. Operation of the air dryers at pressures lower than 4300 psi can cause saturation of the desiccant with subsequent breakdown desiccant carryover and permit saturation air to be sent to the air banks.

The normal range values for the 4th stage were selected in the following manner:

a) Lower range values- 4300 psi- this value is required again for proper operation of the air dryers.

b) Higher range value- 4800 psi- values this high may be required to charge the air banks to 4500 psi.

TEMP 1st STAGE SUCT- Values outside the listed normal range need not be circled as the operator his little or no control over the compartment air temperature. Values outside the normal range will not adversly affect compressor operation.

INTERSTAGE LOWER VALUE TEMP VALUES- Temperature values below the listed lower normal range for the various stages within the compressor need not be circled as these are a function of the circulating water inlet temperature and this value is recorded elsewhere on the log form. In addition these values are not normally an indication of problems developing within the compressor.

FIGURE 16. Standard log form summary

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:05Z
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SAMPLE

SYSTEM	COMPONENT	MRC CARD
Propulsion	Main Circulating Pump	E-5 Q-1
SUBSYSTEM	**RELATED MAINTENANCE**	**RATES** / **M/H**
Circulating and Cooling Water	None	MM3 2.0 FN 2.0
		TOTAL M/H 4.0 **ELAPSED TIME** 2.0

MAINTENANCE REQUIREMENT DESCRIPTION

1. Clean sump and renew oil.
2. Clean lube oil filter.

SAFETY PRECAUTIONS

1. Observe standard safety precautions.
2. Wire steam inlet and outlet valves shut and tag "Do not Open."

TOOLS, PARTS, MATERIALS, TEST EQUIPMENT

1. 13/16" Combination wrench
2. 1/2" Drive socket set
3. Flashlight
4. Lint-free rags
5. Bucket
6. Safety tags
7. Wire, 24 gauge
8. 6" Slip-Joint Pliers
9. Oil, Symbol 2190 TEP
10. 1/32" Gasket material, Symbol 2290

PROCEDURE

Preliminary

a. Wire steam inlet and outlet valves shut and tag "Do Not Open".

1. Clean Sump and Renew Oil
   a. Drain oil sump.
   b. Remove inspection plate.
   c. Clean out sump.
   d. Ensure sump is clear of foreign material.
   e. Using new gasket on clean surfaces, reinstall inspection plate.
   f. Fill oil sump with oil.

2. Clean Lube Oil Filter
   a. Turn cleaning handle on top of lobe oil filter three or four turns.
   b. Remove plug from bottom of filter and drain the sediment.
   c. Reinstall the drain plug.
   d. Remove wire and tags from the steam inlet and outlet valves.

PAGE 1 OF 1

65

8565

Q

LOCATION	DATE
	1 June 1965

MAINTENANCE REQUIREMENT CARD (MRC)
OPNAV 4700-1 (B) (REV. 3/69)

FIGURE 17. Maintenance requirement card (MDC)

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RADIO RECEIVER R-XXX/URR
PUBLICATION NUMBER

INSTALLATION STANDARDS SUMMARY

Input Voltage_____ Vac
Input Frequency_____ Hz
(When reference standard tests are made)

Date_____
Serial number_____
of Model _____
Installed in (ship or station)

_____
Length of transmission line
_____

Record on this summary sheet the test indications which have been obtained during the installation test.

Paragraph No.	Ref. Std.	Paragraph No.	Ref. Std.
5-10	a._____Check	5-46	a._____ uV
			b._____ uV
5-21	a._____Vdc		c._____ Check
	b._____Vdc		d._____ uV
	c._____Vdc		e._____ uV
	d._____Vdc		f._____ uV
		5-51	a._____ Sec
5-33	a._____Check		b._____ Check
	b._____Check		c._____ Check
	c._____Check		d._____ Hz
	d._____Check		e._____ Hz
			f._____ Check
			g._____ Check

FIGURE 18. Installation standard summary sheet

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(Insert Classif. of TMDER Here and At Bottom of Page) **CLASSIFICATION:**

NAVSEA (USER) TECHNICAL MANUAL DEFICIENCY/EVALUATION REPORT (TMDER)
(NAVSEA S0005 – AA – GYD – 030/TMMP & NAVSEAINST 4160.3A)

INSTRUCTION: Continue on 8 – 1/2" x 11" paper if additional space is needed.

1. USE THIS REPORT TO INDICATE DEFICIENCIES, PROBLEMS, AND RECOMMENDATIONS RELATING TO PUBLICATION.
2. BLOCKS MARKED WITH "*" ARE TO BE FILLED IN BY THE CONTRACTOR BEFORE PRINTING.
3. FOR UNCLASSIFIED TMDERS, FILL IN YOUR RETURN ADDRESS IN SPACE PROVIDED ON THE BACK, FOLD AND TAPE WHERE INDICATED, AND MAIL. (SEE OPNAVINST 5510.1H FOR MAILING CLASSIFIED TMDERS.)
4. FOR ADDITIONAL INFORMATION, CALL AUTOVON 551 – 2976/2968 OR COMMERCIAL 805 – 982 – 2976/2968.

1. NAVSEA TECHNICAL MANUAL NO.*	2. VOL. PART*	3. TITLE*

4. REV. NO./DATE OR TM CH. NO./DATE	5. SYSTEM/EQUIPMENT NOMENCLATURE	6. SYSTEM/EQUIPMENT IDENTIFICATION/(MK/MOD/AN/PART NO.)

7. USER'S EVALUATION OF MANUAL *(Check Appropriate Blocks)*

	A. EXCELLENT		B. GOOD		C. FAIR		D. POOR		E. COMPLETE		F. INCOMPLETE

8. GENERAL COMMENTS

9. RECOMMENDED CHANGES TO PUBLICATION

PAGE NO. A.	PARA-GRAPH B.	LINE NO. C.	FIG. NO. D.	TABLE E.	F. RECOMMENDED CHANGES AND REASONS TYPE OF PROBLEM (INDICATE SAFETY (S), MAJOR (M), OR MINOR (P))

10. ORIGINATOR'S NAME AND WORK CENTER *(Please Print)*	11. SIGNATURE OF 3 – M COORDINATOR	12. DATE SIGNED	13. AUTOVON/COMM. NO.

14. SHIP HULL NO. AND/OR STATION ADDRESS (DO NOT ABBREVIATE)

15. THIS SPACE ONLY FOR NSDSA

A. CONTROL NO.	B. COG ISEA	C. DATE			D. PRIORITY	E. TRANSMITTED TO
		RECEIVED	FORWARDED	DUE		

NAVSEA 4160/1 (Rev. 10 – 89) (FRONT) *(REPLACES NAVSEA 9086/10, DESTROY STOCK)*

FIGURE 19. Typical user activity comment sheet (NAVSEA)

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SPAWAR 4160/1A (10-85)

USER ACTIVITY TECHNICAL MANUAL COMMENT SHEET (UATMCS)

1. UATMCS SEQUENCE NUMBER (SITE)		2. DATE
3. PUBLICATION NUMBER	4. VOLUME NO.	5. TITLE NOMENCLATURE
6. ORIGINAL PUBLICATION DATE	7. CHANGES	

8. CLASSIFICATION OF UATMCS ☐ UNCLASSIFIED ☐ CLASSIFIED _____

9. THIS COMMENT IMPACTS: ☐ SAFETY ☐ EQUIPMENT OPERATION ☐ PREVENTIVE MAINT. ☐ CORRECT MAINT ☐ TRAINING ☐ PARTS LIST ☐ OTHER

10. UATMCS TYPE ☐ PROBLEM ☐ QUESTION ☐ SUGGESTION ☐ COMMENT

11. RECOMMENDED CHANGE TO PUBLICATION

PAGE NO.	PARA-GRAPH	LINE NO.	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASON

12. ORIGINATOR NAME 13. RANK/RATE/GRADE AND TITLE

14. WORK CENTER 15. TELEPHONE (AUTOVON/COM'L)

16. SHIP HULL NO. AND/OR DUTY STATION (DO NOT ABBREVIATE)

17. FOR SPAWAR USE ONLY

a. CONTROL NO.	b. COG NO.	c. DATES			d. PRIORITY	e. TRANSMITTED TO
		REC	FWD	DUE		

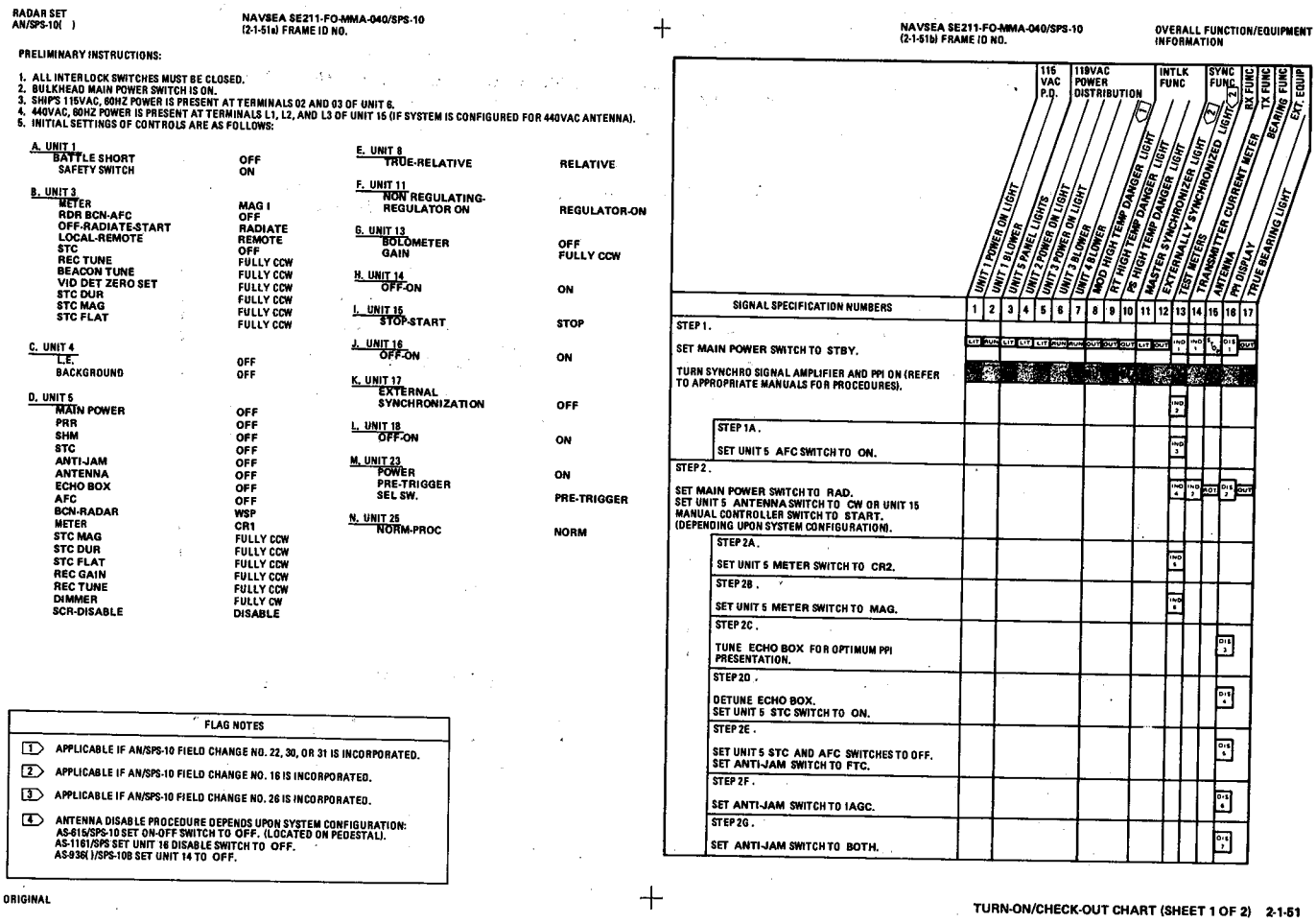
INSTRUCTIONS.

1. USE PLAIN WHITE PAPER FOR CONTINUATION SHEETS.
2. FOR UNCLASSIFIED UATMCS FOLD ON DOTTED LINE ON REVERSE SIDE STAPLE/SEAL AND MAIL TO COMMANDING OFFICER. NAVELEXCEN PORTSMOUTH, P. O. BOX 55, PORTSMOUTH,VIRGINIA 23705-0055 ATTENTION SPAWAR TECH DATA CENTER. MARK AND MAIL CLASSIFIED UATMCS IN ACCORDANCE WITH OPNAVINST 5510.1 (CURRENT VERSION)
3. FOR USE WITH NAVELEX/SPAWAR PUBLICATIONS ONLY.

PAGE 1 OF _____

FIGURE 19. Typical user activity comment sheet (SPAWAR)

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Check the source to verify that this is the current version before use.



RADAR SET AN/SPS-10( )

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-51a) FRAME ID NO.

PRELIMINARY INSTRUCTIONS:

1. ALL INTERLOCK SWITCHES MUST BE CLOSED.
2. BULKHEAD MAIN POWER SWITCH IS ON.
3. SHIP'S 115VAC, 60HZ POWER IS PRESENT AT TERMINALS 02 AND 03 OF UNIT 6.
4. 440VAC, 60HZ POWER IS PRESENT AT TERMINALS L1, L2, AND L3 OF UNIT 15 (IF SYSTEM IS CONFIGURED FOR 440VAC ANTENNA).
5. INITIAL SETTINGS OF CONTROLS ARE AS FOLLOWS:

A. UNIT 1	
BATTLE SHORT SAFETY SWITCH	OFF ON
B. UNIT 3	
METER	MAG I
RDR BCN-AFC	OFF
OFF-RADIATE-START	RADIATE
LOCAL-REMOTE	REMOTE
STC	OFF
REC TUNE	FULLY CCW
BEACON TUNE	FULLY CCW
VID DET ZERO SET	FULLY CCW
STC DUR	FULLY CCW
STC MAG	FULLY CCW
STC FLAT	FULLY CCW
C. UNIT 4	
I.F.	OFF
BACKGROUND	OFF
D. UNIT 5	
MAIN POWER	OFF
PRR	OFF
SHM	OFF
STC	OFF
ANTI-JAM	OFF
ANTENNA	OFF
ECHO BOX	OFF
AFC	OFF
BCN-RADAR	WSP
METER	CR1
STC MAG	FULLY CCW
STC DUR	FULLY CCW
STC FLAT	FULLY CCW
REC GAIN	FULLY CCW
REC TUNE	FULLY CCW
DIMMER	FULLY CW
SCR-DISABLE	DISABLE
E. UNIT 8	
TRUE-RELATIVE	RELATIVE
F. UNIT 11	
NON REGULATING-REGULATOR ON	REGULATOR-ON
G. UNIT 13	
BOLOMETER GAIN	OFF FULLY CCW
H. UNIT 14	
OFF-ON	ON
I. UNIT 15	
STOP-START	STOP
J. UNIT 16	
OFF-ON	ON
K. UNIT 17	
EXTERNAL SYNCHRONIZATION	OFF
L. UNIT 18	
OFF-ON	ON
M. UNIT 23	
POWER	ON
PRE-TRIGGER SEL SW.	PRE-TRIGGER
N. UNIT 25	
NORM-PROC	NORM

FLAG NOTES
1 APPLICABLE IF AN/SPS-10 FIELD CHANGE NO. 22, 30, OR 31 IS INCORPORATED.
2 APPLICABLE IF AN/SPS-10 FIELD CHANGE NO. 16 IS INCORPORATED.
3 APPLICABLE IF AN/SPS-10 FIELD CHANGE NO. 26 IS INCORPORATED.
4 ANTENNA DISABLE PROCEDURE DEPENDS UPON SYSTEM CONFIGURATION: AS-615/SPS-10 SET ON-OFF SWITCH TO OFF. (LOCATED ON PEDESTAL). AS-1161/SPS SET UNIT 16 DISABLE SWITCH TO OFF. AS-936( )/SPS-10B SET UNIT 14 TO OFF.

ORIGINAL

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-51b) FRAME ID NO.

OVERALL FUNCTION/EQUIPMENT INFORMATION

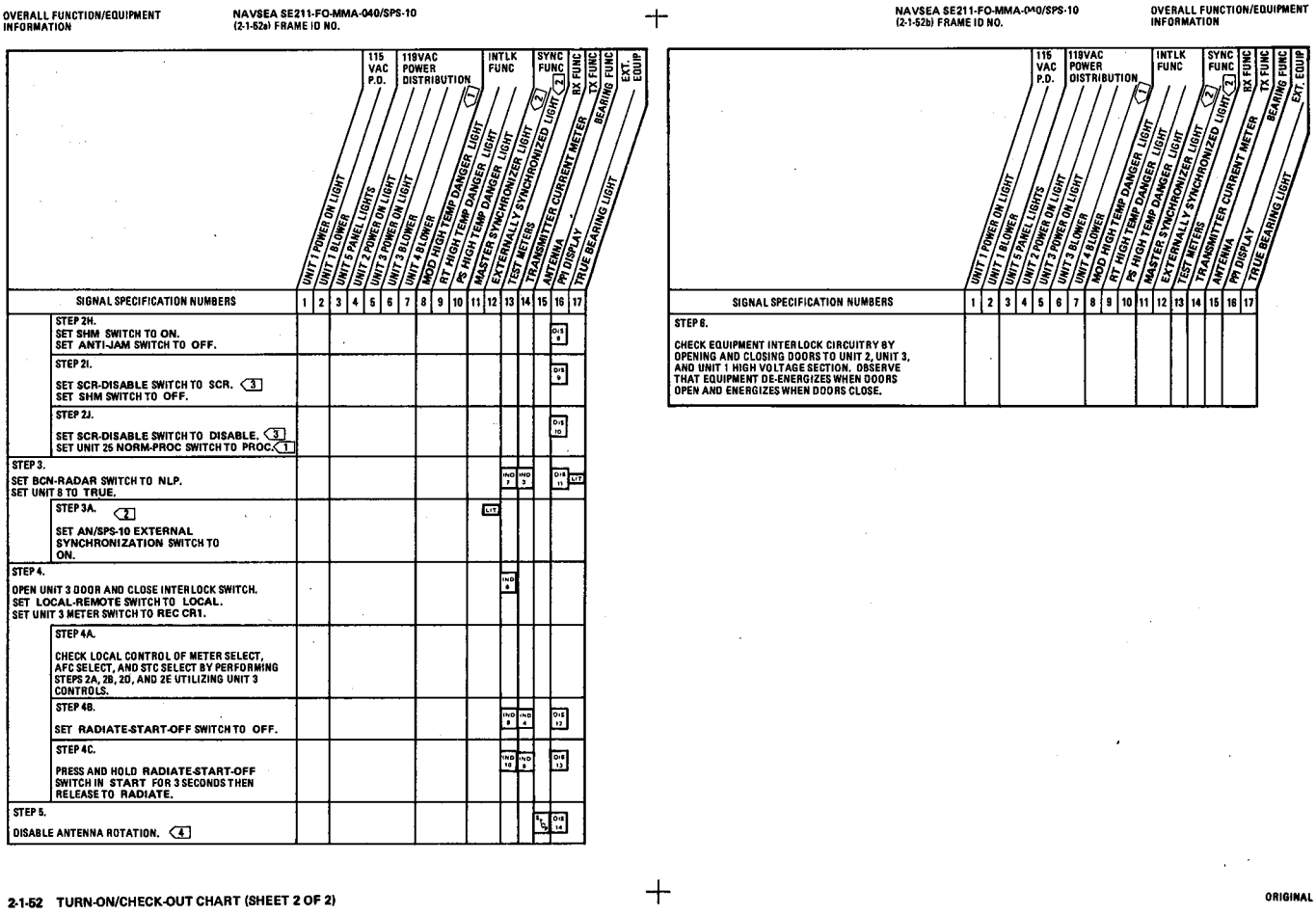
SIGNAL SPECIFICATION NUMBERS	1 UNIT 1 POWER ON LIGHT	2 UNIT 1 BLOWER	3 UNIT 5 PANEL LIGHTS	4 UNIT 2 POWER ON LIGHT	5 UNIT 3 POWER ON LIGHT	6 UNIT 3 BLOWER	7 UNIT 4 BLOWER	8 MOD HIGH TEMP DANGER LIGHT	9 RT HIGH TEMP DANGER LIGHT	10 PS HIGH TEMP DANGER LIGHT	11 MASTER SYNCHRONIZER LIGHT	12 EXTERNALLY SYNCHRONIZED LIGHT	13 TEST METERS	14 TRANSMITTER CURRENT METER	15 ANTENNA	16 PPI DISPLAY	17 TRUE BEARING LIGHT
	115 VAC P.D.					119VAC POWER DISTRIBUTION					INTLK FUNC		SYNC FUNC				RX FUNC / TX FUNC / BEARING FUNC / EXT. EQUIP
STEP 1. SET MAIN POWER SWITCH TO STBY. TURN SYNCHRO SIGNAL AMPLIFIER AND PPI ON (REFER TO APPROPRIATE MANUALS FOR PROCEDURES).	LIT	RUN	LIT	LIT	LIT	RUN	RUN	OUT	OUT	OUT	LIT	OUT	IND 1	IND 1	DIS 1	DIS 1	OUT
STEP 1A. SET UNIT 5 AFC SWITCH TO ON.													IND 2				
STEP 2. SET MAIN POWER SWITCH TO RAD. SET UNIT 5 ANTENNA SWITCH TO CW OR UNIT 15 MANUAL CONTROLLER SWITCH TO START. (DEPENDING UPON SYSTEM CONFIGURATION).													IND 3	IND 2	OUT	DIS 2	OUT
STEP 2A. SET UNIT 5 METER SWITCH TO CR2.													IND 4				
STEP 2B. SET UNIT 5 METER SWITCH TO MAG.													IND 5				
STEP 2C. TUNE ECHO BOX FOR OPTIMUM PPI PRESENTATION.																DIS 3	
STEP 2D. DETUNE ECHO BOX. SET UNIT 5 STC SWITCH TO ON.																DIS 4	
STEP 2E. SET UNIT 5 STC AND AFC SWITCHES TO OFF. SET ANTI-JAM SWITCH TO FTC.																DIS 5	
STEP 2F. SET ANTI-JAM SWITCH TO IAGC.																DIS 6	
STEP 2G. SET ANTI-JAM SWITCH TO BOTH.																DIS 7	

TURN-ON/CHECK-OUT CHART (SHEET 1 OF 2) 2-1-51

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 20. Turn-on/check-out chart (sheet 1 of 2)

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2-1-52 TURN-ON/CHECK-OUT CHART (SHEET 2 OF 2)

ORIGINAL

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 20. Turn-on/check-out chart (sheet 2 of 2)

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AN/SPS-10 OVERALL FUNCTION DIAGRAM

The AN/SPS-10( )is a surface-search radar which operate between 5450 and 5825MHz. It detects, ranges, and tracks surface targets and, to a limited extent, air targets. It supplies target range and bearing intelligence to a standard Navy Plan Position Indicator (PPI). The PPI provides a presentation of surrounding surface area as well as low altitude airborne objects. The OVERALL FUNCTION DIAGRAM depicts the major functions and their interfacing signal lines.

(1) POWER DISTRIBUTION

Refer to page 2-1-65 ( ) for text.

(2) INTERLOCK AND OVERTEMPERATURE FUNCTION

Refer to page 2-1-100 ( ) for text.

(3) CONTROL FUNCTION

Refer to page 2-1-103 ( ) for text.

(4) MODULATOR FUNCTION

Refer to page 2-1-108 ( ) for text.

(5) SYNCHRONIZE FUNCTION

Refer to page 2-1-128 ( ) for text.

(6) TRANSMIT FUNCTION

Refer to page 2-1-132 ( ) for text.

(7) RECEIVE FUNCTION

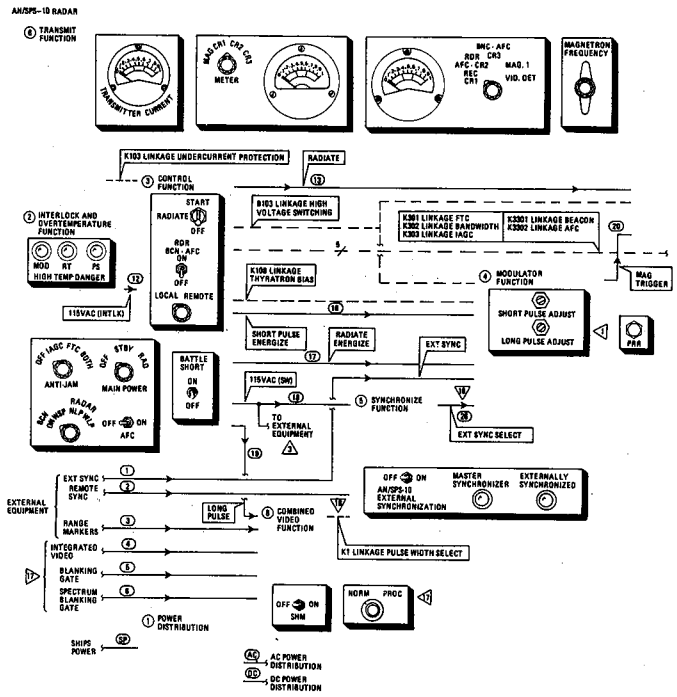
Provides a video output of target echoes. The Receiver has two bandwidths: 5MHz for wide band operation and 1MHz for narrow band operation. AN/SPS-10 Field Change No. 26 incorporates an Interference Blanker (controlled by the SCR-DISABLE switch) that suppresses off-frequency interference from radars operating in the same frequency band as the AN/SPS-10. An Automatic Frequency Control (AFC) automatically tunes the Radar Local Oscillator. For anti-jam purposes, the receiver incorporates an Instantaneous Automatic Gain Control (IAGC) circuit and a Fast Time Constant (FTC) circuit. These circuits may be used together or separately. A Sensitivity Time Constant (STC) circuit reduces receiver sensitivity at close ranges. An Echo Box enables testing of overall radar performance. The Receiver Tune control allows Local Oscillator manual tuning.

(8) COMBINED VIDEO

Sums target echo video, range markers, and SHM signals to provide COMBINED VIDEO at the proper level and time for PPI presentation. ITA DISPLAY VIDEO, from the Incoming Target Alarm, is also combined if AN/SPS-10 Field Change No. 22 and 30, or 31 and 24 are incorporated. Video Clutter Suppressor (AN/SPS-10 Field Change No. 22 and 30, or 31) installation provides a choice of DELAYED COMBINED VIDEO (normal video) or CLUTTER SUPPRESSED VIDEO (heavy sea and weather echoes suppressed) for PPI presentation. The video may be blanked either by a BLANKING GATE from the AN/SPA-42 or AN/SPG-55B RADAR. The Video Clutter Suppressor also provides LOG PROCESSED VIDEO and LOG VIDEO for use with the Incoming Target Alarm.

(9) BEARING FUNCTION

Motor and geartrain rotate the Antenna at 16RPM to provide a full 360 degree surveillance. A synchro transmitter, coupled to the Antenna drive shaft, provides relative antenna position information to the Synchro Signal Amplifier and PPI's, if selected. The Synchro Signal Amplifier references the relative position information to True North and is available for the PPI's, if selected. A RELATIVE WARNING signal is applied to the PPI's when the AN/SPS-10 has selected True Bearing information. A PRF CHANGE signal is provided for the AN/SPS-58 Radar system, if the AS-936(C)/SPS-10B Antenna Assembly is installed.



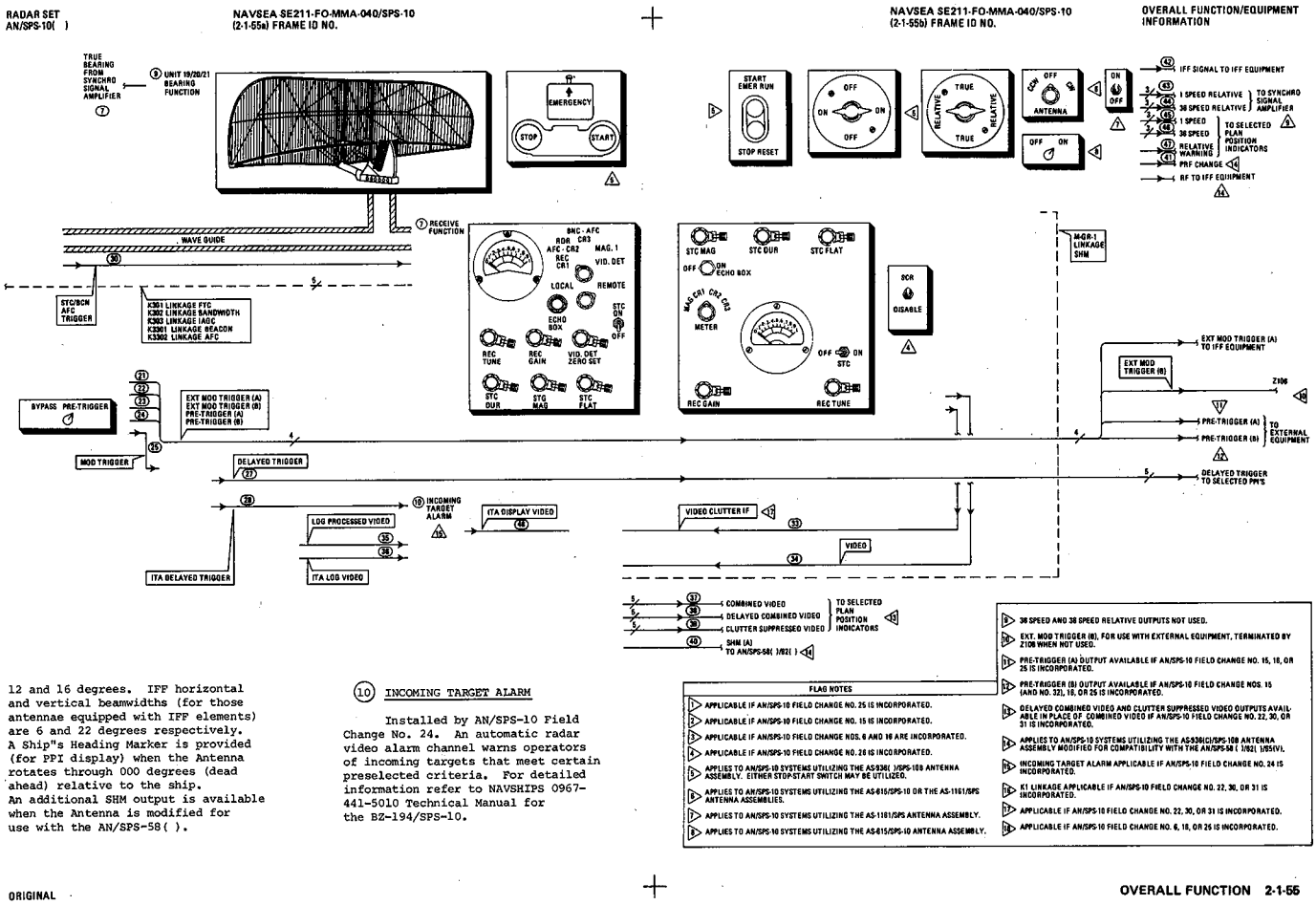
The Antenna rotates at 16 RPM powered by a motor geartrain. Rotary joints couple RF between the waveguide and the radiating elements of the Antenna. The Antenna serves for both transmitting and receiving. The Antenna is directional with a horizontal beam width of 1.5 degrees and a vertical beam width between

DATA-MT-05-07-125

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 21. Overall function diagram (sheet 1 of 2)

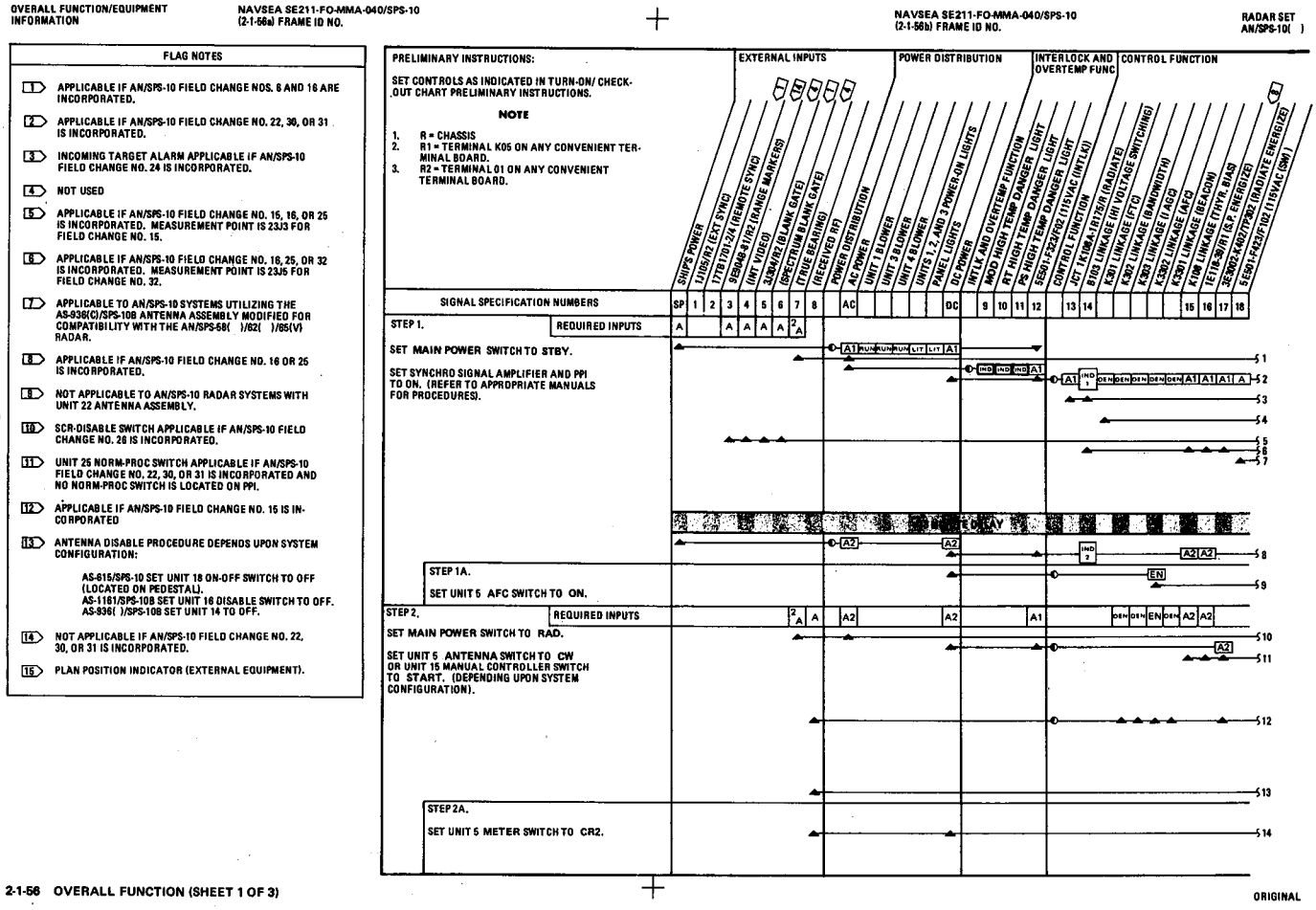
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Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 21. Overall function diagram (sheet 2 of 2)

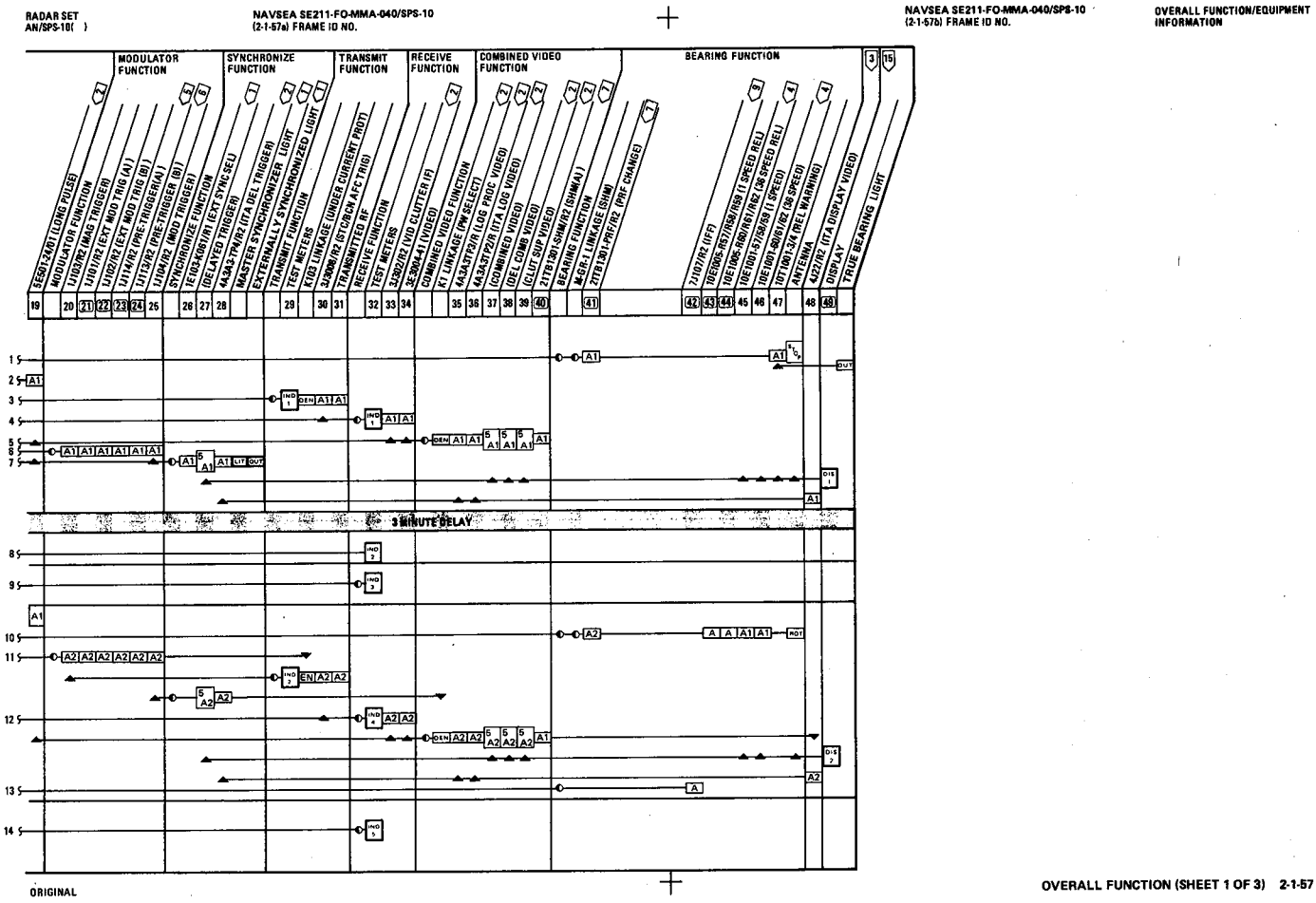
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FIGURE 22. Overall MDC (sheet 1 of 2)

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OVERALL FUNCTION (SHEET 1 OF 3) 2-1-57

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FIGURE 22. Overall MDC (sheet 2 of 2)

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MAJOR FUNCTION (4) INFORMATION

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-108a) FRAME ID NO.

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-108b) FRAME ID NO.

RADAR SET AN/SPS-10( )

MODULATOR FUNCTION (4)

The Modulator function pertains to the basic Modulator and all other configurations of the modulator except those with AN/SPS-10 Field Change No. 25 incorporated. A free running variable frequency oscillator produces the synchronizing triggers for the entire system. The Pulse Repetition Rate (PRR) is 625 to 650 pulses per second. Pulses thus produced trigger the Modulator which produces a high voltage pulse that turns on the Transmitter (one of three pulse widths may be selected). The Modulator supplies:

a. Trigger to initiate transmission.
b. Trigger to synchronize the Indicator Adapter (or Video Clutter Suppressor if incorporated.
c. Trigger for IFF equipment synchronization.
d. A synchronizing pulse to be used with other external equipment, such as test equipment.

If AN/SPS-10 Field Change No. 15 is incorporated, the Modulator will also provide a Pre-Trigger pulse for use with the AN/SLA-10( ). (An additional Pre-Trigger pulse is available for use with AIMS-IFF equipment when AN/SPS-10 Field Change Nos. 15 and 32 are both incorporated.

If AN/SPS-10 Field Change No. 16 is incorporated, the Modulator also provides Pre-Trigger pulses for the AN/SLA-10( ) and the AN/SPA-42( ). The Modulator can also be synchronized from an external source. The Modulator is disabled during pulse width switching and when the equipment is in Standby.

(1) PRR SWITCH

Provides the PRR CONTROL signal that varies the output of V-OSC-1 while the switch is pressed. If AN/SPS-10 Field Change No. 16 is incorporated, the PRR CONTROL signal is applied to the Pre-Trigger Generator Sub-Function (FC-16) to control the PRR of a Unijunction Oscillator.

(2) REPETITION RATE GENERATOR

Produces a square wave output variable between 625 and 650 Hz to trigger:

a. The Blocking Oscillator if AN/SPS-10 Field Change No. 15 or 16 (Pre-Trigger Generator) is not incorporated.
b. Unit 23 Pre-Trigger Generator if AN/SPS-10 Field Change No. 15 is incorporated.

NOTE

1A1 Repetition Rate Generator circuitry is not applicable if AN/SPS-10 Field Change No. 16 is incorporated. Refer to Pre-Trigger Generator Sub-Function (FC-16).

The resonant frequency of V-OSC-1 is controlled by the PRR switch. V-OSC-1 produces a sine wave voltage. V-SHP-1 shapes the sine wave output of V-OSC-1 into a square wave of the same repetition rate and applies it to either the Blocking Oscillator or a Pre-Trigger Generator depending upon the Modulator configuration.

If the Modulator is externally synchronized the output of V-SHP-1 will be a square wave at the repetition rate of the external signal. The Modulator can be externally synchronized by one of the following methods.

a. AN/SPS-10 Field Change No. 16 not incorporated: External synchronization signal, applied to Pin H25 of E107, is jumpered to H125 and applied to V-SHP-1.
b. AN/SPS-10 Field Change No. 16 incorporated: For external synchronization refer to 1A4 Pre-Trigger Generator Sub Function (FC-16).

(3) BLOCKING OSCILLATOR

Provides THYRATRON TRIGGER to fire the Thyratron. The source of the square wave input to the Blocking Oscillator depends upon the configuration of the Modulator.

a. Field Change No. 15 incorporated: The input to the Blocking Oscillator is the SELECTED TRIGGER from the Pre-Trigger Generator Sub-Function (FC-15).
b. Field Change No. 16 is incorporated: The input to the Blocking Oscillator is the DELAYED TRIGGER from the Pre-Trigger Generator Sub-Function (FC-16).
c. Field Change Nos. 15 and 16 not incorporated: The input to the Blocking Oscillator is the Trigger from V-SHP-1 in the Repetition Rate Generator.

The input signal is differentiated and the positive trigger (coincident with the positive leading edge of the input) triggers V-BO-1. V-BO-1 is momentarily disabled, by K108B (High Voltage Switch Assembly 1A3), when pulse widths are being changed. This prevents transmitting during pulse width switching. Since the Beacon pulse width is twice the width of the radar long pulse, the repetition rate must be decreased by a factor of 2 in order to keep the duty cycle less than 0.001 sec. This is the specification requirement for the Thyratron and Magnetron when considering dissipation and deionization time. The recovery time of V-BO-1 is increased during BEACON operation by adding R177 (High Voltage Switch Assembly, 1A3) to the RC recovery time network of V-BO-1. This cuts the Blocking Oscillator repetition rate in half. V-CF-1 matches the positive trigger pulse from V-BO-1 to the Thyratron.

(4) THYRATRON

Provides a discharge path for the Pulse Forming Network (PFN). The positive THYRATRON TRIGGER applied to V-THY-1 causes the Thyratron to conduct which effectively applies a ground to the Pulse Forming Network causing the Network to discharge. The Thyratron is momentarily disabled by K108B (High Voltage Switch Assembly 1A3), while pulse widths are being changed. This prevents transmitting during pulse width switching.

(5) PULSE FORMING

Produces the high negative pulses that turn on the Magnetron Oscillator for transmitting. The High Voltage Supply charges N-PF-1 via V-CD-1. (V-CD-1 also prevents discharge of N-PF-1 until the Thyratron fires). When V-THY-1 fires N-PF-1 discharges to produce the MAG TRIGGER (modulating pulse). L103 reduces the rate of current rise in the Thyratron output pulse to protect the Thyratron and improve the operation of the Magnetron. R105 and R108 are Bleeder resistors for N-PF-1.

(6) VOLTAGE DIVIDER

Divides part of the MAG TRIGGER pulse to provide low amplitude trigger pulses for the Indicator Adapter (or Video Clutter Suppressor if incorporated) and for synchronizing external equipment with the AN/SPS-10 Radar Set. T110 inverts EXT MOD TRIG(B) to produce MOD TRIGGER and EXT MOD TRIG(A).

(7) PULSE INHIBIT

Inhibits the Blocking Oscillator and Thyratron, during pulse width switching, also decreases the Blocking Oscillator repetition rate during Beacon operation. K108B connects return to V-BO-1 and V-THY-1 bias line when contacts are closed (de-energized), enabling both V-BO-1 and V-THY-1. During pulse width changes K108B is energized and return is removed from the bias line. This causes the bias line to go highly negative inhibiting V-BO-1 and V-THY-1. S105B shunts R177, except during Beacon operation. During Beacon operation S105B opens, connecting R177 to the Blocking Oscillator recovery circuit. This increases the Blocking Oscillator recovery time and its repetition rate is halved.

(8) PULSE WIDTH SELECT

Connects the terminals of N-PF-1 as necessary for Short Pulse, Long Pulse and Beacon operation. Short pulse utilizes the 0.25 microsecond terminal. Long pulse utilizes the 0.25 and 1.05 microsecond terminals (1.3 microseconds). Beacon utilizes 0.25 and 2.0 microsecond terminals (2.25 microseconds).

(9) HIGH VOLTAGE SUPPLY

Provides the high voltage to charge

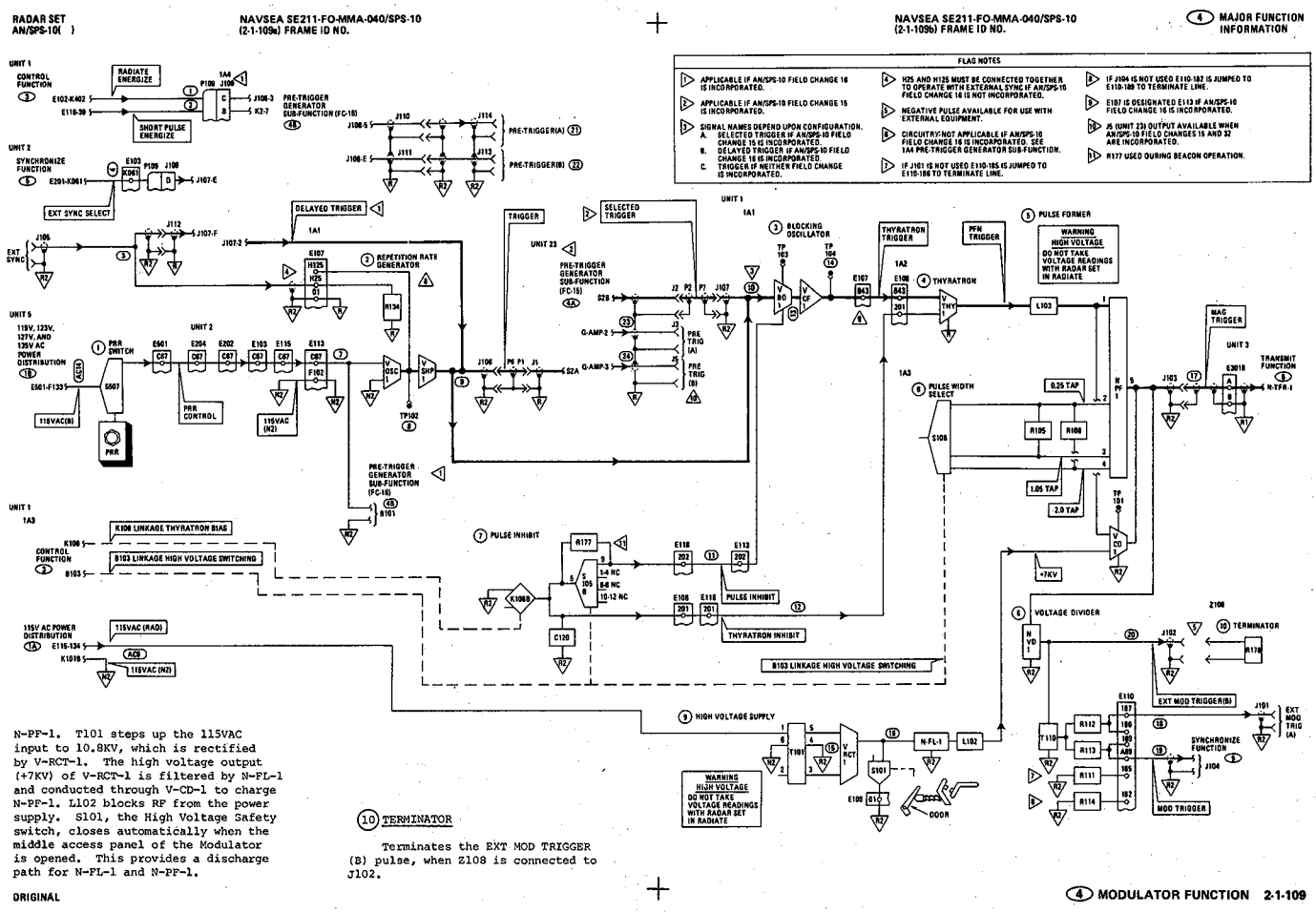
2-1-108 MODULATOR FUNCTION (4)

ORIGINAL

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 23. Major function diagram (sheet 1 of 6)

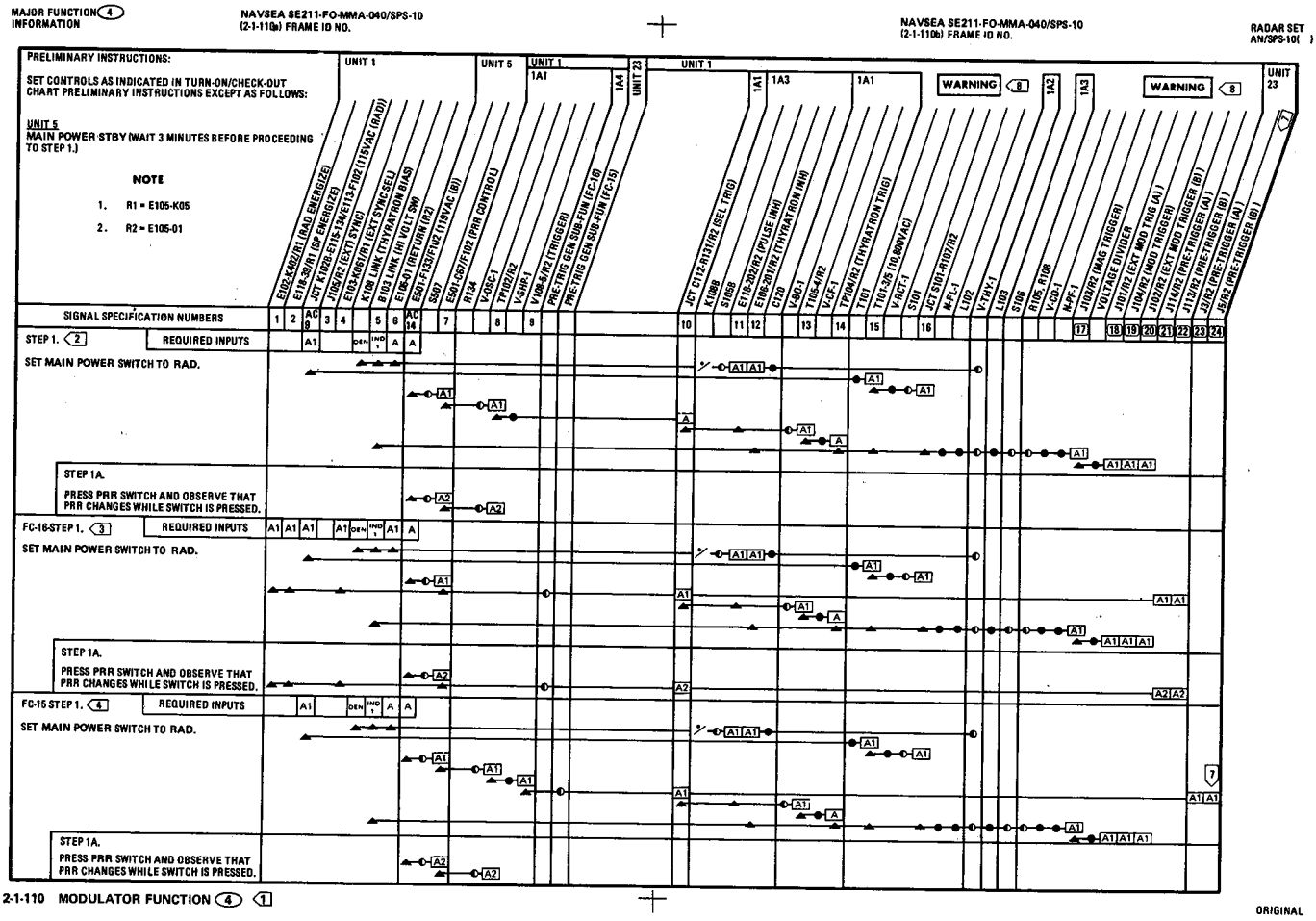
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FIGURE 23. Major function diagram (sheet 2 of 6)

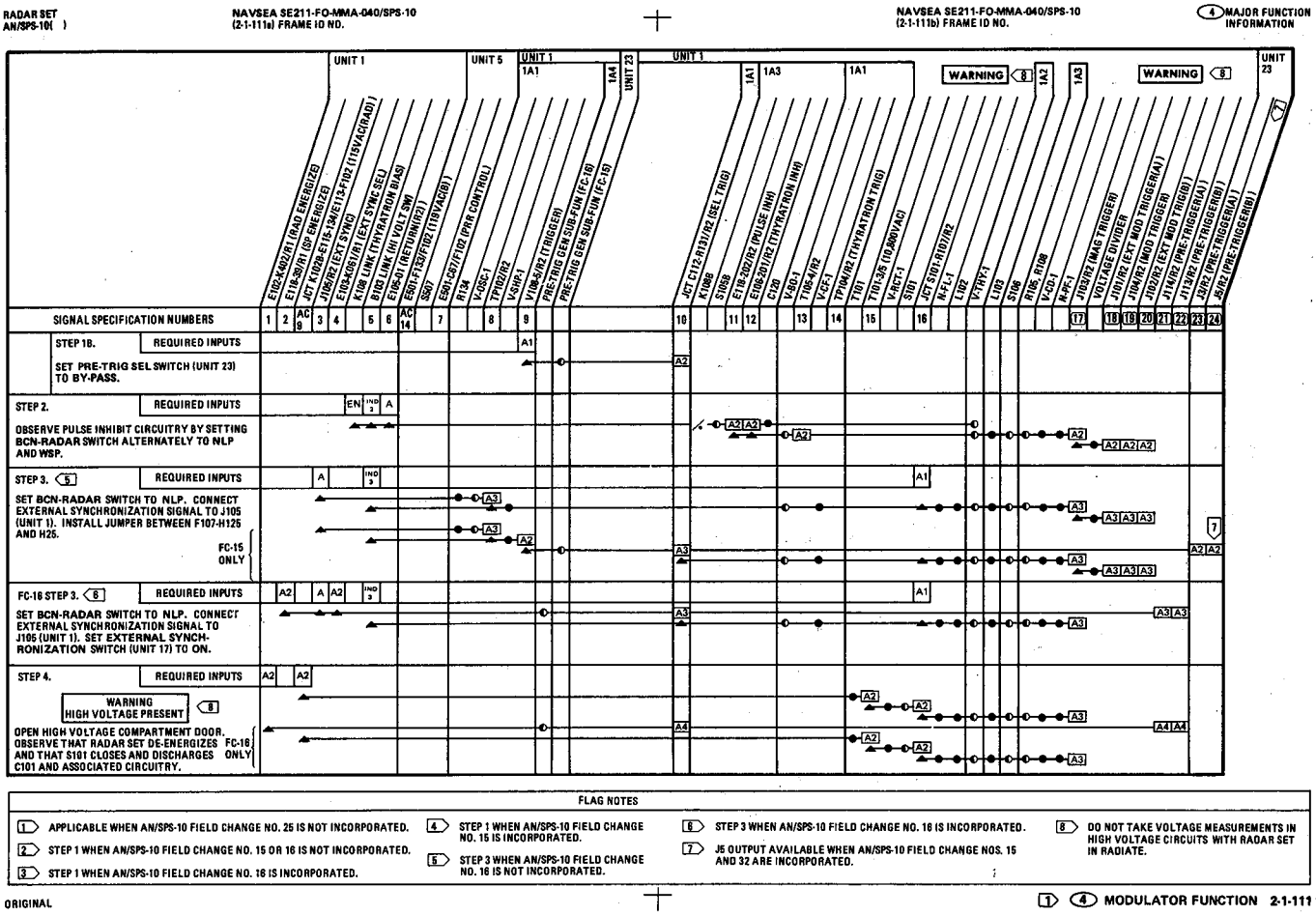
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FIGURE 23. Major function diagram (sheet 3 of 6)

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FIGURE 23. Major function diagram (sheet 4 of 6)

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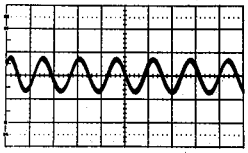
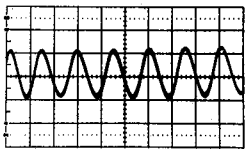
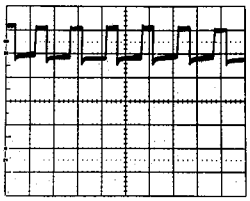
MAJOR FUNCTION ④ INFORMATION

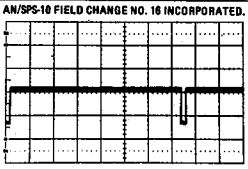
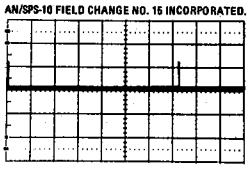
NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-112a) FRAME ID NO.

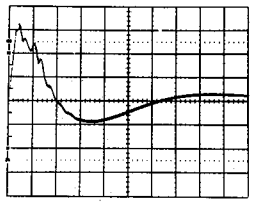
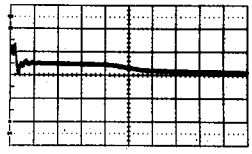
NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-112b) FRAME ID NO.

RADAR SET AN/SPS-10( )

SPEC NO.	SIGNAL SPECIFICATION
	GENERAL NOTES UNLESS OTHERWISE SPECIFIED: 1. VALUES ARE ± 10%. 2. OSCILLOSCOPE VERTICAL IS DC COUPLED; BASELINE IS OFFSET 1 MINOR DIVISION ABOVE GRATICULE CENTER. 3. PROBE: X1.
	TEST EQUIPMENT REQUIRED (OR EQUIVALENT) MULTIMETER AN/USM-311 OSCILLOSCOPE AN/USM-281A
	NOTES (1) +24V (UNREGULATED) VARIES BETWEEN +20V AND +47V DEPENDING UPON SYSTEM LOAD.
AC9A1 A2	115VAC 0VAC
AC14	119VAC
1A1 A2	(1) +24V (UNREGULATED). 0V
2A1 A2	(1) +24V (UNREGULATED). 0V
3	SIGNAL DEPENDS UPON EXTERNAL SYNCHRONIZATION SIGNAL INPUT.
4A1 A2	0V (1) +24V (UNREGULATED).
5	IND 1: S103 RUNS TILL S105B AND S106 ARE POSITIONED FOR SHORT PULSE OPERATION. IND 2: S103 RUNS REPOSITIONING S105B AND S106 BETWEEN LONG PULSE AND SHORT PULSE POSITION. IND 3: S103 RUNS TILL S105B AND S106 ARE POSITIONED FOR LONG PULSE OPERATION.
6	RETURN
7A1 A2	0VAC 119VAC WHILE PRR SWITCH IS PRESSED.

SPEC NO.	SIGNAL SPECIFICATION
8A1	AN/SPS-10 FIELD CHANGE NO. 16 NOT INCORPORATED.  VERT: 50V/DIV SYNC: DC – EXT HORIZ: 1MILLSEC/DIV SYNC POINT: 1J102
A2	AN/SPS-10 FIELD CHANGE NO. 16 NOT INCORPORATED.  VERT: 50V/DIV SYNC: DC – EXT HORIZ: 1MILLSEC/DIV SYNC POINT: 1J102
A3	AN/SPS-10 FIELD CHANGE NO. 16 NOT INCORPORATED. SAME AS 8A1 EXCEPT FREQUENCY DEPENDS UPON EXTERNAL SYNCHRONIZATION SIGNAL.
9A1	AN/SPS-10 FIELD CHANGE NO. 15 INCORPORATED.  VERT: 100V/DIV SYNC: DC – EXT HORIZ: 1MILLSEC/DIV SYNC POINT: 1J102
A2	AN/SPS-10 FIELD CHANGE NO. 15 INCORPORATED: SAME AS 9A1 EXCEPT FREQUENCY DEPENDS UPON EXTERNAL SYNCHRONIZATION SIGNAL.
10	AN/SPS-10 FIELD CHANGE NO. 15 AND FIELD CHANGE NO. 16 NOT INCORPORATED: SAME AS 9A1.

SPEC NO.	SIGNAL SPECIFICATION
10A1	AN/SPS-10 FIELD CHANGE NO. 16 INCORPORATED.  VERT: 50V/DIV SYNC: AC – INT HORIZ: 200USEC/DIV
A2	AN/SPS-10 FIELD CHANGE NO. 16 INCORPORATED. SAME AS 10A1 EXCEPT PRR HAS CHANGED.
A3	AN/SPS-10 FIELD CHANGE NO. 16 INCORPORATED. SIGNAL DEPENDS UPON EXTERNAL SYNCHRONIZATION SIGNAL.
A4	AN/SPS-10 FIELD CHANGE NO. 16 INCORPORATED. 0V
10A1	AN/SPS-10 FIELD CHANGE NO. 15 INCORPORATED.  VERT: 100V/DIV SYNC: DC – EXT HORIZ: 200USEC/DIV SYNC POINT: 1J102
A2	AN/SPS-10 FIELD CHANGE NO. 15 INCORPORATED. SAME AS 9A1.
A3	AN/SPS-10 FIELD CHANGE NO. 15 INCORPORATED SAME AS 9A1 EXCEPT FREQUENCY DEPENDS UPON EXTERNAL SYNCHRONIZATION SIGNAL.
11A1 A2	0V –200V DURING PULSE WIDTH SWITCHING.
12A1 A2	0V –200V DURING PULSE WIDTH SWITCHING.

SPEC NO.	SIGNAL SPECIFICATION
13A1	 VERT: 100V/DIV SYNC: DC – EXT HORIZ: 2USEC/DIV SYNC POINT: 1A1TP104
A2	PULSE DISAPPEARS DURING PULSE WIDTH CHANGING.
14	 VERT: 100V/DIV SYNC: DC + EXT HORIZ: 5USEC/DIV SYNC POINT: 1A1TP102
15A1 A2	DO NOT MEASURE. 10,800 VAC. 0VAC
16A1 A2	DO NOT MEASURE. +7KV. 0V
17A1	NOT MEASUREABLE: –5KV SHORT PULSE TO MAGNETRON.
A2	NOT MEASUREABLE: PULSE INHIBITED DURING PULSE WIDTH CHANGING.
A3	NOT MEASUREABLE: –5KV LONG PULSE TO MAGNETRON WITH PRR DEPENDENT UPON EXTERNAL SYNCHRONIZATION SIGNAL INPUT.

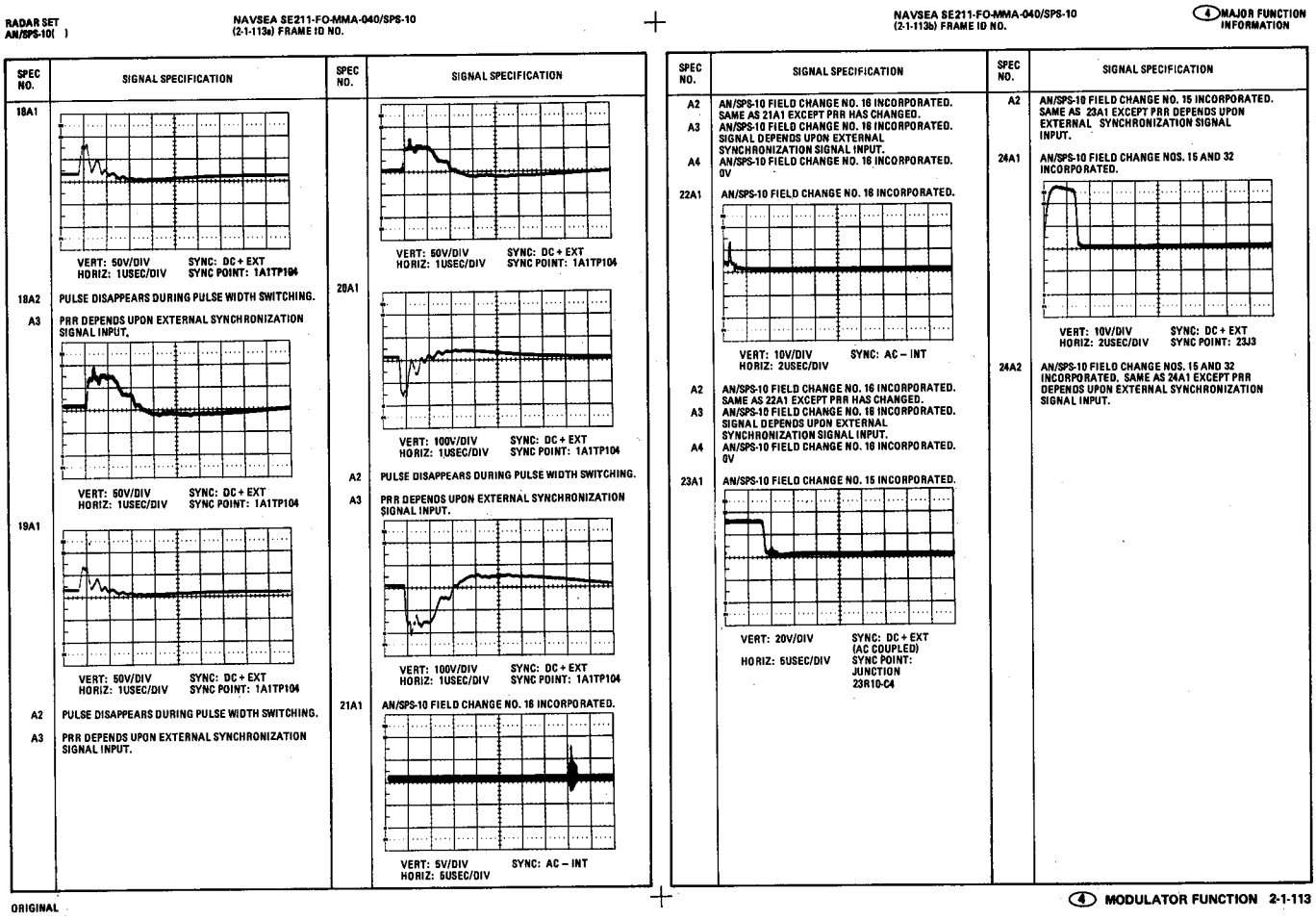
2-1-112 MODULATOR FUNCTION ④

ORIGINAL

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 23. Major function diagram (sheet 5 of 6)

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Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 23. Major function diagram (sheet 6 of 6)

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MAJOR FUNCTION (1AA) INFORMATION

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-74a) FRAME ID NO.

AC VOLTAGE REGULATOR SUB-FUNCTION (1AA)

The Voltage Regulator maintains a regulated output between 113 and 117VAC with an input voltage between 103 and 127VAC. Regulation is accomplished primarily by the use of an autotransformer, which has a fixed center tap in addition to a motor positioned wiper, and an isolation transformer with its secondary in series with the input power. The isolation transformer either adds to or subtracts from the input voltage to provide the regulated output within tolerances. The complete loop for regulation is as follows:

Condition 1. Input Voltage is 115VAC. Output is 115VAC.

The voltage through T1101 and V-RCT-2 to N-BRG-1 is such that the bridge is in balance. The two outputs from the bridge to V-AMP-1 and V-AMP-2 keep both amplifiers conducting sufficiently to keep both switches, V-SW-1 and V-SW-2, cut off so that neither K1101 nor K1102 are energized. N-MOT-1 is not energized and its brake is applied. The output of T1103 is 0 volts so there is no voltage induced in isolation transformer T1102 to be added to or subtracted from the input power to T1101.

Condition 2. Input voltage decreases below 113VAC. Output voltage attempts to drop below 113VAC.

The regulated voltage as fed back through T1101 and V-RCT-2 to N-BRG-1 is such that the bridge is unbalanced. The input to V-AMP-1 becomes more positive so that V-AMP-1 remains conducting, V-SW-1 remains cutoff and K1102 remains de-energized. The input to V-AMP-2 becomes more negative so that V-AMP-2 is cutoff and V-SW-2 is conducting to energize K1101. Contacts K1101A close 3 to 4 to apply power to, and provide a path to neutral for X-RCT-1 through, the 4-2 windings of N-MOT-1. The voltage from X-RCT-1 releases the mechanical brake, N-MOT-1 runs in a forward direction, and moves the wiper of T1103 so an output voltage is present from T1103. This output is coupled through isolation transformer T1102 and the voltage induced in the 3-4 windings of T1102 is in phase with the input voltage and adds to the input to make the output regulated voltage higher than the input. This condition will remain until the output reaches 115VAC at which time the bridge will be in balance and the other circuits return to the state described in Condition 1.

Condition 3. Input voltage increases above 117VAC. Output voltage attempts to rise above 117VAC.

The regulated voltage as fed back through T1101 and V-RCT-2 to N-BRG-1 is such that the bridge is unbalanced. The input to V-AMP-2 becomes more positive so that V-AMP-2 remains conducting, V-SW-2 remains cutoff and K1101 remains de-energized. The input to V-AMP-1 becomes more negative so that V-AMP-1 is cutoff and V-SW-1 is conducting to energize K1102. Contacts K1102A close 3 to 4 to apply power to the 1 winding of N-MOT-1. This makes N-MOT-1 run in a reverse direction and move the wiper of T1103 so that an output voltage is present from T1103. This output is coupled through isolation transformer T1102. The voltage induced in the 3-4 windings of T1102 is out of phase with the input voltage to subtract from the input to make the output regulated voltage lower than the input. This condition will remain until the output reaches 115VAC at which time the bridge will be in balance and the other circuits return to the state described in Condition 1.

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-74b) FRAME ID NO.

RADAR SET AN/SPS-10( )

(1) S1103A

In the event of a Voltage Regulator malfunction, S1103 can be manually switched to the nonregulating position. S1103A contacts 1,7 to 2 are made, bypassing the Voltage Regulator to permit operation of the Radar Set.

(2) ERROR SENSOR

Monitors the regulated output voltage and provides voltage correction signals if the 115VAC output is not within its ± 2% tolerance. T1101 provides both stepped up and stepped down outputs from the 115VAC input. V-RCT-2 rectifies the output of T1101. C1101 filters the output of V-RCT-2. N-BRG-1 maintains a balanced output to the DC amplifier stages if the Voltage Regulator output is within operating limits. R1101 enables adjustment of the rectifier output for 115VAC and has a range of 114VAC to 120VAC. R1103 is adjusted so that K1101 becomes energized when the regulated output decreases to 113VAC. R1108 is adjusted so that K1102 becomes energized when the regulated output increases to 117VAC. V-AMP-1 amplifies any change in voltage on its input from N-BR-1. An unbalanced bridge condition creates a differential output as described for N-BRG-1, when the input to T1101 exceeds 117VAC. V-SW-1 passes a relay energizing current when the input to T1101 exceeds 117VAC. S1102 limit switch opens the power path to the motor when the moveable arm of the autotransformer reaches its mechanical limit of travel. V-AMP-2 operates the same as V-AMP-1 for conditions when the input to T1101 drops below 113VAC. V-SW-2 passes a relay energizing current when the input to T1101 drops below 113VAC. S1101 limit switch operates the same as S1102.

(3) CORRECTION DIRECTION CONTROL

Controls the rotation direction of N-MOT-1 which adjusts the autotransformer in the proper direction to bring the AC line voltage within tolerance. X-RCT-1 rectifies the 115VAC, applied through either K1101A or K1102A, to provide a DC voltage for operation of N-MOT-1. K1102 energizes when V-SW-1 conducts corresponding to an input to T1101 above 117VAC. K1102A connects one side of X-RCT-1 to N-MOT-1 for rotation of the motor, when in the energized condition. K1101 is energized when V-SW-2 conducts corresponding to an input to T1101 below 113VAC. K1101A connects the other side of X-RCT-1 to N-MOT-1 for opposite rotation of N-MOT-1, when in the energized condition. N-MOT-1 drives the autotransformer to correct for errors in the Voltage Regulator output. Direction of rotation of N-MOT-1 and correction of T1103 is dependent upon the condition of relays K1101 and K1102. When the error voltage has been compensated for both relays are de-energized, removing excitation voltage from N-MOT-1.

(4) ERROR CORRECTION

Corrects the Regulator output to 115VAC ±2%. When the source voltage is 115VAC the output from T1103 to T1102 will be 0 volts so the regulated output will be 115VAC. When the regulated output attempts to change, either as a result of load change or source voltage change, T1103 provides an output voltage to T1102. The phase and amplitude of this output to T1102 will offset the attempted change to maintain the regulated output at 115VAC ±2%. T1102 provides an output, in the regulated line, whose phase and amplitude either add to or subtract from the source voltage to maintain the regulated output within tolerance.

2-1-74 AC VOLTAGE REGULATOR SUB-FUNCTION (1AA)

ORIGINAL

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 24. Subfunction diagram (sheet 1 of 2)

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RADAR SET AN/SPS-10( )

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-75a) FRAME ID NO.

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-75b) FRAME ID NO.

(1AA) MAJOR FUNCTION INFORMATION

(5) S1103B

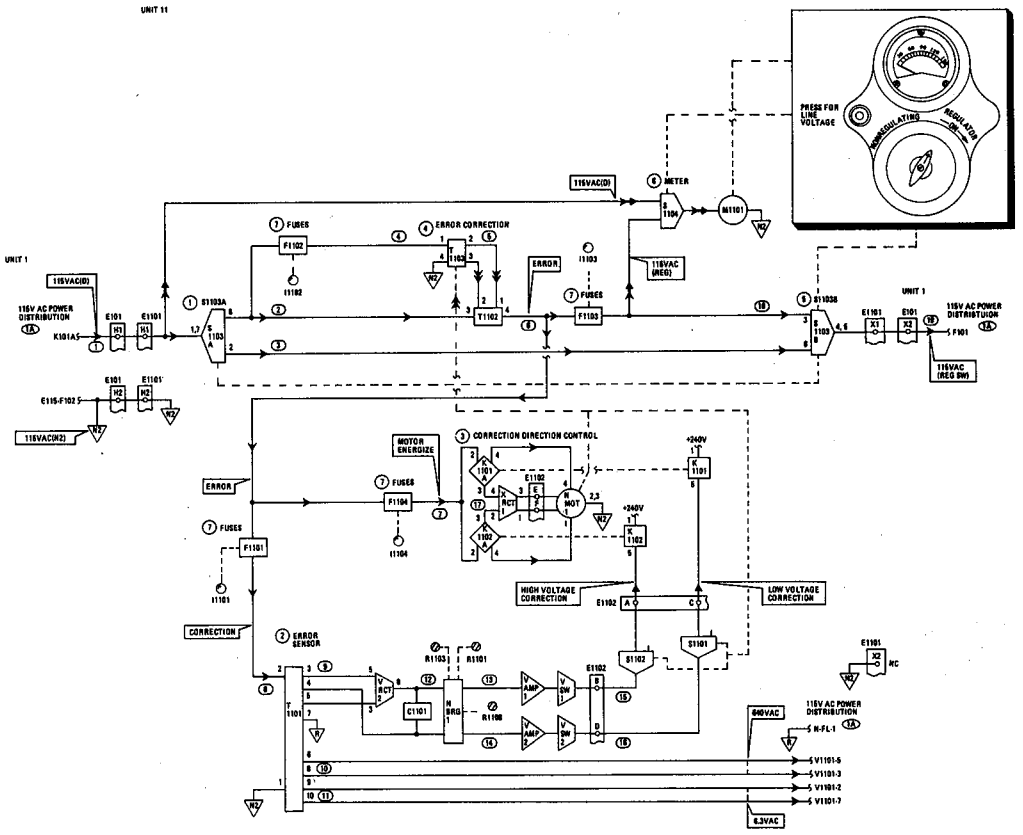
Same as S1103A except contacts 4, 5 to 6 are made.

(6) METER

Enables visual monitoring of either nonregulated source voltage or the regulated output voltage of the unit as selected by S1104.

(7) FUSES

Provide overcurrent protection and blown fuse indication.



ORIGINAL

(1AA) AC VOLTAGE REGULATOR SUB-FUNCTION 2-1-75

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 24. Subfunction diagram (sheet 2 of 2)

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MAJOR FUNCTION (1) INFORMATION

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-66a) FRAME ID NO.

NAVSEA SE211-FO-MMA-040/SPS-10 (2-1-66b) FRAME ID NO.

RADAR SET AN/SPS-10( )

POWER DISTRIBUTION (1)

Depicts the overall AC and DC power distribution of the AN/SPS-10 Radar Set. Ships Power, 115VAC, 60HZ is distributed and stepped up or down as necessary for use throughout the system. An auto-transformer steps up the 115VAC input to provide the 119V, 123V, 127V, and 135V AC Power Distribution. 440VAC is utilized in systems having the AS-936( )/SPS-10 Antenna Assembly incorporated. The Power Supply (Unit 2) generates regulated and unregulated DC voltages required for proper operation of various units throughout the Radar Set. However, many assemblies have self-contained DC Power Supplies.

(1) 115VAC POWER DISTRIBUTION

Refer to page 2-1-69 ( ) for text.

(2) CONTROL FUNCTION

Refer to page 2-1-103 ( ) for text.

(3) SYNCHRONIZE FUNCTION

Refer to page 2-1-128 ( ) for text.

(4) 119V, 123V, 127V, AND 135V AC POWER DISTRIBUTION

An auto-transformer steps up the 115VAC(REG) input to provide Thyratron filament pre-heat voltage, Magnetron selectable filament voltage and 119VAC. The 119VAC powers various unit blower motors, the Control Box panel lights, equipment status lights, the Transmitter waveguide shutter and various unit DC power supplies. Time delays are provided so that filaments can be properly heated before high voltage is applied to vacuum tube plates. Fuses protect the equipment from damage resulting from short circuits or other over-current conditions. Various individual DC power supplies are powered by the 119VAC.

(5) INTERLOCK AND OVERTEMPERATURE FUNCTION

Utilizes 115VAC(C) for the Interlock and overtemperature circuitry. Interlocks are provided in the Power Supply Unit, the Receiver-Transmitter Unit, and the high voltage compartment of the Modulator to prevent energizing the equipment should their respective doors be open. When all the doors are closed the circuit is complete and the Radar Set can be energized.

Overtemp Indicators illuminate if operating temperatures in the Receiver-Transmitter, Power Supply, or Modulator exceed 70 degrees C. Additionally, if the blower in the Receiver-Transmitter fails, the RT HIGH TEMP DANGER light will illuminate.

(6) 3A1 INTERFERENCE BLANKER DC POWER DISTRIBUTION

Utilizes 119VAC(A) to provide DC operating voltages within the 3A1 Assembly. The 3A1 provides +12V for application to Unit 5 (Radar Set Control) for use with the SCR-DISABLE switch (see Flag Note 2).

(7) UNITS 2, 5, AND 17 DC POWER DISTRIBUTION

Unit 2 (Power Supply) generates the regulated and unregulated DC voltages required for proper operation of various units throughout the Radar Set. Unit 5 (Radar Set Control) distributes DC voltages necessary for various control and adjustment functions of the Radar Set. Unit 17 distributes +24V, necessary for External Sync Control if AN/SPS-10 Field Change No. 16 is incorporated.

(8) 1A5 DC POWER DISTRIBUTION

Provides regulated +5V, +15V, +55V, -15V, and -170V for distribution within the 1A5 (Pre-Trigger Generator FC-25) assembly (see Flag Note 1).

(9) UNITS 3, 9, 11, AND 13 DC POWER DISTRIBUTION

Unit 3 (Receiver-Transmitter) utilizes 119VAC(A); stepped up by a transformer to provide the input to the keep alive circuitry; and stepped down to provide various filament voltages, and an input to a negative bias supply. Various DC operating voltages, from Unit 2 (Power Supply), are filtered and distributed throughout Unit 3. Unit 9 (Indicator Adapter) and Unit 11 (Voltage Regulator) are powered by self-contained power supplies. Unit 13 (Standing Wave Ratio Indicator) utilizes +270V REG from Unit 3 for operating voltage.

(10) UNITS 1 AND 23 DC POWER DISTRIBUTION

Unit 1 utilizes +24VDC, from Unit 2, for distribution in the 1A3 (High Voltage Switch) assembly. +105VDC, from Unit 2, is applied to the 1A4 (Pre-Trigger Generator, FC-16) assembly when AN/SPS-10 Field Change No. 16 is incorporated. 1A1 (Trigger Pulse Generator) and 1A2 (Thyratron Assembly) are powered by self-contained power supplies. Unit 23 (Pre-Trigger Generator FC-15) is also powered by a self contained power supply.

(11) BEARING FUNCTION

Utilizes 115VAC(F) to provide the RELATIVE WARNING signal for external equipment. 115VAC(I) is excitation voltage for the Bearing Synchro. 115VAC(H) is power for the Antenna Drive Motor if Unit 18 or Unit 22 is installed. 440VAC is power for the Antenna Drive Motor if Unit 19, 20, or 21 is installed. 115VAC(N2) is the other side of the various 115VAC voltages applied.

(12) MODULATOR FUNCTION

Utilizes 119VAC(B) as the control voltage for varying the Pulse Repetition Rate (PRR) via the PRR switch. 115VAC(RAD) enables the high voltage circuitry during RADIATE operation. 115VAC(N2) is the other side of the AC lines.

(13) RECEIVE FUNCTION

Utilizes 119VAC(B) as power for the Echo Box Tune Motor, as controlled by the ECHO BOX switch. A Shutter closes the waveguide entrance to the Receiver when power is removed from the system.

(14) COMBINED VIDEO FUNCTION

Utilizes 119VAC to enable the video blanking circuitry, when AN/SPS-10 Field Change No. 22, 30, or 31 is incorporated. 115VAC(N2) is the other side of the AC line.

(15) TRANSMIT FUNCTION

Utilizes the MAG FILAMENT (123/127VAC as applicable) to heat the Magnetron filaments. 115VAC(N2) is the other side of the AC line.

(16) 4A3A1, 4A3A2, 4A3A3, AND 4A4 DC POWER DISTRIBUTION

Produces +13V, -13V, +5V regulated, and +55V unregulated voltages, and distributes them within 4A3A1, 4A3A2, 4A3A3, and 4A4. Additionally, distributes these voltages to the 4A1, 4A2, 4A3A4, 4A3A5, 4A3A6 DC Power Distribution (see Flag Note 3).

(17) 4A1, 4A2, 4A3A4, 4A3A5, AND 4A3A6 DC POWER DISTRIBUTION

Filters and distributes +13V, +5V, and -13V regulated DC within 4A1, 4A2, 4A3A4, 4A3A5, and 4A3A6 (see Flag Note 3).

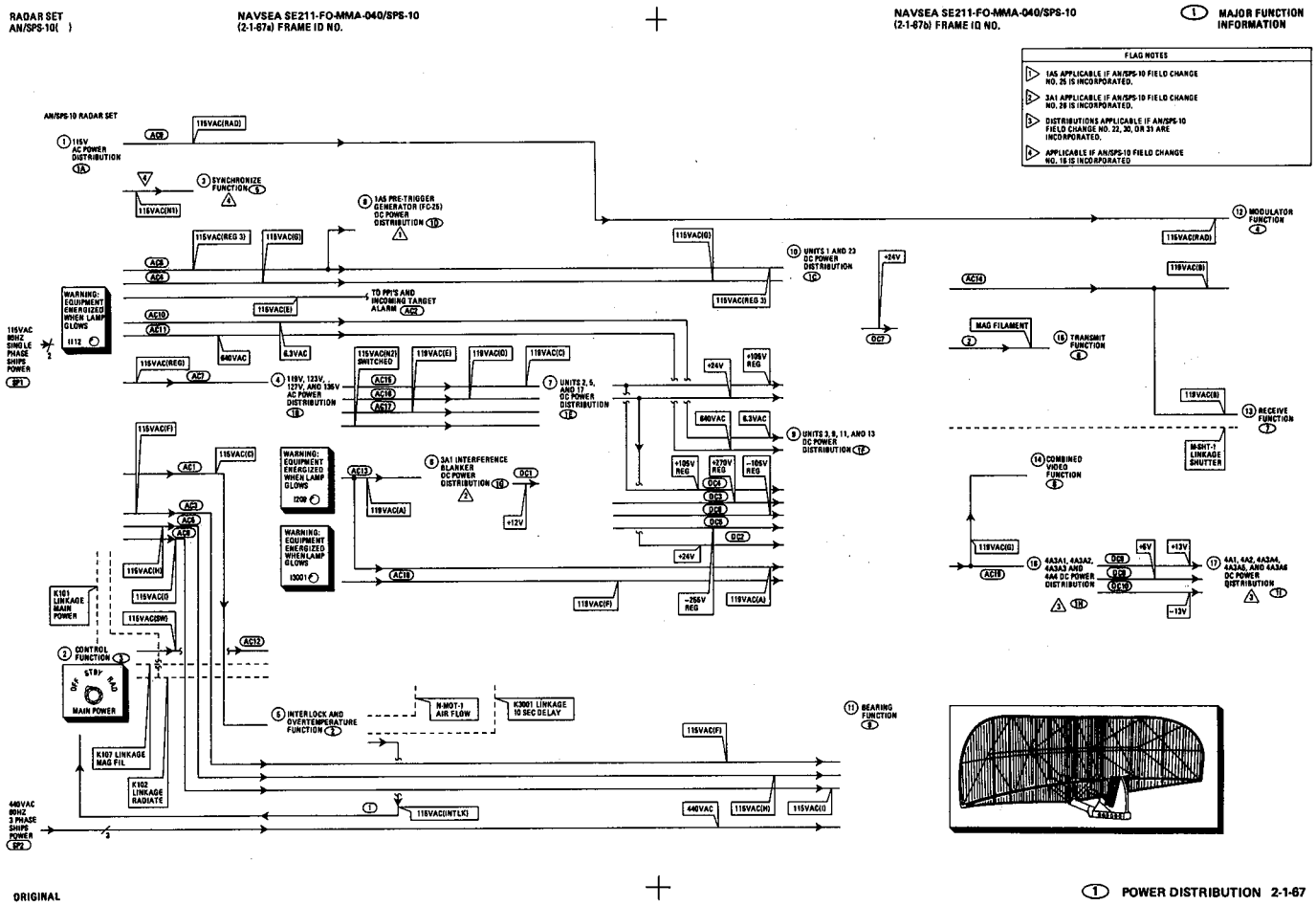
2-1-66 POWER DISTRIBUTION (1)

ORIGINAL

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 25. Overall power distribution diagram (sheet 1 of 2)

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Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 25. Overall power distribution diagram (sheet 2 of 2)

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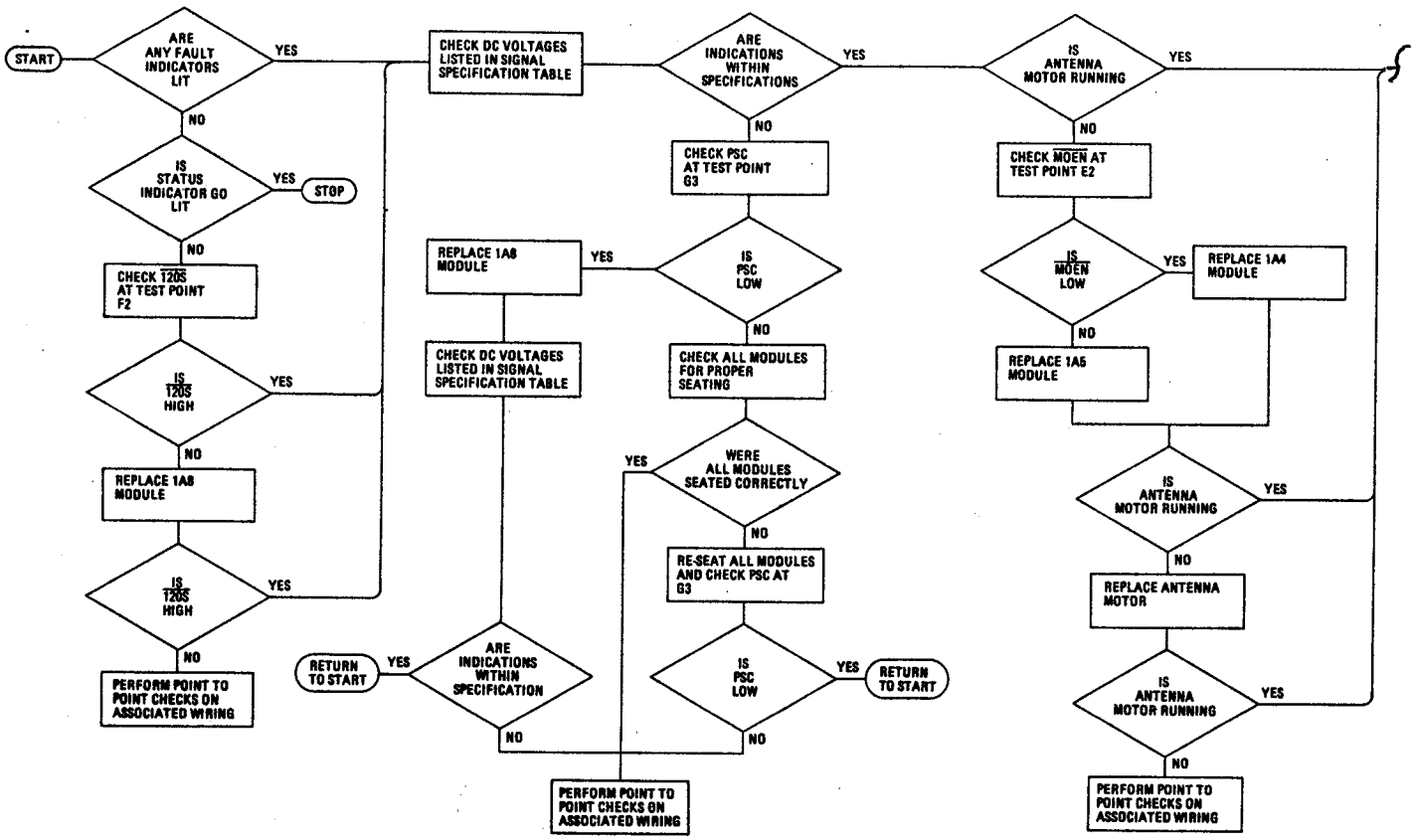


FIGURE 26. Fault logic diagram (sheet 1 of 2)

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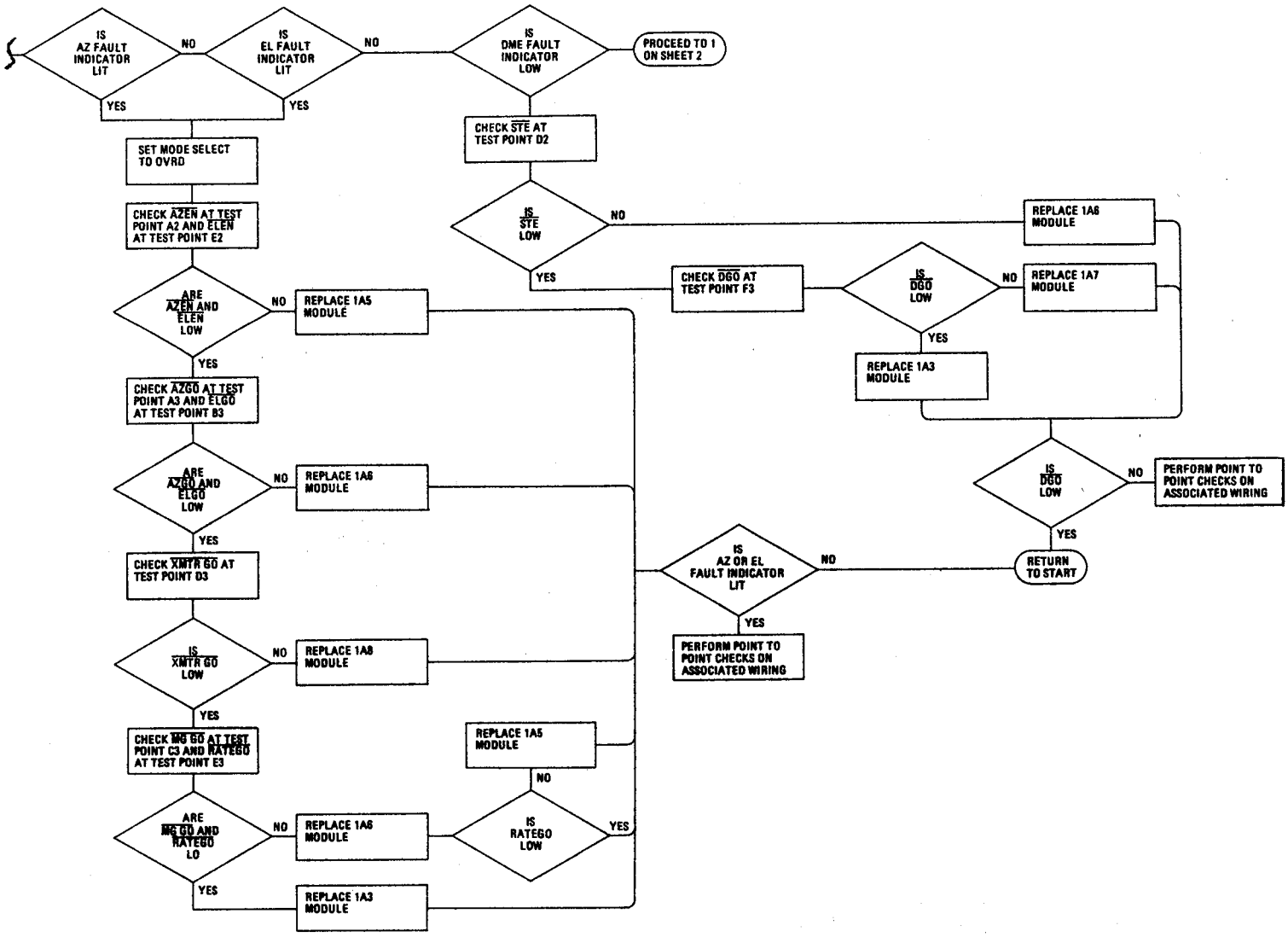


FIGURE 26. Fault logic diagram (sheet 2 of 2)

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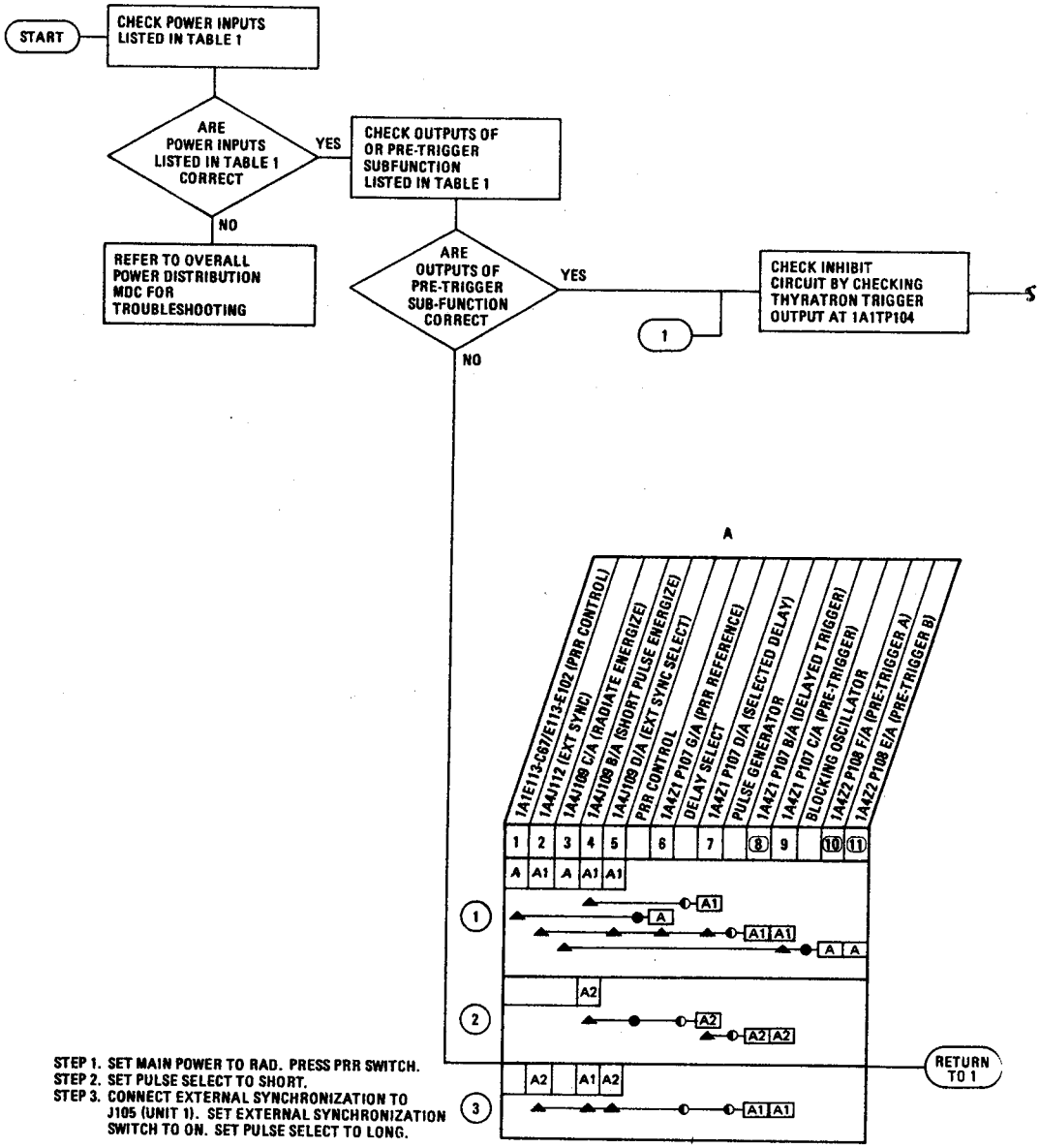


FIGURE 27. Combine fault logic and MDC diagram (sheet 1 of 2)

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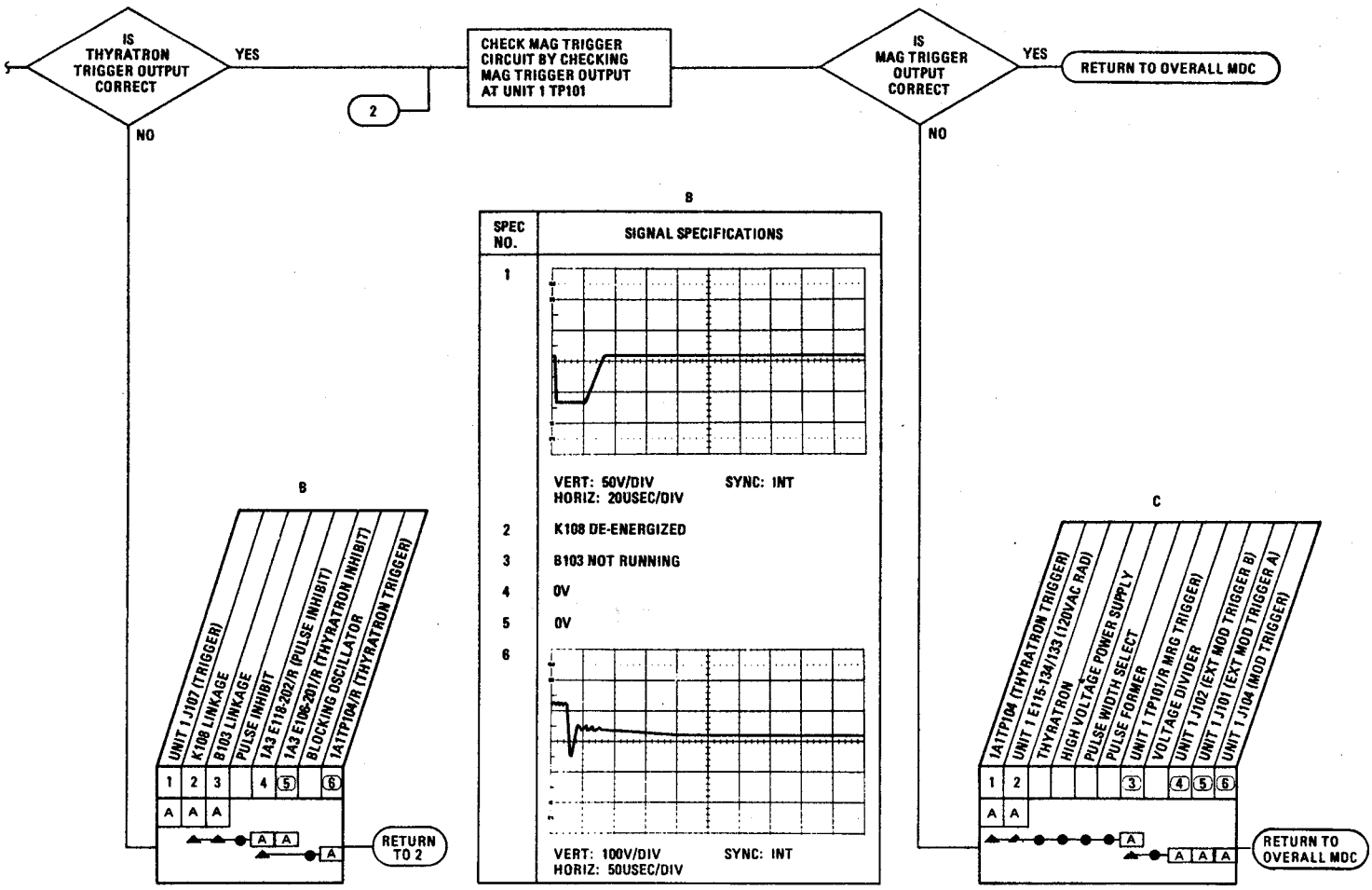
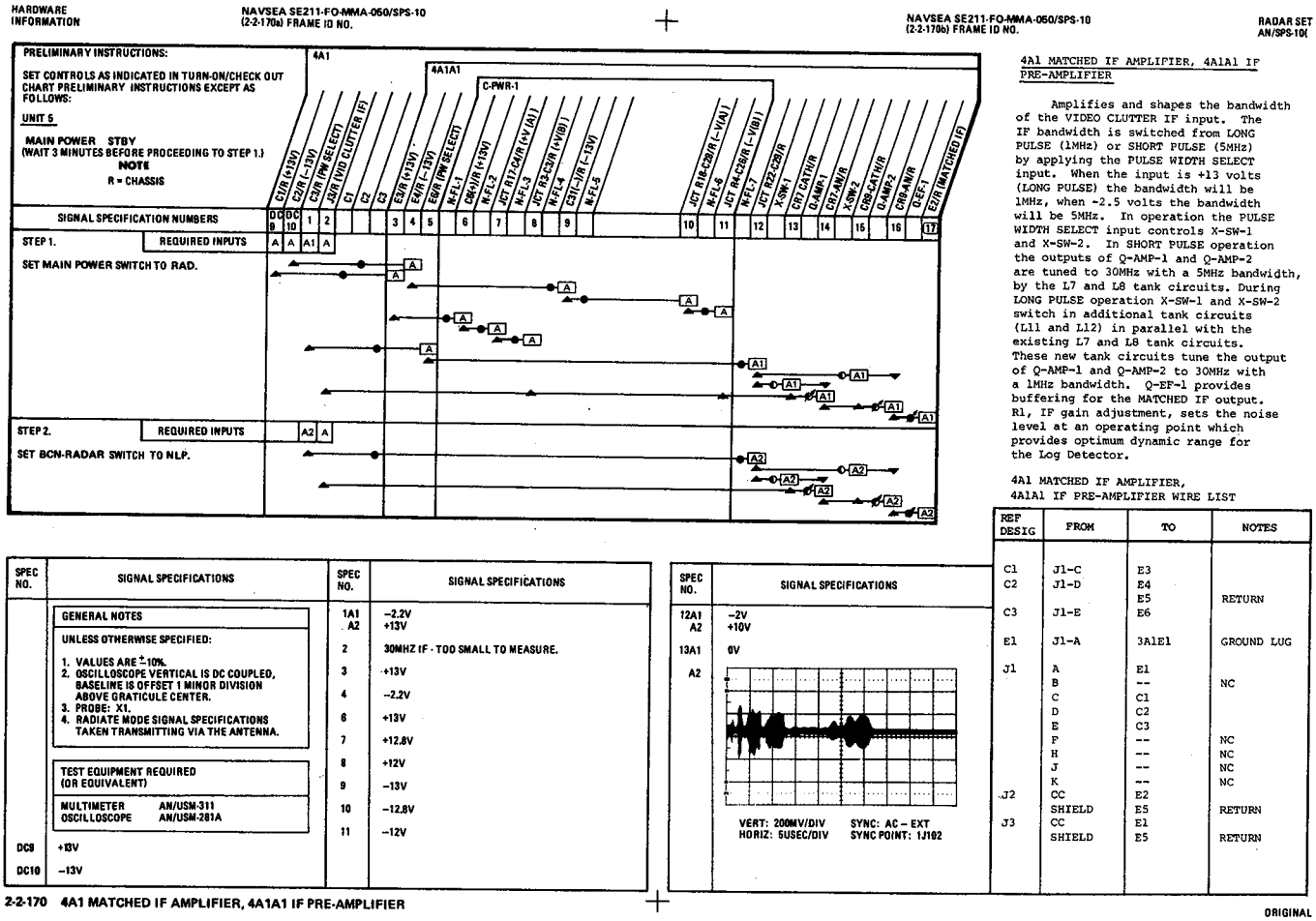


FIGURE 27. Combine fault logic and MDC diagram (sheet 2 of 2)

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Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 28. Hardware information package (sheet 1 of 4)

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RADAR SET AN/SPS-10( )

NAVSEA SE211-FO-MMA-050/SPS-10 (2-2-171a) FRAME ID NO.

NAVSEA SE211-FO-MMA-050/SPS-10 (2-2-171b) FRAME ID NO.

HARDWARE INFORMATION

(1) C-PWR-1 POWER

Distributes and filters the +13V and -13V power.

(2) C1, C2, C3

Feed through capacitors provide filtering for their respective inputs.

(3) N-FL-7 FILTER

Provides filtering for the PULSE WIDTH SELECT input.

(4) Q-AMP-1 AMPLIFIER

Amplifies the VIDEO CLUTTER IF input. The tank circuit (L7 and C1) connected to the collector of Q2 is tuned to 30MHz. In Short Pulse mode of operation the L7 tank circuit provides a 5MHz bandwidth for Q-AMP-1. In Long Pulse mode of operation X-SW-1 switches in an additional tank circuit (L11 and C22). The additional tank circuit decreases the band width to 1MHz.

(5) X-SW-1 SWITCH

Switches an additional tank circuit (L11 and C22) into Q-AMP-1 to decrease the bandwidth to 1MHz when in the Long Pulse mode of operation. When in the Short Pulse mode of operation disables the L11 tank circuit in Q-AMP-1.

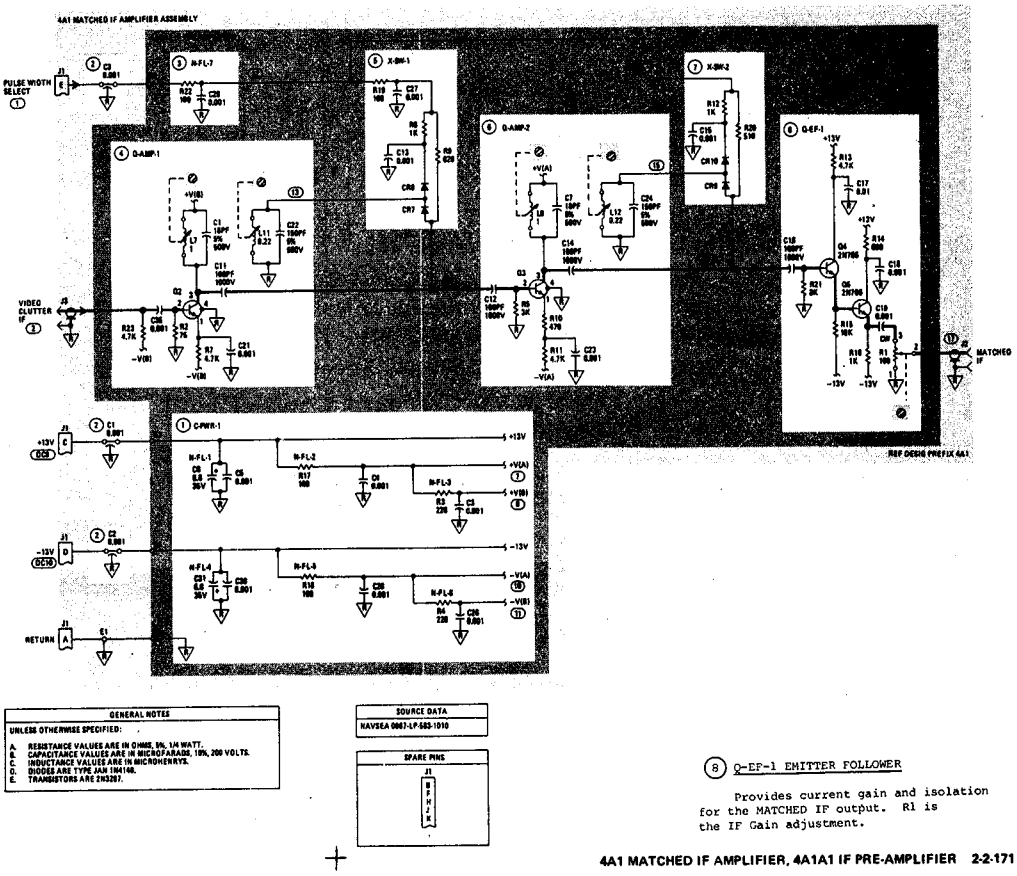
(6) Q-AMP-2 AMPLIFIER

Amplifies the input from Q-AMP-1. The tank circuit (L8 and C7) connected to the collector of Q2 is tuned to 30MHz. In Short Pulse mode of operation the L8 tank circuit provides a 5MHz bandwidth for Q-AMP-2. In Long Pulse mode of operation X-SW-2 switches in an additional tank circuit (L12 and C24). The additional tank circuit decreases the bandwidth to 1MHz.

(7) X-SW-2 SWITCH

Switches an additional tank circuit (L12 and C24) into Q-AMP-2 to decrease the bandwidth to 1MHz when in Long Pulse mode of operation. When in the Short Pulse mode of operation disables the L12 tank circuit in Q-AMP-2.

ORIGINAL



(8) Q-EP-1 EMITTER FOLLOWER

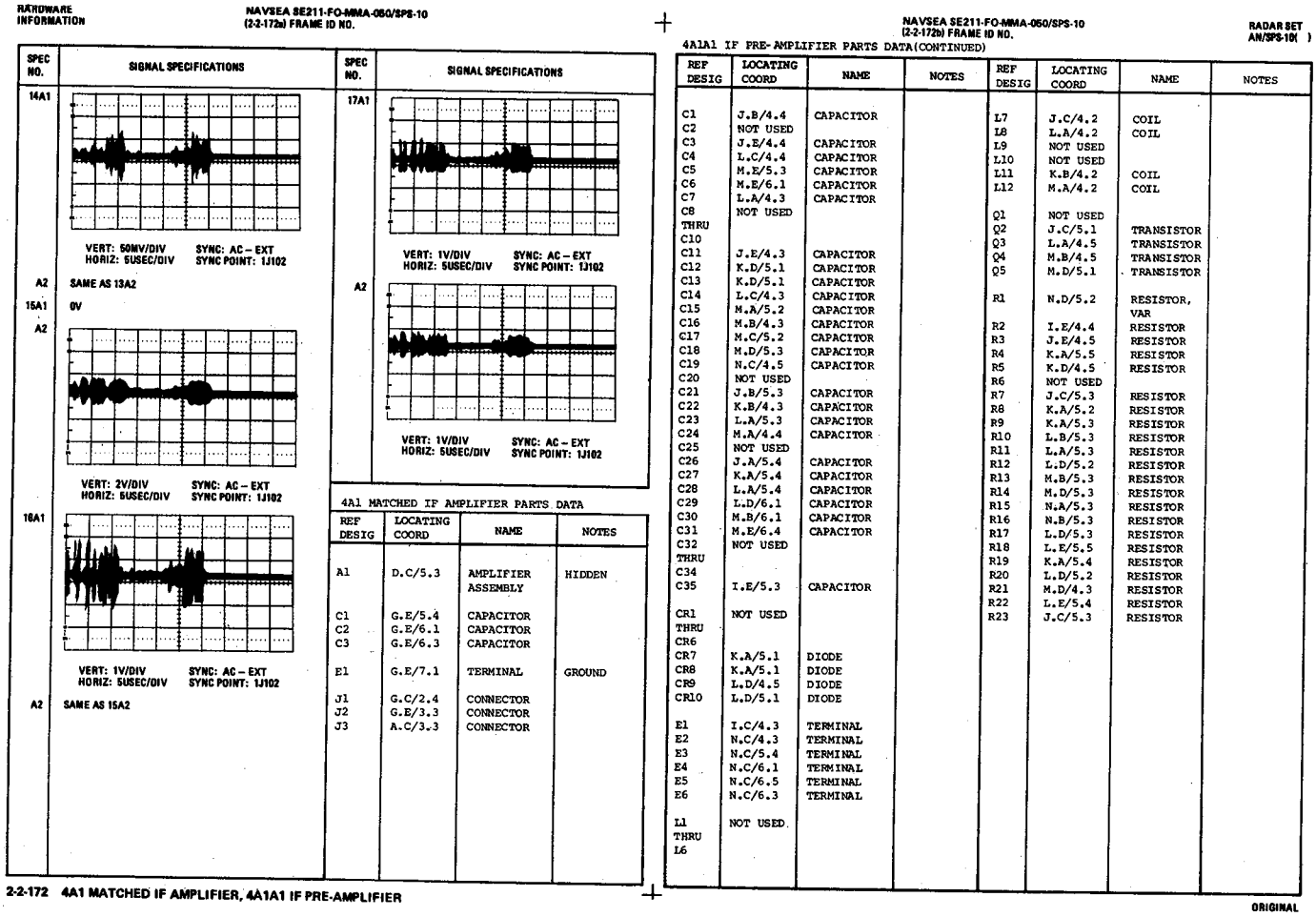
Provides current gain and isolation for the MATCHED IF output. R1 is the IF Gain adjustment.

4A1 MATCHED IF AMPLIFIER, 4A1A1 IF PRE-AMPLIFIER 2-2-171

Sample arrangement only. Type size does not conform to minimum specification requirements.

FIGURE 28. Hardware information package (sheet 2 of 4)

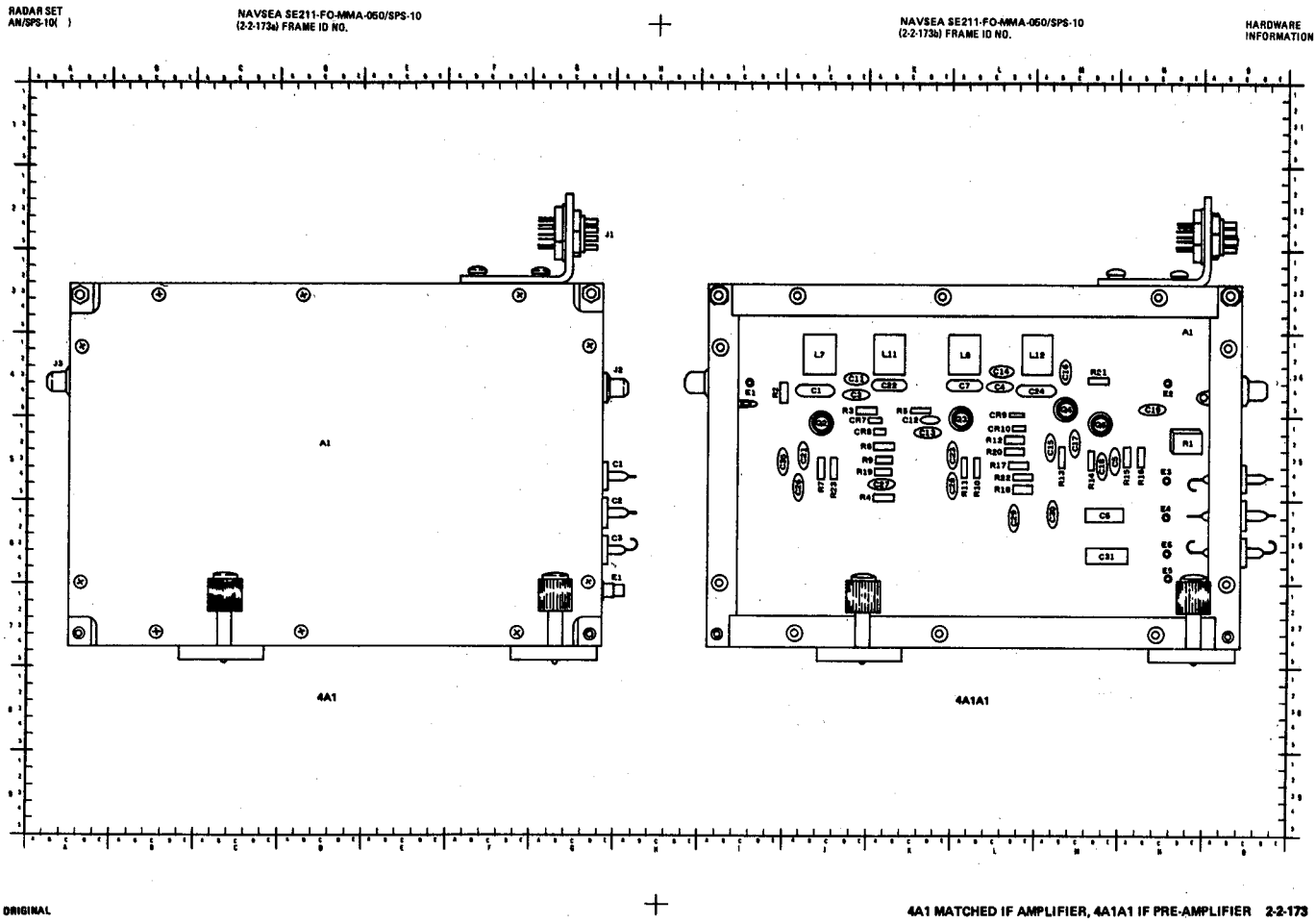
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FIGURE 28. Hardware information package (sheet 3 of 4)

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FIGURE 28. Hardware information package (sheet 4 of 4)

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HARDWARE INFORMATION — NAVSEA SE211-FO-MMA-050/SPS-10 (2-3-208a) FRAME ID NO. — NAVSEA SE211-FO-MMA-050/SPS-10 (2-3-208a) FRAME ID NO. — RADAR SET AN/SPS-10( )

UNIT 18 ANTENNA ASSEMBLY PARTS DATA (CONTINUED)

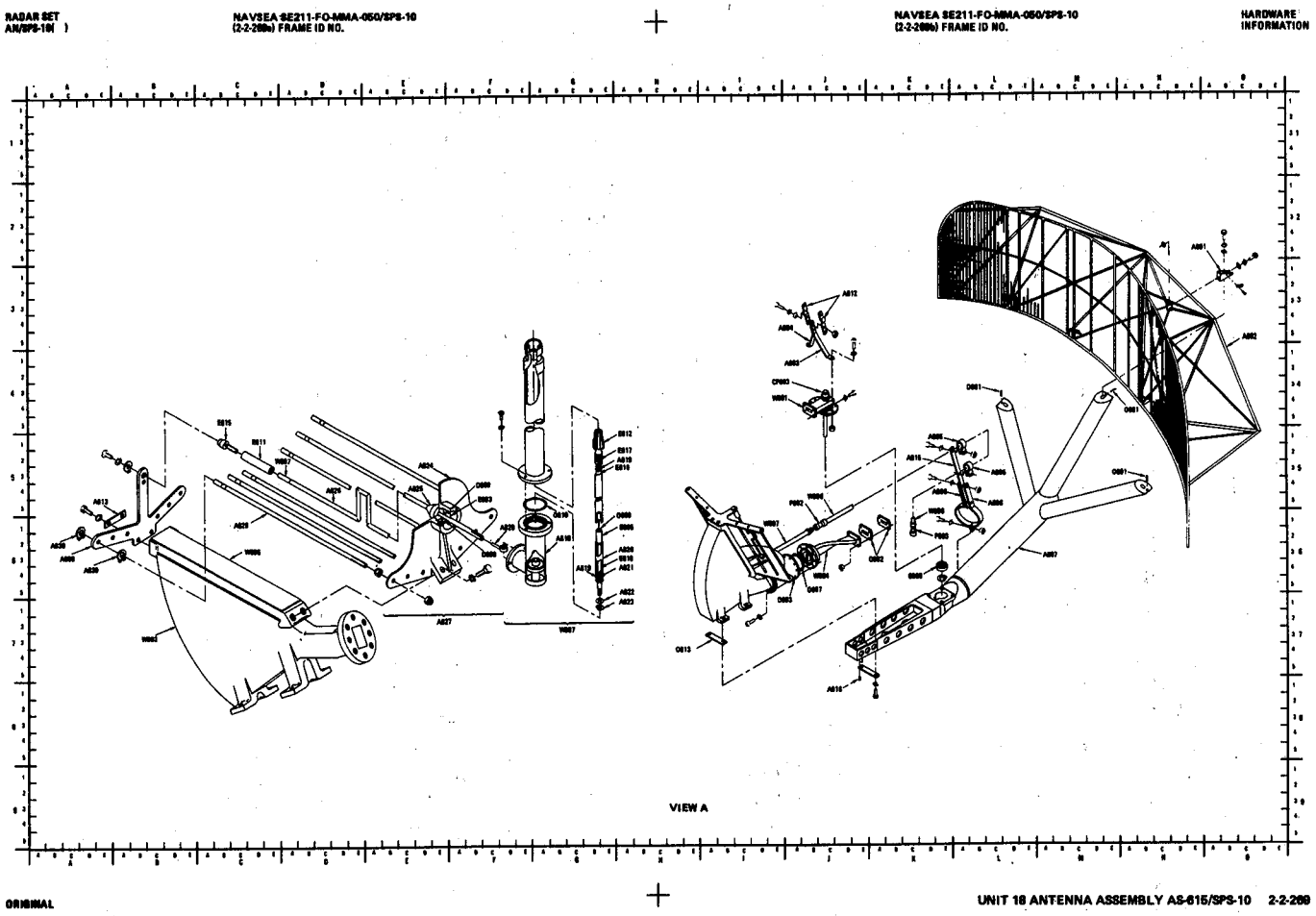
REF DESIG	VIEW	LOCATING COORD	NAME
MP820	B	K.A/5.5	NIPPLE
MP821	B	I.D/8.1	RETAINER, "O" RING
MP822	B	G.B/5.5	EYE, LIFTING
MP823	B	G.B/6.3	SHAFT, DRIVE MOTOR
MP824	B	H.A/7.1	STAND-OFF
MP825	B	H.A/7.3	MOUNTING STRIP
MP826	B	G.B/7.2	TERMINAL STRIP
MP827	B	F.E/7.2	MARKER STRIP
MP828	B	E.C/7.4	BRACKET, MOTOR MOUNT
MP829	B	E.B/7.5	PLATE, MOUNTING
MP830	B	D.E/8.1	BELL, END
MP831	B	D.D/8.3	COVER, ACCESS
MP832	B	D.C/8.2	SEAT, BEARING
MP833	B	D.A/8.3	STATOR
MP834	B	D.D/7.4	COVER, ACCESS
MP835	B	F.B/7.1	KEY
MP836	B	D.E/5.5	COVER, ACCESS
MP837	B	D.E/6.1	BELL, END
MP838	B	E.A/6.2	SEAT, BEARING
MP839	B	E.B/6.1	BEARING, BALL
MP840	B	E.E/5.4	ARMATURE
MP841	B	F.C/5.2	BEARING, BALL
MP842	B	D.E/6.4	COVER, ACCESS
MP843	B	D.A/6.5	FAN
MP844	B	F.C/2.3	COVER
MP845	B	D.C/2.2	KEY
MP846	B	D.C/2.3	SPIDER, KEY
MP847	B	D.B/4.1	WASHER, LOCK
MP848	B	D.B/5.3	WASHER, LOCK
MP849	B	I.A/6.5	CLOSE, NIPPLE
MP850	B	I.A/6.5	CAP PLUG
MP851	B	I.E/2.2	KEY
0801	A	L.C/4.3	PIN, DOWEL
		M.E/4.3	
		N.B/5.4	
0802	A	K.A/6.3	GASKET
0803	A	J.A/6.4	GASKET
0804	A	J.C/4.5(H)	GASKET
0805	B	D.A/1.4	KEY, MACHINE
0806	A	K.E/6.4	NUT, PLAIN, ROUND
0807	A	J.A/6.3	PACKING
0808	A	E.E/6.1	GASKET
0809	A	F.A/5.5	PACKING
		G.E/6.1	
0810	A	G.A/5.5	PACKING
0811	A	J.C/4.5(H)	GASKET
0812	A	J.C/4.5(H)	PACKING
0813	A	I.A/7.3	SHIM ASSEMBLY
0814	A	J.C/4.5(H)	SHIM ASSEMBLY
0815	B	I.D/8.1	HOUSING
0816	B	I.C/8.2(H)	BEARING, BALL
0817	B	I.C/8.5	BEARING, BALL
0818	B	I.C/8.5	PACKING
0819	B	I.C/8.5	SHIM ASSEMBLY
0820	B	I.C/8.5	ADAPTER, BEARING
			SUBASSEMBLY
0821	A	J.C/4.3(H)	GASKET
0822	A	J.C/4.3(H)	PACKING
0823	B	D.B/3.1	BEARING, ROLLER
0824	B	D.B/5.2	BEARING, ROLLER
0825	B	I.E/2.3	PACKING
		I.E/3.4	
0826	B	K.B/3.5	BEARING, BALL
		L.A/3.3	
0827	B	F.E/6.5	BEARING, BA
		G.E/6.1	
0828	B	H.C/7.3	SWITCH DRIVE
			SUBASSEMBLY
0829	B	M.A/2.2	FITTING, TUBING
0830	B	M.C/6.3	COUPLING, FLEXIBLE
		N.E/5.4	
0831	B	L.D/3.1	COUPLING, FLEXIBLE
0832	B	F.B/7.4	COUPLING, FLEXIBLE
		F.A/7.2	
0833	B	I.E/3.4	GASKET
0834	B	F.C/6.5	GASKET
0835		--	WASHER
0836		--	GASKET
0837	B	M.D/1.5	GASKET
0838	B	K.C/6.3	GASKET
0839	B	F.D/2.3	GASKET
0840	B	B.C/7.2	GASKET
0841	B	D.B/3.4	GEAR
		I.B/4.5	
0842	B	D.B/4.4	GEAR
		I.C/5.1	
0843	B	J.D/4.1	GEAR
0844	B	G.A/6.4	GEAR
0845	B	J.B/4.3	NUT, PLAIN, ROUND
0846	B	D.B/5.4	NUT, PLAIN, ROUND
0847	B	D.B/5.1	NUT, PLAIN, ROUND
0848	B	D.B/4.2	NUT, PLAIN, ROUND
0849	B	H.C/7.4	NUT, PLAIN, HEXAGON
0850	B	M.C/2.4	PUMP, LIQUID ROTARY
0851	B	I.C/7.3	PACKING
0852	B	I.E/2.2	PACKING
0853	B	I.C/7.4	PACKING
0854	B	H.C/8.1	PACKING
0855	B	I.B/2.5	PACKING
0856	B	I.B/3.3	PACKING
		I.B/3.2	
0857	B	I.E/3.5	SEAL, OIL
0858	B	F.D/6.5	SEAL, OIL
0859	B	D.B/2.2	SHAFT
		I.C/5.4	
0860	B	I.E/2.5	SHAFT
0861	B	G.B/6.3	SHAFT
0862	B	L.B/3.2	SHIM ASSEMBLY
0863	B	K.A/4.1	SHIM ASSEMBLY
0864	B	H.A/6.1	SHIM ASSEMBLY
0865	B	F.D/6.5	SHIM ASSEMBLY
0866	B	L.B/3.2	SHIM
0867	B	L.B/3.2	SHIM
0868	B	H.A/6.1	SHIM
0869	B	D.B/4.3	SHIM
0870	B	D.B/4.3	SHIM
0871	B	M.C/6.4	BUSHING
		N.E/5.5	
0872		--	NOT USED
0873		--	HOLDER, WRENCH
0874	B	H.E/4.2	VALVE, CHECK
		M.B/2.5	
0875		--	WASHER
0876	B	D.B/4.5	WASHER, LOCK
0877		--	NOT USED
0878	B	H.C/7.5	BUSHING
0879	B	J.A/3.2	GEAR ASSEMBLY
0880	B	J.E/5.2	STRAINER
0881	B	I.E/4.2	COUPLING, FLEXIBLE
0882	B	M.C/5.1	PANL
0883	B	K.C/3.5	GEAR
0884	B	M.C/6.2	BEARING, BALL
		N.A/5.5	
		N.A/6.4	
		N.E/4.5	
		N.E/5.3	
0885	B	M.C/5.2	BEARING, BALL
0886	B	M.B/6.2	PIN, GROOVED
0887	B	N.E/5.2	GEAR
0888	B	N.A/6.3	GEAR
0889	B	M.C/5.3	SHAFT
0890	B	M.C/6.1	GEAR ASSEMBLY
0891	B	N.B/6.1	GEAR ASSEMBLY
0892	B	L.C/5.2	SYNCHRO SYSTEM SUB-ASSEMBLY
0893	B	M.C/5.5	CAMSHAFT ASSEMBLY
0894	B	K.E/5.4	MOUNTING
0895	B	F.E/6.4	WASHER, THRUST
0896	B	L.A/3.3	WASHER, THRUST
0897	B	F.C/6.5	RETAINER, OIL SEAL
0898	B	H.B/5.5	RETAINER, BEARING
0899		--	GEAR ASSEMBLY, SPEED DECREASER
PB01		--	NOT USED
PB02	A	J.C/6.1	CONNECTOR, PLUG
PB03	A	K.C/6.2	CONNECTOR, PLUG
PB04		--	NOT USED
PB05		--	CONNECTOR, PLUG
S801	B	G.B/5.1	SWITCH, THERMOSTATIC, SPST
S802	B	H.D/7.2	SWITCH, TOGGLE, DPST
S803	B	K.D/5.4	SWITCH, SENSITIVE, SPDT
S804	B	K.D/5.5	SWITCH, SENSITIVE, SPDT
W801	A	J.B/4.4	UPPER ROTARY JOINT ASSEMBLY
W802	B	I.B/8.3	LOWER ROTARY JOINT ASSEMBLY
W803	A	B.E/7.2	HORN, ANTENNA
W804	A	J.D/6.2	WAVEGUIDE ASSEMBLY
W805	A	C.C/6.4	COVER, ANTENNA, HORN
W806	A	J.C/6.1	CABLE ASSEMBLY, RADIO FREQUENCY
W807	A	G.C/7.3	ANTENNA SUBASSEMBLY
W808		--	LINE SECTION, RADIO FREQUENCY TRANSMISSION
W809		--	ANTENNA SUBASSEMBLY
Z801		--	GEAR BOX SUBASSEMBLY

2-3-208 UNIT 18 ANTENNA ASSEMBLY AS-615/SPS-10 — ORIGINAL

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FIGURE 29. Repair illustration (sheet 1 of 2)

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FIGURE 29. Repair illustration (sheet 2 of 2)

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FROM	TO	NOTES	FROM	TO	NOTES
C1 +	S10-C8		S3-B2	S4-2	BLK
C1 -	S10-C6		S3-B2	S1-A4	
C2+	S10-C11		S3-B2	S5-COMM	
C2-	S10-C3		S3-B3	XA5-15	WHT
DS8-A	S10-C12	WHT/YEL	S4-1	XA7-25	WHT
DS8-B	S14-C2	WHT/RED	S4-2	S3-B2	BLK
J1-1	XA2-39		S4-2	XDS6-2	BLK
J1-2	GRD BUS	CABLE SHIELD	S5-A	XA7-65	
R1-1	GRD BUS	CABLE SHIELD	S5-B	XA7-59	
R1-2	XA2-41		S5-C	XA7-53	
R1-3	S10-B1		S5-D	XA7-41	
R2-1	NC	NOT USED	S5-COM	S3-B2	BLK
R2-2	TB1-A4	BLK	S5-COM	S6-COM	BLK
R2-3	TB1-A3	BLK	S5-0	NC	
R4-1	S14-A1	BLK	S6-A	XA7-31	
R4-1	GRD BUS	CABLE SHIELD	S6-B	XA7-19	
R4-2	XA2-60		S6-C	XA7-13	
R4-3	XA2-61	WHT	S6-D	XA7-5	
R5-1	GRD BUS	CABLE SHIELD	S6-COM	S5-COM	BLK
R5-2	XA2-54		S6-COM	S7-COM	BLK
R5-3	NC	NOT USED	S6-0	NC	
S1-A1	XA5-53		S7-A	XA6-65	
S1-A2	NC	NOT USED	S7-B	XA6-59	
S1-A3	NC	NOT USED	S7-C	XA6-53	
S1-A4	S3-B2		S7-D	XA6-41	
S1-B1	XDS2-1	ORN/WHY	S7-COM	S6-COM	BLK
S1-B2	TB1-A6	WHT	S7-COM	S8-COM	BLK
S1-B2	XA5-55	WHT	S7-0	NC	
S1-C1	NC	NOT USED	S8-A	XA6-31	
S1-C2	NC	NOT USED	S8-B	XA6-19	
S1-C3	NC	NOT USED	S8-C	XA6-13	
S1-C4	NC	NOT USED	S8-D	XA6-5	
S2-A1	TB1-A11	WHT/GRY	S8-COM	S7-COM	BLK
S2-A2	NC	NOT USED	S9-1	XA4-29	WHT
S2-A3	NC	NOT USED	S9-2	S8-COM	BLK
S2-A4	XF-1		S9-2	TB1-A6	BLK
S2-B1	TB1-A5	ORN/WHT	S10-A		NOT USED
S2-B2	S2-C3		S10-B1	S10-C2	
S2-B2	TB1-A3	BLK	S10-B1	R1-3	
S2-C1	TB1-A12	GRY	S10-B2	S10-B4	JUMPER
S2-C2	S11-A1		S10-B2	XA1-46	
S2-C3	S2-B2		S10-B3	S10-B5	JUMPER
S2-C4	XF2-1		S10-B3	XA2-62	
S3-A1	S3-A3	WHT/BRN	S10-B4	S10-B2	JUMPER
S3-A2	XA3-17	WHT/BRN	S10-B5	S10-B3	JUMPER
S3-A3	S3-A1	WHT/BRN	S10-B6	XA1-16	
S3-A3	S12-4	WHT/BRN	S10-B7	TB1-B6	BLK
S3-B1	XA5-14	WHT	S10-B8	XA4-19	WHT

FIGURE 30. Wire list

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Check the source to verify that this is the current version before use.

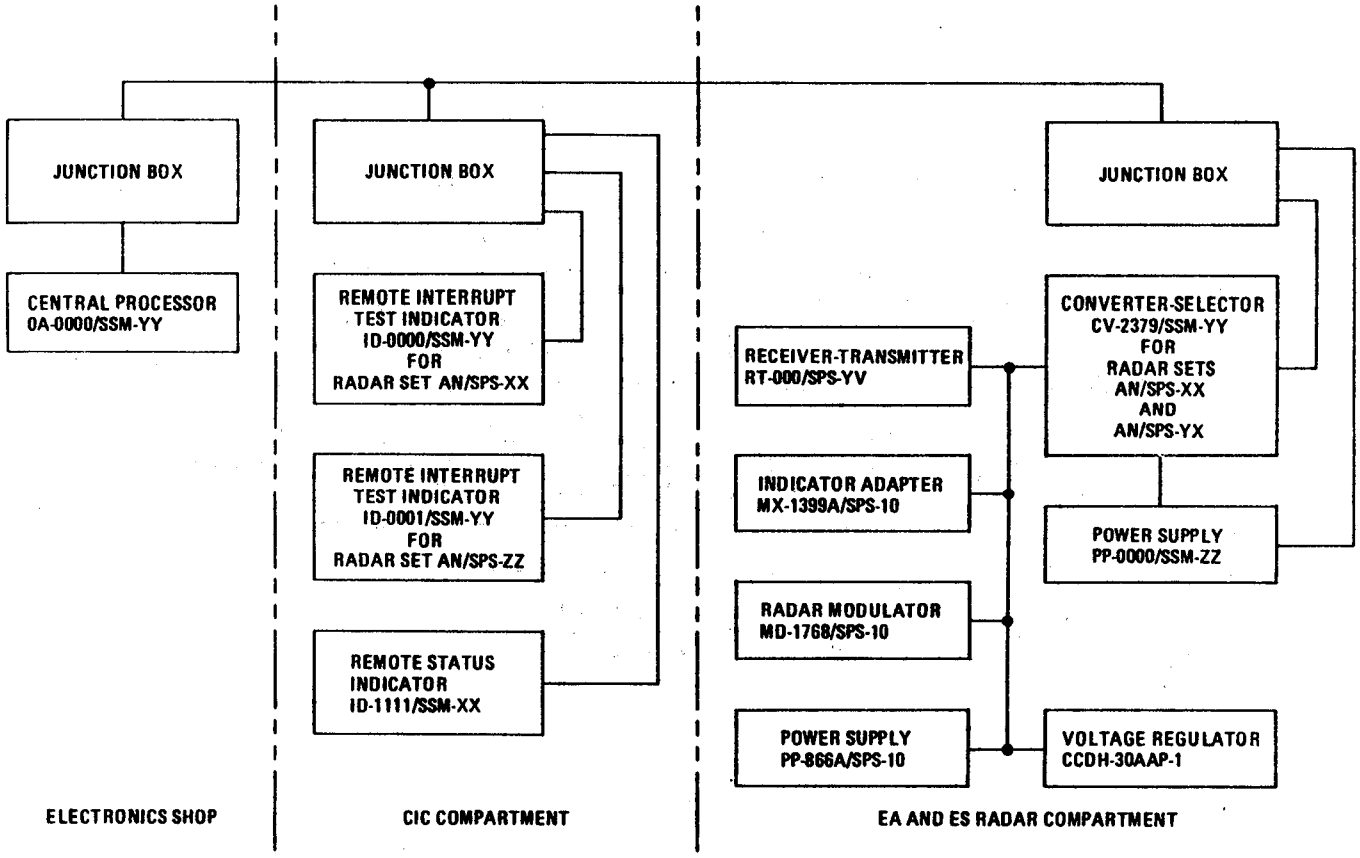


FIGURE 31. Interrelationship of system equipment

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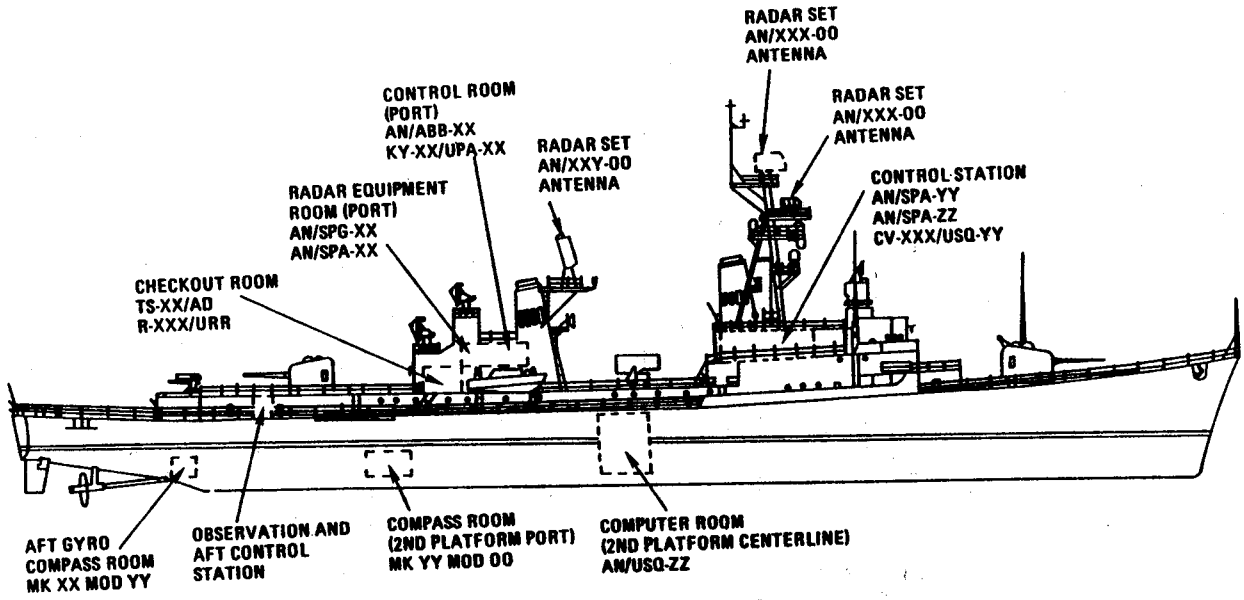


FIGURE 32 Outboard profile drawing

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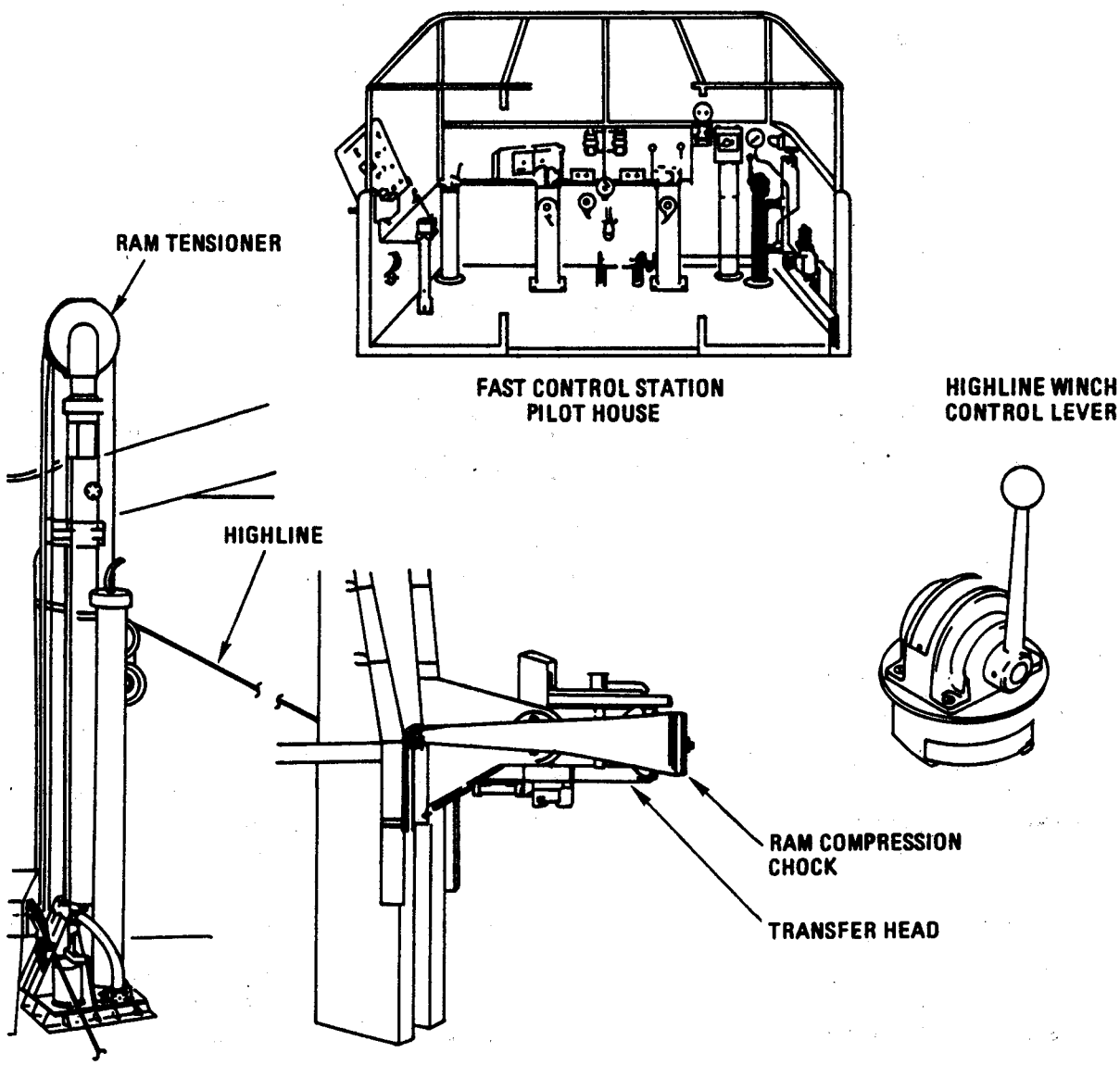


FIGURE 33 Compartment area drawing

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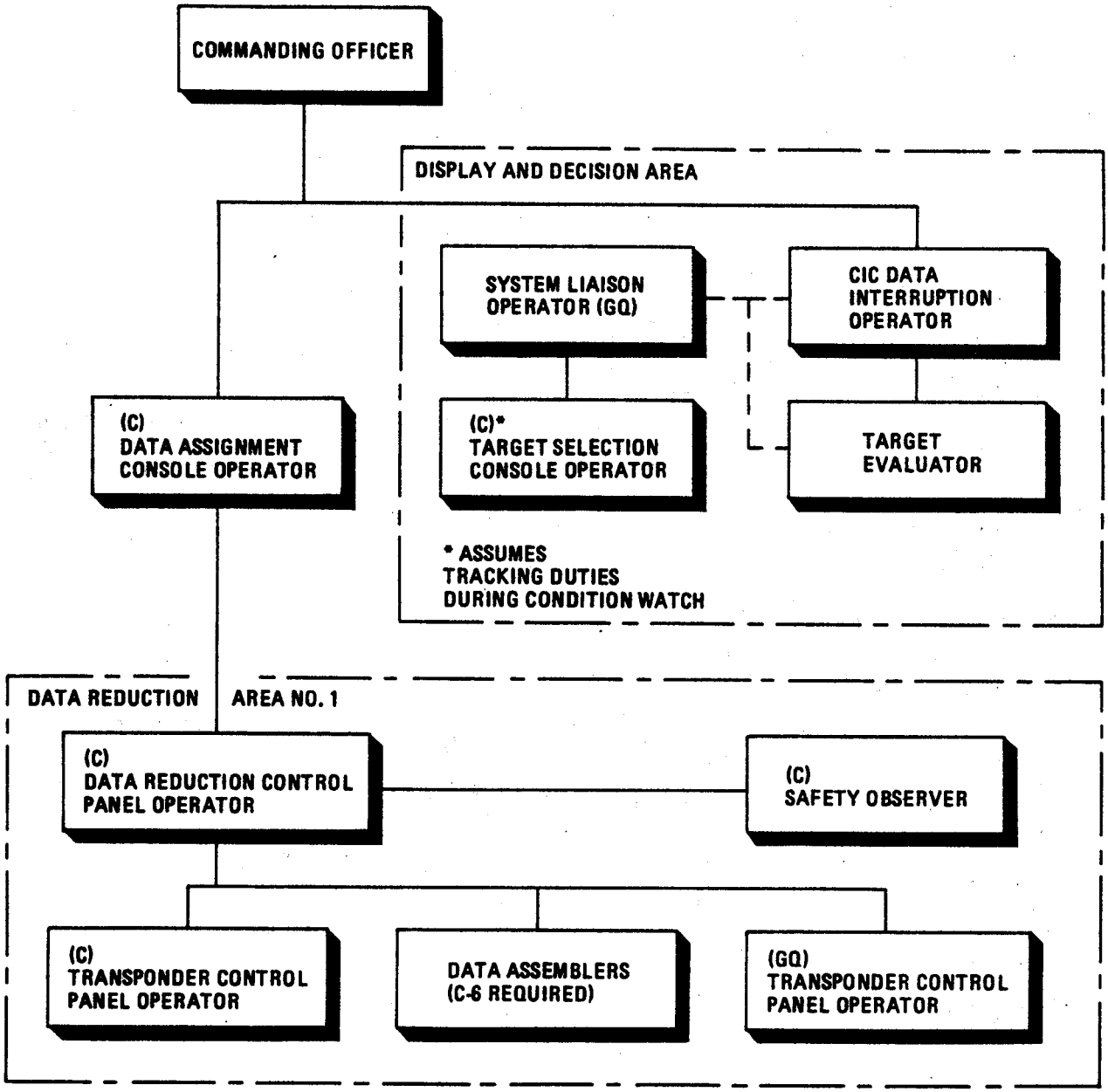


FIGURE 34 Conditions of readiness

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		SUPPORT VOLUME/ MANUAL									TROUBLESHOOTING/REPAIR VOLUME/MANUAL												
TYPE I EQUIPMENT PARAGRAPH NO.		3.2.1	3.2.1.1	3.2.1.2	3.2.1.3	3.2.1.4	———	3.2.1.5	3.2.1.6	3.2.1.3	3.2.1	3.2.2.1.3	3.2.2.1.4	3.2.2.1.5.1	3.2.2.1.5.3	3.2.2.3.1	3.2.2.3.3	3.2.2.3.4	3.2.2.3.5	3.2.3.3.1	3.2.3.3.3	3.2.3.3.4	3.2.3.3.5	3.2.3.3.6
TYPE II SYSTEM PARAGRAPH NO.		3.2.1	3.2.3.1.1	3.2.3.1.3	3.2.3.1.4	3.2.3.1.5	3.2.3.1.6	3.2.3.1.7	3.2.3.1.8	3.2.3.1.2	3.2.1	3.2.3.2.1.5	3.2.2.2.1.4	3.2.3.2.1.2	3.2.3.2.1.4	3.2.3.2.2	3.2.3.2.2.4	———	———	3.2.3.2.3.2	———	———	———	———
		FRONT MATTER	GENERAL INFORMATION	OPERATION	THEORY OF OPERATION	SCHEDULED MAINTENANCE	ALIGNMENT	INSTALLATION	PARTS LIST	SAFETY SUMMARY	FRONT MATTER	CABLE/PIPING DIAGRAM	TURN-ON/CHECKOUT CHART	OVERALL BLOCK DIAGRAM	OVERALL MAINTENANCE	MAJOR FUNCTION DIAGRAM	MAJOR FUNCTION MAINTENANCE	MAJOR FUNCTION REPAIR	MAJOR FUNCTION PARTS DATA	HARDWARE SCHEMATICS	HARDWARE MAINTENANCE	HARDWARE REPAIR	HARDWARE PARTS DATA	WIRING DATA
USING SERVICE	MAINTENANCE LEVEL																							
A, F, MC	OPERATOR	●	●	●		●				●														
A, F, MC	ORGANIZATIONAL	●			●	●	2/	2/	1/	●	●	●	●	●	●									
A, F, MC	INTERMEDIATE 3/										●		●			●	●	●	●					
GG, N	ORGANIZATIONAL	●	●	●	●	●	2/	2/	1/	●	●	●	●	●	●	●	●	●	●					
GG, N	INTERMEDIATE	●	●	●	●	●	2/	2/	1/	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

1/ INCLUDED ONLY WHEN SPECIFIED IN THE CONTRACT OR ORDER.

2/ INCLUDED UNLESS OTHERWISE SPECIFIED IN THE CONTRACT OR ORDER.

3/ INFORMATION PROCURED WITH ORGANIZATIONAL MAINTENANCE INFORMATION.

FIGURE 35 Maintenance levels

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CHECKLIST FOR
FOMM TECHNICAL MANUAL MANUSCRIPT

EQUIPMENT_____ SUBMITTAL DATE_____

NOMENCLATURE_____ SECURITY CLASSIFICATION_____

CONTRACT NO._____ PUBLICATION NO._____

CONTRACTOR_____ TYPE MANUAL_____

	YES	NO	NOT APPL
1. The manuscript consists of the following subdivisions in the order indicated:			
(a) Volume 1, Support Volume			
Cover/Title Page	( )	( )	
Front Matter	( )	( )	
Section 1, General Information	( )	( )	
Section 2, Operation	( )	( )	
Section 3, Theory of Operation	( )	( )	
Section 4, Scheduled Maintenance	( )	( )	( )
Section 5, Installation	( )	( )	( )
Section 6, Parts List	( )	( )	( )
(b) Volume 2, Troublshooting/Repair Volume.			
Cover/Title Page	( )	( )	
Front Matter	( )	( )	
Family Tree	( )	( )	
Interconnecting Cabling/Piping Diagram	( )	( )	( )
Overall Function Information	( )	( )	
Major Function Information	( )	( )	
Hardware Information	( )	( )	( )

2. The following is a detailed list of subdivision of Volume 1:,

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FIGURE 36 Certification check-off list (sheet 1 of 5)

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		YES	NO	NOT APPL
(a)	Front Matter			
	List of effective pages	( )	( )	
	Change Record	( )	( )	
	Table of Contents	( )	( )	
	List of Illustrations	( )	( )	
	List of Tables	( )	( )	
	Functional Index	( )	( )	
	Foreword	( )	( )	
	Safety Summary	( )	( )	
(b)	Section 1, General Information			
	Introduction	( )	( )	
	Equipment Description	( )	( )	
	Relationship of Units	( )	( )	( )
	Reference Data	( )	( )	
	Equipment, Accessories, and Documents Supplied	( )	( )	
	Equipment and Publications Required but not Supplied	( )	( )	
(c)	Section 2. Operation			
	Operators Data Table	( )	( )	
	Turn-On/Check-out Procedures	( )	( )	
	Operating Procedures	( )	( )	
	Operators Maintenance Actions and Schedule	( )	( )	( )
(d)	Section 3. Theory of Operation	( )	( )	
(e)	Section 4. Scheduled Maintenance			
	Introduction	( )	( )	( )
	Scheduled Maintenance Action Index	( )	( )	( )
	Preventive Maintenance Procedures	( )	( )	( )

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FIGURE 36 Certification check-off list (sheet 2 of 5)

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	YES	NO	NOT APPL
Scheduled Performance Test	( )	( )	( )
Plan for Maintenance of Mechnical Components	( )	( )	( )
(f) Section 5. Installation			
Installation Drawings	( )	( )	( )
Site Information	( )	( )	( )
Reference Publications	( )	( )	( )
Tools and Materials Required of Installation	( )	( )	( )
Unpacking, depreservation, represervation, and repacking	( )	( )	( )
Preparation of foundations	( )	( )	( )
Input Requirements	( )	( )	( )
Installation Procedures	( )	( )	( )
Installation checkout	( )	( )	( )
(g) Section 6. Parts List			
Introduction	( )	( )	( )
List of Major Units and Components	( )	( )	( )
Parts List	( )	( )	( )
List of Common Item Description	( )	( )	( )
List of Attaching Hardware	( )	( )	( )
List of Manufacturers	( )	( )	( )

3. The following is a detailed list of the subdivisions of Volume 2:

(a) Front Matter		
Cover/Title Page	( )	( )
List of Effective Pages	( )	( )
Functional Index	( )	( )
Foreword	( )	( )

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FIGURE 36 Certification check-off list (sheet 3 of 5)

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		YES	NO	NOT APPL
	Description of symbols, codes, abbreviations and shading	( )	( )	
	Safety Summary	( )	( )	
	Integrated Circuit Data	( )	( )	( )
	Explanation of MDC Usage	( )	( )	
	Family Tree	( )	( )	
(b)	Overall Function Information			
	Interconnecting cabling/Piping Diagram	( )	( )	( )
	Turn-On/Check-out Chart	( )	( )	
	Overall Function Diagram	( )	( )	
	Text	( )	( )	
	Overall Maintenance Data	( )	( )	( )
	MDC	( )	( )	( )
	Fault Logic Diagram	( )	( )	( )
	Equipment Alignment	( )	( )	( )
(c)	Major Function Information			
	Schematic Diagram	( )	( )	( )
	Text	( )	( )	( )
	Maintenance Data	( )	( )	( )
(d)	Hardware Information			
	Schematic Diagram	( )	( )	( )
	Text	( )	( )	( )
	Maintenance Data	( )	( )	( )
	Assembly Repair Data	( )	( )	( )
	Repair Illustration	( )	( )	( )
	Adjustment Procedure	( )	( )	( )
	Parts Data	( )	( )	( )
	Wiring Data	( )	( )	( )

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FIGURE 36 Certification check-off list (sheet 4 of 5)

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We certify that the Manuscript conforms to the statements as checked above, and meets the requirements of MIL-M-24100 and/or Contract No._____

_____ DATE_____
Lead Writer

_____ DATE_____
Quality Supervisor

_____ DATE_____
Manager, Publications Department

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FIGURE 36 Certification check-off list (sheet 5 of 5)

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INDEX

	PARAGRAPH	PAGE
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Acquisition requirements	6.2	105
Active system tests	3.2.3.1.7.6	61
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Alignment and repair	3.2.2.2.3.3.4	51
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